

THE ARCHITECTURE OF THE ENGLISH LEXICON

A Dissertation

Presented to the Faculty of the Graduate School

of Cornell University

in Partial Fulfillment of the Requirements for the Degree of

Doctor of Philosophy

by

Jonathan B. Alcántara

May 1998

© 1998 Jonathan B. Alcántara
All Rights Reserved

THE ARCHITECTURE OF THE ENGLISH LEXICON

Jonathan B. Alcántara, Ph.D.

Cornell University 1998

This study provides an analysis of stress assignment and vowel alternation in the Latinate vocabulary of English, using the mechanisms of Optimality Theory to express the entire phonology, including lexical information, through the medium of the constraint hierarchy.

Generative solutions depend crucially upon the underlying lexical forms which feed the grammar. A computational study of the English lexicon has been undertaken, identifying the distribution of stress, syllable weight and vowel length throughout the English lexicon. Forms have been classified on morphological and prosodic grounds, and comparative pattern frequencies have been calculated. This reveals a series of majority and minority patterns, each with significant distributions, rather than a large set of regular forms with a residue of exceptions.

The majority patterns are enforced by prosodic constraints, while the minority stress patterns can also be accounted for, without recourse to unstructured exception marking, by proposing additional structure for their underlying forms. Such structure is restricted to constituents already found in the grammar. The distribution of vowel quantity in the data suggests that long vowels seen in alternating forms should be understood as resulting from a morphologically conditioned vowel lengthening. Reassessing these forms as underlyingly short accounts for both vowel alternation and certain otherwise inexplicable stress retractions. Previously exceptional forms, which failed to “shorten”, are hence understandable as underlyingly long. Stress in morphologically complex forms is treated via subcategorization constraints, and all stress and vowel alternation patterns are accounted for with a single constraint hierarchy.

To account for the now small residue of exceptional forms, such as suppletive stems or irregular allomorphs, a new conception of the lexicon within OT is proposed. Morpheme selection is understood to be governed by

lexical selection constraints. While “regular” forms result from the interaction of generally applicable constraints, exceptions are enforced by high-ranking specific constraints, which take complex structures as arguments. The contrast between general and specific constraints can be used to explain competing trends within the lexicon, such as type vs. token frequency. This proposal moves the entire grammar into the realm of the constraint hierarchy, and allows for a principled evaluation metric.

BIOGRAPHICAL SKETCH

Jonathan B. Alcántara was born in New York City in 1965. He received his undergraduate degree in History from Cornell University in 1987, where he specialized in Ancient History. Having gotten a taste of historical linguistics, he decided to undertake graduate studies in linguistics, returning to Cornell University in 1989, where he began to specialize in Phonology. After receiving a Jacob Javits Fellowship in 1990, he took the opportunity to study abroad, doing a course at the Centre for Cognitive Science at the University of Edinburgh in 1991-92 and joining a working group in Computational Linguistics at the Universität Freiburg in 1992-3. He fulfilled the requirements for a Masters in Linguistics from Cornell during 1993. He finished his doctoral dissertation in 1997, two years after entering into full-time employment in the language technology field. He is currently employed by Entropic Cambridge Research Laboratory in Cambridge, England, a speech recognition software company.

To my whole family

ACKNOWLEDGEMENTS

I am extremely grateful to the members of my committee, not only for the guidance and encouragement they have given me throughout my work on this topic, but especially for the patience they had with a doctoral research student who was for the most part out of the country. I need to especially thank my chair, Abby Cohn, not only for all the email, comments, help and feedback, but also for keeping my dissertation and candidacy on track by taking care of the many administrative things that I couldn't do for myself. Her patient advice and guidance were invaluable, and this thesis would not exist without Abby's perseverance and help. Many thanks also go to Draga Zec, who gave me much useful advice, and always had time to discuss the core issues of my thesis. I need to thank Linda Waugh for starting me on this topic, and I will always be grateful to Lin for how she taught me to look at every side of a problem, and not to accept even long-established assumptions on faith. Special thanks go to Nick Clements, who has remained on my committee all these years despite having moved on from Cornell. Nick was always there to provide feedback, giving me many suggestions to improve my analysis, and he even was present at my exam (by speakerphone) during his holidays. Nick was my first phonology teacher at Cornell, and it was he who got me really hooked. I continue to be grateful for his advice and help.

I would also like to thank the rest of the excellent faculty with whom I had the pleasure of studying under at Cornell, especially Jay Jasanoff, who gave me the Linguistics bug in the first place, Sue Hertz, who introduced me to the world of language technology, and Allard Jongman, who turned me on to the Celex database, my primary source of data. Special thanks also go to Sheila Haddad and Angie Tinti, our administrators without equal, who saved my bureaucratic skin many a time.

During a crucial phase of my education the US Congress and the Department of Education supported me with a four-year Jacob Javits Fellowship, which enabled me to travel abroad. I am also grateful for the Cornell graduate fellowship which supported me during my first year of graduate school. Thanks are also due to the hospitable folks who took me in while I studied abroad. The Cognitive Science Program at the University of Edinburgh introduced me into my first serious computational environment. I'm grateful to Robert Dale, who taught me Prolog, Steven Bird and Mark

Ellison, who invited me to phonology seminars and gave me some insight into the interesting constraint-based work they were doing, and Ewan Klein, for giving me access to some English databases.

I must thank and acknowledge my gracious host at the Universität Freiburg, Prof. Dr. Udo Hahn, who made me feel at home in his computational linguistics group. I will always look back fondly at the friendly group in Freiburg. I should also thank Dr. Lilo Mössner for giving me access to various English corpora. Back in the US, I was also fortunate to enjoy the hospitality of Len Schubert and James Allen at the University of Rochester , who let me sit in on their computational project. I should also thank Prof. Steve Young of the University of Cambridge and Entropic Ltd. for giving me access to English pronouncing dictionaries, as well as for bringing me into the language technology business.

I'd like to thank all the friendly and helpful colleagues with whom I have worked at Cornell and elsewhere, but especially my friend Kevin Connelly, whom I met in my first classes as a Cornell graduate. Kevin was kind enough to similarly extend his studies, and as a result I would always have a friendly face to see upon my visits to Ithaca. I also want to thank Michael Bernstein and Niken Adisasmito-Smith for helping me test my research software. Eric Evans was amazingly helpful, downloading and printing out drafts for my committee, installing and checking fonts and generally making sure that I could continue to work remotely without problems. And I appreciate Rob Podesva doing all the final editing and "legwork" on my behalf.

My greatest thanks go of course to my family, whose love, patience, encouragement and assistance were vital to me as I endeavored to complete what I had begun. And my everlasting gratitude belongs to my wonderful wife, Michaela, whose fortitude, helpfulness and understanding were essential and who kept me going during stressful and difficult times. Yes, this means it's really finished!

TABLE OF CONTENTS

	<u>Page</u>
BIOGRAPHICAL SKETCH.....	iii
DEDICATION	iv
ACKNOWLEDGEMENTS.....	v
Chapter	
1. INTRODUCTION	1
2. ALTERNATIONS IN ENGLISH.....	48
3. ENGLISH STRESS PATTERNS.....	80
4. THE ENGLISH CONSTRAINT HIERARCHY AND STRESS.....	119
5. LENGTHENING AND CONSTITUENCY.....	157
6. STRESS RETRACTION AND CONSTITUENCY.....	205
7. THE ARCHITECTURE OF THE LEXICON.....	246
APPENDIX - Stress patterns in suffixed words.....	300
BIBLIOGRAPHY.....	346

1. Introduction

1.0 Goals of the study

This study will address the role of the lexicon within the phonology and morphology of English. As the repository for idiosyncratic information within generative linguistic theory, the lexicon is a powerful theoretical component, but has tended to lack sufficient formalization. The absence of an agreed upon evaluation metric for lexical complexity has made it difficult to determine when particular analyses rely too strongly on the lexical component to yield the proper forms. Here, the analysis of English phonology will be expressed in terms of the principles of **Optimality Theory** (Prince & Smolensky 1993, McCarthy & Prince 1993a, b). While built on a generative base, Optimality Theory offers a methodology which, by restricting the set of theoretical concepts which may be used to account for the relationships between forms, has the potential to be more explicit and more constrained than any previous linguistic theory. The purely structural notation of Optimality Theory allows for both a cogent evaluation metric and a formal representation of the lexicon in the same terms as other grammatical components, ending the formally problematic conception of the grammar as structurally heterogeneous.

This study continues a series of theoretical threads which trace their beginnings to the pioneering work of Chomsky and his concept of **generative grammar**. A generative grammar attempts to account for an observed system (in this case spoken language) through processes which change some internal symbolic representation of information into the observed “surface” expression. Describing the phonology of a language in traditional generative terms, underlying or **input** forms consisting of segment strings are processed by rules which change them into phonetic **output**. This concept continues into OT, where “input” lexical entries are passed to the processes (*Gen*, *Eval*) which eventually determine the optimal “output” candidate. Since generative systems rely crucially on the identity of the input form to yield the correct output as well as to determine the processes which properly describe the data found in a language, it is

important that the method of arriving at these input forms be constrained and consistent.

One method of expressing this is the idea, common to both OT and other generative approaches, that surface forms should correlate as closely as possible to underlying forms ascribed to the lexicon, which is codified in OT in the **Faithfulness** family of constraints (§ 1.4.3). However, derivational approaches to English have never been able to present a concrete evaluation metric for measuring complexity either in the lexicon or of the rules. To consistently relate lexical and surface forms, these theories have always had to resort to diacritical exception marking in the lexicon to account not only for isolated cases but for entire classes of forms inconsistent with general formulations of the grammar. This kind of unstructured exception marking, which in effect allows for completely unconstrained analyses, can render any theory inadequate and should be made formally inexpressible in Optimality Theory. One way to achieve this is to limit both constraints and lexical items so that only formal structural constituents may play any role in their formulation. If this principle is followed, lexical items may only be marked with structures that form part of the constituent inventory, which includes only a small set of morphological and phonological constituents.

Similarly, the constraints which account for the general phonological patterns seen in a language may only refer to such constituents. Another goal of a constrained Optimality Theory would be to capture the entire grammar of a language through the use of a single constraint hierarchy which candidates are evaluated against, rather than requiring multiple hierarchies or “cophonologies” (§ 1.4.3) to explain all surface forms. However, although regularity is a feature of many relationships seen between English words, there are sets of “irregular” words which contravene the general patterns, and these must also be represented in a principled way, preferably using the same theoretical mechanisms. Ideally, the set of “input” lexical items will be able to be evaluated by a single constraint hierarchy, yielding up the correct “output” surface forms, the more general patterns being represented by the constraints which enforce the constituent structures underlying these patterns, and the idiosyncrasies of the “irregular” words being represented via structures in the lexicon.

The primary phonological focus of this study will be the stress patterns of English, with special attention given to problems of vowel alternation. A

computational study of a large lexical database of English words has been performed in order to provide a corpus of data from which to draw observations (some results of this investigation are presented in the Appendix, while others appear in the following chapters). Phonological effects can be inferred based upon alternations seen in this corpus, for example stress contrasts or vowel alternations found between “related” words. The following study of the English data does not in fact reveal general rule-based patterns and a residue of exceptions (as predicted by the usual rule-based derivational approach), but rather majority and minority patterns which must both be expressed and explained via the constraint hierarchy. Such a finding speaks further against analyses which regard all structures which do not conform to the most general patterns as “exceptions” to otherwise absolute “rules”.

Since constraint hierarchy evaluations in OT are highly dependent, as was noted above, upon their “input” lexical forms, a principled way of determining the structure of these underlying forms must be arrived at before any analysis can be proposed. Determining which parts of the data are relevant for the understanding of general patterns in English phonology, and which cases are simply idiosyncratic, is a central task for such an investigation. For the alternation known as **vowel shortening**, the environments and examples set forth by Myers (1987), which are largely drawn from preceding classifications by Chomsky & Halle (1968), will be contrasted with evidence from the investigation of the English lexicon described above (chapter 2). Stress pattern groupings will be approached via the framework of Kager (1989), who has distilled a series of previous approaches into a comprehensive summary of the Lexical Phonological approach towards English stress. The statistics and data drawn from the corpus of English data will be compared to the generalizations contained in these works (chapter 3). Based upon statistical generalizations in the corpus, suitable lexical forms will be proposed and an appropriate constraint hierarchy will be formulated, accounting for both majority and minority patterns using only structurally driven mechanisms (chapters 4-6).

Finally, in chapter 7, the role of the lexicon within Optimality Theory will be re-evaluated. An existing OT constraint type, the subcategorization constraint (McCarthy & Prince 1993a) will be expanded in its application to account for both morphological selection within related word sets and

allomorphy. Subcategorization constraints, which can refer to specific affixes, are interpreted as an instantiation of what will be termed **specific constraints**, which can take specified structures as their arguments. These have a place in the constraint hierarchy in addition to the **general constraints** normally encountered in OT analyses, which refer only to constituent categories. Other kinds of specific constraints can have properties which would be traditionally referred to as lexical. The distinction between specific and general constraints will in turn be related to Bybee's (1995) contrast of type frequency and token frequency within the grammar. Her conception of the grammar as a set of networked lexical nodes will be reinterpreted as formally equivalent to an OT constraint hierarchy, which leads directly to the conception of the lexicon as a set of specific OT constraints, an idea first proposed by Russell (1995). The proposal of the entire grammar as an OT constraint hierarchy unifies and constrains a previously heterogeneous theoretical system and allows for both a consistent representation of all grammatical structures as well as an explanation, in the recognition of specific and general constraints as clear subtypes, for the typological split between the rules and the lexicon found in previous approaches.

In the remainder of this chapter, some theoretical background for the following chapters will be sketched out. The basic concepts underlying Optimality Theory will be discussed, as well as its relation to the derivational theories which preceded it. The structures referred to in OT constraints, such as the constituents of the prosodic and morphological hierarchies will be introduced and issues of relationships between words within the grammar will be discussed. Finally, some discussion of the issues relevant to previous studies will be undertaken and the general conclusions about the grammar and lexicon arrived at in the following chapters will be outlined.

1.1 Alternation in derivational theories

Linguistic data exists pre-theoretically, and all explanatory theories of language are ultimately grounded in this data. The corpus used in this study (Baayen, Piepenbrock & van Rijn 1993, see § 2.3) presents tens of thousands of words, recorded as transcriptions of pronunciations, and marked both for stress and melody. If the phonology of English were completely transparent, the theoretical lexicon could simply consist of lists of words based on various

stress and melodic patterns. This has been the assumption behind certain structuralist studies, and works such as Vennemann (1974), Mössner (1978). However, generative studies of English since Chomsky & Halle (1968) have used the various patterns and correspondences in structure seen across the English data to propose a **derivational** model, which uses **rules** to convert one set of structures into another, usually conceived of in terms of **input** and **output**. While Chomsky regards such relationships as bi-directional in theory, in practice most studies have been concerned with changing postulated **underlying** structures into the attested **surface** forms.

This generative approach has the advantage of aiming for descriptive adequacy, accounting for the correspondences and patterns in the data, rather than simply being observationally adequate. From this perspective, the underlying, unseen structures take on **basic** status, while the concrete evidence of surface forms becomes regarded as **derived**. A theoretical principle of **economy** operates over the system, which endeavors to express the necessary rules in the “simplest” form required to produce the correct, attested output (where “simplest” suggests the maximal reduction of terms). Thus any data used for a study of this kind can only provide the surface forms, the “output” of a constraint evaluation or derivational process, and the underlying basic forms belonging to the lexicon must be inferred from the larger grammatical system, in this case, from the phonology and morphology.¹

Utilizing a corpus of data to determine the structure of the lexicon presents a series of difficulties, because neither the lexicon, the phonology nor the morphology have any independent existence, but are all aspects of a relation between sound and meaning which stretches across the “branches” of linguistics, from semantics to phonetics, each presenting mutual dependencies. These dependencies require each subsystem to be defined in terms of other subsystems, which can lead to circular assumptions. To an especially great extent, those subsystems defined as “phonology”, “morphology” and “the lexicon” are interdependent, and competing

¹“Phonology” and “morphology” are usually applied to the range of structures found between the word level and the phonetic level of linguistic analysis, although phonological effects can also be seen at the syntactic level in English, and other languages may differently align phonological constituents to syntactic units, revealing some of the usual terminology, such as “word”, “phrase” and “morpheme”, to be potentially confusing when applied universally.

linguistic theories on these topics tend to simply reapportion aspects of explanatory power between these three conceptual labels.

Chapter two explores in this context the specific issue of vowel alternation in English. The concept of alternation itself carries with it a series of assumptions, for example, that the items being regarded as “alternating” share some common element. Structurally different words can be impressionistically identified as “related” to each other due to similarities in form and meaning which recur throughout the language. Phonological alternation can be identified when structures identifiable as “the same”, due to parallel structure and meaning, are nevertheless not identical, as in the pair *sáne ~ sánity*. When similar alternations can be seen across a range of items in the data (e.g., *váin ~ vánity*, *gráve ~ gráivity*, *cáve ~ cáivity*), under similar conditions, these may be described as patterns and can be conceptualized as resulting from the application of a general, conditioned process to some “basic” lexical entry. The existence of such patterns led Chomsky & Halle (1968) to propose phonological **rules** to account for such alternations. These rules would change the basic form of a lexical item consistently in a given **environment**.

The identification of phonologically contrastive forms as instantiations of “the same” word or morpheme can lead to phonological abstractions that are quite removed from the surface acoustics. For example, the words *electric* and *electricity* are intuitively perceived as closely related in form and meaning, and parallel a host of other pairs like *elastic ~ elasticity*, *authentic ~ authenticity*. These forms present an alternation between [k] and [s] in the suffix /-ic/, usually characterized as palatalization or coronalization. It is very difficult to formulate such a palatalization rule using Chomsky & Halle’s distinctive feature system, yet the correspondence appears to be a real part of the grammar. Another striking and equally problematic correspondence, which will be one focus of the following investigation, is the so-called **Great Vowel Shift** of English. The following words illustrate the fact that words which are clearly related in form and meaning show, in various environments, an alternation between pairs of syllable nuclei which are acoustically quite different from one another, and which represent distinctive segments in other contexts:

(1.1) <u>Surface contrast</u>	<u>Abstract phonology</u>
[aɪ] - [ɪ]: divine / divinity	/ ī / - / i /
[iː] - [ɛ]: deep / depth	/ ē / - / e /
[eɪ] - [æ]: nation / national	/ ā / - / a /
[oʊ] - [a]: cone / conic	/ ō / - / o /

Chomsky & Halle (1968: 240-41) propose a series of **laxing** rules to account for these alternations, presenting the pairs of vowels as **tense** and **lax** versions of “the same” vowel. Acoustically, the members of these vowel pairs are quite different, the tense versions surfacing as diphthongs, with no segments and only a few if any features in common with their lax counterparts.

Accordingly, any rules used to account for this alternation must be quite complex in form and far from intuitive.

Subsequent studies, such as Halle (1977), have characterized this alternation in more historically accurate terms, on the basis of a vowel quantity difference rather than a tenseness/laxness distinction. Thus, as represented in (1.1) above, as well as in the orthography, such word pairs may be conceived of as containing identical sets of segments, contrasting solely in terms of abstract phonological vowel quantity. The fact that the alternants each have a distinctive status suggest that vowel length is yet another feature that participates in the phonological system of English, and rules proposed to account for this alternation assume underlying long vowels which shorten in certain environments (see chapter 2). This tactic yields a relatively simple phonological rule which changes just one feature, and allows for the positing of a single “basic” underlying root which accounts for both alternant types.

Chapter two focuses on the work of Myers (1987), currently regarded as the standard derivational account of English vowel alternation or “shortening”. Myers identifies a variety of morphologically conditioned “shortening” environments, and attempts to account for them phonologically by generalizing a process of “Closed Syllable Shortening” (§ 2.2.3). His explanation is problematic not only because it entails the proposal of a number of unlikely phonological processes, but also because it fails to jibe with the English data itself. Many of the “shortening” environments he defines are not general, while the rules he posits would in fact overapply beyond these specified shortening environments. There is also

no way, besides exception marking, to account for words which fail to undergo expected shortenings under his system.

Underlying this general problem is Myers' assumption, taken over from earlier studies, that the process he is trying to explain is one of vowel shortening in certain environments. Other approaches, utilizing prosodic categories such as the foot (Halle & Vergnaud 1987, Prince 1990, Burzio 1993) and Optimality Theoretic analyses (Prince & Smolensky 1993: 210-11) still maintain this assumption without question. The corpus investigation undertaken here suggests that while some (but not all) of Myers' environments tend to show short vowel alternants in corresponding stems, there is always a strong minority group which fails to conform to any "shortening" rule, whatever its formulation. Such minority groups ought to be formally and inclusively accounted for in a robust phonological theory, rather than being relegated to exception status, and the violable, ranked constraints used in Optimality Theory offer a promise of resolution for this problem unavailable in derivational rule-based theories. Furthermore, a consistent account of the alternation phenomena needs to be arrived at. Since alternation appears to be tied to a prosodic category, the **foot**, a necessary treatment of English stress and the prosodic structures which produce stress effects will be undertaken in chapter three. In the following sections, some background issues relevant to this treatment are discussed further.

1.2 Stress and prosodic constituents

The investigation of English stress in chapter three is built around the study of Kager (1989), who uses standard derivational accounts of English stress assignment to construct a system of groupings designed to categorize English words based on their stress patterns. Kager's three principle groups differ in terms of the type of extrametricality effects seen at the right edge of the word: his first (and largest) group shows final syllable extrametricality (e.g., *órigín*, *appéndix*), his second group final consonant extrametricality (e.g., *devélop*, *expéct*), and his third, "idiosyncratic" group, no extrametricality whatsoever (e.g., *guitár*, *acquiésce*). He also tries to broadly connect these groups with syntactic categories (although there are clearly many exceptions to this generalization), linking the first group to nouns and the second to verbs and adjectives. Kager also deals extensively with the issue of "retraction", which is based on the derivational assumption that stress in

complex words is laid down iteratively as word formation progresses, and stresses can be assigned and removed during this process.

Kager's decisions about identifying extrametricality and retraction stem from both the observed positions of main stress in the words as well as assumptions about foot-formation and derivation. His analysis conforms generally to the view that English displays the bimoraic trochaic foot (Hayes 1995), and the data in chapter three is presented in terms of such feet. Since the foot appears to play a role in both stress assignment, vowel alternation and many other phonological processes in English, it is relevant to briefly discuss the role of the foot and other members of the prosodic hierarchy in phonological theory. These prosodic constituents figure centrally in accounts of stress in both derivational theories and in OT, where they can act as arguments to various constraints.

1.2.1 The prosodic hierarchy and word-level stress

The prosodic hierarchy was originally developed to account for a wide range of phonological effects at the phrasal level (Selkirk 1980, Nespor & Vogel 1982). It was then extended to account for sub-word structure (Inkelas 1989, Cohn 1989), where it helped explain aspects of natural language such as stress which were difficult to express in terms of the purely segmental rules of Chomsky & Halle (1968). Prosodic constituents such as the prosodic word, foot, syllable and mora play critical roles in the various accounts of English stress discussed herein, both within derivational and Optimality theories.

In Chomsky & Halle's system segments were the only available structural components and thus all features, including stress, were formally part of a segment's feature inventory. Stress, however, was unusual in that it was the only non-binary feature, and furthermore it did not alter the identity of a phoneme when added to its feature matrix, in the way that other features like [voice] or [coronal] would. Stress also showed long-distance effects, the entire stress pattern of a word changing when certain suffixes were added. Stress rules were disjunctive and contained variables, which were not necessary for conventional feature-adjusting rules (Lieberman & Prince 1977: 262).

The introduction of hierarchically organized tree structures and grids to account for stress acknowledged that "certain features of prosodic systems are not properties of segments (or syllables), but rather reflect a hierarchical

rhythmic structuring that organizes the syllables, words, and syntactic phrases of a sentence.” (Lieberman & Prince 1977: 249) Following the crucial insight of Chomsky & Halle (1968) that heavy syllables tend to be stressed in English, syllable nodes were labeled “strong” if they “branched”, i.e., if they contained a long vowel or were closed by a consonant (the classical metrical definition of “heavy” syllable weight). Thus, the same insight that had heretofore been expressed segmentally (by a disjunctive environment: VC or V[+tense]) could be now formulated more elegantly in terms of branching in tree structure.

This approach was successful in eliminating the need for the complex stress rule system of Chomsky & Halle (1968). The n-ary feature [stress] was replaced by the binary feature [$\pm$ stress], needed only to explain surface contrasts in unstressed vowels (e.g., *ábsolute* vs. *éxcellent*), while the role of disjunctive ordering in the application of the stress rule was tied to morphological structure. Lieberman & Prince’s (1977: 274-6) division of English words into (primarily) morphologically-based categories for “stress retraction” (following application of their universal English Stress Rule) is maintained by Kager (1989) (§ 3.1.4). Membership in a particular retraction mode determined how foot-formation would take place, maintaining the link between stress assignment and extra-phonological information such as morphological structure and syntactic category contained within the analysis of Chomsky & Halle (1968).

The foot was introduced into the prosodic hierarchy as a formal prosodic constituent by Selkirk (1980), allowing for the successful jettisoning of the feature [stress], explaining all levels of stress via prominence relationships. Unstressed syllables were understood to be the weak members of feet, while full unstressed vowels were members of weak feet. Stressed syllables were the strong members of strong feet, and metrically heavy syllables could comprise feet by themselves. Primary stress was defined as a property of the **prosodic word**, which dominated the feet.

Selkirk’s ternary feet, used to explain the stress of words like *éxcellent*, in which two syllables follow the main stress, were eliminated (at least for English²) by the proposal of **extrametricality** by Hayes (1981, 1982). This mechanism, which followed from the observation that ternary feet typically

²See Hayes (1995), chapter 8, however, for a discussion of languages whose apparently ternary feet cannot be explained by extrametricality alone.

occurred at word-edges, allows for the elimination of peripheral phonological material from consideration by prosodic rules. This restricted the foot inventory to two types, binary and unbounded feet. The limited inventory of bounded foot types could be further distinguished according to specific parameters (Halle & Vergnaud 1978, 1987, Hayes 1995). These included various qualities which differed cross-linguistically across a limited range, such as the question of which syllable was dominant within the foot (i.e., an iambic or trochaic pattern) or whether the criteria for creating the feet from the underlying representation was **quantity-sensitive**. Quantity-sensitive systems, such as that of English, assign feet on the basis of syllable weight, heavy monosyllables forming feet of their own.³

Hayes (1985, 1987, 1995) further formalized the parameters for mapping the lexical entry onto a prosodic system, which included directionality, iterativity and persistence of parsing. His subsequent cross-linguistic studies (Hayes 1995) show that Hayes' vision of the foot appears to be applicable over a wide range of languages. However, his use of extrametricality in deriving English stress has a large morphological component: many suffixes are marked as extrametrical, and extrametricality appears as a categorial feature of nouns but not verbs. Other rules, such as "Late Extrametricality", serve to mark the final syllables of unsuffixed adjectives, when "preceded by a branching foot" (Hayes 1987: 273) as extrametrical (e.g., *difficult*), effectively stating that adjectives with more than two syllables rather behave like nouns.

While Liberman & Prince's (1977) concept of strong-weak branching constituents led to foot-based theories such as that of Hayes, their alternate representation of stress as a metrical grid led to the development of another approach, the rhythm-based framework used in Prince (1983), Selkirk (1984), and Nespor & Vogel (1986). These attempted to discard foot-like constituency and represent stress solely with a prominence grid. Rather than labeling members of a prosodic constituent as "weak" or "strong", Prince (1983) lays down grid marks which align to **moras**. These units of weight correspond to the presence of a heavy syllable (a long vowel or closed syllable) in other approaches, and so follow the insight previously captured by Chomsky & Halle's stress assignment rule and Liberman & Prince's "branching nodes."

³Although what counts as "heavy" may vary cross-linguistically. See Hayes (1995: 50-51), Zec (1988, 1994).

The latter's insight that the grammar attempts to place stresses in a rhythmically alternating manner is represented by Prince's rule of "perfect gridding", which iteratively assigns grid marks at a binary distance, modeling alternation.

The same principle can lead to **destressing in clash**, one instantiation of which was called "the rhythm rule" in Liberman & Prince (1977), which eliminates adjacent grid marks; i.e., when stresses fall too near to one another, one may be removed. Thus, syllables which we would expect to have a certain degree of stress fail to show it, due to the adjacency of another, stronger stress:

(1.2) thirtéén ~ thirteen mén párent paréntal

In Prince's model, main stress was assigned by the **End Rule**, which added a superior grid level that encompassed the entire word, and he maintained Hayes' principle of extrametricality to produce the proper environments for grid-mark assignment.

While this "pure grid" approach, which purposely eliminated constituency, was continued by Selkirk (1984), Hayes (1985) demonstrated that metrical constituency was crucial in accounting for phrasal rhythmic phenomena such as stress shift and rhythmic strengthening. Hammond (1984) proposed a model which placed the grid marks in a tree structure to represent constituency; this was further developed by Halle & Vergnaud (1987) who representationally conflated grids and feet into "bracketed grids". This formalism was later taken over into Hayes (1987), and his subsequent work. Thus, Halle & Vergnaud (1987) could represent both constituency and the hierarchical levels of grid marks indicating prominence, using a single representational schema.

To account for English stress, Halle and Vergnaud (1987: 277) assigned grid marks to syllables with a "branching rime", thus again capturing the insight that heavy syllables tend to be stressed. In the cases of some words which have unexpectedly stressed light syllables, such as *medúlla*, *cèrebéllum*, they proposed lexical grid marks. Their "Alternator" rule performed the role of Prince's "perfect gridding", modeling rhythmic alternation. Marks assigned at the first level of the grid acted as "heads" to the constituents which are thus defined. Halle & Vergnaud also relied heavily on

extrametricality to eliminate final syllables from consideration, in the case of nouns and many affixed words. Further benefits of a grid-like analysis, such as destressing in clash and other rhythmic adjustments, were retained, but constrained by the use of constituency, thus incorporating the advantages of both the foot/tree and grid approaches.

1.2.2 Moras and minimality constraints

Hayes (1987) adopted the bracketed grids formalism of Hammond and Halle & Vergnaud, but also incorporated Prince's use of moras, reinterpreting foot-assignment as the rhythmic grouping of bisyllabic (in quantity-insensitive languages) or bimoraic (in quantity-sensitive languages) units. In the latter type, heavy syllables were thus interpretable as feet in themselves, monosyllabic but bimoraic, whereas in Halle & Vergnaud's formalism, they were assigned a grid mark simply on the basis of their branching rime. This placed the mora into the prosodic hierarchy, mediating between the syllable and the segments supplied by the lexicon and the morphology.

More evidence for constituency and a moraic analysis came out of the arguments for **moraic conservation** presented in Hayes (1989) and McCarthy and Prince (1986, 1988, 1990ab). Hayes demonstrated that moras, rather than timing tier slots,⁴ could representationally best account for the process known as **compensatory lengthening**. In many languages, moras whose melodic counterparts are deleted in phonological processes are unaffected by this deletion, but rather link to an adjacent melodic unit, which surfaces lengthened, conserving the moraic count within the word. Hayes (1995) gives further evidence for prosodic constituency, such as bounded stress migration, the retention of stress within the foot which occurs when stressed vowels are deleted during morphological processes.

McCarthy and Prince (1986, 1990ab) have shown that many reduplicative and templatic processes of affixation are only explicable through the use of prosodic constituents such as the syllable, foot and prosodic word. Furthermore, their observation of various lengthening processes in the context of such constituents led them to propose **minimality constraints**. They propose minimal word, foot and syllable sizes for languages like Arabic, Mokilese and Ponapean to account for a number of morphological processes

⁴That is, skeletal C's and V's (cf. Clements & Keyser 1983) or X's (cf. Levin 1985).

cross-linguistically. Such constituents are then required by the grammar to contain a certain number of moras (usually two) or risk being ill-formed. The grammar consequently lengthens vowels or consonants inside those constituents when appropriate to achieve the necessary bimoraicism.

For example, as discussed by McCarthy & Prince (1990b: 21), in Modern Standard Arabic, vowels in monomoraic loan words from English are lengthened to achieve the bimoraic limit, e.g., *baar* 'bar', *jaaz* 'jazz', *gaaz* 'gas'. A Saudi Bedouin dialect geminates final consonants in the same situation, e.g., *baṣṣ* 'bus', *natt* 'nut', *rigg* 'rig'. In a different type of case, the reduplicative prefix in Mokilese is obligatorily a bimoraic syllable, regardless of the moraicism of the initial root syllable. Thus, light roots show a lengthened vowel in the reduplicative prefix; *wad-wadek* 'read', *pok-poki* 'beat', and *caa-caak* 'bend' might appear to simply repeat the form of their initial syllables, but *paa-pa* 'weave' and *wii-wia* 'do' show the lengthening required by the minimality constraint (McCarthy & Prince 1986: 20-1). Minimality is a good example of a constraint used within derivational theory, and its underlying concept is maintained in OT constraints such as **FtBin** ("Foot Binarity"), which will play a significant role in accounting for the lengthening processes described in chapter four.

Hayes (1987, 1995) incorporated the concept of quantitative minimality to his foot typology. The various types of binary feet are constrained to consist of two sub-foot units, either syllables or moras. For example, the moraic trochee, the foot-type found in English, is required to be bimoraic. Hayes' typology of bounded feet⁵ is as follows (1995: 86):

- | | |
|--------------------------|------------------------|
| (1.3) Syllabic Trochees: | (ś s) |
| Moraic Trochees: | (ḿ m) = (ś ś) or (ś́) |
| Iambs: | (m ḿ) = (ś ś) or (ś́) |

Hayes bases his principles of foot typology both on the psychology of the perception of iambic versus trochaic rhythms (Hayes 1985) and on the

⁵Hayes' theory also covers unbounded feet, which actually appear to be instances of stress being applied to one end of a word. Hayes regards these as feet which may be of any length, and in many cases they appear to be simply the application of an End Rule (§ 1.2.1), or word-level stress assignment, without foot-level constituency. As they are not relevant to English, they will not be discussed further here.

distribution of the various foot-types among the world's languages (Hayes 1995). Hayes (1995) excludes from his inventory two types of feet allowed in previous theories, such as Hayes (1981) and Halle & Vergnaud (1987), the “even iamb” (s ś) and the “uneven trochee” (ś ś), the latter of which is relevant for English. For cases where this foot appears to be necessitated by surface forms, such as in Latin *domesticus*, Hayes argues for parses like *do(mést)ic<us>* rather than *do(méstic)<us>*, the penult being left unparsed. Other theories, such as Hayes (1981) and Halle & Vergnaud (1987), required all non-extrametrical syllables to be parsed into foot structure.

1.2.3 Stress assignment as foot-formation

Using metrical units like feet reduces the problem of stress assignment to that of foot-formation. Following the principles of foot-formation, based on a few parameters, underlying forms are parsed into constituents, primary and secondary stresses surfacing on those syllables that fall into the relevant head positions in the prosodic structure. Thus, the properties of stress are completely inherent in the prosodic constituents, and removed from the segmental level of analysis. Using such a prosodic hierarchy of constituents to account for stress within the word allows not only for a principled model for the apparent long-distance relationships seen in stress effects, but also provides environments, in the prosodic constituents themselves, for other phonological processes to occur, such as the vowel alternation effects discussed in chapter two.

The survey of foot-formation types in chapter three (§ 3.2, § 3.3) demonstrates that Kager's (1989) ideas about the connections between syllable structure, syntactic category and foot-formation are simply rough generalizations which sometimes fail to hold over sizable groups of words. Evidence from sets of suffixed words (§ 3.3.1) suggests that Kager's extrametricality-based explanation of the contrast between his three major stress types is incorrect, and that instead, lexical structural marking on words of Kager's third group (which, in his terms, fail to show any extrametricality effects) provide a better explanation of their behavior. This is similar to Halle & Vergnaud's (1987) idea of lexical grid marks for words like *medúlla* (§ 1.2.1), but here these are interpreted here as lexical moras, i.e., underlying geminate consonants. The presence of this additional mora in the lexical “input” form, which has no other surface effects, accounts for the unexpected stress patterns

of these words, which display primary stress on apparently light final or penultimate syllables.

For Kager's second group, identified with verbs and adjectives, the apparent lack of final syllable extrametricality will be accounted for herein by the proposal of a verbal/adjectival suffix /-ə/, whose behavior is parallel to that of a class of Latinate suffixes including /-al/, /-or/, /-ous/. Identifying words of Kager's second group as underlyingly suffixed with a non-surfacing final syllable allows for the apparent lack of extrametricality in these forms. It also accounts for the parallel stem shapes of pairs like *propóse* ~ *propósal*, which can now both be understood as suffixed with metrically light monosyllables; these may be contrasted with the prosodic effects seen in other suffixed types (e.g., *resíde* ~ *résident*, *insáne* ~ *insáanity*). This is explained not by a blanket extrametricality restriction, but rather by proposing (§ 4.2.2) that the suffixes /-al/, /-ə/ subcategorize for attachment to morphological **stems** differently than suffixes such as, e.g., /-ity/, and this subcategorization has a concomitant effect on foot-formation. In chapter four, the relation between the foot and the stem will similarly be shown to have an effect on whether the stem vowel in words showing vowel alternation appears long or short in the output surface form.

1.3 Applying Optimality Theory

In chapter four, the observations and conclusions reached concerning the stress pattern of English in chapter three are interpreted using an Optimality Theoretic analysis. In this work, the OT approach is based upon Prince & Smolensky (1993), McCarthy & Prince (1993a, b, 1995) and subsequent work, but is constrained by a further theoretical goal: that the grammatical system should be limited, in the structures it can refer to, to members of the constituent hierarchies. This limitation is aimed at removing unstructured exception marking, special provisos, and other unconstrained extratheoretical mechanisms from a formal theory, forcing linguistic accounts to succeed or fail solely on the basis of whether the observed data can be accounted for using the formal structures and methods available to the theory. Subsets of data which cannot be accounted for by the theory should be understood as demanding improvement of the theory, rather than simply being labeled as exceptional.

While this study makes it abundantly clear that “exceptions” not only exist plentifully in the grammar of English, but in some cases embody sizable minority patterns, the theoretical interpretation of these minority groups needs to change. Rather than being seen as “exceptions” to absolute rules, these subsets can be understood as indicative of structural variations in the lexicon, and need to be accounted for using the same processes as the “regular” forms. This is possible using the novel approach found in Optimality Theory. In this work, OT will not simply be regarded as a “plug-in” replacement for the derivational cycle in phonology, but rather as the vehicle for providing an entirely new, unified understanding of how the grammar works. Many entrenched assumptions of previous analyses, such as the autonomous lexicon, the “basic” status of unaffixed forms, and the concept of “regular” words and “exceptions” are tied very closely to the mechanisms of derivational theory itself. OT provides a chance to re-evaluate these old assumptions and reinterpret the entire phonological system, from input to output form, using the rigorous principles of Optimality Theory.

1.3.1 Principles of Optimality Theory

There are very few theoretical mechanisms in Optimality theory, only constraints, candidates, and the ordering relationship captured by the hierarchical ranking of constraints. In terms of processes, a function termed **Gen** produces a set of all possible “output” candidates from the “input”, which is usually taken to be a lexical entry, while a function **Eval** evaluates this set in respect to the constraint hierarchy. The ranking of a constraint hierarchy defines a particular grammar, and the candidate which minimally violates the constraint hierarchy constitutes the “output”, a surface form (Prince & Smolensky 1993, McCarthy & Prince 1993a, b, 1995). As is conventional, Optimality analyses will be represented here as constraint tableaux, and will adhere to the notations used commonly in the OT literature.

McCarthy & Prince (1994: 3) offer a set of Optimality Theory principles, paraphrased here:

- (1.4) **Universality:** Universal Grammar provides a set of universal constraints present in all grammars.
- Violability:** Constraints are violable but violation is minimal.
- Ranking:** Constraints are ranked on a language-particular basis; this ranking defines minimal violation.
- Inclusiveness:** The constraint hierarchy evaluates a set of candidate analyses that are admitted by general considerations of structural well-formedness.
- Parallelism:** Best-satisfaction of the constraint hierarchy is computed simultaneously over the whole hierarchy and the whole candidate set.

These principles sketch out the theoretical bases of OT, and can be addressed individually.

Universality refers to the set of constraints which may be used for evaluation of the candidates. For OT to be effective as a theory, this set must be limited and evidence for such constraints must be seen cross-linguistically. In this study, the constraints used all appear in some form in previous OT studies.⁶

Violability refers to the fact that constraints are not absolute, but can be violated by surface forms. This violation is mediated by **ranking**; the constraints are arranged in a hierarchy, and the candidate which least violates this constraint hierarchy is considered to be optimal.

Inclusiveness refers to the generation of all possible candidate forms by a function *Gen*. This aspect of the theory is especially relevant to the issue of the lexicon, as the candidate forms are based on an “input” form usually taken to correspond to the underlying lexical form of derivational approaches.

Parallelism refers to the simultaneous evaluation of candidates by the constraint hierarchy, in contrast to the serial derivation process of Lexical Phonology, in which words are built up step by step. While not all OT analyses maintain parallelism, not only can parallelism account for all effects

⁶A brief discussion of the universality of constraints and suggestions for limiting and formalizing the acceptable types of constraints will be pursued in § 7.5.1.

in English usually ascribed to serial derivation, but theories based on serial derivation make incorrect predictions which parallel systems avoid. Simultaneous constraint evaluation provides a far more constrained and explanatory system than that of serial derivation, and parallelism will be maintained as necessary for a constrained Optimality Theory.

As a rigorous theoretical framework, Optimality Theory has the potential to produce more constrained and explanatory grammars than any competing linguistic approach. The complexity of natural language implies that a real-world grammar requires a very large (but finite) constraint hierarchy referring to all the relevant constituent structures seen in a given language. A single mis-ordering of constraints within a proposed constraint hierarchy will yield an incorrect grammar and make the wrong predictions for at least some words. Unlike the relatively risk-free addition of new, limited rules to a proposed grammar (in a derivational context) to account for difficult forms, adding new constraints to a system (or reordering constraints) for the same ends will also impact upon the evaluation of other words in the language. Only by presenting the precise constraint ranking which yields all the correct candidates will an OT analysis be a completely accurate representation of the grammar. Because of this precision, when such an ordering is arrived at, the arguments in favor of it being the correct characterization of the grammar are very strong. However, like any theoretical framework, OT is only useful when its structures and mechanisms can be clearly identified with and related to the data being evaluated.

1.3.2 Constraints, rules and parameters

OT relies on the constraint hierarchy to provide the effects ascribed in the derivational approach to rules. OT constraints present some similarities to the rules, parameters, and constraints familiar from derivational theory, as all of these concepts involve evaluating linguistic structures with regard to specified properties. As should be clear from Hayes' (1995) treatment of foot-formation, constraints had been used in previous rule-based approaches, but were never formalized in the way that rules were. In contrast to rules, they dictated what could not happen, limiting the effect of rule-based processes that would otherwise overgenerate, producing incorrect or non-intuitive results, or prohibiting certain structures which could otherwise provide

unwanted input or output representations. Constraints also motivated processes which otherwise had no impetus discernible from the rules.

One example of a constraint type used in derivational theory is McCarthy & Prince's (1986) **bimoraic constraint**. This constraint is simply the codification of an observation that feet or words in the languages observed necessarily contained two moras, and the further realization that this generalization held explanatory power for the phonology. Constraining the rules, which were applied in an inherently very open-ended and powerful system, and the representations, which were similarly unlimited in regard to possible characterizations, improved the expressiveness of the phonology, although constraints did not have any formal position in the theory and were at root descriptive stipulations. They were often described in terms of "natural" conceptions of linguistic "well-formedness". Some constraints familiar from the early literature of Lexical Phonology include the Obligatory Contour Principle of Leben (1973), Kiparsky's (1973) Elsewhere condition, and the Strict Cycle Condition of Mascaró (1976).

Parameters, introduced by Vergnaud & Halle (1978) and developed further by Hayes (1981, 1995), were similar to constraints, but were understood as varying between languages, which would each have their own "settings". Hayes used parameters to indicate, for example, foot-type and directionality of syllabification. Such parameter settings were crucial to supply information necessary to Hayes' model of metrification. However, there was no formal way to express these parameters within the traditional formulation of phonological rules. Rather, the parameters resembled a somewhat more rigorous form of the **conditions** that had accompanied rules in some previous approaches. When rules were seen as having a limited, rather than universal application, conditions were appended which specified the limiting circumstances. While descriptively helpful, such appended conditions are not the ideal mechanism with which to achieve the formal explicitness demanded by the generative argument.

The term "constraint", like "rule," seems to imply universal application, and this is how constraints were typically interpreted. However, like rules, there were constraints that seemed to apply only sometimes, even within the same grammar, e.g., the extrametricality of Hayes (1987) which applies only in certain cases. Constraint-based theories, such as the "constraints and repair strategies" theory of Paradis (1988) or the "declarative

phonology” of Scobbie (1991), Bird (1990) and Bird & Ellison (1992), dispensed with rules seen in earlier generative theory completely and made the constraints themselves the basis of the grammar. However, the absolute enforcement of constraints presented the many of the same problems as rule-based theories. Only the hierarchy of violable constraints of Prince & Smolensky’s (1993) Optimality Theory formally accounts for the apparent non-universality of certain rule- or constraint-based regularities. The candidate which **minimally violates** the constraint hierarchy (McCarthy & Prince 1994) is considered optimal and will surface. The candidate which survives the constraint evaluation is not the “perfect” candidate, but the least imperfect candidate. This ranking of violable constraints precisely captures in a fully principled manner the linguistic phenomenon first described by the classical Indian grammarian Paṇini, and only approximated by Chomsky & Halle’s disjunctive rules and Kiparsky’s Elsewhere Condition.

Constraints in Optimality Theory not only reprise their more limited role in derivational theory, but also provide, in a unified theoretical mechanism, the functionality divided in derivational theory between rules, constraints and parameters. Like rules, they define environments, and what should appear in those environments. Like derivational constraints, they can restrict the possible set of structures which can surface. And like parameters, they can account for apparent language-specific “settings”, such as the directionality of foot-formation, through references to properties of constituents, such as their edges, as in the Alignment constraints of McCarthy & Prince (1993a). The constraint hierarchy, made up of ranked, violable constraints, allows for the expression of the entire grammar while avoiding the pitfalls of universal constraints, rules and parameters, which always suffered from the previously inexplicable fact that they did not apply universally.

1.3.3 Prosodic constituents in Optimality Theory

The set of structures referred to by constraints will be limited in this study to the constituent members of grammatical hierarchies. Violable, ranked constraints describing relationships between members of these hierarchies will provide all the mechanisms necessary to produce the surface forms of the language. One such set of constituents is found in the prosodic hierarchy, which was developed in the context of derivational theory. (§ 1.2.1)

In chapter four, the OT constraints which are used to account for English stress patterns refer to the same prosodic constituents which yielded stress effects in derivational theories such as Prosodic Phonology (Nespor & Vogel 1986, Inkelas 1989) and Metrical Phonology (Prince 1983, Hayes 1981, 1982, 1995).

The prosodic hierarchy, as noted above, was originally used to model structure above the word level (Selkirk 1980, 1982a, 1984). Nespor & Vogel (1986) describe the clitic group, the phonological phrase, the intonational phrase and the intonational utterance as higher level phrasal constituents. Inkelas (1989) accepts all of these apart from the clitic group, which she believes can be replaced by one of the other phrasal categories in the languages where it purportedly appears. She also notes (p. 43) that compounds, “which form one single morphological constituent, behave as two constituents for the purpose of phonological rules.” and sees compounds as iterative combinations of prosodic words.

Hayes (1995: 367) states that phrasal stress assignment is markedly different from the assignment of word stress, at least in languages where metrical structure has already been assigned to all syllables up to the word level, such as English. Stress rules at these levels usually apply an End Rule which promotes one of the prosodic word heads from the previous level. The most common type of adjustment at these levels is clash adjustment, wherein a head is demoted and an adjacent head within the same constituent is promoted due to the adjacency of a stronger stress, as seen in the *thirteen mén* example given above, (1.2).

At the word-level and below, the relevant level for the discussion of word-stress, the prosodic word dominates the foot, syllable, and mora. This category has been defined by Selkirk (1980, 1982a, b, 1984) and Nespor & Vogel (1986) as the category which encompasses the feet it dominates.

- (1.5) PrWd
 |
 Ft
 |
 s
 |
 m

The word is the domain for “word stress,” and the assignment of this main word stress to one of its constituent feet is accomplished in derivational studies by the application of an End Rule. End Rules usually assigned prominence to a foot on the right or left edge of the word. Although the word in Hayes’ (1995) system includes extrametrical and unfooted material, the assignment of prominence is only possible for feet, thus eliminating unfooted and extrametrical syllables from consideration for receiving word stress.

Using a prosodic system composed of the constituents and mechanisms described above, the assignment of stress to proposed lexical entries of words, in a derivational context, proceeds according to rule. Thus the assignment of stress to the word *extrametricality* could proceed as follows:

(1.6) Assignment of syllable structure:⁷

```

s      s  s  s  s  s  s
|\      |  |  |  |  |  |
mm  m  m  m  m  m  m
||  ///|  /|  ///|  /|  /|  /|
ekstrametrikaliti

```

The final syllable of such a word is extrametrical in English (according to Hayes 1982), and quantity-sensitive trochaic foot assignment proceeds from right to left. Below, using the bracketed grid notation of Hayes (1995), grid marks indicate the heads of syllables, while feet are indicated by parentheses. Syllable notation has been replaced for convenience by the segments themselves, separated by white space:

(1.7) (x) (x .) (x .)
(ek) stra (me tri) (ka li) <ti>

⁷The question of whether the onsets link to the mora or the syllable nodes (Zec 1988, Hayes 1989, 1995: 53) is irrelevant here. However, linking them directly to the syllable node removes the difficulty of explaining the distinction between links indicating weight and links which are irrelevant to weight, such as onsets.

Following Hayes (1995), the second syllable would be left unparsed, rather than creating an uneven trochee. In this work, uneven trochees will be allowed, which would result in the following parse:

$$(1.8) \quad \begin{array}{ccccccc} (x & .) & (x & .) & (x & .) & \\ (ek & stra) & (me & tri) & (ka & li) & <ti> \end{array}$$

Finally, an End Rule which inserts a grid mark on (or promotes the head of) the rightmost constituent head yields the final prosodic word:

$$(1.9) \quad \begin{array}{ccccccc} & & & & x & & \\ (x & .) & (x & .) & (x & .) & \\ (èk) & strə & (mè & tri) & (ká & li) & <ti> \end{array}$$

In English, stressed vowels maintain their quality, while vowels in “weak” position tend to reduce to schwa, especially in open syllables.

The prosodic hierarchy provides a structural framework which allows for the representation of the relative and iterative properties associated with stress as a series of purely local relations. Alternations can be understood as occurring in the context of the prosodic categories which contain them, eliminating the need for rules with long-distance effects. Relations which under previous frameworks had to be accounted for using a number of conditions can be presented in a more succinct and coherent fashion. As in other theories of stress assignment, prosodic phonology relies upon the morphology to provide the underlying forms which are parsed into the prosodic constituents.

1.3.4 Stress in Optimality Theory

In chapter four, building on previous OT treatments (McCarthy & Prince 1993), the stress effects previously modeled in derivational accounts such as Hayes (1995) are treated in an OT framework, using constraints which refer to prosodic constituents. One common English stress pattern described by Hayes (1995), Halle & Vergnaud (1987) and others (see chapter 3) is similar to that generally found in Latin. For trisyllables of this kind, final syllables are treated as extrametrical, and, due to the trochaic form of English feet, stress depends upon the weight of the penult. The contrast is seen between words of

the form (LL)s and L(H)s, where ‘L’ and ‘H’ indicate metrically light and heavy syllables respectively, for example *órigín* and *appéndix*. These can be represented by the following simplified constraint tableaux:

(1.10a)

/origin/	FtBin	Non-Fin	Edgemost
☞ (óri)gin			s
o(rí)gin		!*	
o(rí)gin	!*		s
(órigín)	!*	*	

(1.10b)

/apendiks/	FtBin	Non-Fin	Edgemost
☞ a(pén)diks			s
a(pen)(díks)		!*	
(ápen)diks	!*		s
a(péndiks)	!*	*	

The three constraints used here are parallel to mechanisms used in Hayes (1995). **FtBin** as formulated in Prince & Smolensky (1993) strictly enforces binary feet, just like Hayes’ bimoraic principle. Violations are incurred by feet which do not contain two (and only two) moras. **Non-Finality** states that the primary stressed foot cannot be final in the prosodic word, similar to extrametricality. **Edgemost**, in contrast, demands that the primary stressed foot stands as far to the right edge of the prosodic word as possible, similar to Hayes’ “End Rule: Right”. One advantage of the OT account offered here is that the constraint hierarchy formally represents the dependencies between the three constraints via the ranking relationship. In Hayes’ system, the End Rule was said to be unable to “see” the extrametrical syllable, thus the stress would fall near to the right edge of the word, but never upon it. In other cases, however, extrametrical syllables would become visible. Here, there is no “invisibility”; the presence of the final syllable is indicated by the violation of Edgemost seen in the two optimal candidates. It is the higher ranking of **Non-Fin** which ensures that candidates such as *appendix fail to surface.⁸

⁸In this work, attested words will appear in italics, but unattested hypothetical forms will appear in plain text, with an asterisk.

Another advantage of Optimality Theory is that it can explain apparently iterative effects seen in stress assignment without using the iterative rule application required by derivational theories. Constraint rankings can also be used to explain cross-linguistic typologies. Nowhere is this more striking than in the explanation for Hayes' (1995) typology of stress types, as demonstrated by McCarthy & Prince (1993a), Cohn & McCarthy (1994), and Crowhurst & Hewitt (1995), among others. In chapter four, the framework of **Generalized Alignment**, introduced by McCarthy & Prince (1993a), is presented. This framework allows for the formal representation of a relationship between two constituents, taking two constituent edges as its arguments. Various orderings of constraints aligning the foot to the prosodic word and vice versa can produce the entire typology of stress patterns described in Hayes (1995).

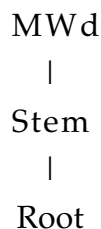
McCarthy & Prince (1993a: 16) regard the violation of Alignment constraints as “gradient, not Boolean”, and quantify violations in terms of each constituent in the universal (first) argument position. Thus, for example, as feet get farther away from the designated aligned edge, they incur more violations of the constraint. This enforces foot-formation in what appears to be an iterative or sequential fashion starting at the designated edge, but which is actually the expression of an alignment constraint within a parallel framework. In addition to the alignment framework, another more general framework for expressing constituent relationships, the **No-Intervening** constraint of Ellison (1995) and Zoll (1996) is also presented in chapter four, and many descriptive constraints (such as **FtBin**, **Edgemost** and **Non-Fin**) are formalized within using one of these frameworks.

1.3.5 Subcategorization and morphology

All treatments of English stress have needed to refer to morphological structure to account for the stress patterns seen in suffixed words. Some suffixes, those described as “level II” in derivational theories like Lexical Phonology (Kiparsky 1982a, b), behave with regard to stress assignment as if they were not present, e.g., *límit* ~ *límitless*. Other suffixes, described as “level I”, appear to participate in the stress assignment of their words, and the suffixed forms take main stress on different syllables than unsuffixed stems, e.g., *átom* ~ *atómic*, *ínstrument* ~ *ínstrumental* ~ *ínstrumentality*. This implies that feet are being applied to these two types in different ways. In

chapter four, such differences are captured by **subcategorization constraints**, which relate prosodic constituents to members of a parallel **morphological hierarchy** (McCarthy & Prince 1993a, b). This morphological hierarchy consists of items previously used in derivational theories, the morphological word, the **stem** and the **root**:

(1.11) The Morphological Hierarchy



The root is simply a lexical item or morpheme, composed of segments. The morphological word is parallel to the prosodic word and represents a root or series of roots together with all affixes. The stem is a necessary intermediate category. These three constituent types are somewhat analogous to the labels /a, b, g/ used by Inkelas (1989), and provide all the complexity necessary to represent the subcategorization possibilities for English.

Inkelas (1989) uses subcategorization frames to mark lexical items, and each item has two frames, one for morphology and another for phonology. Inkelas adopts the three-level morphological system of Lexical Phonology (the third level being the word level), representing these by the symbols /a, b, g/. An affix like /un-/ would have a morphological subcategorization frame like this (p. 104):

(1.12) un- : [_ []_b]_b

This means that the affix combines with level II items to create a new level II item. The underscore indicates the space that the affix itself would fill (defining it as a prefix) and the inner bracket indicates that it combines with a level II item. Its level I analog /in-/ would have the following frame:

(1.13) in- : [_ []_a]_a

This indicates that it combines with level one items to form another level one item. These frames have the advantage of both accounting for morphological “level” effects as well as explaining why some morphemes only combine in certain ways. For example, “bound” roots like /-ceive/ (p. 66) can be marked to require a level I prefix before they can surface:

(1.14) [[]_a ceive]_a

While such a bound root requires a subcategorization frame morphologically, it is no different phonologically from any other stem of its type, and Inkelas does not give it a specific prosodic frame, underlining the potential independence of morphological and phonological structures. Using subcategorization frames, selectional restrictions in either the prosodic or morphological domains may be lexically encoded for the various morphemes which demand them. Subcategorizations of this kind have been reformulated in Optimality Theory as alignment constraints referring to specific affixes (McCarthy & Prince 1993a: 22-4):

(1.15) Align([um]_{Af}, L, Stem, L)

Here, *um-* is a Tagalog prefix which aligns to the left edge of a stem. A similar constraint could be expressed for the English prefix /ex-/, expressing its subcategorization for level one stems:

(1.16) Align([ex]_{Af}, L, Stem, L)

Such a constraint refers, like Inkelas’ subcategorization frames, both to the morphological constituents formed and the directional placement of the affix.

While Inkelas’ system implies that the morphological and phonological constituents tend to correspond, the contents of a morphological constituent being parsed into a prosodic one, she also allows for these constituents to become misaligned. Inkelas uses this proposed misalignment of morphological and prosodic constituents to try to account for extrametricality (p. 151), prefiguring OT alignment constraints.

Morphological constituents, like Inkelas’ three constituent levels, are subject to constraints aligning or mis-aligning them with both prosodic constituents

and other morphological constituents, allowing for the expression of the full range of constituent patterns found in natural language. Subcategorization constraints of this kind will be used in chapters four, five and six to account for the different behaviors with regard to stress seen between “level I” and “level II” suffixes, as well as between some “level I” suffixes, such as /-al/ and /-ate/.

Upon application of morphological subcategorization to the data, it becomes clear that most instances of long vowels in complex words occur in monosyllabic stems of the melodic shape /CVC-/ (§ 4.3). This observation leads to the conclusion that vowels are lengthened in this context due to the combined action of a bimoraic constraint upon feet and an enforced alignment (in certain contexts) of the stem to a foot. In other words, so-called “vowel shortening” is actually indicative of cases where the stem vowel fails to lengthen due to a different stem structure. This solution allows former “lexical exceptions” to shortening rules (e.g., *obésity*, *nótify*) to be treated regularly, as showing underlyingly long vowels, while “shortening” vowels (e.g., *tónic*, *vílify*) are actually underlyingly short, but phonologically lengthened in the contexts where they appear as long (e.g., *tóne*, *víle*). It also accounts for the otherwise unusual distribution of long and short vowels in lexical stems, wherein long vowels are found plentifully in monosyllables but only rarely in bisyllabic stems (§ 4.3). Of course, arriving at this solution means abandoning certain long-held ideas about the lexicon, such as the assumption that unsuffixed surface forms are equivalent to the “basic” lexical stem. In the case of alternations like *sáne* ~ *sánity*, it is actually the affixed form *sánity* which is a more faithful indicator of the lexical entry /san/, whose vowel is lengthened in the shorter stem of *sáne* due to minimality constraints.

1.4 Derivation and parallelism

In Optimality Theory as defined by Prince & Smolensky (1993), while rules have been replaced by the constraint hierarchy, the lexicon still plays the same role familiar from derivational theory. An “input” lexical item is presented to Gen, which expands it into all possible well-formed candidates. These are then evaluated by the constraint hierarchy, producing an optimally prosodified “output” surface form. While this entire process can be conceived of as occurring in parallel, rather than through a series of steps, as in the

prosodification accounts (e.g., syllabification, foot-formation) of derivational studies such as Hayes (1995), OT nevertheless requires the specification of an “input” item, which will come from the lexicon, presumably via the morphology.

While OT accounts of monomorphemic words can proceed smoothly, avoiding the issue of morphology and referring only to the segments of the lexical item, more complex words require that morphological structure be provided along with the input candidate’s segmental melody. As noted above, various OT analyses, such as Prince & Smolensky (1993), McCarthy & Prince (1993a, b), have proposed constraints which also refer to morphological structures. However, for derivational theories such as **Lexical Phonology**, morphological complexity meant more than just the presence of structure, it indicated a serial process of word construction which proceeded in discrete stages. This was used to explain both word-formation patterns and structural alternations, such as those ascribed to the **cycle**. Nothing inherent in OT itself precludes the retention of serial morphological derivation, the *Eval* process simply replacing the phonological rules of derivational theory. However, this vision of serial derivation carries with it all the problems associated with it from derivational theories, and is based on assumptions about word-formation that are made unnecessary and irrelevant by mechanisms found at the heart of OT. It will be illustrated here that a parallel version of OT can capture the supposed effects of serial derivation, while avoiding its shortcomings.

1.4.1 Cyclic rule application

One intuitively appealing concept captured by serial derivation is the idea that larger words are built up from smaller ones. Many English words appear to support this intuition, e.g., *law*, *lawless*, *lawlessness*, *bylaw* or *instrument*, *instrumental*, *instrumentality*. The alternating character of syllable stress led Chomsky & Halle (1968) to propose the transformational cycle. Noting the relationship between suffixed and unsuffixed forms of the “same” word, they proposed a phonology in which rules were applied after the addition of each new morpheme to the “basic” lexical word. The differences in word stress seen between related forms were thought to be the result of the continued application of the rules, and traces of the earlier “cycles” were thought to be seen in secondary and tertiary stresses that should

otherwise not appear. This cyclic model of affixation, which also applied to compounding, was the principal mechanism of the morphological system in generative phonology.

Chomsky & Halle regarded the primary stress of a “base” form like *instrument* as being retained as a secondary stress in the other two derived forms, while the main stress of *instrumental* reveals itself through the full vowel (rather than a reduced vowel [ə]) seen in *instrumentality*. As a minimal pair proof, they argue that the realizations of the phonologically similar words *cond[ɛ]nsation* and *comp[ə]nsation* show distinct vowels in the antepenult, the former showing a full vowel while the latter shows a reduced vowel. This difference is ascribed to the relation of the former to a base form *condense*, with a stressed root vowel, while the latter has no corresponding stressed base form, being built from the derived verb *compensate*.⁹ This interleaving of morphological and phonological processes became a central tenet of Lexical Phonology, developed by Kiparsky (1982a, b) and Mohanan (1982).

Chomsky & Halle (1968) were well aware that many of their “rules” were less general than the term suggests, and regarded the lexicon as that part of the grammar “which should contain only idiosyncratic properties of items, properties not predicted by general rule.” (p. 12) For words which violated the proposed rules, however, they preferred “assigning a representation in terms of segments and boundaries to each apparently nonregular form in such a way that the correct phonetic form is predicted by rules that are needed on independent grounds.” (p. 147) It is in this spirit that they proposed underlying forms like /nixtəŋāl/ and /mixtəli/ for *nightengale* and *mightily* (p. 234), positing an underlying velar fricative /x/ (absent in spoken English) to explain notable exceptions to their Trisyllabic Laxing rule, one of the rules that accounted for vowel shortenings of the type *sáne ~ sánity*.

Such liberality with the postulation of underlying forms was directly in contrast to the practice of the earlier structuralists, who regarded only surface forms as having a linguistic reality (Anderson 1985: 306-9). Indeed, despite its reputation for introducing great abstractness in underlying representations (Anderson 1985: 325), generative theory as presented in Chomsky & Halle

⁹Although one could point out the related noun *rècompense*, with main stress on the same (unsuffixed) root. Sainz (1992: 124-135) offers an excellent summary of the English words used to support this type of argument.

(1968) proceeds generally from the structuralist assumption that the surface form of words corresponded to their “basic” form (in the absence of evidence to the contrary), and creative suggestions like /nixtəŋgāl/ are relatively rare. The entire morphological system with its transformational cycle takes for granted that unsuffixed words are “primitive”. Any variations seen in related affixed forms are regarded as modifications of that primitive representation, for example the characterization of the alternation seen in *sáne* ~ *sánity* as “shortening”, based on taking the unsuffixed form *sáne* as the primitive.

This assumption moved into a central theoretical position, strengthened and codified into the **Alternation Condition** of Kiparsky (1973) and the **Strict Cycle Condition** of Mascaró (1976). These principles stated that cyclic (for some, lexical) rules could apply only in “derived environments”, that is, only to words undergoing affixation processes. Thus, as they were never subject to phonological rules, unsuffixed forms were necessarily identical to their own underlying lexical entries. This had the advantage of eliminating numerous exceptional forms such as *níghtengale*, which has no internal morphological structure, without any need to propose abstract underlying forms like /nixtəŋgāl/. The very principle that allowed words like *níghtengale* to have simple underlying forms also excepted them from the otherwise violated rule.

The specification of certain rules as cyclic thus allowed for a *de facto* hierarchy of rules. Rules that specified the presence of affixes, such as stress assignment rules, could be applied only to words undergoing affixation. It had long been clear that certain English affixes, such as /-ing, -ness, -ly/, were “stress-neutral”. Chomsky & Halle (1968 : 85) had used a word-boundary (“#”) to separate those affixes from words, while stress-affecting affixes were (otherwise arbitrarily) given a morpheme boundary (“+”). The framework of Lexical Phonology (Kiparsky 1982a, b, Mohanan 1982), which was developed as an explanation for (among other things) the Alternation Condition and the Strict Cycle Condition, allowed such boundary distinctions to be dispensed with, as the rule itself became attached to the affix in question.

As in Chomsky & Halle (1968), derivation in Lexical Phonology rests upon the interleaving of phonological and morphological events: phonological rules are applied after each instance of morphological affixation, producing the effects of the cycle. Although the contrast between the stress-neutral and stress-affecting affixes was still idiosyncratic and lexical, the

suffixes themselves were conceived of as being on different **lexical levels**, labeled I and II. The first level was cyclic, and cyclic rules applied to words when affixes of this level were added, hence the effect on stress placement. The second level was (in most approaches) post-cyclic, and the stress rules did not apply. There was also a third, post-lexical level, at which completely regular surface changes like flapping took place. The lexical levels could afford exceptions to otherwise general rules, which would be recorded in the lexicon as such. By establishing such a hierarchy of rule levels, rules could be focused upon the forms they were meant to apply to, and separated from those forms that would otherwise be problematic exceptions, reducing their number dramatically and increasing the explanatory power of the system. For example, words like *míghtily*, although not underived like *níghtengale*, would still be exempt from the cyclical Trisyllabic Shortening rule, because /-ly/ was a level II post-cyclic affix. The need to posit unusual underlying forms was again avoided, in this case due to level ordering.

1.4.2 Shortcomings of Lexical Phonology

Despite their achievements, the Strict Cycle Condition and level ordering of Lexical Phonology show certain inherent problems. While successful in accounting for words like *còndensátion*, there are far more affixed forms that do not at all correspond to their “bases” in stress placement:

- (1.17) *átom* ~ *atómic* *sólid* ~ *solídity* *pýramid* ~ *pyrámidoid*
 [æ̀təm] [ə̀támik] [sálid] [sə̀líditi] [pírəmid] [pə̀ræmidoɪd]

These affixed words completely lose any trace of the supposedly “basic” primary stress. Thus, along with the stress assignment rules, the need arose for “destressing” and “retraction” rules to get rid of or shift unwanted old stresses expected from the cyclic application of rules (see the excellent discussion of this in Kager 1989: 86-101). Furthermore, explaining the supposed contrast between *còmpensátion* and *còndensátion* was more difficult in the framework of prosodic phonology, where stress was no longer a segmental feature but rather governed by prosodic constituents. While “destressing” could be conceived of via de- and re-footing rules, there was no mechanism for representing leftover traces of stress from previous cycles using prosodic constituents.

Another problem involves the central hypothesis that underived words can be exempted from cyclic rules. Since stress assignment, whose effects are present in all words, derived and underived, was still the archetypal cyclical rule at the time, exemptions had to be made in Kiparsky (1982b) for iterative prosodic rules like syllabification and stress assignment. This ushered in a distinction between “structure changing” and “structure preserving” rules, in order to prevent over- and under-application of certain rules to underived and post-cyclic words. Conversely, the fundamental nature of the distinction between derived and underived words, suggested by the Alternation Condition and its successors, is weakened by the existence of many languages, for example, Latin, in which this distinction is absent, as virtually all lexical words show inflectional or derivational morphology. Moreover, certain underived English words do show “structure changing” effects, e.g., pairs like *sign* / *signature*, *damn* / *damnation*. For these, Kiparsky (1985) suggests a relaxing of the constraint on the last lexical level, a theoretical tactic described by Kaisse & Shaw (1985: 23) as “disappointing”.

Finally, as suggested above (§ 1.3.4), the prosodic hierarchy can account for many of the iterative properties the cycle was created to explain. If the cycle was truly active in the grammar of English, only structural changes arising on earlier cycles, but which are not derivable on the final cycle, would give evidence of true cyclicity, but this is not the case. The area in which cyclic effects seemed most clear involved stress assignment in derived words, where Chomsky & Halle claimed that traces had been left of the “old” primary and secondary stresses, as in their example of *cond[é]nsation* and *comp[ə]nsation* given above (§ 1.4.1), or in *instrument* ~ *instrumental* ~ *instrumentality*. Contrasting with such examples, however, are many affixed forms that do not at all correspond to their “bases” in stress placement, and show no trace of the “basic” stress, such as those given above in (1.17). Such cases are typically explained with complex destressing (or defooting) and retraction rules in theories which maintain the cycle (cf. Kager 1989). Chapters five and six address a number of problems which were accounted for in previous studies using retraction rules.

In later work, a number of objections were raised to the view of the cycle as discussed above. Halle & Vergnaud (1987) proposed to dismantle the Lexical Phonological system and make [+cyclic] an optional feature of individual levels of affixation; this was in response to a number of

difficulties, such as the deletions of segments in underived forms like *sign*, *damn*, noted above, which led them to modify the strict cycle used in Lexical Phonology. Halle & Kenstowicz (1991: 460-1) go further, noting that words like *conversation*, *information*, *transportation* fail to show the expected embedded stresses predicted by the *condense* / *condensation* example; they conclude that the presence or absence of such “subsidiary stress” is “an idiosyncrasy of individual lexical items.” Kager (1989: 38) says that such variations are dependent upon “performance factors such as word frequency and the predictability of words in discourse contexts.” He also questions the motivations for shifts in prominence relations in such cases, e.g., *còndensátion* rather than **condènsátion* (p. 87). Sainz (1992: 124-135) offers a summary of examples debunking the connection between supposed cyclic stress application and the presence of tertiary stresses. She has compiled extensive lists of such words (e.g., p. 118-119), which demonstrate that no phonological or morphological pattern can be generalized from the data, and thus no case can be made for the presumed cyclic explanation.¹⁰ Further words can be found, which show, in some pronunciations at least, full vowels in bound stems, e.g., *cònt[ɛ]mplátion*, *cònc[ɛ]ntrátion*, cases which should be parallel to *comp[ə]nsation*. There is even a pronunciation of the latter which shows the full vowel, while some variants of *cònd[ə]nsátion* show a reduced vowel.¹¹ Thus, while the original claim of Chomsky & Halle (1968) might be theoretically appealing and interesting, the data unfortunately does not seem to support it.

It is easy to see how the cyclic approach could arise in the context of the Chomsky & Halle (1968) framework. All of the features in their system, including stress, were properties of segments. In order to explain how an unexpectedly full vowel, like the [ɛ] in *cond[ɛ]nsation*, became “stressed”, their framework required the application of the feature [stress] to that vowel. Combining their concatenative approach to word formation with the observation that the same vowel was given main stress in the embedded

¹⁰In fact, there is a pattern based, as Kager notes, on frequency. Less frequent words tend to show their full vocalism, presumably to aid the listener in accessing the correct lexical information.

¹¹The Celex English wordlist used in this study listed both *compensation* and *condensation* as having variant pronunciations with /e/ and /ə/, the compilers apparently being undisturbed by the theoretical significance the two words supposedly hold. In my own speech, the words behave identically.

word *condénse*, the cycle was the most logical conclusion. Many of the clearest cases of affixation in English are of the stress-neutral “level II” type, in which the intuitive analysis clearly indicates an embedded, consistently stressed word to which an affix has been added, reinforcing this view of word formation. In fact, even if we disregard the *còmp[ε]nsátion* type, it still appears as though many derived words display cyclic effects, e.g., *ínstrument*, *ínstruméntal*, *ínstrum[ε]ntáltity*. Yet this phenomenon is better accounted for, at least in English, by the inherently hierarchical prosodic system than by the cycle.

The fact that the same syllables are landing sites for stress in all three words is a by-product of foot-formation rather than any cyclic process. This can be demonstrated by a derivation of the words in question, following the foot-formation rules of Hayes (1987, 1995):

$$\begin{array}{lll}
 (1.18) \quad (x) \cdot <(x)> & (x) \cdot (x) <.\> & (x) \cdot (x) (x \cdot) <.\> \\
 \text{ínstrument} & \text{ínstrumént - al} & \text{ínstrument - ál -ity}
 \end{array}$$

In *ínstrument*, the final syllable is extrametrical (indicated by $< >$), following a general rule of final syllable extrametricality in nouns, and so main stress falls on the initial. In *ínstruméntal*, the final syllable *-al* is extrametrical (level I suffixes are extrametrical in Hayes’ framework), and the formerly extrametrical syllable *-ment* is, due to its intrinsic weight, a foot in itself. Continuing from right to left, the remainder of the word is again parsed as in *ínstrument*. Finally, in *ínstrumentáltity*, the suffix *-ity* can join with the formerly extrametrical *-al* to yield a typical trochee *-alit-*. This will again result in the rest of the word following the foot structure of the previous two examples. The consistency of footing results entirely from the shapes of extrametrical syllables and of the suffixes. That this does not always happen should be clear from cases like *atom*, *atomic*, *solid*, *solidity*, *pyramid*, *pyramidoid*. In these cases, the extrametrical syllable in the simplex form is not able to stand alone as a foot. When it is suffixed with a non-extrametrical or heavy affix, the parsing of syllables to feet is thrown off by one syllable:

$$\begin{array}{llllll}
 (x)<(.)> & (.)(x \cdot) & (x)<(.)> & (.)(x \cdot) & (x \cdot)<(.)> & (.)(x \cdot) <(x)> \\
 (1.19) \text{ atom} & \text{atomic} & \text{solid} & \text{solidity} & \text{pyramid} & \text{pyramidoid}
 \end{array}$$

It is clear that this solution requires the prosodic and metrical constituents introduced subsequent to Chomsky & Halle (1968). The non-cyclic approach simply applies the foot-formation principles to the domain, informed by the morphology, and arranges the melodic units into syllables and feet using a consistent iterative principle.

In Optimality Theory, apparently iterative and cyclic structural effects can be captured in Optimality Theory using McCarthy & Prince's (1993a) framework of Generalized Alignment. Just as various orderings of constraints aligning the foot to the prosodic word and vice versa can produce the typology of stress patterns described in Hayes (1995), enforcing foot-formation in what appears to be an iterative or sequential fashion (§ 1.3.4), McCarthy & Prince (1993a: 20) note that "replacing PrWd by one of the morphological categories Root or Stem in the alignment constraint will produce effects of the kind often attributed to the cycle." In other words, level ordering is no longer necessary, because constraints can be applied to candidates with reference to specific morphological constituents, effectively limiting the influence of the constraint to items within those constituents. There is no need to conceive of such processes as applying iteratively in sequential order.¹² Serial derivation, long required to account for the cycle, has no place in an explanatory parallel model of OT and can be dispensed with. Serial derivation is a cumbersome mechanism, and introduces unwanted and unneeded complexity to the grammar. As long as the effects assumed to require serial derivation can be captured under parallelism, the latter should be considered more constrained, less complex and thus preferable.

1.4.3 Cophonologies and similar proposals

Another problematic proposal which has been introduced into OT, and which will be excluded from consideration in possible solutions, is the concept of **cophonologies** (Itô, Mester and Padgett 1994, Inkelas, Orgun & Zoll 1994, Cohn & McCarthy 1995) or lexical "levels". This weakens OT by introducing further serial processes back into the grammar. In McCarthy & Prince's (1993b) account of the phonology of Axininca Campa, this is achieved by proposing three Lexical Phonological levels, each of which consists of an

¹²This also eliminates other sorts of problems inherent in Lexical Phonology, such as bracketing paradoxes, which arise from the unnecessary linking, inherent in the serial model, of morphological subcategorization to level-ordered rule application.

independent constraint hierarchy. The optimal output of each hierarchy is passed on to the next as its “input” form. This tactic appears to be necessary because the required constraints seem to be ordered differently at different morphological levels. This problem is rooted in McCarthy & Prince’s (1993b) conceptualization of constraints and candidates, further developed in Correspondence Theory (McCarthy & Prince 1995). Constraints governing segments, such as **Parse**, **Fill** (later **Max** and **Dep**), **Coda-Cond**, are conceived of as universal, governing the relationship between the “input” and “output” strings. Apparently different rankings of **Parse** and **Fill** for the treatment of suffixes and prefixes suggest that there must be “cophonologies”, independent sets of constraint hierarchies in which these universal constraints are differently ordered.

However, this problem can be eliminated through an alternative understanding of “input” and “output” strings. OT as presented by Prince & Smolensky (1993) and McCarthy & Prince (1993a, b) utilized the constraints **Parse** and **Fill** to govern **faithfulness**, i.e., whether the segments in the “input” string appeared in the “output” string and vice versa. The same constraints were also used to describe relationships between various members of the prosodic hierarchy, for example parsing syllables to feet or feet to words. Correspondence Theory, as defined by McCarthy & Prince (1995), focused upon the faithfulness relationship between “input” and “output” strings, introducing the constraints **Max** and **Dep**. These constraints govern the correspondence between segments in the input and output, or the base and reduplicant (in cases of reduplication). The application of this concept to the latter two constituents suggests, as do subcategorization constraints, that constraints may be applied to specific morphological constituents. The “input” strings are, after all, instances of morphemes, whether individual or in combination. The idea of “input-output” correspondence can be more precisely reformulated as (for example) the correspondence of the Morphological Word to the Prosodic Word, just as “base-reduplicant” correspondence is an instantiation of the correspondence of an affixal morpheme (or its prosodic correspondent) to a stem morpheme.

Returning to McCarthy & Prince’s (1993b) case of Axininca Campa, the opposite relative rankings of **Parse** to **Fill** for suffixes and prefixes can be reinterpreted as the presence of four ranked constraints in the same hierarchy, two referring to the suffix constituent and two referring to the

prefix. Two separate cophonologies are no longer necessary to account for the data. The issue of cophonologies, and other cases explained in the literature by cophonologies, will not be further explored in this work; it will be assumed that the approach suggested here will succeed in other cases, and the burden of proof in OT should be on the need for cophonologies to account for otherwise inexplicable data, rather than the need to prove that they are unnecessary. As Inkelas, Orgun & Zoll (1994: 11-12) point out, the concepts behind using cophonologies are “counter to the spirit of Optimality Theory”, and they go on to illustrate various paradoxes arising from the use of cophonologies, concluding that “the potential for cophonology proliferation is astronomical.” (p. 13) Cophonologies are merely the OT analog of lexical rules; the weakness of the Lexical Phonological system, that it was unconstrained, is maintained by the use of cophonologies. What constrains OT is the requirement that the entire grammar must be represented via a single ranked constraint hierarchy, and the one correct grammar for a given language will be that in which a single ordering of all the constraints, including those referring to morphological constituents (as in the case of Axininca Campa), yields the attested data.

Another example of derivational concepts creeping back into OT involves supposed relationships between multiple “outputs” across the lexicon. This reflects a desire to capture within OT the derivational idea that words are “related”, and that the shape of related forms can influence each other. An example of this in derivational theory is that of “trace” tertiary stresses due to cyclic derivation, an idea shown above (§ 1.4.2) to have no basis in the data. McCarthy & Prince (1995: 262) allow correspondence relations to be extended to “types of cyclic or transderivational relationships within paradigms”. For example, Benua (1995) concludes that truncation processes which shorten names like *Larry* /ləri/ to *Lar* /lær/ are examples of such a transderivational relationship, because of a purported constraint on the tautosyllabic sequence /ær/ which she assumes based on distribution as well as orthography (e.g., *barn*, *car*).

The truncated form *Lar* retains the offending sequence because a “Base-Truncated” correspondence constraint outranks the constraint restricting /ær/, the need for the truncated form to resemble the “full” word outranking the segmental restriction. This type of solution opens the door to all kinds of potential transderivational relationships between words influencing the

constraint hierarchy, again adding unnecessary and problematic complexity to OT. What Benua does not take into account is the principal underlying assumption of her proposal, i.e., that the sequence /æɾ/ in *Lar* is tautosyllabic. Since her solution requires the identification of words like *Lar* as morphological truncations anyway, it is far simpler to constrain such truncations (presumably a morphological category) to end in syllable onsets, the final vowel not being parsed into the “output” representation (see also § 4.1.5). Following this reasoning, the sequence /æɾ/ in *Lar* can be seen as indeed phonologically heterosyllabic, just as in *Larry*. The relevant constraint ranking is the higher position of the constraint which prevents the surfacing of the final vowel relative to the faithfulness constraint which normally requires all segments in a morpheme to surface, and there is no need to resort to special transderivational relationships.¹³

A similar transderivational constraint is used by Pater (1995: 20-21) to account for “classic example[s] of stress preservation [like] *còndensátion*,” and for this he proposes a constraint **StressIdent** which states that “if a is stressed, then $\mathbb{I}(a)$ must be stressed”, the function notation apparently representing the transderivational relation. This kind of explanation will be explicitly rejected here out of hand, not only because the “classic examples of stress preservation” have no basis in fact, as Sainz (1992) has shown, but also because transderivational constraints of this kind are far too unconstrained to appear in a formal and constrained OT. In the version of OT argued for here, common properties of “related” words are explained by common lexical roots, mediated by the constraint hierarchy. The *Eval* process only has access to the lexical “input” items and the constraint hierarchy, and no relationships between “output” items are available to it. Any influence of some member of a paradigm upon another (e.g., “analogy”) comes about via **Lexicon Optimization** (Prince & Smolensky 1993, Inkelas 1994), wherein a new interpretation of the lexical entry is arrived at based upon the **perception** of the surface forms, rather than transderivational surface forms directly

¹³One might question the status of an onset without a following vowel, or syllables without nuclei. However, if onsets, syllable nuclei and syllables themselves are understood as no more (and no less) than prosodic constituents in the prosodic hierarchy, governed by the constraint hierarchy, then unfilled nuclei in certain contexts are just as plausible as unfilled onsets, common in the literature. Both of these potential situations depend solely upon whether the constraint hierarchy allows for them.

interacting with the constraint hierarchy or the *Eval* process (i.e., the **production** side of the grammar). This position again reinforces the constrained nature of the OT grammar and resists both interactions at multiple levels and the (re)introduction of serial mechanisms into the grammar and the theory. Transderivational processes will again serve only to introduce unwanted complexity and further paradoxes into OT (see also Hale & Reiss 1997).

A final derivational-style tactic which will be excluded from OT is again illustrated in Pater (1995). Because his original **StressIdent** constraint fails to account for a subset of words, he posits (p. 23-4) a “lexically specific version” of the constraint which he refers to as **StressIdent-g**. While his goal in doing this, avoiding cophonologies, is a positive one, this particular solution is equivalent to unstructured exception marking or lexical rules in derivational theory. The words Pater is concerned with will be accounted for without such tactics in chapters four, five and six. While constraints that refer to specific morphologically defined structures, such as subcategorization constraints, are supported in the conception of OT offered here, constraints that apply only to an otherwise heterogeneous list of words, yielding in this way the correct surface forms, are not. Words that appear to show exceptional behavior must be accounted for rather by the presence of structures in their lexical entries, and such structures are limited to members of the constituent hierarchies, yielding a highly constrained theory of grammar.

1.5 Exceptions and the lexicon

Regardless of how well the linguist designs a traditional generative grammar for a given language (whether derivational or OT-based), there will always be some words which fail to behave as predicted. Chapter seven investigates the nature of such exceptions, and how they can be accounted for in OT. Some words are regarded as exceptional because of assumptions about the nature of their underlying forms. For example, because the surface forms of underived words is assumed to be identical to its underlying form, words like *vanílla* are understood as showing exceptional stress (cf. *órigín*). The best solution for this type of problem involves what Inkelas, Orgun & Zoll (1994) refer to as **prespecification**. In this case, the lexical entry is understood as containing a geminate /l/, whose effects surface only in the surface stress of

the word. Other cases of long segments (i.e., long vowels) result in a melodic as well as a stress contrast (e.g., *horízon* vs. *óorigin*).

Inkelas, Orgun & Zoll (1994: 28-29) regard prespecification as the best explanation of this type of exception within OT because:

On this theory, exceptions differ from “non-exceptions” ... by possessing some underlying phonological structure causing them to resist or trigger a different class of phonological alternations than morphemes not similarly specified undergo. Crucially, alternations are governed by phonological structure, not by rule or grammar features. The contrast between a “regular” and “exceptional” morpheme is formally indistinguishable from that between any two phonologically distinct morphemes. Membership in a static subregularity and lexical exceptionality are formally identical, a species of simple lexical variety... The second advantage of prespecification theory is that it offers the only workable approach to morphologically complex, phonologically heterogeneous words... A third significant advantage of prespecification is that it can locate and identify the exceptional area of the morpheme. (pp. 28-29)

Following this approach, regular and exceptional morphemes of this kind differ only in respect to structure, and are otherwise treated identically in the *Gen* and *Eval* processes. The exceptional structure can be identified and located in relation to the segments of the morpheme, and should be expected to appear consistently wherever that morpheme appears. Limiting the structures involved to members of the constituent hierarchies limits the types of exceptionality and helps to predict what kind of exceptionality is operating. Following the proposals made in chapters four, five and six, most exceptions to stress placement and vowel shortening can be accounted for by the presence of a lexical mora in the exceptional words, attached either to consonants (unexpected stress on an apparently light syllable) or vowels (unexpected failure of “vowel shortening” environments).

Another, less frequent, type of exception involves unexpected subcategorization for affixation. While most instances of affixation bring predictable prosodic effects attributable to subcategorization, some words display properties which are not expected on the basis of their presumed morphological structures. However, these properties are always identifiable with some other existing structure in the language. For example, the noun *récord* does not show the expected prefixed noun structures seen in e.g.,

repórt or *défect*, which foot both the prefix and the stem, but rather a single foot with unlengthened prefix syllable. Such a structure would be expected for a unitary stem like *légend* or *brígang*, but not for a prefixed stem. The combination of /re/ and /cord/ to produce a noun has arbitrarily (lexically) been governed by an unexpected subcategorization. Other exceptions of this kind include words like *impréginate*, which shows an unexpected stress on the penult, normal in words suffixed by, e.g., /-al/, but almost absent from words in /-ate/, which normally show penultimate stress regardless of syllable weight. This is again accounted for by an unusual subcategorization, the suffix /-ate/ in this case arbitrarily subcategorizing for the stem rather than the morphological word (§ 6.1).

A third type of exception involves words whose “input” morphemes themselves present an unexpected segmental form. For example, the pairs *destróy* ~ *destrúction*, *retáin* ~ *reténtive* cannot be linked to common underlying forms without proposing constraints which will yield erroneous results when applied generally to the lexicon. Furthermore, it should be clear that the selection of English morphemes in combination is in general arbitrary. For example, the verbs *compóse* and *propóse* are both built on the root /pos/, and their derived nouns *còmposition*, *pròposition* show similar morphology, but the further nouns *propósal* and *compónent*, the latter of which shows unusual allomorphy, are not predictable either in their structure or meaning. While words showing a common morphological root are often relatable semantically, their specific denotations and their specific morphological structures are not predictable. Other morphological entities, such as some prefixes (e.g., *ab-*) and roots (e.g., *-céive*), have no consistent meanings associated with them, although they might show a predictable behavior morphologically (e.g., *concéive* ~ *concéption*, *decéive* ~ *decéption*, etc.).

To account for these different types of arbitrariness, subcategorization is again applied. Just as Inkelas (1989) proposed that all morphemes have subcategorization frames, in an OT context all morphemes require at least one constraint associating their melodic segments with some morphological structures. McCarthy & Prince’s (1993a) subcategorization constraints for affixes are just a special case of the subcategorization constraints necessary for all morphemes to combine. Thus, while most roots will be governed by a predictable constraint of the type *Align(/limit/, R; Root, R)*, linking the

appropriate segment string to the root category, some roots will require an additional constraint which demands a prefix:

- (1.20) Align(/ceive/, R; Root, R)
 Align(/ceive/, L; Prefix, R)

Furthermore, the fact that semantically related words with identifiably related but nonidentical “input” strings exist (e.g., *destróy*, *destrucción*) suggests that there is some semantic element involved in selecting morphemes and their subcategorizations. For example, if a semantic constituent, or **lexeme** {destroy} is proposed to represent the concept behind these two words, the following complex subcategorization constraints might be used to account for them:

- (1.21) Link({destroy}{Verb}, /de-stroy/)
 Link({destroy}{Noun}, /de-struct-ion/)

The constraint **Link** here simply refers to an unspecified associative relation, or simply morphological selection. Such a constraint chooses the root morpheme, its content-less prefix, and the nominal suffix in question, based on the semantically defined identity of the word in question. Following the same approach, a “regular” form would require less complex subcategorization constraints:

- (1.22a) Link({act}, /act/)
 Link({Noun}, /-ion/)

The constraint given in (1.22a) is a minimal association between a lexeme and the segmental string which consistently carries its meaning, while (1.22b) is a more generally applicable constraint for creating nominals, which can account for many more forms. Such constraints will do for “regular” words, but irregulars will require more complex and specific constraints such as those given above in (1.21). These constraints will have to be ranked above the more generally applicable constraints to have an effect upon the grammar.

Accepting such a proposal leads to a single conclusion: that all material traditionally ascribed to the lexicon in generative approaches must be encoded, in one way or another, into a series of subcategorization constraints of this type. In other words, the lexicon is entirely contained within the constraint hierarchy itself. This follows a proposal by Russell (1995) that constraints should be used to represent the lexicon itself as well as the processes which govern word-formation. Besides accounting for the various types of subcategorization required above, this approach also makes it easier to treat underspecification effects (§ 7.3.3). Deriving the grammar exclusively from constraints within the hierarchy allows for all grammatical relationships to be expressed through the properties of the constraint hierarchy, such as ranking.

Constraints which refer to complex structures, like (1.21), or subcategorization constraints which refer to certain affixes, are referred to here as **specific** constraints. These can be contrasted with **general** constraints, which refer only to grammatical constituents (§ 7.3.2). Constraints like (1.22b) fall into this category, as do most traditional OT constraints. Exceptions arise when specific constraints outrank their corresponding general constraints; the converse relation means that the specific constraints in question have no effect and do not participate in the grammar. Since the number of potential specific constraints is astronomically high, while the number of general constraints is limited by the number of constituents in the grammatical system, it is assumed that only high-ranking specific constraints, which would have an effect on the grammar, are understood to be present in the grammar. The fall of a specific constraint below its corresponding general constraint in the hierarchy effectively eliminates it from the grammar, and this is how regularization would be represented in this framework.

In §§ 7.2-7.3, such a conception of the lexicon is contrasted with previous thinking. The traditional generative lexicon, which treats exceptions and regular forms through two distinct types of processing, is contrasted with Bybee's (1995) proposal of a lexical network to model the relations between words, which takes into account type (pattern) and token (word) frequencies while rejecting discrete morphemes and process-oriented word formation. Each of these competing models is seen to focus on different sets of grammatical relationships, the generative model (like early OT) focusing on the prosody/morphology relationship, while Bybee's network model focuses

on the morphological and semantic connections between words. It is suggested herein that both models can be superseded by an OT grammar as outlined above, with regular correspondences captured by general constraints, which are indicative of high type frequencies, while exceptions are maintained by highly-ranked specific constraints supported by their high token frequency. The only true “input” forms are lexemes from a third **semantic constituent hierarchy**, which account for both the selection of morphemes and their subcategorization. While such selection is often consistent per lexeme, there are instances in which high-ranking selection constraints will arbitrarily choose allomorphs like /de-stroy/ and /de-struc/ for the same lexeme. It is these selected morphemic structures which comprise what has been until now treated as the OT “input string”. Using the constraint hierarchy to model such relationships makes it logically equivalent to Bybee’s lexical network, but more effective since a wider range of relationships (including purely prosodic ones) can be captured, and all relationships can be described formally through constraints and their constituent arguments.

A further advantage of this approach is the presence of a built-in evaluation metric, which allows, unlike the previous approaches, for the complexity of a proposed grammar to be adequately quantified. The number of general constraints, which define the high-frequency relationships between members of the constituent hierarchies, should remain more or less constant between grammars (both cross-linguistically, and between competing grammars for a single language), as these relationships must always be specified. This follows the idea of McCarthy & Prince (1994) that all constraints are present in all languages, with the added proviso that only general constraints are meant. It is then the specific constraints which capture the idiosyncrasies and complexities of a real language; it is in fact preferable that such constraints are not thought to be present in all grammars. Specific constraints correspond to the unpredictable elements in the grammar, such as subcategorization, morpheme selection, and allomorphy. General constraints, driven by regularity (type frequency), are selected by the process of Lexicon Optimization to maximize the number of forms which can be accounted for by the grammar. In the least complex grammar, only those aspects of the language which are truly unpredictable via the general constraints will require specific constraints, enforced by the high token frequency of the

irregular words which require them. The number of specific constraints necessary to produce the grammar is thus an indication of its complexity, and the grammar which minimizes the number of these specific constraints will be the least complex for a given language.

The final sections of chapter six deal with the further implications of the model outlined above for the theory. This proposed OT allows for the representation of the entire grammar through the constraint hierarchy, dispensing with the heterogeneous system of traditional generative theory, with its independent lexicon and morphological and phonological rules. This promises a more constrained system, but also then depends heavily on the constraints themselves. The need for a formal system of constraint definitions is discussed, although an exhaustive treatment is left to future work. Proposals for dealing with what is traditionally described as the post-lexical level are also introduced, although segmental phonology will not be elaborated upon in this work. Since a single constraint hierarchy must account for all relationships in this approach, the different correspondences captured previously in Lexical Phonology by level ordering must be accounted for instead by the use of hierarchically arranged constituents. A distinction between morphemic phonemes and prosodic segments (and their corresponding features) is proposed to account for various levels of segmental relationships.

The goal of the theory pursued here is the representation of the maximum number of grammatical relationships through the use of the minimum number of theoretical constructs, in this case the members of the three grammatical hierarchies and the mechanisms of the constraint hierarchy within Optimality Theory. By proposing a system in which the majority patterns can be produced by one ranking of the general constraints, while the minority patterns can be generated by lexically (through “lexical” subcategorization constraints) specifying some minimal element of structure (prosodic or morphological) for the underlying morphological forms involved, an argument is made both for the validity, productivity and elegance of the proposed model, and against the disconnected and arbitrary mechanisms of previous approaches.

2. Alternations in English

2.1 Patterns and regularity

As in derivational theory, regular morphology in OT involves those words whose proposed lexical “input” forms yield the correct “output” surface forms, while irregular morphology denotes the set of words for which the expected underlying forms would not produce the correct optimal candidates. Unlike the derivational approach, however, OT does not provide any expression for exception marking.¹⁴ Instead, irregular words will have to provide their surface forms on the basis of some lexical input which will produce the optimal candidate. Under this definition, only regular stems will be mappable to a single “underlying” morpheme, and a word’s irregularity may be based on the fact that its apparent lexical form must necessarily differ from those seen in related words. However, since the constraint hierarchy must produce all surface forms of the language, a proposed hierarchy that would necessarily prevent an attested surface word from appearing cannot be the correct hierarchy for that language. The hierarchy of general prosodic constraints must be by definition minimally violable for all surface forms.

To describe the interactions of constraints that take both morphological and prosodic categories as their arguments, forms showing irregular morphology must be removed from the set of data. Defining a complete constraint hierarchy (or grammar) for English thus involves determining which patterns are general and the result of the interactions of prosodic and morphological constraints, and which forms rather have idiosyncratic morphological structures. Once the irregular forms are removed from consideration, the patterns and sub-patterns found in the regular cases must be categorized and accounted for by known constraint interactions, and a hierarchy must be developed which yields all but the irregular cases.

The issue of phonological alternation in English will be explored in this chapter by looking at a class of moraically-characterized vowel

¹⁴While one could propose a pseudo-constraint which fails to apply to certain diacritically marked words, this makes a mockery of OT’s principles of universality, inclusiveness and parallelism, i.e., the universal application of constraints to all candidates (§ 1.3.1). It also violates a central principle proposed in this work, that constraints may only refer to members of the constituent hierarchies (§ 1.0).

alternations found in the English data. These vowel alternations, which appear to interact with prosodification processes expressible via the constraint hierarchy, will be presented and previous treatments of the data will be discussed. The computational corpus investigation undertaken for this study will be introduced and described, and data from that study will be used to illustrate the problems inherent in previous approaches, including solutions couched in Optimality Theory. The question of vowel length alternation will be shown to be dependent upon stress placement, which in turn suggests that the prosodic category “foot” and the constraints which take it as an argument will ultimately provide the explanation for these alternations. Evidence for patterns in the English stress system and the majority and minority stress patterns seen in the data will be investigated in the following chapter.

2.2 Investigating regular patterns in English

Despite the importance of alternations as evidence for particular constraint hierarchy configurations, there is actually not a great deal of stem alternation in English. The vast majority of words in the sample corpus do not provide examples for vowel alternation or stem allomorphy. Most suffixation in English, viewed in terms of surface phonetic realizations, appears to be simply concatenative. It might seem possible then, that we could simply follow Vennemann (1974) and just list the cases of apparent alternation, writing them all off as due to multiple stems. However, a prosodic analysis of English is necessary to address the issue of English stress patterns, and as will be illustrated below, generalizations seen across the categories required to describe English prosodic structure will provide the means for explaining many apparent phonemic alternations seen across morphemes. Particular alternation will be focused in the sections below upon vowel alternation, of interest because it involves both prosodic (moraic) and morphological components and is intimately connected with the question of stress and the structure of the prosodic system in English. Previous attempts to account for the alternation in a generative context will be discussed; in the course of the discussion some other relevant phonological alternations involving consonant loss and palatalization will be discussed in the context of the theories to be described.

2.2.1 Vowel alternation within the derivational paradigm

Myers' (1987) analysis is useful as a starting point for a description of the problem, as he presents a summary of the traditional environments for vowel alternation in English, as well as an account of the processes involved, from a derivational perspective. The terms used below in the discussion of the traditional derivational approach will be, of course, couched in the language of derivational theory. The surface vowel alternation in question, between diphthongs and simple lax vowels, has been concisely described, in abstract phonological terms, as an alternation between long and short vowels (§ 1.1), that is, vowels distinguished by the presence of phonemic vowel length, which would be described in prosodic phonology in terms of the presence or absence of a prosodic constituent, the mora. Myers identifies the following types of vowel alternation found between morphologically related words. The sample pairs given below are taken from his much longer lists. The first three sets involve regular alternations based on suffix-type:

(2.1a) Consonant-initial suffixes (Myers 1987: 488-9):

(/-t/, /-th/, /-tive/, /-tion/, /-ture/)

keep / kept	wide / width
describe / descriptive	convene / convention
discreet / discretion	revise / revision
scribe / scripture	five / fifth

(2.1b) Multisyllabic vowel-initial suffixes (Myers 1987: 494-5):

(/-ity/, /-ative/, /-itive/, /-atory/, /-ible/, /-able/, /-ify/, /-itude/, /-ual/, /-acy/)

sincere / sincerity	sane / sanity
derive / derivative	appeal / appellative
prime / primitive	define / definitive
exclaim / exclamatory	oblige / obligatory
creed / credible	placate / placable
mode / modify	vile / vilify
sole / solitude	grateful / gratitude
grade / gradual	rite / ritual

supreme / supremacy conspire / conspiracy

(2.1c) Monosyllabic vowel initial suffixes (Myers 1987: 499-500):

(/-ic/, /-id/, /-ish/, /-ule/)

cone / conic

state / static

pale / pallid

vapor / vapid

final / finish

Spain / Spanish

globe / globule

node / nodule

In all these cases, the addition of a suffix to a “base” (unsuffixed) form showing a long vowel apparently causes the vowel of the base form to shorten. These suffixes are, for the most part, those described in Lexical Phonological work as level I, the stress-affecting suffixes (Kiparsky 1982a, b, Halle & Mohanan 1985). These contrast with the level II affixes, such as /-ship/, /-ness/, /-ing/, which do not affect the location of main stress.

The shortening environments described above are familiar from Chomsky & Halle (1968), who proposed three separate shortening (in their terms, “laxing”) rules to cover these cases, namely **CC Shortening** for set (2.1a), **-ic/-id/-ish Shortening** for set (2.1c), and **Trisyllabic Shortening** for set (2.1b). CC Shortening shortened underlying vowels appearing before two consonants, -ic/-id/-ish Shortening shortened vowels before that set of suffixes, and Trisyllabic Shortening shortened stressed vowels that were followed by two unstressed syllables. These rule environments reflected generalizations that Chomsky & Halle drew from their own English data.

Additionally, Myers notes the shortening seen in what have been called the “Latinized” prefixes, in suffixed words of the following kind¹⁵ (p. 500):

(2.2) re-fúte / ré-fut-ation

de-fórm / dé-form-ation

pre-sént / pré-sent-ation

pro-víde / pró-vid-ent

¹⁵It should be noted that some words of this type do not show this shortening in some British pronunciations, e.g., *déformation* / dè-form-ātion/.

In these words, addition of a level I suffix to the base form apparently causes a long vowel in the prefix to shorten; furthermore, the main stress of the word shifts off the base to the prefixal syllable.

The preceding cases may be compared with a series of vowel-initial suffixes of level I which, although they may affect stress assignment, do not produce such vocalic alternations; the suffixed forms contain the same vowel as the base forms:

(2.3) Suffixes producing no length alternations:

(/-al/, /-or/, /-ory/, /-ary/, /-ous/, /-ive/)

scribe / scribal

tone / tonal

advise / advisor

propel/ propellor

advise / advisory

supervise / supervisory

fame / famous

desire / desirous

These suffixes have a similar level of productivity and distribution in the corpus of English to that of the “shortening” suffixes noted above.

2.2.2 Stress assignment and vowel quantity

According to Myers (1987: 486), “vowel quantity in English depends to a great extent on stress.” This statement encompasses a range of notions, from the fact that Chomsky & Halle’s shortening rules referred explicitly to stress, to later observations, described in chapter one, that sound change occurs in the context of prosodic constituents. As was described above, the earliest metrical accounts of English stress, such as Liberman & Prince (1977), Hayes (1982), and Selkirk (1980), moved toward doing away with the segment-based stress rules of Chomsky & Halle (1968), replacing them with prosodic or metrical analyses. Vowel alternation was clearly being related to stress in these analyses, and in the relevant literature that touches upon this problem (Chomsky & Halle 1968, Myers 1987, Halle & Vergnaud 1987, Prince 1990, Sainz 1992, Burzio 1993), stress is always a factor in the formulation of vowel shortening rules. Consequently, such accounts are necessarily dependent upon their theoretical framework for stress assignment in their explanation of vowel alternation.

However, the early metrical accounts of English stress conspicuously did not address the issue of vowel alternation. For example, the influential account of stress given in Hayes (1982) does not treat any cases involving

vowel alternation. Since Chomsky & Halle's vowel alternation rules were dependent upon their stress rules, improving upon their stress rules invited similar improvement for the vowel alternation rules. For the same reasons, discarding Chomsky & Halle's segment-based approach forced later studies to discard their formulation of the shortening rules; Myers (1987) attempted to use a prosodic category, the syllable, to help achieve the same results. Myers followed the prosodic theory of Selkirk (1984), which did not recognize the foot as a category, and based its approach to stress upon syllables and a metrical grid. Thus, the only relevant prosodic category available to the analysis is the syllable, and any rules which affect segments are bound to the syllable domain.

2.2.3 The account of Myers (1987)

Myers' (1987: 488) solution proceeds from an important generalization that he noticed concerning the distribution of vowel quantity in the English lexicon : “within roots, *long vowels appear only in open syllables*. A nonfinal closed syllable must generally have a short vowel...” Single final consonants in English, as in *tape* or *seem*, are regarded as extrametrical, following Hayes (1982), leaving these syllables “in effect open” as well. True closed syllables in final position must therefore end in two consonants, e.g., *predict*. The small class of words, such as *basic*, *deictic*, *danger*, which unexpectedly show long vowels in the shortening environment are regarded by Myers as “idiosyncratic lexical exceptions,” which he lists in an appendix to the article.¹⁶

On the basis of this generalization, Myers (1987: 491) proposes a rule of **Closed Syllable Shortening** which shortens a long vowel before a tautosyllabic consonant, and notes that this simply restates, in slightly different terms, earlier solutions given by Stampe (1972) and Chomsky & Halle (1968) (“CC Shortening”). As this rule also appears to represent a “constraint” on the lexicon itself, Myers (1987: 511) formalizes this restriction by using a syllable template proposed for English by Borowsky (1986), wherein a well-formed syllable has the shape C* V (X), where X can be fulfilled by C or V (C* indicates an optional onset). This template, a mechanism used in pre-

¹⁶Myers excuses a large group of words, such as *pint*, *beast*, *round*, as ending in an extrametrical cluster of final coronals, following Kiparsky (1981).

moraic approaches to syllable structure, allows only for one optional post-vocalic timing slot, which can be either filled by a consonant or the “second half” of the long vowel, but not both.

The consonant-initial suffixes from (2.1a) above, (/ -t/, / -th/, /tive/, /-tion/, /-ture/), when added to underlyingly long “roots” with final, extrametrical consonants, yield a situation in which the root-final consonant closes the syllable. No longer word-final, it cannot be extrametrical, following the peripherality constraints on the application of extrametricality from Hayes (1982), and thus triggers the shortening rule. In terms of the syllable template, once the formerly extrametrical consonant is syllabified as part of the word-final syllable, linking to the final ‘C’ slot of a (C)VC template, the long vowel loses its second timing slot in the template, and thus “shortens”:

(2.4) *convene : convention*

CVV		*CVVC		CVC
con. ve:<n>	+	tio<n>	->	con. ve:n. tio<n>
				->
Suffixation		Loss of Extrametricality		Shortening

For the vowel-initial suffix cases listed in (2.3), such as /-al/, /-or/, /-ory/, the formerly extrametrical final consonant resyllabifies as the onset of these onsetless suffixes, and thus does not affect the shape of the preceding syllable. As a result, no shortening takes place:

(2.5) *fame : famous*

fa:<m>	+	ou<s>	->	fa: mou<s>
Suffixation		Loss of Extrametricality		

Having established a shortening environment, Myers moves on to a less transparent set of cases: those in which a vowel-initial suffix triggers shortening, such as the examples in (2.1b). For such cases, Chomsky & Halle (1968) proposed the rule of Trisyllabic Shortening (or Laxing); Myers instead suggests that these cases are once again instances of Closed Syllable Shortening. He notes that, in every case, “the shortened vowel is always in a stressed syllable immediately followed by an unstressed one.” (Myers 1987: 495) He cites, among others, Stampe (1972), Selkirk (1982b) and Borowsky (1986) for evidence of a “stress-sensitive rule of **Resyllabification**”, which

resyllabifies sequences like V'. CV as V'C. V. Thus, a form like *vilify*, derived from the monosyllabic adjective *vile*, would undergo the following type of change:

$$(2.6) \quad \text{ví:} \langle l \rangle + i. \langle fy \rangle \rightarrow \text{ví:} . li. \langle fy \rangle \rightarrow \text{ví:l.} i. \langle fy \rangle \rightarrow \text{víl.} i. \langle fy \rangle$$

Suffixation Resyllabification Shortening

After Resyllabification, the stressed syllable has become closed, and the long vowel shortens by Closed Syllable Shortening.

The suffixes in (2.1b) condition the Resyllabification rule because they are all analyzed as disyllabic and unstressed, with final extrametrical syllables, following the principle of affix extrametricality (Hayes 1982). A parallel solution applies to the words with monosyllabic vowel-initial suffixes from set (2.1c) above (/-ic/, /-id/, /-ish/, /-ule/), because, again according to Hayes (1982), these suffixes are idiosyncratically exempt from suffix extrametricality. This exemption produces an identical environment for shortening:

$$(2.7) \quad \text{stá:} \langle t \rangle + ic \rightarrow \text{stá:} . tic \rightarrow \text{stá:t.} ic \rightarrow \text{stát.} ic$$

Suffixation Resyllabification Shortening

Although Myers does not explicitly discuss the suffixes that do not cause shortening, set (2.3) above (/-al/, /-or/, /-ory/, /-ary/, /-ous/), presumably these cases are not subject to Resyllabification because the entire suffix remains extrametrical throughout the affixation process.¹⁷ Thus, revising (2.5) above, root-final consonants remain peripheral, and never lose their extrametricality:

$$(2.8) \quad fa: \langle m \rangle + \langle ous \rangle \rightarrow fa: \langle mous \rangle$$

Finally, Myers similarly handles the cases of prefix shortening, set (2.2) above. To create the proper stress environment for Resyllabification, Myers invokes a rule of Sonorant Destressing (cf. Hayes 1982, Kiparsky 1979) which

¹⁷For the treatment of /-ary/ and /-ory/ as monosyllabic suffixes, see, for example, Chomsky & Halle (1968), Liberman & Prince (1977), Kager (1989).

destresses a heavy syllable containing a sonorant when following a stressed syllable:

(2.9)	de: . fé<r> + <ence> ->	dé: . fe. <rence>
	Suffixation	Sonorant Destressing
	-> dé:f . e. <rence>	-> déf. e. <rence>
	Resyllabification	Shortening

Thus, Myers has converted the three shortening rules of Chomsky & Halle (1968), i.e., CC Shortening, -ic/-id/-ish Shortening, and Trisyllabic Shortening, into one rule of Closed Syllable Shortening, relying on extrametricality and Resyllabification to create the proper shortening environments.

2.2.4 Problems with Myers' analysis

While Myers' account is internally consistent, there are reasons even within the terms of a derivational theory to question a number of his mechanisms and assumptions. One problem is noted by Myers himself (1987: 514-5) : for roots ending in two consonants, the following situation would be reached upon Resyllabification:

		CVC		*CVCC	
(2.10)	fálse + ity	->	fál. si. ty	->	fáls. i. ty
		Suffixation		Resyllabification	

At the stage where the “root” syllable is /fáls/, we should expect a compensatory shortening parallel to that found in long-vowel roots, due to the syllable template's failure to license CVCC, in which a consonant would be deleted, e.g., *facity or *fality. To remove this false prediction, Myers proposes for English a **structure preservation** condition which preserves timing units corresponding to separate feature matrices. Thus, consonants will not be left without a timing slot, as in the case of *falsity*, but long vowels may lose a timing slot without completely delinking the feature matrix of the vowel in question. As a similar deletion of consonants does occur at level II (e.g., *sign*, *bomb*, *damn*), Myers' constraint must apply only to level I.

To preserve a consistent treatment of the syllable template,¹⁸ Myers regards this constraint as “undoing” Resyllabification. Since the onset is not lost in these cases, Myers assumes that it returns to its place. The rule is presumably also blocked in the case of exceptional words like *notify*, *basic*, which unexpectedly do not shorten their vowels in the conditioning environment. One way to account for this might be to idiosyncratically mark their final consonants as not being subject to Resyllabification. At this point, it appears that Resyllabification, introduced as a general syllabification rule, seems only to function in those cases where an explanation for vowel shortening is needed.

Myers cites other evidence for Resyllabification, but there are difficulties with these as well. He notes that Selkirk (1982b) claims that the flap [ɾ] is restricted (in American English) to syllable-final position as an allophone of /t/, and appears in ambisyllabic position “by virtue of Resyllabification. The /t/ is attracted out of the unstressed final syllable and tacked onto the end of the preceding stressed syllable, where it is transformed into the syllable-final allophone [ɾ].” (Myers 1987: 496).

However, directly linking Resyllabification and the appearance of [ɾ] is fraught with problems. Firstly, the assumption that [ɾ] is the syllable-final allophone of /t/ does not stand up. Word-final /t/, while also syllable final, is never flapped unless it becomes the onset of a following vowel-initial word, e.g., *get out* [gɛɾaʊt]. In many cases where word-internal /t/ is clearly syllable final, as in *atlas*, *Atkins*, flap is again never encountered. Syllable-final /t/ produces glottalized [tʔ] rather than flapped [ɾ] (Kahn 1976). Selkirk (1982b) needed to introduce a new feature [release] to distinguish between flapped and glottalized syllable-final /t/ in order to maintain her claim. If the unlikely proposal that flap is the syllable-final allophone of /t/ is dropped, following Kahn (1976), Resyllabification becomes an obstacle to flapping, because flapping would apply (as in Kahn) only to ambisyllabic /t/, not to /t/ in syllable-final position.

In addition, flapping extends beyond the stress environment delineated by Myers; words like *capital* [kæpɪɾəl] and *legality* [li:gælɪɾi], for example, show flaps (in some pronunciations, including my own) in the second rather

¹⁸Presumably, words like *function* and *juncture*, which have the illegal template *CVCC in their lexical entries, are somehow allowed to be exceptional.

than the first stressless syllable following a stress.¹⁹ Flapping is a post-lexical rule, occurring also across word boundaries, e.g., colloquial “I wanted to tell you” as [aɪ wàɪnədə télyə]. According to Kahn (1976: 95), flap can even appear across a word boundary before a stressed syllable, as in “ge[D] Ann home on time.” Kahn notes that flap may appear in all unstressed syllables, not just those following a stressed syllable. Thus, Myers' environment for Resyllabification is clearly not identical to that of flapping, and the phenomenon of flapping cannot be used to support Resyllabification.

Furthermore, the post-lexical nature of flapping would then suggest that the Resyllabification rule continues to be in effect beyond level I. But this would produce incorrectly shortened forms in the case of words like *boating*, *floater*, *motor*, unless the syllable template (which governs the shortening effect) simply fails to operate after level I. Similarly, Halle & Vergnaud (1987: 252-3) reject Myers' account of shortening through Resyllabification, due to the lack of any observable difference between the realization of resyllabified and non-resyllabified onsets, such as the /l/ seen in pairs like *phallus* and *phallic*. Additionally, in words suffixed with level I affixes like *mótor*, *fátal*, we should either expect vowel shortening or the absence of flapping. If the extrametricality of the suffixal syllable, as noted above, prevents the Resyllabification that would otherwise shorten the vowel, it should also prevent the Resyllabification that produces the flapping.

Additionally, linking Resyllabification and flapping implies that when Resyllabification is blocked, flapping should also fail to occur. Thus, lexical exceptions to shortening like *notify* should not show a flap, as the idiosyncratic failure of Resyllabification to apply to the /t/ of this root, preventing shortening, leads us to expect *[núɹtʰɪfàɪ] rather than the attested [núɹDɪfàɪ], with flap. Milliken (1988: 254-5) also demonstrates that a Resyllabification rule which feeds flapping cannot occur past level one without predicting incorrect results. If flapping is indeed produced by a resyllabification process, this process has no relation to that which putatively triggers shortening, and indeed seems to occur post-lexically rather than at level I. Positing a separate process of Resyllabification that applies at level I simply to corroborate an explanation for vowel shortening unnecessarily

¹⁹Borowsky(1986) tries to account for forms like *capital* by suggesting that the underlying form is /kapitl/, but admits that this approach cannot deal with forms like *divinity*, *repetitive*.

complicates the grammar, in derivational and generative terms, for the same process will reapply post-lexically in any case.

Myers also refers to Borowsky (1986) for evidence of Resyllabification. Borowsky (1986: 262) claims that deletions of the following type are also the result of Resyllabification:

(2.11) h-deletion	vehícular / vé[Ø]icle	prohíbit / pro[Ø]ibítion
y-deletion	annúity / ánn[i]ual	volúminous / vól[i]ume
palatalization	cónstitute/constí[č]utive	résidue / resíd[j]ual

In each case, stressed syllables show a different phonological treatment than unstressed syllables following a stressed syllable. Borowsky claims that when Resyllabified, /h/ is deleted, /ɪ/ is retained post-coronally (in American English this on-glide, a part of the triphthong often represented as /ɪuə/, is usually deleted after coronals, e.g., *tune* [tuən]), and coronals before such retained glides are palatalized, respectively. However, there are complications. In the first place, any deletion of /h/ or /ɪ/ will violate Myers' structure preservation constraint, even if it is restricted to level I as noted above. Myers' constraint requires all feature matrices to be associated with a timing unit (1987: 515). His shortenings involve deletions of timing units rather than feature matrices. Furthermore, both the palatalization environment and the retention of /ɪ/ in the preceding syllable require an illegal (C)VCC syllable template (e.g., *annual* /ánɪ. u. al/). According to Myers' explanation of words like *fálsity* (2.10), the glide should be returned (or confined) to the onset position, where, following Borowsky, we would expect it to be deleted. Additionally, the proposed Resyllabification of /h/ fails to shorten the preceding vowel.²⁰

Finally, the rule of Resyllabification may be criticized for a more fundamental reason: it removes onsets from syllables, creating onsetless syllables which follow closed syllables. This option is specifically prohibited by the **Onset Principle** of Itô (1989), who also notes that such a syllabification pattern never occurs in any known language. Following this, it should be clear that Resyllabification is not a workable solution; to accept a violation of

²⁰Although one could also argue that the vowel was shortened and then re-lengthened after deletion of /h/.

the otherwise universal Onset Principle, we would require much stronger proof than that offered by these accounts.

Myers, in a footnote (p. 496), allows for an alternative explanation of the facts of Resyllabification via a rule of **ambisyllabicity**, as defined in Kahn (1976). This would allow the onset in question to remain attached to its syllable, while simultaneously intruding into the preceding syllable and shortening the vowel. This has the advantage of removing the objection against violating the Onset Principle. However, Kahn's environments for ambisyllabicity, like the environments for flapping, are not the same as Myers'. If ambisyllabicity causes shortening in preceding syllables, we would again have difficulty explaining commonplace level I exceptions like *motor*, *total*, *votive*, not to mention failures to shorten seen in other strata across words and phrases with ambisyllabic onsets, e.g., *coating*, *eat a potato*. Once the other evidence for Resyllabification proves to be less than persuasive, we are left with a rule which shows no effects apart from the shortening which it was devised to explain.

An objection to another part of Myers' account is found in Kager (1989: 119). He notes that while Myers attempts to characterize the stress shift needed to explain forms like *déference* from *défer* through the "Sonorant Destressing" rule of Kiparsky (1979), this is not supported by the facts. Myers has modified Kiparsky's rule in two significant ways: a long vowel is treated as a "sonorant" for the purposes of the rule, and primary stress can be removed. Kiparsky's principal examples, such as *infirmary*, *dispensary*, contradict the second claim. But the first is also clearly violated by forms with real sonorants, such as *dépendent*, *fraternal*, as well as long-vowel forms like *reprisal*. The fact that no *-ent* / *-ant* words with true sonorant consonants in the relevant position show this destressing is a clear indication that "Sonorant Destressing" is not responsible for the *déference* type. In fact, only roots with long vowels in their unstressed forms undergo this type of stress shift and shortening.

2.2.5 The assumptions of Myers' account

Myers' account of vowel shortening outlined above rests on a number of assumptions. One assumption involves the prosodic theory Myers was using, which relies on the syllable as the principal prosodic constituent. While this improves upon Chomsky and Halle's minimalist notion of the

segment as the only relevant phonological unit, the result is that rule environments are thus couched wholly in terms of segmental positions within and adjacent to syllables. The only phonological units available to do the theoretical work are timing slots and syllable positions like onsets and codas, thus the syllable-based explanation that is Myers' rule of Resyllabification.

Another set of his assumptions involves the representation of lexical items themselves, based on Structuralist notions of unaffixed surface forms as basic, as well as the derivational processes by which words were thought to be "built up". Myers follows the principles of Lexical Phonology (Kiparsky 1982a, b, Mohanan 1982) and its attendant derivational mechanisms, described above (§ 1.4). These assumptions about word-formation explains why Myers must posit, for example, "destressing" and "retraction" rules to account for forms like *résident*, because it is assumed that one must begin with a base form *resíde*, with a stressed long vowel in the final syllable. Discrepancies between "base" and "derived" forms require a complex sequence of both structure-building and structure-removing processes to dispose of unwanted structures present at various stages, regardless of the fact that both surface forms in the example noted above show simple and regular stress patterns.

A third major issue involves the data being used as evidence for both the underlying forms and the rules governing their behavior. Myers proposes sweeping generalizations based on a carefully chosen and limited set of data. The concept of "rules" implies that a generalization drawn from empirical observation can be viewed as the result of a phonological process applying across the grammar. Unfortunately, many of the sweeping generalizations found in Myers' and others' work on this topic are less than comprehensive.

For example, Myers offers *nátion* / *nátiona* as one of his first exemplars of vowel alternation, yet this case is exceptional; the regular pattern for words in /-ion-al/, which involves no shortening, can be seen in *emótion* / *emótiona*, *sensátion* / *sensátiona*. An idiosyncratic alternation such as this is thus not an example of a phonological process that should be explained by rule. In another case, Myers' principle of CC shortening alone relegates hundreds of forms, comprising about 5% of the lexical entries,²¹ to

²¹Figures cited here and below refer to the computational corpus investigation outlined in the Appendix and discussed further in chapter 3.

the status of lexical exceptions. Myers explains vowel shortening as due to the action of the onset of the “consonant-initial suffix” in cases like /-tion/, /-ture/, seen in group (2.1a) above, but does not mention that the consonant-initial suffix /-ment/, which appears to take part in level I suffixation (cf. *argumentation*, *departmental*), regularly forms words like *státément*, *impróvement*, *postpónement*, and never shows any shortening effects.

Myers lists /-able/, /-acy/, /-ative/, and /-atory/ as part of the “multisyllabic vowel-initial suffixes” of group (2.1b) that trigger shortening, along with more reliable candidates like /-ity/, /-ify/ and /-ible/. His example *plácate* / *plácable* is one of only a few forms in /-able/ showing apparent shortening effects; most of these words do not suffer shortening, as seen in *quótable*, *cápable*, *sízable*, etc. Similarly, his examples *conspíracy* and *suprémacy* are unusual in both their shortening (cf. *pápacy*, *prímacy*, *diplómacy*) and their stress, in light of their phonological structure (cf. *cándidacy*, *délicacy*). Citing the form *apéllative* does not in itself bring up the issue of why the same stem shows up in the unusual form *apéllate* (compare *aspíre* / *áspir[ə]te*, *désign[ə]te*), or why some words with this suffix come out stressed like *cònnótáitive*, *lègisláitive* or like *restóratíve*. A similar comparison can be made between his examples *explánatory* and *réspiratory*, not to mention the differently stressed British forms like *respíratory*.

Likewise, the shortening in prefixes noted in his examples *províde* / *próvident*, *defórm* / *déformation* must be compared with the different realizations seen in other words like *decédent*, *propónent*, *émérgeñce*, *précedent* (British pronunciation), and *prèsentátion*, *pròtestátion*, *rèlocátion*, etc., before a general rule can be proposed. That his use of “sonorant destressing” to account for the stress shift and shortening seen in words like *résident*, *próvident*, would overapply to forms like *reprísal* while failing to apply to words like *depéndent*, *propónent* has been already noted. The fact that so many words in the cases discussed above do not behave as predicted would thus force us to mark as exceptional the stems of a significant number, perhaps even a majority, of the words with the aforementioned suffixes.

Even in the case of suffixes that display shortening more often than they don't, there are a significant number of exceptions, about 12-15% of the stems involved overall, and a much higher percentage in the case of certain suffixes, that fail to conform to the “rule”. Some forms, as can be seen from the pair *strátégý* / *stratégitic*, even put a long vowel into a shortening

environment, when there was no long vowel in the unsuffixed “base” form. When such a significant number of exceptions exists, the question must be raised as to whether they can all be dismissed as “idiosyncratic lexical exceptions”, marked for exclusion from a general rule.

Even for an approach couched in a lexical phonological, derivational framework, which tolerates a degree of idiosyncrasy, generative principles still require a lexicon containing morphemes from which the attested forms can be reliably and economically generated. Marking entire sets of stems and suffixes as somehow “exceptional” raises the question of whether this kind of generative lexical phonology is at all viable. In any case, Myers’ particular analysis of vowel alternation fails to satisfy even the demands of its own quite loose theoretical framework.

2.2.6 Foot-based approaches

Myers’ account was constrained by the prosodic theory he adopted from Selkirk (1984), which focused upon the syllable as the relevant rule domain. Other accounts, such as Halle & Vergnaud (1987), Burzio (1993) and Prince (1990), acknowledge Myers’ generalizations about the distribution of vowel alternation, but recast the focus environment for the rule in terms of another prosodic category, the foot. Regarding the foot, like any prosodic constituent, as a potential rule environment, it is possible to better characterize, for example, the phenomena explained above using Borowsky’s Resyllabification rule, in terms of the foot.²² The generalization Borowsky and Selkirk had attempted to capture with their syllable-based rules was extended directly from Kahn’s insights, which first articulated, in the context of generative theory, the need for certain rules to refer to syllabic (and therefore prosodic) structure. However, in the cases mentioned above (§ 2.2.4, fig. 2.11), the prosodic constituent which best describes the domain for these rules is, rather, the foot, a theoretical entity not available to Kahn.

Regarding the cases outlined above in (2.11), /ɪ/ can now be seen as deleted in the foot-onset, bleeding the assimilatory palatalization which may occur elsewhere in the foot. Aspiration is shown to be a feature which marks

²² It should be clear from the flapping facts noted above that flap should not be treated as parallel with the other cases.

both the foot and word onsets. The environments for the phenomena noted by Borowsky can thus be restated descriptively as follows:²³

(2.12) aspiration: appears in the onset of the foot or prosodic word

ve (hícu) <lar>	vs.	(véhi)<cle>
[vɪhíkɹələr]		[víɹəkəl]
pro(híbi) <t>	vs.	(próhi)biti<on>
[prəhíbit]		[pròʊəbíʃən]

y-deletion: occurs after a coronal in the onset of the foot

an(núi)<ty>	vs.	(ánnu)<al>
[ənúɹitiɹ]		[æɹɹɹuəl]
vo(lúmi)<nous>	vs.	(vólu)<me>
[vəlúɹminəs]		[válɹəm]

Using the categories “foot” and “prosodic word”, processes previously explained by a questionable “Resyllabification” rule can thus be represented as conditioned by position within prosodic constituents.

2.2.7 Foot-based accounts of vowel alternation

Halle & Vergnaud (1987: 252-4) do not devote much space to vowel alternation, but, rejecting Resyllabification, regard the shortening effect as occurring in two separate environments: in closed syllables and in “the heads of binary metrical constituents”, i.e., feet. They see a parallel between the two cases in that both shortenings take place inside metrical constituents, either the foot or the syllable. Halle & Vergnaud, unlike Hayes (1995), do not regard closed syllables as feet in themselves, but as simply receiving a “grid mark” through their inherent weight. Burzio (1993) also recognizes that vowels appear to shorten in bisyllabic feet. However, he views this as part of a general principle (with many exceptions) that “vowels shorten in word formation”

²³Below, feet are denoted by parentheses. Feet not relevant to the discussion are not marked. Extrametrical syllables are marked with angle brackets, as above.

(p. 364), massively extending Kiparsky's (1982a, b) notion that underived forms are exempt from structure-changing cyclic rules (see § 1.4.1 above).²⁴

Prince (1990: 367-70), in a pre-OT analysis, provides a clearer motivation for the alternation, regarding the shortening seen in (2-3) above as the result of a tendency to optimize foot shape. Since the “best” trochaic foot shape, under his definition, is composed of two light syllables, the less optimal shape “heavy-light”, as seen in possible forms like *c^hónic, is shortened, resulting in the better “light-light” pattern of attested *cónic*. For monosyllabic feet, the heavy syllable of the type *c^hone* is the only possible foot, as the foot in such a quantity-sensitive system is defined as minimally composed of two moras (McCarthy & Prince 1990a, b, Hayes 1995). Prince thus improves upon Myers’ account, retaining the scheme of a common environment (the foot) for the words noted in (2.1-2.2) above, while avoiding problematic rules like Resyllabification. Utilizing the concepts of quantitative constraints and the optimization of foot-shape, he is able to present the various vowel alternations as natural and well-motivated. Recall also that Hayes (1995) explicitly rejected the foot type “heavy-light”, which he termed the “uneven trochee” (§ 1.2.2). Prince’s adjustment rule results in what is for Hayes the only permissible foot in English, the moraic trochee, and avoids the underparsing that Hayes would resort to in such cases. Prince & Smolensky (1993: 210-11) present a solution for English vowel shortening in Optimality Theoretic terms, using the constraints *_{smmm}, **Parse-Seg** and **Parse-m**:

(2.13)

/keep-t/	Parse-Seg	* _{smmm}	Parse-m
☞ ke<m>pt			*
keep<t>		!*	
kee<p>t	!*		

Here, the syllable is restricted to two moras by a negative constraint, and the low ranking of **Parse-m** allows for the underparsing of one of the underlying moras. Consonant deletion to yield the bimoraic syllable is blocked by the high ranking **Parse-Seg**. A similar solution, which follows the argument of

²⁴Burzio’s position is challenged in § 4.3.4 below.

Prince (1990) could be used to account for the shortening seen in bisyllabic feet:

(2.14)

/vilify/	Parse-Seg	*Ft _{mmm}	Parse-m
☞ (vi<m>li)fy			*
(viili)fy		!*	
vii fy	!**		

Changing the negative constraint to be based on the foot rather than the syllable (which would yield identical results in the case of monosyllabic heavy feet), the shortening seen in words like *vílify* can be accounted for.

Yet the argument given in Prince (1990), as presented in the above OT analysis, leave a number of questions unanswered. While the foot is undoubtedly the domain for these adjustments, it is not clear how all the required feet would be laid down in the first place, especially in cases like those seen in (2.2) above, the *résident* type, where a stress shift from the unsuffixed form is not explained. If the feet were laid down according to, for example, the foot-formation principles of Hayes (1995) discussed above, we would expect *re(síd)<ent>*, parallel to *re(cít)<al>* and *pro(pón)<ent>*. Additionally, the mechanism behind exceptional forms like *notify* and *basic* is not covered; presumably, these ill-formed “heavy-light” feet are exceptionally maintained, regardless of their non-optimal form. Thus, while the foot appears to be the correct prosodic constituent for descriptively locating stress in the word, the issues of how the feet are laid down onto the underlying representations and what subsequent processes result in shortening once again appear to be issues that can only be solved in the lexicon itself, that is, in the complete system encompassing the prosodic, morphological and semantic hierarchies.

2.3 The corpus investigation

As was mentioned above, Myers’ analysis, and those which preceded it, suffered from an imprecise treatment of the data. Very basic assumptions, such as the one that rule-based vowel shortening is supported by the data, are given little evidence other than citations of previous works, many featuring

varying assumptions of their own, and short lists of forms which provide the “evidence” for what is claimed to be a general process applying to the entire language. A look into the various studies reveals that the same short lists of words have been used in all the studies on vowel alternation, but the status of these words within the corpus of English data has not been rigorously investigated. Very broad terms, such as the claim that certain alternations are “regular”, always supported by positive evidence and ignoring the negative, have been offered and accepted. No statistical or computational investigative methods have ever been applied to measure the degrees of correspondence involved, the status of the exemplary words within the language, or the possibility that the apparent shortening effect was an epiphenomenon of some other process. The structuralist assumptions driving this view of the vowel alternation taken in works like Myers (1987) has been discussed above (§ 1.4.1).

This study attempts to address that deficiency through an exhaustive study of the corpus of English words, investigating both the usual wordlist used as evidence for vowel shortening, as well as their status within wider subsystems of the language. Using a corpus of English data to investigate the phonology of that language sounds like a straightforward proposition, but is of course a very complex undertaking. The need to carefully define what “the phonology” entails has been discussed at length above. Another entirely relativistic and potentially misleading body of information is a “corpus” of linguistic “data”. Using a list of words, or dictionary, as the source material for a study of a language assumes a linguistic status for a list of prosodically and semantically defined structures, taken out of their syntactic context. They provide only an indirect and somewhat abstracted model of the language itself, as it is actually used by speakers.

By using a specific corpus as a model of English, one also assumes that the corpus is sufficiently representative of the English language, in other words, that whatever material that fails to appear in that corpus is not necessary for the description of English phonology. The corpus used in this study, the Celex database of English (Baayen, Piepenbrock & van Rijn 1993), contains 52,446 lemmas (words regarded as basic forms, subsuming predictable variants such as plurals, preterites, participles), drawn from the COBUILD corpus, which consists of 160,594 individual words. The unabridged version of the Oxford English Dictionary, for example, contains more than

half a million words. This might suggest that samples such as the Celex database are not complete enough representations of the language.

However, the question then arises as to what the term “the English language”, the descriptive phonology of which is the aim of the investigation, signifies. No English speaker has a mental lexicon approaching the size of the OED; a significant number of its entries are archaic words that are no longer in use, or foreign terms that never attained currency among most speakers. Other words appear only in certain English dialects. Thus it could be said that such words certainly do not inform the phonology of the modern English speaker, who has likely never heard or used them. For example, Pinker (1994) cites a maximal figure of 84,000 “listemes” (lexical bases, equivalent to Celex’s “lemmas”) for English, with 45,000 being the “average” number for a given speaker. Such figures then suggest that the Celex corpus might suffice as a representative sample of the “mental lexicon” of an English speaker, or perhaps be even somewhat larger. This highlights the next assumption: the phonology sought is a representation of the language as it exists in the mind of a “typical” modern native speaker of English.

This is again a problematic assumption, however, because this “typical” speaker is a complicated fiction. English is one of the most widespread of the world’s languages, with a series of standards based on national and regional identities, and affected by dialects and regional variation. The Celex corpus contains data from British English, while the author of this study is an American speaker (Long Island dialect) who has lived in the United Kingdom (Edinburgh and Cambridge). I have decided to treat the phonology of English as broadly as possible, covering both the English and American standards. These standards are certainly mutually intelligible subsets of the “same” language, and they both must rest on a large base of common lexical items and phonological processes. Where they disagree, the differences will be highlighted, and attempts will be made to account for these differences in terms of the phonology and the lexical entries specific to these subsets. Other standards and dialects (not to mention other languages), except where individual words of such origin have been purposely included in the data as common variants in standard pronunciations or established borrowings, are

necessarily excluded. Thus the phonology to be investigated is that of abstract “typical” modern speakers of American and English standards.²⁵

This specification narrows down the types of English that are to be represented by the data. Another criterion that could be applied to the data necessary to such an investigation is that of word frequency. Words that frequently occur in a speaker’s experience must inform the phonology of the language (as in Bybee’s (1995) proposals, § 7.2.1), while words that are extremely infrequent might not even be known to large numbers of individual speakers. However, that being said, it is also necessary that the system used to describe the language can also account for the production of low frequency words. The Celex corpus appears to contain most, if not all, English words with appreciable currency in the modern language as used by the “typical” speaker. Any word not appearing in this corpus is not represented in the data, and at that point, one must yield to practicality and hope that using such an enormous corpus as the basis of the study is sufficient to reach its goal.

2.3.1 Research methods and data

To process such a large database, I designed and programmed two applications, implemented in Prolog on an Apple Macintosh computer. The first application was designed to convert wordlists of any shape, for any language, into a Prolog database consisting of structured objects. This flexible design was especially important as I used a number of other corpora at various stages of the research, eventually settling on the Celex corpus for its size and detail well after the project had begun. The second application allowed for complex searches of such converted databases according to morphological and phonological criteria including stress and syllable weight patterns, and the generation of contrastive tables cross-referencing the various patterns that were encountered. Thus, using the front-end application, I could, for example, collect the set of all words suffixed in /-al/ or /-ent/, the set of all words with three syllables, words with certain orthographic sequences, phonemic structures, or syllable shapes, etc.

²⁵The database contains various British pronunciations of the corpus; for the most part some or all of these correspond phonologically to American pronunciations. When they clearly do not, as in the case of words in /-atory/ and a few other sets, distributions for American English pronunciations have been calculated separately. See the Appendix for more detail.

For such subsets, I could then create tables outlining the various stress or weight patterns presented, or the morphological structures, complete with information about the distribution of such features across the word set or corpus, as well as the frequencies of the words in question. The program also allowed me to view the words classified as containing various characteristics, enabling me to remove spurious cases, and to create new subsets based on that information, combining or contrasting various other subsets. Information drawn from those tables will be presented in the Appendix, and cited throughout the analysis herein. Specific treatments which took account of idiosyncrasies found in the Celex data are documented in the Appendix. Some relevant issues with regard to using data of this type for a phonological investigation are spelled out in the following sections.

In the Celex list, as in most dictionaries and lexical databases, the entries are arranged as individual (prosodically defined) word forms, abstracted out of the many syntactic contexts in which they might be produced. They are accompanied by a quasi-phonemic “pronunciation”, indicating in some cases variant pronunciations. This is another variable which must be somewhat smoothed over in treating the data. Some of these variant pronunciations are dialectal, and must be regarded as parts of different sub-phonologies of English. Other listed variations in the pronunciations of single lexical entries are due to differing speech rates or the appearance of words in different syntactic contexts, as well as unsystematic alternate pronunciations of the same word within a single dialect system. Since these heterogeneous pronunciations are listed in a uniform manner, the material must be carefully sorted through before being used as evidence for any phonological processes.

The quasi-phonemic symbols used in the Celex database are a subset of the actual phonetic variations encountered in real data, but are more fine-grained than a phonemic system. The level of abstraction in these quasi-phonemes eliminates, for example, predictable “post-lexical” surface contrasts between allophones, such as aspiration of stops or release features, while it maintains other, potentially phonemic differences, such as the voicing of obstruents in certain positions, even though this voicing might be predictable (e.g., *exist* [egzɪst] ~ [eksɪst]). Variant pronunciations of this type are listed separately for each lexical entry. Another realm in which variants are explicitly listed is vowel quality. The vowel alternation described

morphophonologically in terms of length (the “Great Vowel Shift” set of alternations) is listed in terms of the component vowels and glides which make up the surface sounds, thus /ī/ is given a pronunciation [ai], etc. For cases of vowel reduction in unstressed syllables, the symbol [ə] is used, and syllabic liquids and nasals are often denoted by either representations like [ən], [ŋ], or both types listed as alternants. Sometimes variant pronunciations list the deletion of the schwa vowel in some forms, e.g., *delivery* /dəlivəri/ ~ /dəlivri/. In general, the representation of vowels is treated more “phonetically” than that of the consonants, perhaps indicative of the greater linguistic interest vowel alternations hold for investigators of English.

Another variation marked in individual database entries is that of stress pattern, also related to vocalic issues. For many words alternate stress patterns are listed, some conditioned by the position of the word in the sentence or phrase, for example the well-known contrast of the word *thirtéen* to the phrase *thirteen mén*, noted by Liberman and Prince (1977). The shift to initial stress in the latter case is the result of the word’s position in the phrase, but the Celex lexical entry, as with most dictionary entries, simply lists both variants without indicating the context. The stress levels noted in the database are main stress (“1”), secondary stress (“2”) and no stress (“0”). Unstressed syllables may show full or reduced vowels, suggesting another level of contrast, although there is sometimes little difference between the actual pronunciations (in many dialects) of some syllables listed with secondary stress and others listed with no stress but full vowels. Many words are listed with variant pronunciations that show full unstressed vowels in one form, and schwa or no vowel in another, suggesting these alternations are simply conditioned by other factors such as speech rate. To summarize, the raw material of the data, as it appears in such a database, is already idealized and abstracted out of the sentence context into “independent” words, and any phrasal, speech rate, and other conditioning effects originating above the word (or outside the phonological sphere, as in dialect issues) are necessarily left out of the descriptions. The only categorical contrast remaining in such cases is between words with multiple pronunciations and words without.

2.3.2 Vowel quality and vowel reduction

An investigation of vowel alternation must first define what the surface alternants of “the same” vowel, seen across the corpus, represent. The segments used in a list of lexical entries should ideally be defined by their correspondence with the surface sounds, based on conclusions about the morphology and phonology to be drawn by the researcher. These relationships should be grounded in recurring correspondences seen in the corpus of attested words. Thus the proposal, for example, of an underlying segment /o/ which shows prosodically contrasting variants /o/ and /ō/ for the forms *tōne* and *tōnic*, derives from the recognition of range of similar correspondences across the corpus in semantically and morphologically related words, as well as the understanding of the quasi-phonetic descriptions [ɑ] and [oʊ] (in American English) as prosodically conditioned reflexes of a more abstract melodic segment /o/.

This abstract /o/, with its moraic alternants, is supportable because a series of other words appears to show the same alternation, with the same surface vowel alternants. However, there are other alternations such as *abound* ~ *abundant*, in which a simple vowel also appears to alternate with a surface diphthong, but which are infrequent enough to call the proposal of a parallel prosodically based alternation of an abstract phoneme /u/ into question. The same point applies to cases like *join* ~ *junction*, in which the short vowel [ʌ] also appears to alternate with the diphthong [ɔɪ]. Must this be a “different” [ʌ] than in *abound*? Because this alternation occurs in so few words, the researcher may decide that it is simply idiosyncratic and demands an allomorphic rather than a phonological explanation. As the number of words suggesting an abstract segment increases, however, the correspondence moves into the realm of phonological alternation. There is certainly a connection between the number of examples and the type of explanation which must be used to account for the data, but in the absence of strict evaluation metrics, it is difficult to draw the line. There is no concrete way available in current derivational theories to measure whether an allomorphic or abstract explanation of such limited alternations is appropriate. In this study, the correspondences seen in the corpus will be noted and presented in term of percentages seen across the data. Only abstract correspondences which are strongly represented in the data (such as the

vowel alternation seen in *tóne* ~ *tónic*) will be accepted without detailed comment.

Another point, not clear-cut in the literature, becomes important for any study trying to use a set of data to draw conclusions about vowel alternations while working with nebulous terminology such as “vowel shortening”. This involves **vowel reduction**, which also appears to be conditioned by stress and should be regarded, in English, as distinct from true correspondences between phonologically “long” and “short” vowels. Some studies, such as Burzio (1993), conflate vowel shortening with vowel reduction (see § 4.3.4). However, vowel reduction appears to be conditioned by a different set of prosodic environments than traditional vowel shortening. Both “short” monophthongs and “long” diphthongs can be reduced in unstressed position, surfacing as [ə] or [ɪ]:

- (2.15) *cònnōtātive* /kànətéɪtɪv/ *átom* /ætəm/
 rèputātion /rèpjətéɪʃən/ *atómic* /ətámɪk/

Note that the triphthong /juu/, usually corresponding to orthographic ‘u’, retains its onglide when reducing, indicating that it has retained some of its phonological characteristics and is a surface-reduced triphthong rather than being phonologically shortened. Vowels like the /ō/ in *cònnōtātive* can appear both reduced and with quality, a variation which seems to be conditioned only by non-phonological factors such as speech rate or social register.

There are a wealth of examples for phonologically long vowel segments reducing (in terms of losing their quality) but not being phonologically shortened, a great many of these being seen in prefixes. It is clear from the attested English data that monosyllabic open prefixes with long vowels show variant pronunciations, in the position before the main stress, with either a reduced or a long vowel but never a phonologically short vowel:

- (2.16) *deférral* /dè-fár-əl/ or /də-fár-əl/ but not /*dɛ-fár-əl/
 retríeval /rè-trév-əl/ or /rə-trév-əl/ but not /*rɛ-trév-əl/

This contrasts with cases in which the prefix is stressed, in which case it appears short:

- (2.17) *déference* /dé-fər-əns/
 résident /ré-sid-ənt/

Here, for the sake of accurate description, only the correspondence between stressed short vowels with stressed long counterparts will be regarded as representing the traditional “shortening” relationship. Since some heavy closed syllables, which should receive their own feet under the bimoraic principle, nevertheless can appear with reduced vowels in unstressed position (e.g., *còm[pə]nsáte*, etc.), it appears prudent to regard the reduction of long vowels in unstressed position as parallel in the absence of other evidence for a “shortening”.

With regard to the data as listed in the Celex database, there is one further note to be made regarding this issue. While most cases in which full vowels are reduced to [ə] are listed as such, reduced, unstressed variants of /o/ and /ō/ are listed in the database as alternatively an unstressed surface [o] segment similar in its phonetic realization to the sound corresponding to phonemic /ō/. This apparently unstressed long vowel can always alternate with [ə], e.g., *dèsolátion*, *ídiolēct*. Since this descriptively “long” vowel surfaces in a position where it is unstressed, and where other vowels tend to reduce, it is likely that this [o] should also be regarded as a reduced vowel. A similar phenomenon is found to a lesser degree with another rounded vowel, the triphthong [juɤ]. This can appear in unstressed position variably as [ju] or [jə], the retention of its on-glide indicating the identity of the segment even when the vowel quality has been reduced to a surface schwa. 90% of the apparent “long vowels” in unstressed position listed in the Celex corpus are cases of /ō/ and /ū/ in this context, which suggests that this apparent vowel length can be safely abstracted away, and understood as a reduced vowel. The remaining few cases apparently do show long vowels in unstressed positions, e.g., *réndezvous* /róndəvùɤ/.

2.3.3 The shortening environments of Myers (1987)

Conducting a detailed investigation of the corpus as described above, Myers’ evidence for vowel shortening can be evaluated in the context of the

entire phonological system. Only then can generalizations be drawn about the mechanisms necessary to account for the forms which appear to be involved.

What the corpus reveals is that while many identifiable stems do show an alternation in the “shortening” environments described by Myers (1987) and discussed in the previous sections, a consistent percentage of stems, about 12%, nevertheless maintain long vowels in such environments, rather than shortening. As the majority of the short vowel stems in the sets involved never take part in alternations, only a small subset of the 48% of stems showing short open syllables have long-vowel unsuffixed counterparts, i.e., can be analyzed as having been “shortened”. The stems that do alternate belong primarily to a particular structural type, which shows a monosyllabic long-vowel stem closed by a single consonant in its “bare” form. Furthermore, as was commented on above (§ 2.2.5), a number of the environments cited by Myers do not show shortening as a majority effect. A review of the various types discussed by Myers is as follows:

(2.18a)

Shortening environments:

	<u>Examples</u>	<u>Exceptions</u>
Consonantal suffixes:		
/-t-/	descri <u>b</u> e ~ descri <u>p</u> tive convéne ~ convén <u>t</u> ion scri <u>b</u> e ~ scri <u>p</u> ture discréet ~ discrét <u>i</u> on	
Monosyllabic suffixes:		
/-ic/	cóne ~ cónic míme ~ mímic	básic scénic
/-ule/	glóbe ~ glóbule gráin ~ gránule	
/-id/	pále ~ pállid	
/-ish/	fínal ~ fínish	
Multisyllabic suffixes:		
/-ity/	sáne ~ sánity profóund ~ profúndity	obésity
/-ible/	crédo ~ crédible	féasible
/-ify/	týpe ~ týpify móde ~ módify	nótify glórify

/-it-ive/	defíne ~ definitive
/-itude/	sóle ~ sólitude
/-ual/	gráde ~ grádual

(2.18b)

Inconsistent shortening:

/-J/	fíve ~ fífh	nínth
	wíde ~ wídh	éighth
	héal ~ héalth	-téenth
	déep ~ dépth	
	stéal ~ stéalth	
/-atory/	excláim ~ exclámatory	expíre ~ expíratory
	expláin ~ explánatory	explóre ~ explóratory
	infláme ~ inflámmary	appróve ~ appróbatory
	decláre ~ decláratory	respíre ~ respíratory
	prepáre ~ prepáratory	excíte ~ excítatory
	oblíge ~ oblígatory	mígratory
	órate ~ óratory	rótatory
/-ative/	vócal ~ vócatíve	connótative
	lócal ~ lócatíve	restórative
	provóke ~ provócatíve	ìnnovátíve (etc.)
	decláre ~ decláratíve	
	compáre ~ compáratíve	
/-acy/	deríve ~ derívative	
	supréme ~ suprémacíe	diplómacy
	conspíre ~ conspíracy	prívacy (AmE)
/-able/		píracy
		pápacy
		prímacy
	fláme ~ flámmable	quótable
	próbe ~ próbable	cápable
	plácate ~ implácable	chángeable
		maintáinable
		advísable
		bréakable
		unspéakable

There are also stems which nearly always show “long” vocalic elements such as diphthongs (e.g., /aʊ, ɔʊ, juʊ/) in this position. Only words with long stem vowels of the type that usually alternate, i.e., /a, e, i, o/, have been listed above, although occasionally diphthongs alternate, as in *profóund* ~ *profúndity*.

Regarding the inconsistent cases, stems taking the suffix /-J/ show so much unusual allomorphy that their surface forms are better represented by multiple “suppletive” morphemes than by a prosodically conditioned process, e.g., *stróng* ~ *stréngth*, *lóng* ~ *léngth*, *bróad* ~ *bréadth*, *béar* ~ *bíρθ*, *díe* ~ *déad* ~ *déath*. Words in /-able/, /-acy/ only show a few shortened forms, most stems not being altered, and so these can hardly be considered as “shortening” suffixes. The situation of /-ative/ and /-atory/ is more complex, involving stress variations and many alternants, some dialectal. Among /-atory/ words the non-shortening type predominates; for /-ative/ words, shortening is primarily found among stems which also surface in unsuffixed verbs, e.g., *provóke* ~ *provócative*, while stems of verbs in /-ate/ largely tend to retain the long vowel, at least for the dominant variant pronunciation in /-àtive/ (e.g., *ínnovàtive*).²⁶ However, it is clear that neither suffix can be regarded as regular shortening environments as Myers’ listing suggests.

Myers also treated the “Latinize prefixes” as an environment for shortening under certain conditions. His cases may again be contrasted with words that fail to shorten the prefix, as well as those which do not show the “stress retraction” he poses to account for the first set:

(2.19) <i>resíde</i> ~ <i>résident</i>	<i>cohérent</i>	<i>compóment</i>
<i>províde</i> ~ <i>próvident</i>	<i>detérrent</i>	<i>complácent</i>
<i>abstáin</i> ~ <i>ábstinent</i>	<i>précedent</i> (Br.)	<i>survéillance</i>
		<i>excítant</i>

About 23% of the penultimately stressed forms of this type show long vowels, violating Myers’ retraction rule (§ 2.2.5). Myers’ other example of this phenomenon involved words in /-ation/. Many of these cases appear to

²⁶ A discussion of this complex problem is found in § 6.4 below.

have variants in which the prefix vowel may appear long, although there are others in which this is never the case, and some more wherein the vowel always appears long. Furthermore there are dialectal variants. Here are some examples:

(2.20) <u>Variable</u>	<u>Short</u>	<u>Long</u>
présent ~ présentation	rèformation	dèportation
defórm ~ dèformation	rèfutation	dèmarcation
defáme ~ dèfamation	démonstration	dènotation
evóke ~ èvocation	prèparation	rècantation
prótest ~ pròtestation	sèparation	rètardation

What all this evidence suggests is that the processes described unitarily by Myers and others as “vowel shortening” apply in varying degrees over different morphologically defined sets of words. With certain suffixes, such as the /-ic/ group seen in (2.18a) above, the shortening examples outnumber the exceptions, and short vowels tend to generally appear in this position in such words. In other groups singled out by Myers, however, the shortening effect is as frequent as non-shortening, or itself the clearly exceptional phenomenon.

As will also be shown in the following survey of generalizations about stress presented in chapter five, there tend to be both majority and minority patterns which appear in the data. The majority patterns are often strong enough to be classed as tendencies but are rarely so overwhelming as to be considered “rules”, while the minority patterns are significant enough to have a status other than simple lexical exception marking. An acceptable phonology must be able to account for all attested forms, and must have mechanisms that can account for the weak correlations as well as the strong.

To determine the workings of the prosodic system of English, which presumably provides the constituent environments in which the “sound changes” that result in surface alternations reside, items showing individual lexical idiosyncrasies must be separated from true minority patterns. In an Optimality Theory framework, these minority patterns will need to be accounted for by the same constraint hierarchy governing the majority patterns, and require some consistent variation in their “input” structures to mark them out from the majority group. It must be then shown how these variations interact with the constraint hierarchy to produce the attested

forms. Classifying English words into a number of morphologically and prosodically grouped sets on the basis of their presumed “input” forms helps to determine the identity and ranking of the constraints in the hierarchy, which must account for all identifiable phonological patterns on the basis of those lexical “input” forms.

3. English stress patterns

3.1 The prosodic system and stress

Both Chomsky & Halle (1968) and Myers (1987) noted that vowel alternation was dependent upon stress. As was discussed above in chapter 1, this dependency is part of a larger relationship: the dependency of all phonological processes upon the prosodic (and morphological) constituents which contain them, in the sense that phonological processes are defined by the constraints taking these constituents as their arguments. Word stress, a phenomenon of the prosodic word and its component units the foot, the syllable and the mora, is just one phonological variable whose presence is defined via the prosodic hierarchy. Vowel length alternation, which simply expressed indicates the presence or absence of a prosodic category, the mora, is also dependent on these structures. Ultimately, all sound change involves the relationship between “input” morphological categories (including “input” segments and features) to “output” prosodic constituents. To describe the constraints which reflect the phonology of a language, one need only describe the relationships found in its prosodic and morphological systems.

The following chapter will begin to do this by investigating English word stress, helping to define and identify, on the basis of the surface forms, the prosodic and morphological constituents which will ultimately supply explanations for the phonologically predictable surface forms of the language. At this stage in the presentation of this data, the lexicon will be taken to be the repository of phonologically idiosyncratic information, in the form of morphemic “input” strings. Below, it will be proposed that the majority pattern seen in the English data represents the regular English stress pattern, and that words showing the minority patterns contrast minimally with the regular forms. The two principle minority patterns seen in the data owe their unusual stress either to lexical moraic markings (i.e., underlying segment length, or moras) and/or a particular kind of suffixation.

3.1.1 Describing English stress patterns

The various stress patterns of English words and the conditions under which each pattern is encountered have been classified and discussed in

various works, most notably Chomsky & Halle (1968), Liberman & Prince (1977), Kiparsky (1979), Hayes (1982), Selkirk (1980, 1984), Halle & Vergnaud (1987), Burzio (1993). (see chapter 1) An extensive survey of these stress patterns and derivational accounts of the rules that produce them is found in Kager (1989), who nicely summarizes the work that precedes him. His outline will be followed here in presenting the various stress patterns of English words. While the solution to be offered below in chapter four will be framed in Optimality Theory, descriptive terminology familiar from derivational accounts, for example the term “extrametricality”, will be used below, for both historical reasons and because these terms are essentially descriptive and do not fundamentally conflict with the forthcoming explanations of their effects. Thus, while suffix extrametricality can be described and accounted for in OT as a misalignment of morphological and phonological constituents, rather than as a lexical marking on a suffix, descriptively we are still dealing with “extrametrical suffixes,” regardless of the theory used to account for this phenomenon.

The organization of Kager’s outline dates back, in part, to premetrical analyses, and is framed descriptively. Kager’s division of the words into groups, as well as the rules and processes used to describe the phonology, are based upon expectations concerning foot shape and (primarily morphological and categorial) extrametricality effects, and hinges ultimately on the position of the main stress within the word. As will be shown, Kager’s system, like that of his predecessors, is driven too strongly by assumptions about morphological structure and underlying representations. Kager crucially assumes, following many earlier frameworks, that underlying forms should be maximally identical to their corresponding surface forms. In OT terms, this is equivalent to assuming that Faithfulness constraints should always be high-ranking in the hierarchy. Many of his subsequent rules arise as a consequence of this assumption. The danger of reaching circular conclusions by basing both the theoretical input (underlying forms) and the process (phonological rules) on assumptions based heavily on the output (surface forms) can be seen in his approach. Here, in the description and in the analysis to follow, broad prosodic patterns across the corpus of English words are sought before conclusive lexical and morphological information is brought to bear.

In the discussion below, a prosodic foot structure will be proposed for words according to their surface stress patterns. What this essentially amounts to is designating stressed syllables as foot-heads, and plausibly constructing feet around them on the basis of the trochaic foot of Hayes (1995),²⁷ which is assumed to be the correct general foot type for English. Issues of derivational complexity or underlying representation will be left out of these descriptions, as the goal is to first arrive at a coherent assessment of stress patterns and apparent prosodic constituency at the surface level. By representationally inserting the prosodic structures suggested by these stress patterns into the examples given below, they may be contrasted with our expectations about foot-formation, based on assumptions drawn from prosodic and metrical phonology, and how it should interact with underlying forms and morphological constituency. Comparing the data with principles established by prosodic theory, conclusions may be reached concerning the shape of any proposed underlying representations.

Kager (1989), following typical Lexical Phonological assumptions about English stress, divides the lexicon into a number of subgroups for ease of exposition. This tactic will be followed here as well. To determine what the basic stress patterns found in simplex words are, a large number of other words can be disregarded, as secondary factors will interfere and cloud the data. For example, words suffixed with the “stress-neutral” affixes (e.g., /-ness/, /-hood/), usually described as “level II” in Lexical Phonology, are not used to determine stress placement, as the stress of these words falls on the same syllables stressed in the corresponding unsuffixed forms (e.g., *háppy* ~ *háppiness*, *néighbor* ~ *néighborhood*). Likewise, compounds like *áircraft* involve multiple word-level constituents, which behave differently from simplex forms and will be discussed elsewhere (§ 5.4.2). Monosyllabic content words can also be removed from consideration, as all maintain stress on their lone syllable and do not inform us further. Finally, words showing so-called “level I” affixation, including the type showing vowel alternation discussed in the preceding chapter, are morphologically complex and will be deferred for later discussion. The remainder of the lexicon should thus serve to inform us about the stress patterns of multisyllabic simplex content words.

²⁷Although the issue of the uneven trochee will be minimized.

3.1.2 Stress patterns for simplex words and "stress retraction"

Kager further subdivides this set of words based upon another assumption, that English words must first be stressed (assuming a derivational account) on one of the final three syllables. Any words (e.g., *articulatòry*) unexpectedly showing stress further back in the word are said to suffer "stress retraction" later in their derivation, a concept that dates back to Liberman & Prince (1977). The "original" main stress ultimately surfaces as a secondary stress near the end of the word. While the abandonment of derivational explanations renders the logic behind this distinction meaningless, it is descriptively still relevant. The first section of this classificatory exposition will treat the words regarded by Kager as non-retracted.

Kager also divides this part of the lexicon on the basis of an assumption concerning final syllables with long vowels. He claims (p. 30) that "we find an asymmetry between two types of final heavy syllables: those with long vowels are invariably stressed...while closed syllables with short vowels can be either stressless...or stressed." Kager's claim here is a bit circular: under his approach the definition of secondary stress is determined by vowel quality, and thus an unreduced, unshortened long vowel is by definition at least secondarily stressed. He does not discuss in this context long vowels that appear to reduce finally, for example, the verb/adjective alternation of the type *móderāte* ~ *móder[ə]te*. Following this assumption, he sets forth a principle that all final long vowels are stressed (although not always main-stressed, via retraction²⁸), and this effectively leads to a fourth stress group in addition to the three basic groups he posits.

Kager presents three main groups (pp. 27-30) of multi-syllabic, non-retracted simplex words. This classification is based upon two factors: the relationship between stress and syllable weight also seen in all previous accounts, and the degree of extrametricality present in the word. His first two groups are said to be common and are associated with lexical categories, while the third is labeled "idiosyncratic" and may contain words from any grammatical category.

²⁸The distinction of words with long vowels in their final syllables from other words with final heavy syllables also stems from the fact, discussed below (§3.1.4), that most words said to undergo retraction end in such syllables. The automatic assignment of "original" stress to final long-vowel syllables is necessary to account for this retraction process.

The largest and most prevalent, Kager's group one, shows extrametricality of the final syllable as its unifying feature. Members of this group with antepenultimate stress are said to have, as a rule, light penults. Those with penultimate stress have heavy penults, containing either a long vowel or a syllable-closing consonant. This contrast suggests Hayes' moraic trochees, and gives these words the stress pattern established for Latin (see Hayes 1995: 91), where main stress falls, as a rule, on the rightmost non-extrametrical foot in the word. The words used below are drawn from Kager (1989: 28) and represent structurally distinct subclasses. In presenting them, for the sake of readability I have inserted the foot-structures suggested by the stress onto the orthography, and mark stressed syllables and long vowels (Kager himself does not overtly mark metrical or prosodic structure, although he discusses stress on the basis of prosodic and metrical assumptions). For the sake of uniformity, proper names will not be capitalized. As elsewhere, parentheses indicate where Hayesian feet would be expected, while angle brackets mark Hayesian extrametricality:

(3.1) Group I

(3.1a) <u>Light penults</u>	<u>Long vowel penults</u>	<u>Heavy penults</u>
a(méri)<ca>	a(rǒ)<ma>	a(gén)<da>
(cínna)<mon>	ho(rǐ)<zon>	as(bés)<tos>
(láby)<rinth>	massa(chǔ)<setts>	ap(pén)<dix>

Kager also includes bisyllabic nouns in this group, many of which show light initial syllables:

(3.1b) v́illa	/ (v́i) <lə> /	vénice	/ (vé) <nis> /
vénom	/ (vé) <nəm> /	cábin	/ (cá) <bin> /
hérald	/ (hé) <rəld> /	éffort	/ (é) <fərt> /

If final extrametricality, the mark of this group, is to be maintained, then these forms violate Hayes' bimoraic constraint on feet. They would be understood as stressed according to Hayes' principle that whatever remains of the stress domain after the application of extrametricality should receive stress (§ 1.2.1).

Kager additionally notes that this pattern is shared by (level I) suffixed nouns and adjectives (morphological structure is indicated here by hyphens):

(3.1c) <u>Light penults</u>	<u>Long penults</u>	<u>Heavy penults</u>	<u>Light Bisyll.</u>
(pré-si)<d-ent>	com-(pǒ)<n-ent>	de-(tér)<g-ent>	(pré)<s-ent>
mu(níci)<p-al>	anec(dǒ)<t-al>	fra(tér)<n-al>	(jéa)<l-ous>
le(gál-i)<ty>	(pǒ)<t-ent>	(nér)<v-ous>	

The relationship of these words to the simplex members of pattern one noted above will be taken up below, in § 3.3.1.

Kager's second stress pattern is identical to his first, but the extrametricality is restricted to the final consonant rather than the final syllable. Words in this group are, according to Kager, primarily verbs and underived adjectives, sets described by Hayes (1982) as exempt from final syllable extrametricality (words from Kager 1989: 29):

(3.2) **Group II**

(3.2a) de(vélo)<p>	main(tái)<n>	tor(mén)<t>
as(tóni)<sh>	su(pré)<m>e	ex(péc)<t>
il(líci)<t>	al(lów)	o(vér)<t>

Note that verbs and adjectives with final long vowels that conform to this pattern are also included here. Kager (p. 30) also allows for the inclusion of certain nouns into this group, as well as derived adjectives in /-ic/, /-id/:

(3.2b) ter(rífi)<c>	pro(fésso)<r>	ce(mén)<t>
ala(báma)	va(nílla)	re(súl)<t>

Not included in this group, however, are verbs and underived adjectives whose final syllables are stressed regardless of the fact that they are light (e.g., *omít*), although these are completely parallel in formation to the *vénom* type noted above in the group I, in the sense that they are stressed on a light, metrically final syllable (since final syllable extrametricality must be suspended) :

- (3.3) (vé)<nom> : (bán)<dit>
o(mí)<t> : o(vér)<t>

That is, the relationship of group one words like *vénom* to words like *bándit* (both show a stressed penult, regardless of its weight, because the final syllable is eliminated by extrametricality) is parallel to that between words like *omít* and *ovért* (both show a stressed final), but only *ovért* is considered to belong to group two (since it can be conceived of as showing final consonant extrametricality, while this does not work for *omít*). For words like *omít*, Kager has devised another group, which he labels idiosyncratic, into which words of any grammatical category may fall; they share the distinction of stressing a final short syllable (data from Kager 1989: 30):

(3.4) **Group III**

gui(tár)	ha(ráss)	bi(zárre)
chif(fón)	o(mít)	co(quétte)
ciga(rétte)	acqui(ésce)	

The motivation for proposing a separate group probably lies in the fact that their unexpected stress is not explicable via Hayes' idea of stressing the reduced domain (applicable to the *vénom* type because of extrametricality). Although other syllables are available to take the main stress and produce typical trochaic feet, these words nevertheless show stress on final light syllables,²⁹ in contravention to both extrametricality and foot binarity.

Kager suggests (p. 30) that the "final consonant is also taken into account," that is, he suspends the usual final consonant extrametricality in these cases. This avoids the problem of unexpected monomoraic feet. Kager uses the fact that there are no English multisyllables with a stressed open short final syllable to extend quantity sensitivity to this group; thus, words of this group with final light open syllables (that is, with no final consonant) are said to take stress on the penult, while closed syllables take stress on the final. This neatly accounts for the distinction between the three groups, which thereby undergo foot formation identically but differ according to the degree

²⁹Note that some of these words also have regularized forms, e.g., *guítar*, *cigarette*.

of extrametricality each shows. However, while this would potentially bring vowel-final words like *vanílla* and *vílla* into group III, Kager maintains them in groups II and I, respectively, perhaps wishing to reinforce the marginal status of group III.

As was mentioned above, Kager's assumption concerning the intrinsic stress of final long vowels leads to a *de facto* fourth group, also showing final stress. Since the type of extrametricality seen in group two obliges verbs and adjectives with long finals to also be finally stressed, only finally stressed nouns are considered to necessarily belong in this group:

(3.5) Group IV

tippeca(nóe)	kànga(róo)	màga(zíne)	po(líce)
tènnè(sée)	chìmpàn(zée)	bùcca(néer)	ca(réer)

Such stress suggests a suspension of extrametricality, although it is not possible to say to what degree extrametricality is suspended. Apart from Kager's claim about final long vowels, these words are prosodically identical to the nouns which fell into group II, such as *cemént*. Furthermore, group I forms like *cáthode*, *sátire*, *quátrain*, *cólleague* illustrate that nouns with final long vowels are not all finally stressed; final long syllables can be extrametrical.³⁰ The price of eliminating group IV, of course, is the further contamination of group II's status as a verbal/adjectival category, already compromised by the presence of forms like *vanílla* and *cemént*. Upon final analysis, all that can be said is that Kager's system associates nouns with final syllable extrametricality, and nouns that fall into any other group must necessarily be lexically marked as exceptional.

3.1.3 The systematic status of Kager's stress patterns

Kager delineates these four groups using rough generalizations based partially on grammatical category and extrametricality, and partially on other issues such as vowel length and lexical distribution. Groups I and II are said to be large, are identified with syntactic categories (noun and verb/adjective

³⁰ Alternatively, such words can be said to suffer "retraction", which is Kager's explanation for these forms. See § 3.1.4 below.

respectively), yield feet which conform to Hayes' trochaic principles, and are distinguished from each other by the presence or absence of final syllable extrametricality. Groups III and IV are small, "idiosyncratic", lack extrametricality (or, alternatively, fail to meet expectations about foot structure), and violate generalizations about the syntactic categories covered in groups I and II. While Kager attempts to distinguish these exceptional words in a principled way, there is no empirical motivation for two such groups to account for finally-stressed nouns; all of these words may simply be seen as exceptions that fail to fit into the two major groupings, apparently lexically unpredictable in their stress. Likewise, verbs or adjectives that show apparent extrametricality effects must be regarded as unexpected members of group I (e.g., the "Late Extrametricality" proposal of Hayes (1982) to account for words like *difficult*). Kager's system ultimately boils down to two sets of words (group I nouns and group II verbs/adjectives) in which an extrametricality contrast supposedly parallels a categorial contrast, plus a set of unpredictable exceptions.

3.1.4 "Primary Stress Retraction"

One method of dealing, in a derivational account, with the many forms that fail to conform to the general patterns outlined by Kager, is so-called "stress retraction". In such cases, the main stress appears further from the end of the word than predicted, and is said to have been "retracted" during the process of derivation, usually leaving the former primary stressed syllable with a secondary stress. All the major derivational accounts of English stress mentioned above (§ 3.1) have included some form of stress retraction. Kager regards stress retraction as one of the major subdividing features of the grammar, and this section will again follow his description of the various kinds of primary stress retraction (pp. 40-43), although I will primarily stick to the set of unsuffixed words, as above.

Kager (p. 40) says that "stress retraction in bisyllabic words is regular in nouns"; since the bisyllabic nouns of group I obligatorily stress the initial syllable, this can only refer to those nouns with final long vowels that fail to join his "group IV" (e.g., *archive*, *microbe*, *sátire*, *colleague*). The status of this particular retraction depends entirely on the reality of Kager's claims concerning such final long syllables. For trisyllabic and longer words, Kager claims that retraction "is more common", and he identifies four types of

primary stress retraction, all of which tend to retract stress from the final syllable, that is, in members of groups II, III or IV. Many of the words described as suffering retraction have a final syllable with a long vowel, hence Kager's decision to regard words ending in such syllables as a distinct group, as well as the need to define all final long vowels as automatically stressed.

The four patterns are "strong retraction", "weak retraction", "long retraction" (terms coined by Liberman and Prince 1977) and "sonorant retraction" (Kiparsky 1979). The first pattern, called "strong" because it retracts the stress over two syllables, is regular for suffixed verbs in /-ate, -ize/, and is seen as well in simplex words of the following type (examples from Kager, p. 40):

(3.6)	ábsolùte	mánifèst	sátisfỳ
	húrricàne	tácitùrn	cáterwàul ³¹
	páradìse	dífficùlt	ánecdòte

Note that if the nouns in the first two columns belonged to group I, rather than IV (due to their long finals), initial stress would be entirely regular. The verbal and adjectival forms, however, fail to show the expected stress of group II; another analysis that satisfies most of the above would be to place them in group I, via the "Late Extrametricality" of Hayes (1982). The heavy penults in *ánecdote*, *cáterwaul* and *sátisfy*, however, should then receive main stress, and this failure indicates that these words do indeed display an unexpected stress pattern. This pattern is also difficult to explain using Hayes' metrical theory, as all possible interpretations for the primarily-stressed feet suggested by this stress violate expectations about properly formed English feet:

(3.7)	(ánec)dote ~ (á)necdote
	(cáter)waul ~ (cá)terwaul
	(sátis)fy ~ (sá)tisfy

³¹Due to the sonorant in the penult, one might wish to regard this word as undergoing "sonorant retraction", described below. However, Kager lists it with the strong retracted forms.

The next pattern, “Weak Retraction”, is so called because retraction occurs only over one syllable. Most of the words of this type are suffixed (e.g., with /-ite, -ide, -oid/), although the morphological status of some of these suffixes is questionable due to their limited distribution and unclear meaning. All of the words suffering weak retraction end in syllables with final long vowels or diphthongs.

(3.8) e(léc)tròde sta(lág)mìte py(rámi)dòid (cá)thòde

Kager describes this pattern as rare, and notes that it matches his stress group III; it is better described as constructing a proper trochaic foot to the left of the final syllable, resulting in the pattern seen in group one. Unlike the “Strong” pattern seen above, this type of retraction respects syllable weight; only in forms with light initials like *cáthode*, parallel to the *vénom* type from group I, is an improperly light foot formed. The only thing that distinguishes this set from group I is, again, that the final syllable contains a long vowel and should thus, according to Kager’s assumption noted above, hold the main stress.

Kager describes “Long retraction”, which retracts the stress over three syllables, as “extremely rare” (p. 42). It can affect suffixed words as well as simplex ones (data from Kager):

(3.9) pégrinàte cáamaràn manípulatòry
 amélioràte húllaballòo hallúcinatòry
 détérioràte rígamaròle artículatòry

The syllables skipped must be light, unlike in strong retraction. These words cannot be accounted for by any of the stress groups from above, and do not suggest canonical feet. Explanations utilizing **ternary feet** have been suggested for such forms (e.g., Selkirk 1980, Halle & Vergnaud 1987), although Hayes (1995) claims that ternary feet do not exist (see § 1.2.1 above).

The final retraction pattern is “Sonorant Destressing”, proposed first in Kiparsky (1979) and referred to by Myers (1987) in his attempt to explain stress retraction and vowel shortening in certain suffixed forms (see § 2.2.4 above, § 6.7 below). This pattern resembles strong retraction, with the additional proviso that the syllable skipped over (usually the penult) must be closed by a

sonorant, i.e., /m,n,r,l/, and there must be only one syllable preceding the skipped syllable. This retraction can apply to words ending in any kind of finally stressed syllable:

- (3.10) hélminthòid cávalcàde árgentìne hóttentòt
háckensàck pálimpsèst álgernòn

When the conditions stated above are not met, retraction fails to occur, and the penult is stressed (following either the pattern of group I, or some other retraction):

- (3.11) No sonorant:

aráchnoid eléctron stalágmite ojíbway

Two syllables preceding:

sàlamándroid àgamémnon èlephántine àdiróndack

As in the other cases of strong retraction over a heavy syllable, the apparent metrical structures suggested by cases showing sonorant retraction fail to conform to the norms of foot-formation; that is, the weak syllable of any proposed bisyllabic trochee is always closed, while the stressed syllable may be light and thus monomoraic, e.g., (cával)<cade> or (cá)val<cade>.

To summarize primary stress retraction, if one disregards Kager's claim of a special status for long final syllables, many cases of strong retraction as well as all cases of weak retraction fall into group I. However, strong retraction over heavy syllables, including sonorant destressing, presents an unusual structure inexplicable in the conventional metrical framework. The very rare long retraction pattern, which at best leaves one word-internal syllable unaccounted for, must also be noted as an idiosyncratic stress pattern. While Kager goes into some detail regarding retraction, his account never makes it clear how generally it is encountered, and in the end it appears to simply be yet another source of idiosyncratic deviation from the general stress pattern.

3.1.5 Possible configurations for bi- and trisyllables

Kager's four groups are meant to account for the stress of English words, but a comparison across all possible permutations of syllable type for bisyllables does indicate some gaps in coverage. Here and in the following sections, all possible syllable weight configurations for bi- and trisyllables will be presented along with Kager's predictions about what stress pattern they should show. This will then be compared to the actual stress patterns as found in the Celex database.

The distribution of all possible weight and stress patterns for two-syllable words can be sketched as follows, using a "CvC" schema to represent the possible syllable structure types. Prosodic structure in the form of Hayesian feet will be applied according to Kager's proposals regarding syntactic categories and the expected position of main stress for each stress group. Note that since onsets are not relevant for the calculation of syllable weight, the onset 'C' stands for any onset, regardless of how many consonants it encompasses (including zero).

(3.12) Initially stressed bisyllables:

	<u>Noun</u>		<u>Verb / Adjective</u>	
C'vCv	(C')<Cv>	Group 1	(C'vCv)	Group 2 (?)
C'vCvC	(C')<CvC>	Group 1	(C'vCv)<C>	Group 2 (?)
C'vCvCC	(C')<CvCC>	Group 1	--exceptional--	(Group 1)
C'CCv	(C'vC)<Cv>	Group 1	(C'CCv)	Group 2 (?)
C'CCvC	(C'vC)<CvC>	Group 1	(C'CCv)<C>	Group 2 (?)
C'CCvCC	(C'vC)<CvCC>	Group 1	--exceptional--	(Group 1)
(etc.)				

Kager's system, which exempts verbs and adjectives from final syllable extrametricality, suggests that initially stressed bisyllables from these syntactic categories must end in light syllables (i.e., syllables closed only by the single extrametrical consonant). Any heavy syllable appearing in final position but not showing main stress must, following the principles of foot-formation, be understood as extrametrical and so such words fall out of group II and exceptionally into group I (e.g., *éxile*, *cúckold*, *árgue*, *véto*), or else must be regarded as "retracted". Conversely, all regular nouns should belong to group I regardless of final syllable weight.

Although Kager does not specifically discuss this, bisyllabic verbs with light finals would presumably be analyzed as having bisyllabic feet, i.e., as members of group II (e.g., (*édi*)<*t*>). On the surface, of course, there is nothing to distinguish a verb like *édit* from a noun like *vílla*, which is presumed, as a member of group I, to have an extrametrical final syllable. Verbs with heavy or long initials, e.g., *sávor*, could, on the other hand, invite an interpretation that includes extrametricality, parallel to the nominal type *ómen*, although other explanations without extrametricality are also available.³²

For finally stressed bisyllables, most of the possible structures for verbs and underived adjectives will correspond to the “regular” group II, while finally stressed nouns with identical weight patterns must exceptionally belong to either groups II, III or IV, depending on the shape of the final syllable. Both nouns and verbs with metrically light final syllables, however, are banished to group III. The non-extrametrical final consonants suggested by Kager for this group are underlined.

(3.13) Finally stressed bisyllables:

	<u>Noun</u>		<u>Verb / Adjective</u>	
CvCvC	Cv(Cv <u>C</u>)	Group 3	Cv(Cv <u>C</u>)	Group 3
CvCvCC	Cv(CvC)<C>	Group 2	Cv(CvC)<C>	Group 2
CvCv́v	Cv(Cvv)	Group 4	Cv(Cvv)	Group 2
CvCv́vC	Cv(Cvv)<C>	Group 4	Cv(Cvv)<C>	Group 2
CvCCv́C	CvC(Cv <u>C</u>)	Group 3	CvC(Cv <u>C</u>)	Group 3
CvCCv́CC	CvC(CvC)<C>	Group 2	CvC(CvC)<C>	Group 2
CvCCv́v	CvC(Cvv)	Group 4	CvC(Cvv)	Group 2
CvCCv́vC	CvC(Cvv)<C>	Group 4	CvC(Cvv)<C>	Group 2
CvvCv́C	Cvv(Cv <u>C</u>)	Group 3	Cvv(Cv <u>C</u>)	Group 3
CvvCv́CC	Cvv(CvC)<C>	Group 2	Cvv(CvC)<C>	Group 2
CvvCv́v	Cvv(Cvv)	Group 4	Cvv(Cvv)	Group 2
CvvCv́vC	Cvv(Cvv)<C>	Group 4	Cvv(Cvv)<C>	Group 2

³²Hayes (1995) disprefers feet of the shape(HL) and regards such structures as (H)L, where the light syllable is simply unparsed into foot structure.

For words of three syllables, the situation is similar (here, for simplicity of representation, 'H' stands for metrically heavy syllables, 'L' for metrically light syllables³³):

(3.14) Initially stressed trisyllables:

	<u>Noun</u>		<u>Verb / Adjective</u>
LLL	(LL)<L>	Group 1	{ All would exceptionally
HLL	(HL)<L>	Group 1	belong to Group 1 }
LLH	(LL)<H>	Group 1	
HLH	(HL)<H>	Group 1	

Only nouns should, among the trisyllables, ever hold initial stress under this system. Additionally, no (non-retracted) group exists to account for any initially stressed trisyllables with heavy penults (i.e., LHL, HHL, LHH, HHH), following the assumptions of bimoraic foot-formation and the regular assignment of main stress to the rightmost non-extrametrical foot in the word. These are in fact the very forms that must be explained by retraction.

(3.15) Penultimate stress:

	<u>Noun</u>		<u>Verb / Adjective</u>
LLL	L(LL)	Group 2	L(LL) Group 2
HLL	H(LL)	Group 2	H(LL) Group 2
LHL	L(H)<L>	Group 1	L(HL) Group 2
HHL	H(H)<L>	Group 1	H(HL) Group 2
LLH	L(L)<H>	-- no analysis --	-- no analysis --
LHH	L(H)<H>	Group 1	-exceptional-(Group 1)
HLH	H(L)<H>	-- no analysis --	-- no analysis --
HHH	H(H)<H>	Group 1	-exceptional-(Group 1)

Verbal/adjectival forms with light finals would fit comfortably into the "regular" group II, but words with heavy unstressed finals suggest an extrametricality only acceptable in group I. Feet of the form (LH) or (HH) would violate both bimoraicity and the regular trochaic form of English feet;

³³Heavy syllables would be word internal CvC, Cvv, CvvC and word-final (if final consonant extrametricality is in force) CvCC, Cvv, CvvC, CvvCC. Light syllables would be word-internal Cv and word-final CvC.

more seriously, forms with the pattern HLH or LLH are inexplicable using any of Kager's groups, as they violate foot bimoraicity. It is not clear how Kager would, apart from proposing some sort of unmotivated weak retraction, handle nouns (e.g., *mulátto*, *Achíles*, which are structurally similar to *vanílla*, but cannot be handled by loss of extrametricality since the stressed syllable is light) or verbs (e.g., *contínue*, *contribúte*, which should show final stress) with this pattern. None of Kager's previous devices for avoiding monomoraic feet via extrametricality can account for this type.

(3.16) Final Stress:

	<u>Noun</u>		<u>Verb / Adjective</u>	
LLL	LL(L)	Group 3	LL(L)	Group 3
HLL	HL(L)	Group 3	HL(L)	Group 3
LHL	LH(L)	Group 3	LH(L)	Group 3
HHL	HH(L)	Group 3	HH(L)	Group 3
LLCvCC	LL(H)	Group 2	LL(H)	Group 2
LLCvvC	LL(H)	Group 4	LL(H)	Group 2
LHCvCC	LH(H)	Group 2	LH(H)	Group 2
LHCvvC	LH(H)	Group 4	LH(H)	Group 2
HLCvCC	HL(H)	Group 2	HL(H)	Group 2
HLCvvC	HL(H)	Group 4	HL(H)	Group 2
HHCvCC	HH(H)	Group 2	HH(H)	Group 2
HHCvvC	LL(H)	Group 4	HH(H)	Group 2

Here again, membership in group III depends solely on possessing a final stressed syllable closed by a single consonant, while nouns may find themselves in the "verbal" group II or in group IV depending on their final heavy syllable's vowel length. It must also be noted that retaining stress on the final means resisting retraction.

3.2 Pattern frequencies from the corpus

The previous sections suggest that Kager's categories fail to cover some potential structures. However, the relevance of such proposed structures depends upon whether they actually occur in the corpus. For example, Kager's account cannot explain multisyllabic words of the type *CvCv́, and such words indeed fail to appear in the corpus. His major categories are generalizations, illustrated by the fact that he requires groups III and IV for words which violate those generalizations. Indeed, even the "regular" stress patterns appear to contain members of the "wrong" syntactic category. Further investigation requires the general patterns suggested by Kager and outlined above to be compared to the distributions of the actual forms in the corpus of English. A summary of the findings regarding unsuffixed words will be presented in the following sections. Separate sections treating bisyllables and trisyllables will be presented below.

3.2.1 Bisyllables

The first table presented below in (3.17) treats two-syllable unsuffixed words. Represented here is syllable stress and the distribution of syllable structure in the stressed syllable; the sub-groupings refer to whether the stressed syllable contains a long vowel,³⁴ is closed by a non-extrametrical consonant ("heavy"), or is not closed by a non-extrametrical consonant ("light"). Vertical columns add up to approximately 100%.

(3.17)	<u>All Bisyllables</u>	<u>Nouns</u>	<u>Verbs</u>	<u>Adjectives</u>
Initial Stress:	<u>75%</u>	<u>85%</u>	<u>42%</u>	<u>68%</u>
long (Cvv):	25%	24%	16%	26%
heavy (CvC):	35%	38%	33%	42%
light (Cv):	39%	37%	50%	31%
Final Stress:	<u>25%</u>	<u>15%</u>	<u>57%</u>	<u>31%</u>
long (Cvv):	49%	46%	48%	49%
heavy (CvC):	36%	33%	42%	40%
light (Cv):	15%	20%	10%	9%

³⁴Long closed syllables (as in *compound*), not specifically treated by Kager, compose only about 5% of the syllables in the sample and are treated as "long" in this table.

Overall, words with initial stress outnumber words with final stress by about three to one. Kager's group I, which would fit into the set of initially stressed words, was specified for nouns, of which 85% belong, leaving 15% to show exceptional final stress. Of these, nearly half show final stressed long vowels ("group IV"), while a third end in a closed syllable (group II) and a fifth show the unusual stressed light final of group III. Although Kager tries to distinguish nouns with final stressed long vowels from other words of this form, the distribution of stressed long vowel finals is nearly identical across the syntactic types.

While the monomoraic initial foot (e.g., (*ví*)<*lla*>) seems to appear in 31% of the nouns (where the final syllable would be extrametrical), this strongly contrasts with the final apparently monomoraic foot (e.g., *gui*(*tár*)), found in only 3% of nouns and 4% of all bisyllables. Half of the initially stressed verbs take stress on a light syllable (e.g., *é**dit*), suggesting the pattern lacking final syllable extrametricality (group II), i.e., (*é**di*)<*t*>. However, the adjectives, which should pattern with the verbs, better resemble the nouns in the distribution of syllable weight and stress. Although the verbs and adjectives would ideally all be expected to belong to group II, the distribution of weight in the final syllables of initially stressed words is as follows:

(3.18) Final syllables of initially stressed bisyllables

	<u>All</u>	<u>Nouns</u>	<u>Verbs</u>	<u>Adj.</u>
long (Cv $\bar{v}$ /Cv $\bar{v}$ C):	15%	19%	9%	18%
heavy (CvCC):	14%	17%	5%	23%
light (Cv/CvC):	70%	63%	86%	58%

Since the second syllable of a bisyllabic trochaic foot must be light, this is consistent with a general lack of extrametricality in the verbs; only 14% of them have a form suggesting group one. However, a greater percentage of adjectives actually require extrametricality (e.g., *fécund*, *prólix*, *ríbald*, *hónest*) to eliminate their finals from consideration for stress than do the nouns, suggesting that more than a quarter of the adjectives must belong to group I (or are for some reason "retracted"). Additionally, the initially stressed adjectives with final light syllables are ambiguously interpretable regarding extrametricality. Again, the distribution of syllable weight over the adjectival

forms resembles that of the nouns, and is completely dissimilar from the verbal distribution.

Although most verbs indeed behave as predicted, there are about 5% of the total that seem to require extrametricality or idiosyncratic retraction as well (e.g., *véto*, *árgue*, *éxile*, *béllow*, *clímax*, *cúckold*, *hérald*), and must be accounted for somehow. Some of these words also function as nouns, and such cases force us to ask whether words like *véto* should be regarded as nouns or verbs (or both); English word-forms often yield multiple syntactic interpretations. Among the initially stressed verbs that show light finals, half also show heavy initials, e.g., *sávor*, *órder*, which could suggest either $(s\bar{a})\langle vor \rangle$ or $(s\bar{a}vo)\langle r \rangle$. While the latter is consistent with the general verbal pattern (group II), Hayes (1995) actually tries to preclude feet of the form (HL). 68% of initially stressed underived adjectives also present this situation (e.g., *sécret*, *sóber*, *quíet*), again suggesting that they belong rather to group I.

Regarding Kager's claim that final long vowels should always take main stress, almost half of the bisyllabic nouns with final long syllables are nevertheless stressed on the initial (e.g., *créole*, *cýclone*, *bóvine*, *décoy*, *prónoun*, *rélay*):³⁵

(3.19)

	<u>Initial Stress</u>	<u>Final Stress</u>
vvCC:	26%	74%
vvC:	30%	70%
vv:	66%	33%
All long finals:	46%	54%
vCC:	57%	43%

As can be seen, although Kager groups them together, there is a marked contrast between the behavior of final long syllables closed by consonants and those that are open. As a whole, however, there is only a small difference between the distribution of stress for long vowel finals versus heavy (vCC) finals in bisyllables. The tendency to stress the final also appears to depend upon the weight of the initial syllable, as well as the syntactic category of the

³⁵While Kager regards all long vowels as secondarily stressed, the Celex database lists most of these with the pattern [1,0], i.e., with unstressed (or tertiarily stressed) finals. This distinction appears to be entirely subjective.

word, as the following table contrasting syllable structure and stress illustrates:

(3.20)	<u>All Bisyllables</u>		<u>Nouns</u>		<u>Verbs</u>		<u>Adjectives</u>	
	(1.0)	(0.1)	(1.0)	(0.1)	(1.0)	(0.1)	(1.0)	(0.1)
CvCv	100%	0%	100%	0%	100%	0%	100%	0%
CvCvC	92%	7%	91%	9%	84%	15%	86%	13%
CvCvv	64%	35%	70%	30%	38%	61%	66%	33%
CvCvCC	46%	53%	57%	42%	10%	89%	40%	60%
CvCvvC	16%	83%	29%	71%	2%	98%	12%	87%
CvCvvCC	9%	90%	20%	80%	3%	97%	0%	100%
CvCCv	100%	0%	100%	0%	100%	0%	100%	0%
CvCCvC	90%	9%	96%	3%	79%	20%	82%	16%
CvCCvv	69%	30%	82%	17%	27%	71%	70%	29%
CvCCvCC	61%	38%	83%	16%	13%	87%	55%	43%
CvCCvvC	43%	56%	62%	37%	10%	90%	38%	60%
CvCCvvCC	32%	67%	73%	26%	4%	96%	46%	53%
CvvCv	100%	0%	100%	0%	100%	0%	100%	0%
CvvCvC	88%	11%	93%	6%	77%	21%	100%	0%
CvvCvv	63%	36%	84%	15%	83%	17%	91%	8%
CvvCvCC	72%	27%	82%	17%	15%	84%	75%	23%
CvvCvvC	41%	58%	71%	28%	6%	94%	50%	50%
CvvCvvCC	28%	71%	57%	42%	0%	100%	--	--

Of words with a closed final long syllable, e.g., *políce*, *domáin*, a large majority shows final stress; this tendency grows in proportion to the lightness of the initial syllable.³⁶ However, this preference for final stress is not displayed by words with final open long vowels (e.g., *éssay*, *bambóo*), which in fact have an even greater preference for initial stress than do words with final heavy syllables (e.g., *álmond*, *colláps*e). Nouns do have a tendency to prefer initial stress, verbs prefer final stress, and adjectives fall somewhere in

³⁶Kager (1989: 40) mentions that “in disyllabic nouns the location of primary stress is correlated with the strength of the initial cluster”, i.e., the shape of the initial syllable, ascribing this finding to Oehrle (1972).

between. Words ending in light syllables are initially stressed 92% of the time, and final open light syllables are never stressed. In other words, syllable weight is directly correlated with stress, but the degree of correlation varies according to syntactic category, and, contrary to expectations, long vowels and closed syllables, both indicated in prosodic theory by a single representation, the presence of an additional mora in the syllable, appear to show different behaviors.

The following table offers a final summary of the distributions anticipated by Kager's idealized system compared to the actual data from the corpus of unsuffixed bisyllables, as seen in (3.20). Percentages which appear to clash with Kager's predictions are indicated in boldface:

(3.21)	<u>Nouns</u>				<u>Verbs & Adjectives</u>					
	<u>Grouping</u>		<u>Nouns</u>		<u>Grouping</u>		<u>Verbs</u>		<u>Adjectives</u>	
	<u>(1.0)</u>	<u>(0.1)</u>	<u>(1.0)</u>	<u>(0.1)</u>	<u>(1.0)</u>	<u>(0.1)</u>	<u>(1.0)</u>	<u>(0.1)</u>	<u>(1.0)</u>	<u>(0.1)</u>
CvCv	1(all)	Ø	100%	0%	2(all)	Ø	100%	0%	100%	0%
CvCvC	1(most)	3(rare)	91%	9%	2(most)	3(rare)	84%	15%	86%	13%
CvCvv	1	4(most)	70%	30%	Ø	2(all)	38%	61%	66%	33%
CvCvCC	1(most)	2(rare)	57%	42%	Ø	2(all)	10%	89%	40%	60%
CvCvvC	1	4(most)	29%	71%	Ø	2(all)	2%	98%	12%	87%
CvCvvCC	1	4(most)	20%	80%	Ø	2(all)	3%	97%	0%	100%
CvCCv	1(all)	Ø	100%	0%	2(all)	Ø	100%	0%	100%	0%
CvCCvC	1(most)	3(rare)	96%	3%	2(most)	3(rare)	79%	20%	82%	16%
CvCCvv	1	4(most)	82%	17%	Ø	2(all)	27%	71%	70%	29%
CvCCvCC	1(most)	2(rare)	83%	16%	Ø	2(all)	13%	87%	55%	43%
CvCCvvC	1	4(most)	62%	37%	Ø	2(all)	10%	90%	38%	60%
CvCCvvCC	1	4(most)	73%	26%	Ø	2(all)	4%	96%	46%	53%
CvvCv	1(all)	Ø	100%	0%	2(all)	Ø	100%	0%	100%	0%
CvvCvC	1(most)	3(rare)	96%	3%	2(most)	3(rare)	77%	21%	100%	0%
CvvCvv	1	4(most)	84%	15%	Ø	2(all)	83%	17%	91%	8%
CvvCvCC	1(most)	2(rare)	82%	17%	Ø	2(all)	15%	84%	75%	23%
CvvCvvC	1	4(most)	71%	28%	Ø	2(all)	6%	94%	50%	50%
CvvCvvCC	1	4(most)	57%	42%	Ø	2(all)	0%	100%	--	--

The distributions presented above suggest that some of Kager's claims do reflect real trends among the bisyllables, yet there are clearly substantial segments of the lexicon which fail to conform to them. Kager's alignments of syntactic categories and stress patterns can only be regarded as generalizations, rather than rules. There are significant minority groups, often apparently not distinguishable structurally or syntactically from "regular" forms. Noun and verb groups roughly correspond to his first two sets, but underived adjectives do not line up neatly with the verbs as expected. Long finals do prefer stress, but only in those long finals closed with a consonant. Long finals as a whole show almost the same distribution between the two stress patterns as do short-vowel closed finals. The weight of both syllables in the word appears to interact, affecting stress, but along a gradient. Only the absence of stress on a final open short vowel (e.g., *commá) is an absolute.³⁷

Although long closed finals thus appear to pattern differently from long open finals and heavy finals, under Hayes' system of final consonant extrametricality we would not expect the presence or absence of a final extrametrical consonant to cause any such effects. Nor does moraic theory distinguish between long vowel syllables and closed syllables; each are said to have two moras.

3.2.2 Trisyllables

Kager primarily uses trisyllables to illustrate his stress groups, as the presence of a third syllable helps to clarify the contrast between his major groups I and II. Trisyllables also present the clearest examples of primary stress retraction. The table below illustrates the distribution of the three possible stress patterns across the syntactic categories, and the plausible structure of the stressed foot in each case (the sub-groupings refer to the stressed syllable and the syllable following, a domain relevant for determining the structure of the stressed foot). More tables including the effect of word-final syllable weight will be offered below in (3.23) and (3.24). Here, 'H' and 'L' again indicate metrically heavy and light non-final syllables; only among stressed finals, which presumably compose feet in themselves, is the internal structure of the syllable noted. The headings on the left edge of

³⁷A handful of borrowed words, e.g., *bourgeois* [buʒwá], *elán* [eɪlá] can show final stress on a qualitatively short vowel /a/. These words are often regularized, by stress shift or the addition of a perceived final consonant: [búʒwa], [buʒwár], [eɪlán].

the table indicate whether words show antepenultimate, penultimate or final main stress. As before, these tables refer to unsuffixed words only.

(3.22)	<u>All Trisyllables</u>	<u>Nouns</u>	<u>Verbs</u>	<u>Adjectives</u>
Antepenult:	<u>51%</u>	<u>53%</u>	<u>22%</u>	<u>41%</u>
<u>HH</u> s:	8%	10%	6%	11%
<u>LH</u> s:	8%	8%	8%	9%
<u>HL</u> s:	38%	38%	36%	40%
<u>LL</u> s:	45%	42%	49%	39%
Penult:	<u>30%</u>	<u>31%</u>	<u>37%</u>	<u>26%</u>
s <u>HH</u> :	22%	28%	5%	26%
s <u>LH</u> :	6%	6%	5%	10%
s <u>HL</u> :	49%	48%	49%	41%
s <u>LL</u> :	22%	16%	40%	22%
Final:	<u>18%</u>	<u>16%</u>	<u>38%</u>	<u>29%</u>
ss <u>Cvv</u> :	19%	16%	9%	6%
ss <u>CvvC</u> :	27%	25%	33%	43%
ss <u>CvvCC</u> :	2%	2%	2%	1%
ss <u>CvCC</u> :	30%	30%	35%	43%
ss <u>CvC</u> :	21%	26%	19%	6%

For trisyllables, the percentage of words with final stress drops to 18%. Such words have secondary stress on the initial. A fifth of the words with final stress end with a metrically light syllable, indicating Kager's "idiosyncratic" group III (e.g., *cigarette*). About half of the words with final stress end in a long vowel syllable.

Across the other two patterns, stress does correlate strongly with the weight of the penult, although about one quarter of the words with penultimate stress have light penults (the *vanilla* type) and some 16% of the initially stressed words skip heavy penults (e.g., *réconcile*, *círcumspect*), indicating apparent cases of strong retraction. Regarding Kager's syntactic distinctions, a fifth of the verbs clearly require extrametricality (or retraction) to explain their initial stress (e.g., *bárbecue*, *bénéfit*, *cónstitute*, *cóunterfeit*, *jéttison*, *sácrifice*), as do 41% of the adjectives (the *dífficult* type). For initially stressed trisyllables, the distribution of weight across the syntactic categories is nearly identical, suggesting membership in group I for all these cases;

distributions across the other two stress pattern show a clear contrast between nouns and verbs. Only 10% of verbs unexpectedly skip a final heavy syllable (e.g., *defórest*, *unbáalance*, *embárgo*), as opposed to higher percentages among nouns and adjectives (e.g., *impérfect*, *dishónest*, *misfórtune*). Adjectives again pattern more like nouns than verbs; for words ending in two light syllables, the expected group II forms (e.g., *explícit*, *decrépít*) are outnumbered by initially stressed words like *génuine*, *ínterim*, *víolet*. About a sixth of nouns appear to show pattern two, e.g., *apóstle*, *appárel*, *persímmon*, *molásses*, as was also true among the bisyllables above.

Examining all combinations of heavy and light syllables in unsuffixed trisyllables, the following stress pattern distributions can be seen:³⁸

(3.23)	<u>All Trisyllables</u>			<u>Nouns</u>			<u>Verbs</u>			<u>Adjectives</u>		
	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>
HHH	25%	36%	36%	33%	54%	11%	4%	11%	84%	34%	35%	30%
LHH	37%	52%	10%	36%	53%	9%	58%	17%	25%	47%	52%	0%
HLH	61%	5%	33%	77%	3%	17%	24%	6%	69%	56%	3%	40%
LLH	72%	7%	20%	72%	7%	18%	62%	2%	34%	71%	8%	19%
HHL	25%	66%	5%	35%	62%	1%	3%	91%	5%	3%	97%	0%
LHL	22%	74%	3%	23%	73%	2%	6%	85%	8%	31%	67%	0%
HLL	60%	26%	12%	74%	15%	8%	12%	60%	27%	51%	42%	6%
LLL	73%	19%	7%	78%	15%	5%	36%	45%	18%	82%	17%	0%

Final stressed light syllables (group III) are again a small minority, as predicted; verbs appear to be most likely to show such a stress. Final heavy syllables are stressed more often than final light syllables, although main stress nevertheless appears about twice as often on a preceding syllable in words ending with a heavy syllable. Failure to stress a heavy final appears to be correlated with syntactic category, but the interaction is complex and is also influenced by the structure of the preceding syllables in the word.

³⁸Note that no distinction is made here between the stress patterns [100] and [102], or [010] and [210]; there was not enough evidence from the corpus to warrant separate categories for these, and many instances of these stress patterns in the database were simply variant pronunciations of the same word.

About a quarter of the verbs with heavy finals stress the initial when it is heavy (e.g., *intérêt*, *exécute*, *multiplie*) and more than half show this pattern when the initial is light (e.g., *cátapault*, *sácrifice*, *réprimand*). This trend, even more prevalent among the adjectives, has been attributed to the so-called “Late Extrametricality” of Hayes (1982) or to strong retraction. However, retraction over a heavy syllable, which indicates true strong retraction rather than extrametricality, is seen less than a third of the time; the other cases are better explained by the purely metrical phenomenon of extrametricality; that is, these forms fall into group I.

The data thus suggest that there is no plausible correlation between the exemption from final syllable extrametricality (Kager’s second group) and the trisyllabic adjectives, for only about a third of them must be regarded as lacking extrametricality. This is only slightly less than the distribution of nouns in group II, which are regarded as exceptional. Together with the evidence from the bisyllables, this indicates that it is not possible to correlate underived adjectives as a category with any general tendency towards the presence or absence of extrametricality. While verbs show group II characteristics more regularly, there is a sizable minority of around 20% (e.g., *jéttison*, *régister*, *bénéfit*, *discipline*) which also apparently belong to group I.

Among the nouns, the expected patterns for group I forms appear, although about 15% appear to belong to group II. Between 30%-40% of nouns and adjectives with the structure /sHs/ skip over heavy penults to stress initial syllables, both heavy and light. 20%-30% of nouns and adjectives with final heavy syllables stress those finals, although, as the following table illustrates, nouns with final long vowels do not appear any more likely than those ending with two consonants to take final stress:

(3.24) Stress according to final syllables:

	<u>All Trisyllables</u>			<u>Nouns</u>			<u>Verbs</u>			<u>Adjectives</u>		
	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>	<u>100</u>	<u>010</u>	<u>201</u>
v v	31%	38%	28%	42%	38%	17%	32%	19%	47%	25%	61%	12%
vvC	65%	3%	30%	73%	2%	22%	38%	2%	57%	64%	7%	27%
vvCC	40%	0%	59%	47%	0%	52%	0%	0%	100%	100%	0%	0%
vCC	45%	20%	31%	48%	26%	22%	28%	9%	62%	52%	12%	32%
vC	46%	38%	13%	65%	21%	11%	14%	67%	16%	45%	51%	3%
v	18%	80%	0%	19%	79%	0%	42%	58%	0%	57%	42%	0%

This table also suggests that the final consonant closing a long syllable has a significant effect upon stress placement. Significantly, words ending in /-vvC/ rarely show penultimate stress (e.g., *philhellene*, *recóndite*); stress appears either on the final or the initial. Words with heavy closed short finals show penultimate stress about a fifth of the time, and more often in nouns than in verbs or adjectives. For nouns and adjectives, words ending in a final open long vowel present similar patterns to those ending in a light closed syllable; only for verbs does the open long final resemble the closed long final in distribution.

3.3 The data contrasted with Kager's stress groups

Among all the types proposed by Kager in his description of English stress patterns, the most prevalent is certainly group I, the set showing final syllable extrametricality. This group not only covers most nouns, but seems to encompass many adjectives as well as some verbs. While a majority of verbs appear to conform to Kager's generalization for group II (final consonant extrametricality only), there are structural patterns in which long syllables, final and otherwise, are skipped over more often than not. Furthermore, the data for adjectives simply does not correlate with their supposed description as members of group II. There are additionally the circa 15% of nouns that fall unexpectedly into group II and the 4% of words that are stressed on final light syllables, suggesting group III and its lack of all extrametricality. Added to this may be the small groups of words whose form is not explicable by any

permutation of extrametricality effects and must be regarded as somehow “retracted”. Kager’s claim of a special status for final long syllables (versus final vCC syllables) has not been significantly supported by the data, although the composition and weight of the syllables, including that of extrametrical consonants, appears to have a gradient effect on the distribution of final versus initial stress.

As a set of broad descriptive generalizations, Kager’s three patterns are informative. However, even as a blueprint for a rule-based account of English stress, they are problematic. While the category-based generalizations may be accurate enough for a majority of nouns and verbs, too many exceptions appear for them to form the basis for a set of rules, and the situation of the adjectives is completely unpredictable. Furthermore, although the distinction of the three categories on the basis of variations in extrametricality is clever and interesting, it requires an inordinate amount of lexical marking and ultimately this type of incomplete correlation is likely to prove to be a side effect of some other process rather than the source of the variation.

In fact, in Optimality Theory, such an explanation is not possible. Unlike a rule-based system, where extrametricality is a condition that can be imposed on various forms at will, and whenever necessary to derive the correct result, in OT the “extrametricality” effect appears only when a misalignment of (traditionally aligned) constituents produces the optimal, extrametrical form (§ 4.1.1). While in a derivational rule-based theory allowing different sets of words to be marked for different kinds of extrametricality effects requires “only” that lexical entries must carry a diacritical mark, in OT this implies that there are different constraint hierarchies for each set of data. This goes against one of the principal tenets of OT, which is that a single constraint hierarchy shall yield the entire grammar.³⁹ The rigor of Optimality Theory eliminates exactly this kind of explanation from consideration. The apparent difference in extrametricality between the various

³⁹Cophonologies, suggested by McCarthy & Prince (1993b) and Itô, Mester & Padgett (1994), are explicitly excluded from the strong version of OT advocated here (§ 1.4.3). Diacritics marking certain lexical entries as “immune” from certain constraints are also not permitted; only members of the constituent hierarchies may appear in the constraints or the lexicon. The only thing which can exempt a candidate from the action of a constraint would be the absence of the structure governed by the constraint from that candidate, making its application vacuously satisfied.

stress groups can only be a side effect resulting from some other condition, brought about by conflicting constraints in the hierarchy.

This means, in effect, that the stress difference seen between apparently structurally identical words must be due to a difference in the “input” structures of these words. In other words, the answer lies in the lexicon, although not in the form of diacritic markings, which have no status in OT, but rather by using available grammatical constituents to modify the structure of the “input” lexical segment string for such words. In the following sections explanations of these variations will be put forward, and it will be shown that the “group I” pattern, which contained the most examples and covered all the grammatical categories, is in fact that basic stress pattern for English, and that the other patterns result from variations in morphological or prosodic structure in the lexical entries themselves. The first stage of the solution lies with a set of words that without exception fall into this group, the suffixed nouns and adjectives usually described as “level one” in the Lexical Phonological literature.

3.3.1 Suffixed nouns and adjectives

In addition to his classification of unsuffixed words, Kager (1989) pointed out that certain nouns and adjectives formed with Latinate suffixes (e.g., /-al, -or, -ous, -ive, -us, -um, -ar, -an, -ary, -ory/) also exhibit the majority stress pattern, group I, featuring final syllable extrametricality (§ 3.1.2 above). Hayes (1982) also classifies these suffixes as extrametrical. Unlike the case of the unsuffixed nouns discussed above, a portion of which displayed final stress, these words are never stressed on the final syllable, the suffix syllable. Following the classificatory scheme used in the Appendix, the Latinate suffixes showing this stress pattern fall into three distinct groups, based upon stress patterning:

- (3.25) The /-al/ group (/ -al, -or, -ous, -ive, -us, -um, -ar, -an, -ary, -ory /)
 The /-ent/ group (/ -ent, -ant, -ence, -ance, -ency, -ancy /)
 The /-ity/ group (/ -ity, -ify, -ible, -ion, -ic, -id, -ule /)

The first two of these sets will be discussed here with regard to the stress and weight patterns they display, while the /-ity/ group, in which vowel alternation is most often encountered, will be dealt with in § 4.2.2.

Among the bisyllables showing suffixes from the /-al/ and /-ent/ groups, stress falls exceptionlessly on the initial and the distribution of weight in the stressed root syllable is as follows:⁴⁰

(3.26)	Cvv-s	43%	tón-al	fám-ous	cláim-ant
			ráz-or	mót-ive	
	CvC-s	44%	fórm-al	vénd-or	sólv-ent
			nerv-ous	áct-ive	
	Cv-s	13%	mámm-al	érr-or	érr-ant
			férr-ous	páss-ive	

This distribution stands in contrast to that of weight in the bisyllabic nouns, in which 37% of the initially stressed forms began with light syllables, and only 24% showed long vowels in initial syllables. For the trisyllables the following distributions can be seen:

(3.27)					
<u>Initial Stress:</u>		45%			
HH-s		1%	súbstant-ive		
LH-s		2%	cávern-ous	góvern-or	prótest-ant
HL-s		41%	séason-al	órbít-al	méndic-ant
LL-s		56%	pívót-al	pósit-ive	hésit-ance
<u>Penultimate Stress:</u>		55%			
sH-s		84%	triúmph-al	vindíct-ive	
			propós-al	horrénd-ous	
sL-s		16%	submíss-ive	rebútt-al	
			colóss-al	proféss-or	

More strongly than in the suffixless words, stress correlates with syllable weight. Nearly all initially stressed words show light penults, with extremely few cases for “retraction” in evidence. The preference for stress on a heavy penult is also strong, although there is a larger minority, about 16%, that shows stress on a light penult, e.g., *proféssor*, *colóssal*. This can be compared

⁴⁰Below, the dash (‘-’) indicates where the suffix syllable would follow.

to the distribution of weight in the penultimately stressed unsuffixed nouns, 15% of which were stressed on light penults.

3.3.2 A geminate interpretation

Kager classified words like *proféssor* as exceptional members of group II, that is, he claimed that they failed to show final syllable extrametricality. However, this explanation is more problematic in these cases than it was for the unsuffixed words, because these suffixes appear otherwise to be always extrametrical. Since final extrametricality in these cases (in Kager's derivational terms) is understood as a morphological feature, lexically marked on the suffix, suspending it in special cases is problematic for such a rule-based approach. If these suffixes are to be taken as consistently extrametrical, however, we are left with apparently monomoraic feet, in contravention to the assumptions of metrical theory. A solution to this dilemma can proceed from the observation that stems which take stress in this manner appear so in nearly all cases, regardless of suffixation:

(3.28) érr	érr-or	érr-ant
propél	propéll-or	propéll-ant
succéss	succéss-or	succéss-ive

This suggests that the phenomenon is marked on the stem rather than as a property of the suffix. It will be proposed here, as it has been proposed in the past by Chomsky & Halle (1968: 86) and more recently by Burzio (1993: 365), that these words have underlying geminates, which reveal themselves on the surface through a consistently short vowel rather than a long consonant, which in English does not appear distinctively on the surface below the word level.⁴¹ Halle & Vergnaud (1987) proposed a "lexical grid mark" for such cases (§ 1.2.1), but moraic theory provides a simpler explanation, a lexical mora associated with the consonant, parallel to the marking of underlying long vowels in the lexicon (McCarthy & Prince 1986). Thus, the stressed syllables are indeed regular bimoraic feet, structurally identical to those closed by a consonant, and this exceptional minority is accounted for.

⁴¹Surface geminates may appear across the word boundary for some speakers, e.g., *róommate* [rúmmmejt].

The geminate status of these consonants is clearly suggested not only by the orthography and the fact that historical geminates existed in these words in earlier stages of English,⁴² but more importantly by the synchronic behavior of these words. Although suggested by Chomsky and Halle (1968), this geminate interpretation was replaced in later work, for example, Halle & Vergnaud (1987), by diacritic marking, likely as a result of the ascendancy of the theoretical maxim, discussed earlier (§ 1.4.1), that surface forms should be taken as basic whenever possible (see Anderson 1982), codified in Kiparsky's (1973) Alternation Condition. Positing geminates where no geminates surfaced was thus dispreferred theoretically. However, the relationship between abstract phonological structure and its surface correspondents has never been one-to-one; the abstract structures are best understood as capturing the fact that there are distinctions between entities, rather than specifying what those distinctions are. Multiple examples of such abstraction can be found in even very conservative generative approaches.⁴³ That contrasts at the abstract level of phonology often correlate to "different" features on the surface should not be controversial.

For these proposed geminates, the consistent presence of apparently monomoraic feet, in violation of an otherwise undominated foot binarity constraint, is reason enough to propose that the missing mora is in fact present underlyingly, its surface effect being the pattern of foot-formation itself. Any other explanation is hard to maintain within OT. While all constraints are violable in OT, there would need to be clear evidence that the constraint governing foot binarity is crucially dominated in words like *molâsses* but not in *câmelot* or other words with identical /CvCvC/ surface structure, and such evidence is not apparent. Since these words tend to have possessed geminates historically, positing them for the synchronic grammar actually restores to the lexicon a cogent contrast which has evidently been maintained to the present day, although its surface expression has, as in so many other cases, shifted. In other words, the inherited lexicon is identical (with regard to these words); it is the surface realization of these "input"

⁴²These words displayed true geminates in earlier stages of English or in Latin, from which most were borrowed.

⁴³In English, for example, we discuss "long vowels" that are surface diphthongs and a voice/voiceless distinction in obstruents that is represented phonetically as a contrast in the duration of the vowel preceding the consonant in question.

lexical forms which has changed, due to diachronic changes in the grammar. Explaining exceptional behavior utilizing a structural constituent such as the lexical mora, already necessary to account for underlyingly long vowels, is far preferable to the diacritic marking used by previous accounts, which furthermore has no status in Optimality Theory.

Chomsky and Halle suggested the geminate interpretation because it was historically based and had explanatory value. However, it is the factors suggested above which indicate that such a structurally based marking is the best way to account for these forms. The regularity of foot-formation across word-forms for stems ending in a geminate, and the consistent frequency, around 10-15% across all types, that this kind of structure presents, suggests a marked lexical structure for the root morphemes of these words as listed in the lexicon. The geminate can be used to explain the following sets:

(3.29)

Bisyllabic verbs and adjectives

of group three (e.g., *bizárre*, *omít*): 10% of all finally stressed forms

Bisyllabic suffixed forms

(e.g., *mámmal*, *érror*): 13% of bisyllabics

Trisyllabic nouns with light stressed penults

(e.g., *vanílla*, *molásses*, *Achíllés*) 15% of such words

Trisyllables of group three

(e.g., *rètrogréss*, *silhouétté*) 15% of finally stressed trisylls.

The consistent distribution of this marked structure in different categories suggests that it is present throughout the lexicon; its frequency, about one in seven for words with stressed heavy syllables, is too high to be considered idiosyncratic or “exceptional”, but is rather consistent with the low but stable frequency expected from a lexically marked structure.

The markedness of the geminate is also evident from the way it may be lexically simplified over time. For example, the alternate pronunciations of the adjective *cómbative* and verb *cómbat* suggest that an underlying stem /kom-batt/⁴⁴ (cf. *combátive*, *combát*) is being lexically reanalyzed (in some

⁴⁴Here the orthographic convention of the double consonant, representing a geminate, indicates a complex structure where a mora is lexically linked to a consonantal segment, i.e., /bat_m/. This is

speakers and dialects) as the less marked /kombat/. This regularization might stem from a reinterpretation of the regular group one nominal form *cómbat* /kóm-batt/, wherein a form with a final geminate would be indistinguishable a non-geminated form (i.e., /kóm-batt/, /kóm-bat/). The British version of the stem for *hárass* (American /haráss/) shows the same type of treatment (lexical simplification to /haras/), and there are many examples of lexical “evolution” to which this interpretation could be extended.

As was noted above, the proposed geminate explanation simplifies Kager’s system by eliminating group III. Words like *vanílla*, *proféssor*, *molásses* all become regular members of group I, while *rètrogréss*, *omít*, *rebél*, *bizárre* become regular members of group II. Nouns with final stressed geminates like *cìgarétte*, *guitár* still remain exceptional, but move out of group III into group II, parallel to forms like *cemént*, *resúlt*, *políce*. The geminates are not realized on the surface melodically or durationally, but express their weight through stress effects.

3.3.3 Final long vowel syllables, open and closed

The explanation for the contrast in the behavior of final long vowels in open and closed syllables runs parallel to the geminate explanation outlined above. It was noted previously that final syllables with the form CvvC were far more likely to hold final stress (70%) than words ending in Cvv (33%), although moraic theory does not differentiate between both types (each has two moras) and the final consonant is traditionally regarded as extrametrical and thus should have no metrical effect. Kager has indeed noticed this, and mentions in a footnote (fn. 2, p. 102) that

“in spite of the *long* (phonetically *tense*) vowel in their final syllables, many words are stressed according to stress placement I... The standard analysis is to assume underlying short (non-tense) vowels which are lengthened (tensed) by a rule affecting word-final non-low vowels in this position.”

the presentational equivalent of the macron, used herein to represent a similar structure involving vowel segments.

Thus, words like *búffalo*, *bróccoli* are assumed to end in underlying short vowels which are lengthened later in the derivation. This is in spite of the fact that such final vowels are on the surface identical with “true” long vowels, such as those in *chìmpanzée* or *àpropós*. Thus, the distinction drawn here is entirely parallel to that given in the case of the geminates: a moraic contrast is proposed over a set of segments that do not phonetically contrast, to explain a contrast in stress. This explanation will be maintained and serves to contrast, for example, verbs like *véto* /véto/ with initial stress from verbs like *obéy* /ōbǎ/ with final stress, although of course the process will not be rule-based.⁴⁵ Thus the 66% of words with final long vowels that are unexpectedly left unstressed are to be understood as underlyingly ending in short vowels, parallel to forms with light open finals that universally fail to show final stress. The remaining 33% can then be seen as true long vowels, identical to those seen in words with closed long finals.

3.3.4 Accounting for group II

Proposing geminates eliminates group III and moves part of the group II nouns (e.g., *vanílla*, *molásses*) into group I, but the system still retains two major stress patterns which appear to differ according to degrees of extrametricality. The solution to the status of group II, which has a much lower (but still significant) frequency than group I, and which is strongly correlated with the verbs, can again be found in the distribution of the suffixed Latinate forms noted above. For example, the distribution of syllable weight in the suffixed forms of the /-al/ group resembles nothing more than the distribution across unsuffixed verbs:

(3.30)

<u>Bisyllabic Verbs</u>			<u>Suffixed Forms</u>		
<u>Initial Stress:</u>					
HH	9%		HH-	4%	
LH	6%		LH-	2%	
HL	40%	órbít	HL-	46%	órbít-al
LL	43%	édt	LL-	47%	édt-or

⁴⁵An Optimality account will be given below, § 5.3.1.

Final/Penultimate Stress:

HH	45%	advise	HH-	41%	advís-or
LH	42%	appráise	LH-	44%	appráis-al
HL	7%	représs	HL-	7%	représs-ive
LL	5%	propél	LL-	8%	propéll-or

The verbs in most cases show unsuffixed stems that are identical to those seen in the forms suffixed with members of the /-al/ group, as if the verbs were suffixed with a similar light syllable suffix. This is in fact what will be proposed, again following both the history of the language as well as similar suggestions by Chomsky & Halle (1968: 45) and Burzio (1993: 371).

Chomsky & Halle propose that for words like *ellipse*, a lexical representation /elipse/ should be proposed, with a final vowel. They note that the phoneme /e/ never occurs word-finally, and that this is a gap in the system. (p. 147) This exempts the final surface syllable /-lips-/ from their version of an extrametricality rule which would otherwise apply and yield the incorrect */élips/. They delete the unwanted final vowel via an “e-elision rule”. Burzio simply proposes a piece of “phonetically null structure” which he represents as /Ø/. It plays the role of a vowel in regard to syllabification and stress, yet does not surface. Here, a similar construct will be adopted, but it will be represented in this descriptive account as /ə/.⁴⁶ As in previous proposals, this segment is never allowed to surface, but will provide a final syllable responsible for the apparent exception from extrametricality effects.

Such a proposal of a segment that fails to surface might be objected to on the grounds that it introduces complexity into the grammar which is not warranted by surface structure. Additionally, a large number of lexical items are affected; for bisyllabic unsuffixed words alone, 25% show a final stress that might require this interpretation. However, as with the introduction of moraic marking discussed above, there are a number of corroborative reasons to propose such a structure. One is the fact that these forms are marked in any case: for exemption from final syllable extrametricality. It has been shown

⁴⁶One advantage of using a segment /ə/ rather than “phonetically null structure”, among others, is that /ə/ will then be present in morphological structure, and prosodic constituents corresponding to /ə/ can be understood as distinct from prosodic constituents without segmental correspondents, which can incur certain faithfulness constraint violations in OT. There is also evidence for /ə/ surfacing in other contexts (§ 5.4.3).

above that the overlap between categorial membership and extrametricality effects is imperfect at best, and in the case of the adjectives not even general. This would require diacritical markings on every adjective and noun that fell into group II, as well as a marking for every verb that belongs to group I. Additionally, under Optimality Theory, maintaining two extrametricality patterns based on diacritical markings is far worse than proposing a segment which was historically present in the surface form and which, like the geminates, retains its status within the lexicon despite the lack of a direct surface reflex.

In English, there is a more cogent type of lexical marking that produces the same effect on stress as the one discussed here, and which affects a greater part of the English vocabulary than the words affected by this proposal: suffixation. The suffixes of the /-al/ group produce words which, as seen above in (3.30), maintain their stress on the presuffixal stem. Proposing that the elided final schwa offered above is actually a verbal and adjectival suffix /-ə/, parallel in structure and function to the other /-al/ group suffixes, which indeed combine with the same set of stems, adds minimal new information to the lexicon. The marking thus present on the words in question then belongs to the morphology and works like any other instance of suffixation. Morphologically, proposing a suffix /-ə/ also allows for a streamlined account of the most productive type of derivational verbal paradigm in English, which can be exemplified here:

(3.31) <u>Verb</u>	<u>Verbal Noun</u>	<u>Verbal Adjective</u>
emót-e	emót-ion	emót-ive
posséss [-ə]	posséss-ion	poséss-ive
condúct [-ə]	condúct-ion	condúct-ive
décor-at-e	dècor-át-ion	décor-at-ive

By regarding the verbal forms as suffixed, the entire paradigm can be seen as derived, rather than just the noun and adjective. This pattern is regular for the extremely productive /-ate/ class of verbs, which can be seen as also extended by this same suffix.

Once the presence of the suffix /-ə/ is reckoned with, pattern II verbs regularly join pattern I. Those apparently ending in a heavy or geminate

syllable maintain final stress, parallel to trisyllabic nouns with heavy penults, while those ending in a true light syllable take penultimate stress:

(3.32)	advise	/advís-ə/	angína	/anǵína/
	eléct	/elékt-ə/	agéndá	/aǵéndá/
	propél	/propéll-ə/	dilémma	/dilémma/
	límit	/límit-ə/	álgebra	/álǵebra/
	delíver	/delíver-ə/	basílica	/basílika/

However, just as not all adjectives or nouns are suffixed, not all verbs are necessarily formed with the suffix /-ə/, although it is certainly very common among verbs. Unsuffixed verbs account for the verbal forms which showed the first stress pattern on the surface, which amounted to 13% of the bisyllabic verbs and around 25% of the trisyllables:

(3.33)	véto	éxile	lícence	múshroom
	tríumph	hárvest	cúckold	hérald
	ásterisk	bárbecue	cátapult	jéttison

Not surprisingly, most verbs of this type also function as nouns and can be thought of as denominative, formed without any further morphology (Kiparsky 1982a, b). The converse can be seen for nouns formed from verbs. One variety of deverbative noun is formed from prefixed verbs in /-ə/; the nominal form, however, lacks the suffix, yielding a difference in stress:

(3.34)				
Verbs:	protést-ə	conflíct-ə	addíct-ə	rejéct-ə
Nouns:	prótest	cónflict	áddict	réject

Another variety of deverbative noun maintains the same form as the verb, i.e., retaining the suffix, and thus shows the same stress, e.g., *efféct*, *accóunt*, *debáte*. As was the case in the verbs noted above, related nouns and verbs often have the identical form in English. Adjectives can similarly be formed in both these ways, e.g., *perféct*[-ə] ~ *pérfect*, *exémpt*[-ə].

The suffix /-ə/ clearly has an adjectival use as well, although its presence is not as widespread as among the verbs; it appears to have a frequency consistent with that of the major adjectival suffixes such as /-al/ and /-ive/. Thus, the large group of adjectives, e.g., *dífficult*, *ríbald*, *módest*, which showed group I stress patterns need not be explained via “Late Extrametricality” or retraction, but rather simply lacks suffixation. As with most of the adjectival suffixes in the /-al/ group, the suffix /-ə/ can also form nouns, although less frequently, e.g., *caréer*[-ə], *cemént*[-ə]. 12% of the apparently unsuffixed nouns show final stress and thus require such suffixation. Many of these are deverbals such as *resúlt*, *efféct*. Similarly, a final schwa can also account for the stress of words in /-ic/,⁴⁷ as has also been proposed by Burzio (1993), who characterizes this suffix as /-icØ/. An underlying form /-icə/ for the suffix is historically accurate, and parallel to the other Latinate forms discussed above.

This accounts for the remainder of Kager’s group II, and unifies all unsuffixed words under a single stress pattern, the majority group I. The fact that all syntactic categories could display group I stress is the result of cases in which no suffixation takes place, infrequent among the verbs but common among adjectives, thus the difference in the distribution of stress between the two categories. It can also be seen that the extreme rarity of group III, which comprises around 4% of the forms, was due to the fact that members of this group need to show both stem-final geminates and suffixation in /-ə/ to create the non-extrametricity effect (e.g., *guitár* /gitárr-ə/).

Relying only on a lexical moraic difference necessary to describe long vowels, and a suffixation process supported by both clear historical evidence and synchronic surface effects, the principal minority stress patterns seen in both complex and simplex English bi- and trisyllables can be accounted for descriptively. No unstructured diacritical exception marking is necessary, and indeed, is prohibited under a conception of Optimality Theory which restricts both the lexicon and the constraints from referring to anything but the constituents of the relevant grammatical hierarchies. It should be clear that, using this explanation, the stress patterns described above can be accounted for cleanly in OT, without recourse to multiple constraint hierarchies. The underlying forms arrived at in the preceding descriptive account will yield

⁴⁷Stress for the /-ic/ group will be discussed in § 4.3.2, below.

the proper optimal candidates via the evaluation of a single constraint hierarchy that demands both bimoraic feet and final syllable extrametricality for every word. Words that apparently fail to show either on the very surface nevertheless present the relevant structures to the constraint hierarchy, satisfying the pertinent constraints at that level, although the structures themselves fail to surface as anything but stress effects. These effects, however, and their interaction with the general prosodic and morphological systems of English are sufficient to declare their presence.

4. The English constraint hierarchy and stress

4.1 Stress constraints for English

The preceding account of the general stress patterns seen in English bi- and trisyllables was for the most part descriptive, referring only to the surface structure of the words in question. The purpose of this exposition was to determine which stress pattern was the least marked in the data, wherein the surface prosodified forms could be seen to consistently correspond to the plausible morphemic structures underlying them. In this way, the prosodic differences necessary to account for the minority patterns, in this case moraic and segmental differences, could be determined and underlying forms incorporating both phonemic and morphemic constituents were proposed for such words. Since a proper Optimality Theoretic analysis depends on evaluating a candidate set, it is necessary for the selected “input” morphemes to be able to supply the eventual optimal candidate in a principled way.

Following the analysis set out in the preceding chapter, most English words are seen to be composed of clear, discrete morphemes with no lexical prosodic component, such as a lexical mora marking underlying length.⁴⁸ Stress is assigned according to what has been described as Kager’s (1987) first pattern, wherein words appear to display final syllable extrametricality and show main stress on the rightmost foot. Words belonging to the other major patterns show either a lexical specification for prosodic constituents, specifically moras providing consonant length, or the suffix /-ə/, unusual in that it fails to be parsed into the surface form, or both. This chapter concerns itself with expressing these relationships via Optimality Theoretic constraints, and proceeds to account for other phenomena, such as vowel alternation and stress retraction, on the basis of these constraints and the evidence from the English corpus. It is assumed that stems which do not conform to these generalizations are not mappable to single morphemes, but rather show allomorphy, or multiple morphemes which yield the forms with contrasting behavior. However, the constraint hierarchy proposed here should account

⁴⁸Apart from morphemes with underlying long vowels, of course, which are also associated with lexical moras.

for the surface forms of all English words, given some additional morphological structure.

In this chapter, a general constraint hierarchy will be introduced, which will capture the generalizations about stress and syllable weight seen in Kager's first stress group. Constraints which yield the same effects as familiar derivational devices like extrametricality and foot-formation will be used. Principles of syllabification and syllable structure will be discussed, with some relevant constraint types defined and introduced to reflect certain aspects of the English data. It will be demonstrated that the additional lexical structure proposed for words of Kager's groups two and three will allow the same constraint hierarchy to be used to account for all unaffixed words. Subsequently, the role of morphology in the system will be introduced, and the differences between various suffix subcategorizations demonstrated using OT. Finally, distributional evidence for the presence of long vowels in the data will be explained morphologically, and the problem of "vowel shortening" and its lexical exceptions will be presented in a new light, as the result of a morphologically conditioned vowel lengthening process.

4.1.1 Expressing the general stress pattern in OT

The general stress pattern identified above shows as one of its defining features final syllable extrametricality, that is, the word-final syllable is apparently excluded (whenever possible) from joining a foot. This exclusion is especially pertinent because words of this pattern also display main stress on the rightmost foot of the word. Extrametricality allows for the exclusion of the final syllable from consideration for main stress, although it might be a heavy syllable, and thus a potential foot itself. For example, in words like *básilisk*, stress falls on the antepenult, although the final is a footable heavy syllable. A form **bàsilísk* is ruled out by extrametricality restrictions.

Couching this in terms of Optimality Theory, final syllable extrametricality of this type can be enforced via a constraint such as **Non-finality**, which has been used by Prince & Smolensky (1993: 57) to stipulate that the head foot of the prosodic word cannot be word-final:

- (4.1) **NonFin:** No prosodic head of PrWd is final in PrWd

In other words, the final foot in the word cannot be the head (i.e., primary stressed) foot of the word. The presence of the main-stressed foot at the right edge of the word (as in Hayes' (1995) End Rule: Right) can be enforced by using the constraint-type **Edgemost**, here formalized as an alignment constraint (McCarthy & Prince 1993a: 14, 18):

(4.2) **Edgemost:** Align(PrWd, R; Ft(Head), R)

This states that the head foot coincides with the right edge of the prosodic word. The combination of the preceding two constraints yields the extrametricality effects seen in English, so long as **NonFin** is ranked above **Edgemost** (ranking it below **Edgemost** would eliminate any trace of its influence). It is also important to recall that another kind of extrametricality, final consonant extrametricality, is apparently operative in English (Hayes 1982, § 2.2.3). How to express this in Optimality Theory will be discussed below (§ 4.1.5); until then, it should be assumed that word-final consonants are excluded from bearing weight by some constraint, and candidates violating that constraint will not be considered in the tableaux given here.

To enforce proper foot-formation for the words showing the general stress pattern, it is necessary to identify the type of foot used by the grammar. The data suggest that Hayesian bimoraic trochaic feet, i.e., ($\acute{s}_m s_m$) or ($\acute{s}_{mm}$), are the canonical foot-type in English. These can be enforced by a constraint **FtBin(m)**, which requires the presence of exactly two moras in the foot. The trochaic nature of feet is universal in English and we can assume an undominated constraint **FtForm(Trochaic)** (McCarthy & Prince 1993a, Prince & Smolensky 1993).

In figure (4.3) below, this preliminary constraint hierarchy is applied to the possible trisyllabic stem types, defined as combinations of heavy or bimoraic (H) and light or monomoraic (L) syllables. The main-stressed syllable is underlined in each example, and feet are indicated by parentheses. Violations of **Edgemost** are calculated in a gradient manner, each syllable falling between the head foot and the word edge incurring a mark:

(4.3)

/sss/	FtBin	Non-Fin	Edgemost
☞ (LL)L			S
L(LL)		*!	
L(L)L	*!		S
(LLL)	*!	*	
☞ (LL)H			S
(LL)(H)		*!	
L(L)H	*!		S
☞ (H)LL			SS
(H)(LL)		*!	
(HL)L	*!		S
(H)(L)L	*!		S
☞ (H)LH			SS
(H)(LH)	*!	*	
(H)L(H)		*!	
(HL)H	*!		S
☞ L(H)L			S
(LH)L	*!		S
L(HL)	*!	*	
☞ L(H)H			S
L(H)(H)		*!	
L(HH)	*!	*	
(LH)H	*!		S
☞ (H)(H)L			S
(H)HL			!ss
(HH)L	*!		S
☞ (H)(H)H			S
(H)(H)(H)		*!	
(H)HH			!ss
(HH)H	*!		S

Here, it is assumed that syllables will be parsed into feet (suggesting perhaps a constraint **Parse-s**, Prince & Smolensky 1993), except when this causes violations

of more highly ranked constraints, such as those shown above. Thus, word-final syllables fail to be footed in the optimal candidates, due to **NonFin**, while other stray syllables can be left unfooted due to the highly ranked **FtBin**, which demands strict foot binarity.

This constraint hierarchy yields the general stress pattern of English, Kager's first group. For each syllable structure type listed above, the optimal candidate corresponds to the majority stress pattern for unsuffixed nouns of that type, as seen in table (3.26), although of course there are a minority of nouns which fail to show the expected stress pattern. The provisional constraint hierarchy illustrated above raises a number of other questions, such as the status of unfooted syllables, and further investigation of such issues will be taken up later (§ 5.1.1).

In the case of two-syllable unsuffixed words, Optimality Theory succinctly captures Hayes' (1995) "unstressable word syndrome": in cases where extrametricality would yield a monomoraic stress domain, it is suspended by the ranking of **FtBin** over **NonFin**:

(4.4)

/ss/	FtBin	Non-Fin	Edgemost
☞ (LL)		*	
(L)L	*!		S
☞ (H)L			S
(HL)	*!		
☞ L(H)		*	
(L)H	*!		S
(LH)	*!	*	
☞ (H)H			S
(HH)	*!	*	
(H)(H)		*!	

While bisyllables ending in a light syllable will always show initial stress, the effect of **FtBin**, which strictly enforces binary feet, is illustrated by words with final heavies. Due to the ranking of **FtBin** above **Non-Fin**, the stronger requirement for binary feet effectively suspends extrametricality in the L(H) cases, where both (L)H and (LH) would violate **FtBin**. This provisional

constraint hierarchy appears to yield the majority patterns for each weight configuration as seen in the two-syllable unsuffixed words (cf. figure 3.23).⁴⁹

4.1.2 Lexical moras and the minority patterns

For bi- and trisyllables, words showing the two largest minority patterns were ascribed by Kager (1989) to different stress groups, his groups II and III, as discussed in chapter 3 above. I argue instead that these forms are also explicable using the same constraint hierarchy that yields the group I forms, but that words from groups II and III crucially differ from these in their underlying structures. For example, certain words in groups II and III were accounted for above by the proposal of “underlying” geminates as part of the “input” form. That is, for a given stem, a lexically specified mora is associated to one of the consonantal segments that comprises the “input” form. *Gen* must take such underlying moras into account when producing candidates, even though these moras are not motivated by the segmental structure of the “input” morpheme, as in the case of morphemes given to coda consonants.

This raises the issue of how moras are linked to phonemes in ordinary prosodification. In the above examples, the moraicism of the abstract words was a given, based upon the segmental structure of the underlying forms and abstractly represented by symbols signifying heavy and light syllables. Zec (1994: 3) has offered an account of the introduction of moras in the prosodification process, which she refers to as **Moraic Prominence**:

(4.5) Moraic Prominence:

Segment r_i projects a mora iff it is not followed by a more sonorous segment r_j .

m

r_i r_j Condition: $\text{Son}(r_i) \geq \text{Son}(r_j)$

This encapsulates what is likely to be a complex set of constraints governing the relationships between individual segments (or features) and the sub-constituents of the syllable. Every segment in a candidate “input” will

⁴⁹A revised interpretation of the data (§ 5.3) will demand a modified version of this hierarchy, accounting for, e.g., attested cases of (LH), as in *légend*, *brigand*.

correspond to either a mora, or to a non-moraic subsyllabic **margin** entity such as the onset, moras being projected from those segments which meet the sonority condition given in (4.5). This projection of moras should not be disturbed by “underlying” lexical moras; vowels linked to lexical moras will nevertheless project a second mora by (4.5), yielding the expected two moras of a “long” vowel, while consonants linked to lexical moras should nevertheless be parsed into syllable onsets where the sonority relationships make this appropriate, producing a geminate structure. This is illustrated below; subscripts indicate moras which have been projected by *Gen* into the structure, and the segments which correspond to them:

(4.6)

m ₁ m ₂ m ₃	m ₁ m m ₂ m ₃	m ₁ m ₂ m ₃ m ₄	m ₁ m ₂ m m ₃
	/		
k a ₁ l e ₂ r a ₃	a ₁ m e ₂ b a ₃	v e ₁ r a ₂ n ₃ d a ₄	v a ₁ n i ₂ l a ₃
/kalera/	/ame[m]ba/	/veranda/	/vanil[m]a/
‘ cho ra’	‘a me ba’	‘ve ra nda’	‘va ni lla’

The moras that are part of the lexical entry in these cases are not subscripted, and are shown in square brackets in the lexical entries given below the figures.

If the principle summarized by (4.5) did not operate in this way, and if, for example, vowels linked to underlying moras did not project a second mora in this way because of their linked status, then such lexical moras would never distinctively surface. The only alternative way of treating such a situation would then be to specify all moras, for both short and long vowels, in the lexicon. However, this would complicate the lexicon and reduce the role of the grammar in syllabification. Zec’s principle of Moraic Prominence allows for the simpler approach in which only underlyingly long segments require the lexical specification of moras, while also accounting for the projection of moras and the syllabification of segments necessary for any account of prosodification. If this proposal is accepted, it is simple to treat the cases described above in §3.3.2 as possessing lexically marked geminates. When the syllables closed by geminate consonants are treated as heavy, such words then fit into the tables as given above in (4.3) and (4.4). That is, when words like *vanílla* and *molásses* are understood as having a syllable weight

pattern LHL, rather than the apparent LLL, due to a lexical geminate in the penult, then they become interpretable as instances of the optimal candidate L(H)L, just like *veránda* or *amálgam*, rather than as inexplicable suboptimal candidates which surface despite violating **FtBin** (i.e., *L(L)L) or **NonFin** (i.e., *L(LL)).

Similarly, the words proposed (§ 3.3.4) to be suffixed in /-ə/ can also be fit into the set of optimal candidates given above in (4.3) and (4.4), if understood as underlyingly extended by a final light syllable. The suffix /-ə/ projects a mora and syllable like any monosyllabic suffix, but its segmental features are never parsed into the prosodic representation. For the constraint hierarchy evaluation, however, this final syllable is nevertheless present, and allows for the apparent lack of extrametricality exhibited by words like *cemént*, *devóte*. The final consonants in these words, regarded by Kager and others as extrametrical and thus not weight-bearing, can be understood as the onsets of this otherwise invisible final syllable: [se-(mén)-tə], [dē-(vố)-tə].

(4.7)

/sementə/	FtBin	Non-Fin	Edgemost
☞ se(mén)tə			s
(sémen)tə	*!		s
(sé)(men)tə	*!		ss
(sé)mentə	*!		ss

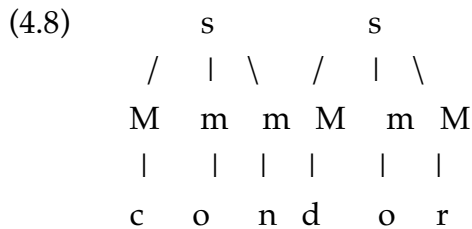
This understanding of a word-final syllable whose vowel fails to surface accounts both for the stress pattern of such suffixed words and the extrametricality of the final consonants. For the small minority of words which show both light stressed final syllables, e.g., *guitár*, the combined explanation of a final geminate consonant and the schwa suffix yields Kager's exceptional third stress group: [gi-(tár)-rə].

4.1.3 Syllable structure: moras and syllable margins

This is an appropriate point to discuss the status of final extrametrical consonants as well as the fate of word-final vowels, such as the non-surfacing schwa proposed above. This requires a more detailed treatment of syllabification and syllable structure. In the previous section, Zec's Moraic Prominence principle was used to account for the projection of moras in

syllabification, and in doing so provided a framework for syllabification. Those segments which projected moras would in most cases be associated with them,⁵⁰ forming the nucleus of a syllable, while segments not projecting moras would become associated with the syllable in other ways, for example joining the onset.

The syllable is different from other prosodic constituents in that it directly dominates more than one kind of sub-constituent. One of these sub-constituents, the mora, has been extensively described in the literature (Hayes 1989, McCarthy & Prince 1986, see also § 1.2.2) and is used as a unit of quantity, impacting upon foot-formation through constraints such as **FtBin**. However, the status of other subsyllabic constituents is less well-defined. The **Onset** constraint (Prince & Smolensky 1993) appears to give the syllable onset some status as a constituent, while their constraint **NoCoda** refers to consonants following the syllable nucleus. Sherer (1994) introduces constraints such as ***Appendix**, referring to a post-nuclear consonantal constituent. Prince & Smolensky (1993: 87) use the term **margin** as a cover term for onsets and codas, contrasting with the **peak** or syllable nucleus. Here the term “margin” will be borrowed and modified to indicate a second subsyllabic constituent type, present on the same tier as the mora, and dominating the nonmoraic segments in the syllable. The coda, as defined in Prince & Smolensky (1993), may or may not belong to the margin, since in some languages (e.g., English) coda consonants are moraic. Onsets and appendices are always margins, occurring at the left and right edges of the syllable respectively.



In English, moras always correspond to (and thus dominate) single segments; additional segments in moraic positions will add to syllable weight and affect foot-formation. In this regard, the status of margins is less clear

⁵⁰One point of Zec (1994) was that segments do not always associate to the moras they project, but that the constraint governing this association can, like all constraints, be violated. In this way she accounts for compensatory lengthening processes (see § 5.2.2).

than that of moras: while English onsets and appendices can contain up to three segments (e.g., *string*, *sixths*), there is no immediate way of telling whether three individual margin constituents are involved, or if a single margin constituent can dominate up to three segments, since the margin count, unlike the mora count, has no effect on other prosodification. There is, however, evidence from other languages which can help to clarify the status of both the mora and the margin. While English allows complex onsets, some languages, like Lardil, only allow a single consonant in the onset (Prince & Smolensky 1993: 98). Prince & Smolensky use the constraint ***Complex** as a “cover term” expressing this limitation, but this could also be regarded as the limitation of the margin constituent to a single dominated segment, like the English mora.

Likewise, it is not necessarily a universal that the mora can only dominate a single segment. Relaxing this constraint would yield the “quantity-insensitive” stress system described in Hayes (1995): for example, in such languages, both the vowels and the coda consonants in a given syllable could be understood as dominated by a single mora. Following this approach, foot binarity would always be enforced by a constraint on the number of moras per foot (and would require two syllables per foot to yield the two moras in such a language), rather than by viewing quantity insensitivity as the result of a different binarity constraint, **FtBin(s)**. Such languages would also be expected not to have long vowels, as the definition of a long vowel involves linkage of the same segment to two moras. Multiply linked moras could also be used to account for phenomena such as short diphthongs.

This proposal does not impede the analysis of other kinds of weight systems. For example, in languages showing weight distinctions only between short and long vowels (but ignoring coda consonants), such as St. Lawrence Island Yupik (Hayes 1995: 51), the **Moraic Prominence** constraints would allow moras to be projected only by vowels, consonants always linking to margin constituents (and thus being non-moraic). The system outlined here allows for the various combinations of constraints on the mora and margin constituents to yield a typology of syllable types with regard to syllable structure and quantity sensitivity.

For English, it is sufficient to note that while moras may only dominate single segments, margins may accommodate a number of segments. For a syllabified candidate to be optimal, its segments must link to

the correct subsyllabic constituents (as governed by **Moraic Prominence**). While it would be possible for a moraic segment, such as a vowel, to project a number of moras rather than one, candidates showing such a structure must be ruled out by constraints of the ***Struc** family (Smolensky & Prince 1993: 25) restricting the proliferation of moras and margins. Such constraints might be formally expressed using **NoIntervening** constraints (Ellison 1995, Zoll 1996) which will ensure that the minimal number of each sub-syllabic category are selected into the representation:

(4.9) Minimize subsyllabics:

*Margins:	NI(Margin, PrWd, Margin)
*m :	NI(m, PrWd, m)

These constraints state that no subsyllabic of a given type should intervene between each subsyllabic and the word-edge.⁵¹ While all (but one) subsyllabics in a given word will incur violations, each additional mora or margin beyond what is absolutely necessary to satisfy **Moraic Prominence** will bring further violations, ensuring that, other things being equal, the minimum number of necessary constituents will be found in the optimal candidate.

4.1.4 Alignment and Intervention

The No-Intervening formalism presented above deserves further explanation. The constraints presented in Smolensky & Prince (1993) were for the most part binary and stipulative, e.g., **NonFin**, which accounts for final extrametricality effects, and is either violated or satisfied by any particular candidate. McCarthy & Prince (1993a) introduced a more formal mechanism for expressing constraints, **Generalized Alignment**, which as noted above in §1.3.4, can be used to capture apparently iterative and cyclical relationships between constituents. Generalized Alignment introduces a constraint type which takes two **constituent edges** as its arguments (McCarthy & Prince 1993a: 2):

⁵¹ In this case, the particular edge in question is not relevant, as the number of violations across a candidate will be identical for each edge.

(4.10) Generalized Alignment:

$\text{Align}(\text{Cat1}, \text{Edge1}, \text{Cat2}, \text{Edge2}) =_{\text{def}}$
 " Cat1 \$ Cat2 such that Edge1 of Cat1 and Edge2 of Cat2 coincide.
 Where
 Cat1, Cat2 $\in$ PCat » GCat (i.e., prosodic or "grammatical" categories)
 Edge1, Edge2 $\in$ {Right, Left}

The key concept expressed here is the relational one of two constituents "sharing an edge" (McCarthy & Prince 1993a: 9). The edges of the constituents are defined by (and can in fact can only be determined in terms of) their terminal nodes; in practice, this usually means segments. In their later reformulation of faithfulness as **Correspondence** (McCarthy & Prince 1995), Alignment is translated into "Anchoring", although the concept remains the same. The grounding of constituents in terminal segmental nodes is formalized, and each constituent defined in the formulation is defined as a string of its corresponding segments.

The Alignment framework can productively account for phenomena which were previously derivable only through the iterative application of rules. For example, McCarthy & Prince (1993a) and Crowhurst & Hewitt (1995) have shown that various possible orderings of constraints aligning the foot to the prosodic word and vice versa can produce the entire typology of stress patterns described in Hayes (1995) (see §1.3.4). McCarthy & Prince (1993a: 16) regard the violation of Alignment constraints as "gradient, not Boolean", and quantify violations in terms of each constituent in the universal (first) argument position. Thus, for example, as feet get farther away from the designated aligned edge, they additively incur more violations of the constraint. This enforces foot-formation in what appears to be an iterative or sequential fashion starting at the designated edge, but which can actually be interpreted as the expression of an alignment constraint within a parallel framework.

One problem with Alignment as presented by McCarthy & Prince (1993a) is that there is no explicit method for determining violation. While exact alignment of constituents is not problematic, misalignment can take myriad forms. Counting segments is one possibility, but this fails to capture generalizations concerning other constituents, such as syllables or feet. It was

customary in analyses like that of Crowhurst & Hewitt (1995) to count violations in terms of the universal constituent, so if syllables were misaligned to feet, violations were measured in terms of syllables (rather than, for example, segments). However, this is an arbitrary generalization which doesn't hold in all circumstances, as noted by Zoll (1996: 110). She adopts the general **NoIntervening** constraint of Ellison (1995: 2), qualifying it with a reference to which intervening constituents incur violations:

(4.11) **NO-INTERVENING**(f; E; D)

There is no material intervening between f and edge E in domain D.

- (i) x(x intervenes between f and edge E)
- (ii) Assess one mark for each value of x

This formalism will be slightly restructured here, and will be presented as follows:

(4.12) NI-Edges(" Cat1, \$ Cat2, I Cat3)

This states that for every constituent of Cat1, no members of constituent Cat3 intervene between an edge of Cat1 and an Edge of some constituent Cat2. When Cat1 and Cat3 indicate identical categories, the typical Alignment effect is achieved, and the following formalisms are equivalent:

(4.13) NI-R(s, Ft, s)
Align(s, R; Ft, R)

In such cases both formalisms may be used interchangeably. The No-Intervening formalism is not identical to Alignment, however, in cases where the intervention category is not identical to the universal. This can be necessary, for example, when dealing with subsyllabics, where more than one type of constituent is present on the same prosodic tier (see § 4.1.5 below), or where it is relevant to specify which intervening subordinate category incurs a violation. Thus, it would be possible to reformulate the **Edgemost** constraint governing the main stressed foot, expressed above as an Alignment constraint (4.2):

(4.14) **Edgemost:** NI-R(PrWd, Ft(Head), Ft)

Under the previous formulation, using Generalized Alignment, it was not clear what would incur a violation of the Alignment constraint: a foot, a syllable, or perhaps an extrametrical unfooted final consonant? Here, it is explicitly stated that the offending intervening category must be a foot, rather than, for example, a syllable or segment.

4.1.5 The margin in word-final position

In English, the margin constituent tends to appear only on the left edge of a syllable as the “onset”, and all other segments in the syllable beyond the onset are dominated by moras. There is one very relevant exception to this statement, the word-final extrametrical consonant, which appears on the right edge of the final syllable. Above, it was suggested that the word-final extrametrical consonant in words showing final stress (Kager’s second type) were the onsets for a final syllable, projected by a suffix /-ə/ which itself fails to surface. However, even in words not explicable by schwa-suffixation, the behavior of word-final consonants is similar. Borowsky (1986: 197) has noted that “a sequence of consonants may be marked extrametrical if and only if they would constitute a well-formed onset.” She regards some word-final syllabic sonorants, such as those seen in *anger*, *crackle* and *wonder* as part of final extrametrical onset-like clusters (p. 225), and demonstrates that these clusters do indeed act like onsets in suffixed forms like *angry*, *crackling* and *wondrous*, suggesting that the proper underlying forms for these words end in final onset clusters, i.e., /wondr/, /krakl/, /anɡr/.⁵² Thus, final extrametrical consonants of this type should be regarded as dominated by the margin constituent, rather than the mora.

Word-final position also appears to show interesting effects with regard to vowels. There is a limited set of word-final vowels in English:

⁵²She also notes (p. 230) the contrast between words of this type, and those that are truly bisyllabic, such as *butter* /bʌtər/ ~ *buttery* /bʌtəri/.

(4.15)	<u>Unstressed</u>	<u>Stressed</u>
[a]	bóurgeois	spá
[ə]	umbrélla	---
[iy]	háppy	decréé
[ey]	ósprey	survéy
[ay]	álly	supply
[ɔy]	álloy	destróy
[uw]	íssue	shampóo
[ow]	hálo	plateáu
[ɔw]	árkansas	guffáw
[aw]	pówwow	allów

Apart from final unstressed [ə] and final [a], all the word-final vowels are phonetically glide-final diphthongs. As was noted above (§ 3.3.3), the standard analysis assumes that some of these word-final long vowels are underlyingly short, but are lengthened or “tensed” by rule in this position. In an OT account, this can be explained using the margin constituent. As with the consonant-final words described above, these words can also be understood as margin-final. Since there is no word-final consonant to fill the margin, the place features of the vowel segments are spread to the margin, resulting in final glides:

(4.16)		s		s	
	/	\	/	\	
	M	mm	M	m	M
		/		/	
	h	a	l	o	

Although this final margin constituent is unnecessary in light of **Moraic Prominence**, and constitutes an extra violation of ***Margins**, it is clear from both these cases and the consonant-final cases above that a word-final margin is necessary to explain the shape of English words. Word-final margin constituents can be enforced by a No-Intervening constraint:

(4.17) **Final Margins:** NI-R(PrWd, Margin, m)

This constraint states that for every prosodic word, no moras intervene between the right edge of the word and a margin constituent.

This constraint has a further use in that it can explain the failure of the word-final suffix /-ə/ to parse. The only word-final short vowels that surface in English are [ə] and [a]. This surface schwa, which is represented orthographically by 'a' and usually understood as the word-final reflex of /a/, presents a violation of the **Final Margins** constraint. Assuming that /a/ has no features which could spread to a margin constituent (e.g., [high]), a margin-final form of a word like *umbrella* would necessarily fail to parse the /a/:

(4.18)

/umbrela/	Final Margins
umbréla	!m
 umbrél<a>	

Since a form *umbrél does not surface in English, we can assume that another constraint enforcing the parsing of this segment, which we can call **Parse-/a/**, is ranked more highly than **Final Margins**:

(4.19)

/umbrela/	Parse-/a/	Final Margins
 umbréla		m
umbrél<a>	*!	

None of the English full vowels are deleted in this way (most are rather lengthened by **FinalMargins**), so the high ranking of constraints that parse vowels into the output is reasonable. However, if **Parse-/ə/**, the constraint which governs the parsing of the segment /ə/ into the output, is ranked below **Final Margins**, the schwa's features will fail to be parsed into prosodic structure and will thus fail to surface:

(4.20)

<i>cement</i> /sement-ə/	Final Margins	Parse-/ə/
☞ cemént-<ə>		*
cemént-ə	m!	

Non-parsing of the final schwa into prosodic structure leads to a margin-final prosodic word, satisfying **FinalMargins**. The low ranking of **Parse-/ə/**, as compared to constraints governing the other vowels, allows for the non-parsing of this segment; also assumed in this account is the inability of any place features in the schwa to spread to a margin constituent. Presumably, the schwa segment does not possess any place features, only vowel height features.

4.2 Stress and morphological constituency

The preceding account of the general English stress pattern was based upon data reflecting stress patterns seen in monomorphemic noun forms. It should be clear from the preceding chapters, however, that English presents a great deal of morphological complexity, which can have a considerable influence on word stress. English affixation has in the standard Lexical Phonological analysis been divided into two sets, known as “level I” and “level II” (Kiparsky 1982a, b, Mohanan 1982, Halle & Vergnaud 1986, Kager 1989). Level I affixation is “stress-affecting”, while level II affixation is not. This description refers to the central assumption of the standard approach that affixed words are extensions of their unaffixed counterparts, built up via a derivational process. Words with level II affixation show stress on the root parallel to that of corresponding unstressed forms, suggesting that the suffix is added “after” stress has been fixed, e.g., *límit* ~ *límitless*. Conversely, words affixed at level I can show a “movement” of the “basic” stress, which is said to apply cyclically, e.g., *átom* ~ *atómic*, *ínstrument* ~ *ínstruméntal* ~ *ínstrumentálicity*. The derivational interpretation, and arguments against it, have been discussed in § 1.4. However, it remains necessary to explain the observed data in terms of Optimality Theory.

4.2.1 Level II affixation

For words showing level II affixation, the generalizations concerning stress and syllable structure noted above apparently do not apply. Although most of the suffixes in this group (e.g., *-less*, *-ness*, *-hood*, *-ship*) are consonant-initial, the closed syllables which result from their addition to the stem nevertheless fail to attract stress:

- | | | |
|--------|----------------|----------------|
| (4.21) | cómeradeship | *comerádeship |
| | schólarship | *scholárship |
| | apprénticeship | apprentíceship |

Instead, the stress remains parallel to that seen in the unaffixed forms *cómerade*, *schólar*, *appréntice*. As word-stress is a feature of the Prosodic Word, it is plausible to understand these suffixes as standing outside the Prosodic Word. This can be expressed using an Alignment constraint, following the formulation for suffixation used by McCarthy & Prince (1993a: 22-4):

- (4.22) Align([*ship*]_{Af}, L; PrWd, R)

This states that level II suffixes such as /-ship/ align to the right edge of a Prosodic Word. While they may be part of the Morphological Word, their corresponding prosodic expression lies outside of the Prosodic Word, and their presence or absence thus has no effect on the position of word-stress. Outside the prosodic word, there is no stress, and no footing; level II suffixes are never stressed. Forms like *wórdlessness* show the following structure, where braces indicate the edges of the prosodic word:

- (4.23) [(wórd)]-less-ness

While these suffixes are parsed into syllables, the syllables are not parsed into feet (or the prosodic word). How these syllables outside the prosodic word might be exempted from footing is discussed below in § 5.2.1. Their position outside the prosodic word also exempts them from the constraints governing stress and footing within the prosodic word, discussed in the next section.

4.2.2 Level I affixation and stress

The differing behaviors of level II and level I affixes suggest that level I affixes are governed by different constraints than those applying to level II affixes, exemplified by the constraint seen in (4.22). The investigation of the English lexicon undertaken here has identified two major groups within the level I rubric. Words showing the /-ity/ group of suffixes (e.g., /-ic, -ity, -ible, -ify/) show consistent antepenultimate stress, while words showing the /-al/ and /-ent/ groups (e.g., /-al, -or, -ous, -ive/ and /-ent, -ence, -ant, -ance/) are stressed according to Kager's group I, that is, on the basis of the weight of the penult:

(4.24)

<u>/-ity/ group</u>	<u>/-al/ group</u>	<u>/-ent/ group</u>
(nóti)fy	ca(thé)dral	pro(pó)nent
(dénsi)ty	ad(ví)sor	per(fór)mance
(mími)c[ə]	de(fén)sive	(írri)tant
(váni)ty	(áddi)tive	(árro)gance
	(dánge)rous	

The /-ity/ forms will always show antepenultimate stress,⁵³ because the first syllable of the suffix itself forms the weak, open second syllable of a bisyllabic foot. That is, the suffix is part of the stressed foot, while in the other sets the suffix is “extrametrical”, standing to the right of the stressed foot, which includes the lexical “root” of the word. If the stressed foot is understood as coterminous in these cases with the morphological category **Stem**, the two types of suffixation seen here can be captured by the following constraints, which govern the **subcategorizations** for these families of affixes:⁵⁴

⁵³Recall that /-ic/ is understood as /-icə/, making stress in words like *mímic* antepenultimate with regard to the underlying form.

⁵⁴See also § 4.3.2 below.

- (4.25) al-Suffixation:⁵⁵ Align(Suf_{al}, L; Stem, R)
 ity-Suffixation: Align(Suf_{ity}, R; Stem, R)

This states that the /-ity/ class of suffixes joins within the stem; the resulting constituent is a stem itself. The /-al/ and /-ent/ suffixes align to the right edge of a stem, and stand outside of it. This can be expressed using the following morphological hierarchy, similar to the prosodic hierarchy:

- (4.26)
- | |
|------|
| MWd |
| |
| Stem |
| |
| Root |

The structure of words suffixed with the level I suffixes described here would be as follows:

- (4.27)
- | | |
|------------|--------------|
| (MWord) | (MWord) |
| (Stem) | (Stem) |
| (Root) | (Root) |
| not - ify | cathedr - al |

Words in the /-ity/ group thus comprise stems in themselves, and are similar in this respect to monomorphemic words, such as *origin*, with which they share a common stress pattern. Affixes of the /-al/ group, on the other hand, will always be excluded from the stem when word-final. This situation is evident from the behavior of the two suffix types: the /-ity/ suffixes are always part of the stressed foot, which includes the morphological stem, while the /-al/ suffixes are always outside the stressed foot, and thus presumably outside the stem. This assumes that there is a direct relationship

⁵⁵Suffixes in the /-ent/ group have the same subcategorization as the /-al/ group and the constraint described here as “al-Suffixation” might be more properly characterized as “al/ent-Suffixation”.

between the stem and the foot, suggesting an Alignment constraint of the following type:

(4.28) **Stem-to-Ft** Align(Stem, R; Ft, R)

This constraint must be ranked below the constraints that enforce final syllable extrametricality (e.g., **NonFin**), as the foot-edge does not coincide with the word-edge in monomorphemic words, but is rather one syllable removed from it.

When regarding the morphological hierarchy and the suffixal subcategorization constraints given above, it is important to note that superordinate morphological constituents such as **Stem** and **MWord** are not inherently linked to any individual morpheme (i.e., string of segments) but are rather projected into candidate morphological structure in the same way that moras, syllables, feet and PrWd's can be projected into candidate prosodic structure by *Gen*, and are likewise evaluated by the constraint hierarchy. Thus, it is not just the case that /-al/ "must" align to a pre-existing stem, but that whatever /-al/ is aligning to should be dominated by a stem constituent. In the same way, suffixes of the /-ity/ group demand that the structure they join to becomes part of a stem, whether or not it possessed any "stem-hood" in its own right. Examples of this phenomenon can be seen in the multi-affixed words *métricality* and *incidental*.

(4.29)

(Stem) (Stem)	(Stem) (St.)
(Root)	(Root)
metr - ic - al -ity	in - cid - ent - al

The stem-hood of the sequence /-al-ity/ in *métricality* is enforced by the constraint Align(Suf_{ic}, R; Stem, R), which requires the suffix /-ity/ to be at the right edge of a stem. The suffix /-al/, while not usually part of a stem, will be incorporated into one to satisfy the constraint. Its own constraint, Align(Suf_{al}, L; Stem, R), requires it to be aligned to the right edge of another stem; this is satisfied by the stem /metr-ic/, which itself satisfies the subcategorization constraint for /-ic/ suffixes by being a stem. In the second example, *incidental*, the suffix /-ent/ appears as a stem in its own right,

enforced by the “al-Suffixation” subcategorization constraint, which requires /-al/ to be adjoined to a stem. As the final stem in the word, /-ent/ can be both footed (due to its intrinsic weight) and given main stress, despite the fact that it is not the lexical root of the word.

These relationships can of course be expressed using OT tableaux; curly braces will be used to signify stem constituents:⁵⁶

(4.30a)

/metr-ic-al-ity/	Align(Suf _{ic} R; Stem, R)	Align(Suf _{al} , L; Stem, R)
☞ {metr-ic}{al-ity}		
{metr}{ic-al}-ity	! **	*
{metr-ic-al}-ity	! **	*
{metr-ic-al-ity}	! *	*

(4.30b)

/in-cid-ent-al/	Align(Suf _{ic} R; Stem, R)	Align(Suf _{al} , L; Stem, R)
☞ {in-cid}-{ent}-al		
{in-cid}-ent-al		! *
{in-cid-ent}-al		! *
{in-cid}-{ent-al}		! *

Because /-ent/ is governed by the same kind of subcategorization constraint as /-al/, candidates like {in-cid-ent}-al are ruled out because /-ent/ must also align to a stem on its left.

4.3 Long vowels and morphological constituency

Above in § 3.3.4, the suffix /-ə/ was classed as a member of the /-al/ family of monomoraic vowel-initial suffixes (e.g., /-al, -ive, -or, -ous/). It would appear from the stress data for this class of words that they also follow the general pattern described above, and that indeed the prosodic form of a vast majority of English words can be accounted for by the simple tableaux

⁵⁶In the following tableaux certain conventions will be used to mark morphological and prosodic structure within candidates. These are not meant to be interpreted as part of the representation, but are rather a means to distinguish the different constituents without adding so many parenthetical markers as to make the forms uninterpretable. Feet will be surrounded by (), stems by {}, morphological words by [], and segments which fail to surface by <>. Furthermore, ‘-’ may be used to divide stem syllables from affixes and affixes from each other, for the sake of further clarification.

offered in (4.3) and (4.4). However, the account given thus far is still incomplete. Although the set of words suffixed with so-called level I suffixes exhibits the same stress pattern (the general pattern described above) as the set of unsuffixed words, there are other adjustments, involving vowel alternation and foot placement, which make the English constraint hierarchy more complex and which, when taken into account, further simplify the English lexicon.

Conclusions were previously drawn (§ 3.3.2) arguing for a geminate explanation for forms like *propéllor* on the basis of the distributions of syllable weight seen across suffixed and unsuffixed forms. The argument was that the proportion of penultimately stressed, short vowel syllables in suffixed words like *propéllor* was much lower than that of stressed short vowels in initial syllables of otherwise similar phonological forms. Without a geminate explanation, the stress on the penult implied either a loss of extrametricality on the suffix, or an ill-formed monomoraic foot, both theoretically problematic. The geminate solution proposed above brings these forms into line with words showing metrically heavy (e.g., *proyéctor*) and long vowel penults (e.g., *arríval*). Trisyllabic suffixed words with truly short penults, like *críminál*, *éditor*, show initial stress.

However, there is another facet to this relationship which was not pointed out in the previous chapter. The distribution seen between words with antepenultimate stress and penultimate stress is not simply conditioned by syllable weight in the penult; there is also a significant morphological component. While the bisyllabic monomorphemic stems which combine with the /-al/ series of suffixes reveal a very regular stress pattern based on the weight of the antepenult, prefixed stems show virtually only one kind of penult, a stressed, bimoraic syllable:

(4.31)

<u>Breakdown</u> ⁵⁷		<u>Showing Initial Stress</u>		<u>Showing Penultimate Stress</u>	
Monomorphemic stems:		<u>88%</u>		<u>11%</u>	
sCvCC-	6%	cávern-ous	5%	triúmph-al	95%
sCvvC-	4%	rhómboid-al	2%	cathédr-al	98%
sCvC-	90%	órbít-al	98%	colóss-al	2%
Prefixed stems:		<u>3%</u>		<u>97%</u>	
s-CvCC-	46%	súb-stant-ive	3%	re-héars-al	97%
s-CvvC-	42%	có-ev-al (alt.) ⁵⁸	1%	re-tríev-al	99%
s-CvC-	12%	ín-fam-ous	5%	re-bútt-al	95%
		<u>Monomorphemic</u>		<u>Prefixed</u>	
Initial Stress:		95%		5%	
Penultimate Stress:		16%		84%	

Ninety percent of the monomorphemic stems exhibit forms with a light penult, but such a shape is nearly absent among the prefixed cases. Of these, practically the only words showing surface light penults are the proposed geminate-final stems, which are nevertheless stressed (these comprise 12% of penultimately stressed prefixed forms). Only a handful of exceptions⁵⁹ show light penults and stress falling on the prefix (e.g., *infamous*), and all these cases have other peculiar morphological characteristics.⁶⁰ All other prefixed stems show heavy stressed penultimate stem syllables.

⁵⁷The breakdown refers to the percentage of each type of bisyllabic stem (monomorphemic, prefixed) that shows the given structure, which focuses on the form of the penult when these stems combine with the /-al/ series of suffixes. The percentages given on the rows indicate what percentage of words with the structural type given on that row show initial or penultimate stress.

⁵⁸This is marked in the lexical database as an alternant to *coéval*, with expected stress.

⁵⁹The very small number of cases noted here includes even questionable examples of prefixed words like *pródigal* and *pósthymous*, which begin with phoneme sequences identifiable with clear prefixes (here, /pro-/ and /post-/), but which are not necessarily morphologically prefixed.

⁶⁰The form *infamous*, for example, being from the long-vowelled *fame* /fām/, is, following the standard view, idiosyncratically shortened.

Once the minority set of stressed monosyllabic geminate-final stems is regarded as also showing a bimoraic penult, we find that among monosyllabic prefixed stem morphemes of this type, practically none (only the circa 3% similar to *infamous*) consist of a single light syllable. Since the surface form of these prefixed stems (e.g., *eléct*, *eléctor*, *arríve*, *arríval*) is assumed to be identical to their lexical entries, it might appear that this class of suffixes never combines with light monosyllabic stems. Following standard assumptions about lexical entries, all stems of this type would be regarded as underlyingly heavy, featuring long vowels or consonant clusters in their lexical entries. This, however, stands in stark contrast to the situation of bisyllabic stems, which overwhelmingly show light syllables. The simple syllable form CV,⁶¹ which makes up 90% of the penults in multisyllabic stems, and which is generally regarded as the unmarked syllable type across the world's languages, is virtually absent from these prefixed stem types.

Turning to monosyllabic roots suffixed by members of the /-al/ group, the results are similar:

(4.32)

CvCC-	44%	cánd-or
CvvC-	43%	tón-al
CvC-	13%	páss-ive

Even if the surface CvC- forms were not regarded as geminates, the contrast with the 90% figure cited above is striking. Among initially stressed unsuffixed bisyllables, 40% show light penults, the most prevalent syllable shape in that position. If the 13% of suffixed monosyllabic stems that do show light surface forms are understood to contain geminates, it appears that the light syllable, which forms at least 80% of syllables found in bisyllabic and longer stems, is present only in traces of less than 3% among monosyllabic suffixed stems that combine with the /-al/ group (i.e., only in words like *infamous*).

Such a distribution is extremely unlikely and suggests that another explanation must be found, based upon the principles of prosodic

⁶¹From CvC- stems, the stem-final consonant acting as the onset to the final syllable, and thus not included in the penult.

constituency discussed previously in chapter 1. Taking into account that the /-al/ group suffixes are always extrametrical, and following the assumption that all feet must be minimally bimoraic in English, the solution to this problem can be presented in a prosodic framework. The stems, both monosyllabic and bisyllabic, are equivalent in shape to bimoraic trochaic feet of the shape (s_{mm}) or (s s). Only in the latter case can light syllables appear in the stem, e.g., *éditeur*. In the cases of both the unprefixed forms like *tónal* and the prefixed forms like *reprisal*, the heavy stem syllable comprises a foot in itself. What is notable is that although the prefixed words contain two non-extrametrical syllables, ample material from which to build a proper foot (as in the exceptional (*ínfa*)<*mous*>), words showing these suffixes apparently “prefer” that their stems alone comprise the head foot of the word. That is, the grammar prefers the prosodic constituent, the foot, to be congruent with the morphological constituent, the morphemic stem. Only a bisyllabic stem, or a heavy monosyllabic stem, can ensure that the morphological stem and phonological foot will coincide, due to the bimoraic constraint on feet.

At this point, it would be possible to claim that these suffixes are restricted by the grammar to combine only with stems that contain at least two moras. However, such a simplistic approach would ignore a number of significant distributional facts. One is that the /-al/ group of suffixes is one of the major groups of formatives available to the (primarily Latinate) stems that participate in level I morphology; the preceding account of verbal morphology (§ 3.3.4) proposed that the suffix /-ə/ responsible for the general verbal stress pattern was also a member of this category. Most verb stems undergoing level I morphology exhibit suffixation with members of the /-al/ group, and show a similar structural pattern, either bisyllabic with light finals⁶² (e.g., *límit*, *édit*, *súffer*, *finish*) or monosyllabic and heavy (*áct*, *tóne*, *re-síde*, *ex-plóde*, *pro-fáne*, *ex-tráct*). Any limitation on combination with the /-al/ group should thus limit verbs to this description as well, and indeed, apart from those displaying the proposed geminates (e.g., *préss*, *o-mít*, *con-féss*), monosyllabic CVC verbs do not appear to take part in any level one word-formation processes, although they are plentiful in the vocabulary that takes part in level II affixation (e.g., *kíck*, *hít*, *táp*, *plúg*).

⁶²Recall that word-final consonants are parsed by Margin constituents and are thus non-moraic, i.e., effectively “extrametrical”.

Thus, while nearly all monosyllabic verbs undergoing level I affixation do show either heavy monosyllables or long stem vowels, the above proviso restricting /-al/ suffixation to heavy stems suggests an otherwise obscure connection between underlying lexical form and affix selection. Or, if the phenomenon is to be linked to “Latinacy” or level I status, it would follow that nearly all Latinate (or level I) monosyllabic stems are inexplicably heavy. The other objection to this proviso is that many of these purportedly underlyingly heavy monosyllabic stems do not always surface with long vowels. In fact, it is these very same morphemes that participate in the so-called vowel shortening described above in chapter 2, for example, *tónic*, *résident*, *profánity*.

4.3.1 Morphologically conditioned vowel alternation

Here a phonological solution to this problem will be proposed rather than a purely lexical one. The vowel length seen in monosyllabic stems combining with these various suffixes (e.g., *tóne*, *tónal*, *profáne*, *resíde*) will be shown to be the result of a predictable lengthening process conditioned by the interaction of the prosodic foot and the morphemic stem in the forms suffixed by members of the /-al/ group. Thus, the lexical stems involved in this process are actually monosyllabic short syllables of the form /CVC/, a canonical stem shape (i.e., /ton/, /fan/, /sid/), rather than underlyingly long forms with the shape /CVVC/, as is usually assumed (i.e., /tōn/, /fān/, /sīd/).

The traditional approach assumed that the lexical stem was identical to the unaffixed surface form of the verb, which in derivational terms was the “basic” word, from which suffixed forms were derived. The interpretation given above, in which most of these verbs are understood as affixed with a suffix /-ə/, means that for most stems, there is no such unaffixed form. The inexplicable patterning in which more than 90% of bisyllabic stems showed short vowels but nearly all monosyllabic stems (apart from geminates) with open syllables showed long vowels is removed by this proposal. Following the suggestion offered here, most lexical stems of both types are understood as possessing underlyingly short vowels.

The lengthening process proposed above can be captured utilizing the constraint **FtBin**, which requires feet to have at least two moras, and an alignment constraint that correlates the morphological stem with the

prosodic foot. Such a constraint, **Stem-to-Ft** (4.28), was suggested above to help govern *ity*-suffixation. Under this approach, the underlying stems for words like *tóne*, *tónal* can be seen as light, i.e., /ton/, rather than the underlyingly long /tōn/ usually assumed for such words. Given a choice between a monomoraic foot (*ton*) and a violation of suffix extrametricality, the grammar instead adds an “additional” (in the sense that it is not required segmentally, by **Moraic Prominence**) mora to the stem, satisfying the binary foot constraint. The process by which an additional mora may be added to the representation will be governed by the constraint *m (4.9).⁶³ This constraint will be violated by the presence of more moras than required by **Moraic Prominence**.

(4.33a)

/ton-al/	FtBin	Stem-to-Ft	*m
☞ (tō)n-al			m
(ton-al)		*!	
(to)n-al	*!		

(4.33b)

/ton-ə/	FtBin	Stem-to-Ft	*m
☞ (tō)n-ə			m
(ton-ə)		*!	
(to)n-ə	*!		

Such a lengthening for the satisfaction of prosodic and morphological constraints is found in other languages and has been described extensively by McCarthy & Prince (1986, 1990a, b). Note that the alignment constraint **Stem-to-Ft** as formulated above in (4.28) can be interpreted as demanding strict alignment at the phone level between the stem and the foot. As the stem-final consonant will form the onset to vowel-initial suffixes like /-al/, this

⁶³One might also wish to understand this as functioning like a constraint **Fill-m**, to use a formalism from Prince & Smolensky (1993). Moras required by the lexical entry (via **Moraic Prominence**), while violating this constraint, will violate it equally for every candidate and thus do not crucially violate it. Only additional moras without lexical segmental correspondents will incur further violations. The concept behind **Fill-m** conflates *m and **Moraic Prominence** in this way.

consonant will be always excluded from the foot syllable. This suggests that the constraint needs to be reformulated more precisely as a No-Intervening constraint:

(4.34) **Stem-to-Ft-R** NI-R(Stem, Ft, s)

This states that no syllables intervene between the right edge of each stem and the right edge of a foot. Thus, the fact that stem-final consonants are excluded from the foot do not incur a violation of the constraint, as no syllables intervene between the right stem-edge and the right foot-edge, only an onset does.

In the previous example, the monosyllabic stem was the only item available for foot-formation, as the suffixal syllable was excluded from consideration. In trisyllabic prefixed cases like *reprisal*, however, the grammar (in the form of *Gen*) has the option of choosing between such a lengthening, and forming a canonical bisyllabic foot which includes the prefix, e.g., a form **reprisal* [(ré-pri)z-əl]. The actual choice made in English disfavors the misalignment of constituents that such a form would demand. It is possible to account for this using a variation of the constraint **Stem-to-Ft**, focusing on the left edge:

(4.35) **Stem-to-Ft-L** NI-L(Stem, Ft, s)

This constraint, together with **Stem-to-Ft-R**, will favor a stem that is aligned on each side to the edge of a foot. For a monosyllabic stem, this means that the stem will be congruent with the foot. The resulting situation is a typical case of competing constraints in Optimality Theory:

(4.36)

/re-pris-al/	FtBin	Stem-to-Ft-R	Stem-to-Ft-L	*m
☞ re-(pri)s-al				m
re-(pri)s-al	*!			
(ré-pri)s-al			*!	
re(prís-al)		*!		

The form *reprísal*, which only violates the lowest-ranked constraint in the tableau above, surfaces as the most optimal candidate, showing as its main-stressed foot the monosyllabic stem /pris/, its vowel lengthened to satisfy the bimoraic constraint on feet.

4.3.2 Stress and vowel quantity in the /-ity/ group

Above in § 4.2.2, it was pointed out that for suffixation with members of the /-ity/ group, one light syllable of the suffix is always available to form the weak second member of a bisyllabic trochee, due to differing constraints governing the suffix groups.

- (4.37) al-Suffixation: Align(Suf_{al}, L; Stem, R)
 ity-Suffixation: Align(Suf_{ity}, R; Stem, R)

As noted above, this states that the /-al/ suffixes adjoin to stems while the /-ity/ suffixes actually comprise stems themselves, adjoining to morphemic roots (the a-level of Inkelas 1989). Such subcategorization constraints are analogous to the subcategorization frames offered in Inkelas (1989) and the specific morphological constraints presented in McCarthy & Prince (1993a) (see § 1.3.5).

Since /-ity/ suffixes join the root morphemes to form a morphological stem, there is no need for lengthening of the stem vowel in monosyllabic roots, and their underlying short vowel quality surfaces, e.g., *pro(fáni)<ty>*. The syllable preceding the suffix will always be stressed, and the stressed foot will always appear to be a bisyllable of the form (HL) or (LL) :

- (4.38) ro(bót-i)c[ə] ve(híc-u)lar so(líd-i)fy
 te(mér-i)ty con(vért-i)ble e(térn-i)ty

The interpretation of /-ic/ as /-icə/, mentioned above (§ 3.3.4), allows this suffix a representation parallel to those of /-ity/, /-ify/, and the other members of the group. Since the initial syllable of these suffixes is onsetless and light, such syllables can always form the weak part of a bisyllabic trochee. Words in this category show an almost universal adherence to this stress

pattern, about 99%.⁶⁴ As words suffixed with the /-ity/ group are, according to the above subcategorization constraints, structurally identical to bare monomorphemic stems, this standard stress pattern is expected. What is also found, similar to bare stems and in contrast with the /-al/ group, is a dearth of long vowels. As suggested by the form *pro(fánit)y* offered above, the true underlying short quality of most English stems is able to surface in these words, due to the bisyllabic (and thus bimoraic) nature of the stems involved:

(4.39a)

/pro-fan-ə/	FtBin	Non-Fin	Edgemost	Stem-Ft-R	Stem-Ft-L	*m
☞ pro-(fā)n-ə			s			m
(pró-fa)n-ə			s		!s	
pro-(fán-ə)		!*		s		

(4.39b)

/pro-faniti/	FtBin	Non-Fin	Edgemost	Stem-Ft-R	Stem-Ft-L	*m
☞ pro-(fáni)ti			s	s		
pro-(fāni)ti			s	s		m!
(pró-fa)(niti)			!ss		s	

There is no need for any lengthening in cases like *profánity* as the resulting multisyllabic stem, extended with members of the /-ity/ group, will always contain at least two moras and thus satisfy **FtBin**. Any candidate showing the lengthened stem vowel will be eliminated by ***m**. Note also that candidates which bring the main stress too far forward in the word (e.g., *prófanity) are excluded by **Edgemost**, which attempts to keep the main stress as close to the right of the word as possible.

4.3.3 Vowel shortening resolved

Regarding stems like those seen in *tónal* and *reprisal* as /ton/ and /pris/, as possessing lexical short vowels rather than long ones revolutionizes the previously intractable problem of “vowel shortening”. Casting it instead in terms of predictable morphologically conditioned vowel

⁶⁴The exceptional words are almost all in /-ic/, such as *héretic*, and suggest stems like /héretik/, without the final /-ə/. Some of these forms have irregular morphology (cf. *héresy*) and could be interpreted as unitary stems.

lengthening in other cases (including, in the case of verbs and many adjectives, the supposed “basic” unsuffixed form), the entire character of the lexicon is changed. The distribution of lexical long vowels, which should be regarded as more marked than short vowels, is suddenly reduced and stems of the form CvC resume their expected frequent position among the lexical items participating in level one affixation:

(4.40)	<u>Stems in /-al/</u>	<u>Initially Stressed Bisyllables</u>
CvCC-	44%	35%
CvC-	43%	39%
CvG- ⁶⁵	13%	
CvvC-	(0%)	25%

While the complete elimination of long vowels suggested by the table is premature (see below), the distribution of short vowels in these stems is brought closer into line with that seen in initially stressed bisyllables, for example. By regarding these stems as lexically short, this portion of the set of morphemes becomes structurally simplified as well, no longer containing a majority of prosodically marked long vowels, each requiring lexical moras to be associated with their morphemes.

Under the traditional approach, which assumes (insofar as vowel length is concerned) that the surface forms of these stems best represent their underlying forms, the lexicon would have to contain a very high percentage of phonologically marked monosyllabic long vowel morphemes, starkly contrasting with the extremely high percentage of short vowel syllables (about 80%) seen in multisyllabic morphemes. Earlier approaches, both in the structuralist tradition and in Lexical Phonology, contained the default assumption that the unaffixed forms were equivalent to the “basic” stem forms, and the special status of unaffixed forms in such theories has already been discussed (§ 1.4.1). However, it should also be clear that such minimally affixed morphemes would be the prime candidates for showing the

⁶⁵G’ standing for geminates. As geminates would have no effect on the surface form of monomorphemic bisyllables in this position, it is not possible to say how many items in the set of these bisyllables are lexically geminates, without checking morphologically related forms where the gemination might make itself known. For this table, that proportion is in any case irrelevant.

incremental effects necessary to achieve minimality constraints, such as the cross-linguistically common tactic of minimal word lengthening, which typically applies in other languages to monomoraic roots for the same reasons (McCarthy & Prince 1986, 1988, 1990a, b). Only in longer forms, where the lengthening is no longer required by the bimoraic constraint, is the short underlying quality of the vowel able to surface.

Although the previous table suggests that a vowel length contrast in the English lexicon is no longer necessary, this is of course not the case. However, the indication of true lexical vowel length in a particular morpheme can now be defined, according to the above analysis, as the presence of a long vowel in environments where a “shortening” was expected under approaches such as that of Myers (1987) (§ 2.2.3). In previous analyses, words such as the following were regarded as “lexical exceptions” to shortening:

(4.41)	obésity	próbity	aphásic	códify
	nótify	glórify	phonémic	deféasible

Under the interpretation offered here, they can rather be regarded as showing true, lexically marked long vowels. As with the lexically marked geminates, these stems will consistently show the same stem vowel regardless of morphological combination (e.g., *nóte*, *nótary*, *nótable*, *connótative*, *nōtátation*, *nótify*). Lexically marked long vowels will not crucially violate the constraint *m used above, as all candidates will have to contain this lexical mora.

Another interesting property of such lexically marked stems is their potential simplification, not in phonological combination, but as a lexical development in individual or dialectal grammars. For example, the variant pronunciations of *cýclic* /síkl-ik/ ~ /síkl-ik/, *próbity* /prōb-iti/ ~ /prób-iti/, *códify* /kōdi-fi/ ~ /kód-ifi/ all suggest a lexical simplification from the more marked stems /síkl/, /prōb/ and /kōd/ to the regularized forms /sikl/, /prob/ and /kod/. Both the marked and unmarked stems will yield the unsuffixed forms *cýcle*, *próbe* and *códe* (due to the bimoraic constraint on feet), and this move towards lexical simplification is parallel to that seen in the geminates such as *harass* and *combat* (§ 3.3.2). Another place where lexical long vowels can be noted is in positions where the morphologically

conditioned vowel lengthening discussed above is not applicable, for example within bisyllabic stems like those found in *cípher*, *séason*, *líbel*, *dánger*, *sécret*. Long vowels in the initial position of such bisyllabic stems occur only 14% of the time, a distribution similar to that of the equally marked geminate consonants. Likewise, the less than 5% of words that show superheavy syllables, such as *dánger*, *ángel*, must also exhibit true long vowels, as there is no need for lengthening in syllables already heavy by consonant weight.

Recognizing true long vowels as distinct from surface long vowels which predictably correspond to short underlying representations completes the syllable inventory, and accounts for a large number of words that were previously inexplicable exceptions to vowel shortening. Forms like *nótify* were especially problematic for all previous explanations. For Myers (1987), this word inexplicably failed to “resyllabify” but nevertheless flapped its /t/; for Prince’s (1990) foot-based approach, despite forming an unbalanced foot, (*nóti*)*fy* failed to optimize its foot shape by shortening the vowel. Such an exception is not possible under Optimality Theory,⁶⁶ as the entire grammar ought to be the result of a single constraint hierarchy. Using the constraints outlined above, the contrast between *nótify* and *vílify* becomes a prosodic one, based on lexical vowel length, rather than the result of lexical exception marking.

Thus, the observation seized upon by Myers (1987: 488) that “within roots, long vowels appear only in open syllables... [while] a nonfinal closed syllable must generally have a short vowel” is shown to be indeed relevant to the problem of vowel alternation. But the solution is not, as Myers proposed, that this means long vowels shorten before consonant clusters (making a whole series of words like *pint*, *beast*, *round*, *dánger*, *ángel*, thus lexical exceptions), but rather the simpler realization that no lengthening is needed in monosyllabic stems that have closed syllables, as these syllables are already bimoraic. Stems with lexical long vowels can freely display them, whether in open (e.g., *nótify*) or closed (e.g., *dánger*) syllables. The observation is the same, but the direction of causality in Myers (1987) is completely the opposite of that presented here, driven as it is by assumptions about underlying forms traditionally left unquestioned, but shown here to require an otherwise

⁶⁶At least not under the “strong” variant of OT pursued here, which rejects “co-phonologies” as a device which weakens the theory and simply imports unwanted strategies from derivational theory (see § 1.4.3).

inexplicable and unusually marked lexical distribution of underlying vowel length.

4.3.4 Vowel reduction and the account of Burzio (1993)

At this point it is relevant to discuss the claims of Burzio (1993) in the context of the above explanation. Burzio (1993: 364) bases his analysis of English phonology on the idea that “vowels shorten in word formation.” This obviously assumes the traditional “long” underlying forms discussed (and argued against) in the previous section. The solution just offered clarifies both the apparent phenomenon of “vowel shortening” in word formation, and the fact that it does not always occur, especially in forms of the /-al/ group, where long vowels are endemic, despite the operation of word formation. To support his claims, Burzio (1993: 362-3) discusses a number of cases of vowel reduction that he groups together with general shortening processes:

(4.42) blàsphéme ~ blásp[hə]mous	vagína ~ vág[ɪ]nal
aspíre ~ ásp[ɪ]rant	excréte ~ éxcr[ə]tòry
chástize ~ chást[ɪ]zement	sátire ~ sát[ɪ]rist
admíre ~ ádm[ɪ]rable	sémite ~ sém[ɪ]tism
píous ~ ímp[ɪ]ous	fámous ~ ínf[ə]mous ~ ínf[ə]my
fínite ~ ínf[ɪ]nite	cýcle ~ bíc[ɪ]cle
thésis ~ antíth[ə]sis	pótent ~ omníp[ə]tent

A few of these forms (e.g., *sémitism*, *áspirant*) represent productive types, but most are rather examples of exceptional behavior for the affixes in question (e.g., *ínfamous*, *ádmirable*). The difference between vowel shortening and vowel reduction was discussed above in § 2.3.2. The stressing environments shown here are either inconsistent ones discussed previously in § 2.3 (e.g., /-able/, /-ist/, /-ism/, /-ment/), or represent the /-al/ class of suffixes (§ 3.3.1), which overwhelmingly do not show shortening effects. In the bottom six examples listed in (4.42), a shift in stress is brought on by prefixation, a phenomenon primarily seen rather in the /-ent/ suffix group (e.g., *áspirant*, *omnípotent* noted here).

The distributional data from the corpus suggests that these examples put forth by Burzio are extremely marginal; in terms of the /-al/ type words

with apparent prefix shortening, relatively simple lexical, rather than transformational, explanations are available. Such words can be seen as having historically undergone a morphological reanalysis of the stem. This type of effect is rare but telling when it occurs; it appears usually to be the result of a misparse based on structural or semantic confusion. Typically, prefixed monosyllables suffixed with the /-al/ set retain the stress on their roots. However, the form of the word *impious*, with its unusually short root /pi/, allows for a false parse as /imp-ious/, a stem with the complex suffix /-i-ous/, e.g., *precarious*, *spurious*, *copious*. In the case of *infamous*, the lack of a clear semantic connection between this word, meaning “badly famous” rather than “not famous” and *famous* has allowed for a reanalyzed stem /infəm/ or /infm/, lacking a clearly analyzable vowel in the root position. This stem was accessible for reanalysis from the form *infamy*, a type in the suffix /-y/ which generally shows antepenultimate stress in trisyllables, e.g., *calumny*, *alchemy*, *autopsy*.⁶⁷

Burzio maintains, however, that this type of reduction is identical to the shortening discussed above in § 2.2, except that what he refers to as “stress preservation” is not maintained. He bases the (shifted) stress of these words on regular patterns of words with short vowels in penultimate position, for example, *américa*, *bárbarous*. This follows from his general principle that “vowels shorten in word formation” (p. 364), which joins with a principle, which is sometimes violated, that stress is preserved in word-formation. This kind of suggestion is akin to the Strict Cyclicity Principle of Lexical Phonology (§ 1.4.1) taken to an extreme: lexical structure is not just only preserved in underived words, it can be dispensed with entirely (at least in terms of length marking) in derived words. The fact that many complex words with surface long vowels do not show any shortening does not appear to hinder Burzio’s proposal of this “rule”, to which there are of course numerous exceptions in which “the choice of which contending principle is satisfied in each case [is] idiosyncratic, although further study may turn up additional principles” (p. 366). For the overwhelmingly frequent cases with long vowels exemplified by the extremely common /-al/ suffixes, Burzio has decided that forms like /tón-al/ do not surface with short vowels, as his theory predicts, because foot

⁶⁷See § 7.1.1 for ways of treating such exceptions to otherwise general subcategorization patterns in the context of OT.

structures with the pattern (LL), that is, two light syllables, are comparatively ill-formed at the edge of the word (p. 412). This unfortunately disregards the thousands of words, forming the largest group among unsuffixed bisyllables, as well as the hundreds more forms in /-ic/, which show this pattern,⁶⁸ not to mention the fact that cross-linguistically this foot type is the most common and has been described as the least marked and most optimal (McCarthy & Prince 1993a, Prince 1990).

While Burzio's work is full of both interesting observations about the English lexicon and interesting proposals to account for the forms, it should be now clear that the generalizations he cites are simply epiphenomena of the very different processes discussed and explained here, on the basis of patterns seen in an exhaustively researched corpus of English data. Such a small percentage of stems actually engage in vowel shortening in English that it is implausible to introduce "General Shortening" as a general word-formation principle. The generalization that apparently drives Burzio's approach, that affixed words (in fact most multisyllabic words) tend to show a lot of short vowels, is valid and important, but is rooted in a system completely at odds with Burzio's proposed explanation, a system arrived at by applying cross-linguistic universal concepts of markedness to patterns of data in a large corpus, using a constrained and explicit linguistic theory.

4.4 Summary of stress and vowel alternation

In this chapter, most English words have been shown to follow a basic stress pattern characterized by final syllable extrametricality and quantity-sensitive trochaic feet. The minority of words apparently not conforming to this generalization can be accounted for by structural adjustments in their lexical form, such as the recognition of geminates in positions where light monosyllables seem to comprise feet, or the understanding of a suffix /-ə/ which fails to surface but whose syllable affects stress placement. The reinterpretation of vowel length as a constituency-enforced marker in certain suffixed forms, enforced by alignment and minimality constraints, allows for the overall simplification of the lexical entries, vastly reducing the number of underlying long vowels which need to be proposed, explaining the

⁶⁸Burzio evidently posits for these an empty final syllable, characterized as /Ø/; thus the (LL) foot is no longer word-final.

phenomenon of “vowel shortening” and accounting for exceptions to this “shortening” in a principled way, through true lexical vowel length. By delinking issues of structure and meaning, the form of the morphemes in question could be focused upon, and unmotivated assumptions concerning the underlying forms could be disregarded in the face of the distributional prosodic evidence.

5. Lengthening and constituency

5.0 Refining the constraint hierarchy

The OT account of English stress and vowel alternation presented in the previous chapter, while consistently reflecting the theory of the English lexicon offered in this study, was somewhat simplified in terms of the constraint hierarchy offered. There are a number of further interactions between stress and structure in English which call for a fuller analysis, using a more detailed constraint hierarchy. The principal insight from the last chapter, that many long vowels seen in English words are predictably morphologically lengthened, will be maintained in this chapter, as will the general principle that only variations in lexical structure, rather than exception marking, can and should be used to explain forms which display minority patterns on the surface.

Below, the supposed stress retraction seen in /-ent/ words such as *résident*, is discussed, leading to a refinement of the constraint hierarchy and the redefinition of some constraints used in the last chapter (§ 5.1). The environments for morphological lengthening will be further explored, and lengthening is restricted to the small set of constituents in which evidence for it can be seen from the data (§ 5.2). All other environments are prevented from displaying lengthening in optimal candidates by various constraints, to prevent the grammar from overgenerating (§ 5.2.1). This is followed by an account treating other types of lengthening found in the English data (§ 5.2.2). Refinements to the existing constraint hierarchy are proposed and further constraints are introduced to fully account for extrametricality effects and stress clash (§ 5.3). Following this, the bare bisyllables and trisyllables will be re-evaluated with the updated constraint hierarchy (§ 5.3.1), and the role of prefixes will be explored (§ 5.4).

5.1 Constituency and the /-ent/ suffixes

The third major suffix group introduced in chapter 2 was the /-ent/ group, which displays a series of vowel-initial heavy monosyllabic suffixes. The stress pattern seen in this group again follows the general pattern for English (Kager's group I), but like the /-al/ forms, the prosodification of stems

in combination with /-ent/ suffixes is not completely straightforward. Myers tried to ascribe a type of “shortening” seen in these words (e.g., *résident*) to “sonorant destressing” (§ 2.2.3), a tactic that has been dismissed by Halle & Vergnaud (1987), Kager (1989) and others. Taking into account the explanation of monosyllabic stems given above, the resolution of this type of shortening is now possible:

- (5.1) re(síd)e (résí)dent
 re(spónd) re(spón)dent

The underlying form of *resíde* can now be understood as /re-sid-ə/, with a light stem syllable, while the stem /-spond-/ seen in *respónd* and *respóndent* is intrinsically heavy due to its segmental structure. Thus, the difference in stress between these forms is directly linked to the underlying weight of the root syllables. In terms of the current constraint hierarchy, the violation of **Stem-to-Ft-L** (§ 4.3.1), crucial for the resolution of forms like *reprísal*, seems to be insufficient to prohibit *re(sí)dent as opposed to the attested (résí)dent.

It might simply be suggested, as was the case for the /-ic/ group, that the /-ent/ suffixes form part of the stem, whereby {(résí)dent}_{stem} would be the expected output, similar to {(sáni)ty}_{stem}. However, this is contradicted by contrasts such as *sónant* ~ *sónic*. The former evidently shows a lengthened stem syllable, which would not be the case if this was a bisyllabic stem of the shape /LH/; a lengthened monomoraic stem {son} is more likely:

(5.2)

/son-ant/	FtBin	Non-Fin	Stem-to-Ft-R	*m
☞ {(sō)n}-ant				mmmm
{(son)-ant}	*!	*	s	mmm
{(so)n}-ant	*!			mmm

A stem of the form /LH/ would have resulted in *sōnant, parallel to *sónic* or *brígrand*.⁶⁹ Only when more syllables precede the stem syllable does lengthening fail to take place, e.g., *résonant*.

⁶⁹Note that the current hierarchy cannot account for words like *brígrand*, which will be dealt with below, § 5.3.

The correct optimal form for trisyllabic /-ent/ words like *résonant*, *résident*, contrasting with heavy stems words like *respóndent* and /-al/ forms like *reprísal*, will not be arrived at with the current set of constraints used in (5.2), but as was mentioned above, these constraint tableaux are preliminary. Additional issues need to be addressed, such as how unfooted syllables are regarded, and how foot binarity is to be interpreted. In the preceding analysis, the high-ranking descriptive constraint **FtBin** was used to enforce strict binarity of feet, a tactic which is too powerful as it would exclude many attested forms. Below, the constraint hierarchy will be further refined to address such issues. Some of the constraints introduced above, although capturing clear linguistic insights, are too general in their coverage. A notable case is **FtBin**, which captures a number of generalizations in one constraint. A reformulation of **FtBin**, splitting it into a number of component constraints, provides some of the framework needed to account for the retraction seen in the /-ent/ forms.

5.1.1 Foot binarity and the Lapse constraint

Green & Kenstowicz (1995) have suggested that the constraint **FtBin** can be separated into two independent constraints, **Lapse** and **Min-2**. **Min-2** ensures that every foot contains at least two moras, and is violated only by feet containing a single mora. Green & Kenstowicz define the **Lapse** constraint as follows (p. 1):

- (5.3) **Lapse-m, Lapse-s:** adjacent unstressed moras or syllables must be separated by a foot boundary.

This constraint is used to handle syllable sequences (for quantity-insensitive systems) of the following type (Green & Kenstowicz 1995: 1):

- (5.4)
- | | | <u>Lapse</u> |
|--------------------|--------------|--------------|
| i. too large: | ('ss)s | ÷ |
| | ('sss) | * |
| ii. too far apart: | ('ss)('ss) | ÷ |
| | ('ss)s('ss) | ÷ |
| | ('ss)ss('ss) | * |

While each well-formed foot will incur at least two violations (the syllable at the right edge of the foot is not at the left edge and vice versa), any adjacent weak syllables that are not separated by a foot edge will incur at least one extra violation. These constraints, like other alignment constraints, are gradient rather than Boolean, so a trisyllabic foot (sss) will always set the edge syllables too far from the opposite edge to ever be more optimal than a bisyllabic structure. When only trisyllabic output candidates are available, one syllable must be excluded from the foot in the optimal candidate. Excluding two syllables from footing (even if permitted by **Min-2**) would still produce a sub-optimal candidate according to **Lapse**. In the example above (5.6), the choice between the two candidates that equally minimally violate the **Lapse** constraint is decided by another constraint, in this case **Non-Fin**.

Below, other general consequences of this interpretation of the **Lapse** constraint will be presented. The various **Lapse** environments given by Green & Kenstowicz can similarly be produced, the constraint again being expressed using the alignment formalism:

(5.7)

/sssss/	A(s, L; Ft L)	A(s, R; Ft, R)
☞ (śs)s(śs)	sss	sss
(śs)(śss)	! ssss	sss
(śss)(śs)	! ssss	sss
(śs)(śs)s	! ssss	sss

For the five-syllable candidate, the favored candidate (other things being equal) is the so-called “initial dactyl”. Again, ternary feet are impossible due to this constraint, and for the unfooted syllable, the position in the middle of the word brings it closest to the relevant foot-edges.⁷¹ Of course, other constraints, such as a higher-ranking **Non-Fin**, could eliminate the dactyl candidate. Candidates with an even number of syllables would be optimally fully parsed according to the **Lapse** constraint.

⁷¹In example (5.7) the difference between the optimal candidate and the ternary foot candidate is of course purely theoretical, as both would surface identically. Differences can be seen in longer words, e.g., (5.8).

(5.8)

/SSSSSS/	A(s, L; Ft L)	A(s, R; Ft, R)
☞ (śs)(śs)(śs)	SSS	SSS
(śs)s(śss)	! SSSSS	SSSSS
(śss)s(śs)	! SSSSS	SSSSS
(śs)ss(śs)	! SSSSS	SSSSS
s(śs)s(śs)	! SSSS	SSSSS

But again, a higher-ranking constraint like **Non-Fin** can eliminate otherwise optimal candidates; even then, bisyllabic feet are preferred and ternary feet will never occur:

(5.9)

/SSSS/	Non-Fin	A(s, L; Ft L)	A(s, R; Ft, R)
(śs)(śs)	!* SS	SS	SS
☞ s(śs)s		SSSS	SSSS
(śss)s		!SSSSSS	SSSS
(śs)ss		!SSSSSS	SSSS
s(śss)	!* SSSS	SSSS	SSSSSS

In this final example (5.9), the fully parsed candidate (śs)(śs) is clearly optimal solely on the basis of the **Lapse** constraints. However, the candidate s(śs)s, which Green & Kenstowicz use as an example of an acceptable candidate, is the most optimal of those that show unparsed syllables. A superordinate constraint, such as **Non-Fin**, preventing the final syllable from belonging to a foot, could eliminate the fully parsed candidate and yield the selected form as optimal.

Breaking **FtBin** into **Min-2** and **Lapse-s** allows for two of the insights conflated into **FtBin**, minimal foot binarity and maximal foot binarity, to be treated independently. This is necessary because while feet in English never have less than two moras, they may have more, and while they never have more than two syllables, they may consist of less, i.e., one multimoraic syllable. Restructuring the hierarchy to account for this, the tableau for *résident* might appear as follows:

(5.10)

/re-sid-ent/	Min-2	Non-Fin	Stem-Ft-R	Lapse-s	Stem-Ft-L	*m
re-(sī)-(dent)				! sss		mmmmn
re-(sid-ent)		!*	*	sss		mmmm
☞ (ré-si)-(dent)					*	mmmm

The **Lapse-s** constraint enforces the correct choice of *résident*, which foots all available syllables, from among these candidates. Since the suffix /-ent/ is a heavy syllable, it can be regarded as forming a foot in itself;⁷² not footing the suffix syllable would yield more **Lapse** violations.

Note however that these two constraints are insufficient to perform all the work that **FtBin** did previously; this hierarchy produces a similar (and incorrect) result in the case of *respóndent*, namely *(ré-spon)-(dent). What is needed is a further constraint, **FtForm**, which dictates the possible moraic shapes for the bisyllabic foot and will penalize trimoraic feet of the form (LH), as in (ré-spon):

(5.11) **FtForm:** NI-R(s, Ft, s_{mm})⁷³

This restricts the right-hand member of a multisyllabic foot to a monomoraic syllable; a bimoraic syllable in this position would intervene between the left-hand syllable and the right edge of the foot, incurring a violation. This constraint treats the possible feet of English in the following ways:⁷⁴

⁷²This final syllable can receive a secondary stress in some pronunciations; however it is normally unstressed. For the issue of footed yet stressless syllables, see § 5.4.1, § 6.5.

⁷³While I would prefer a formulation of this constraint which would not include a complex argument like s_{mm}, I was unable to arrive at any other correct formulation. In any case this complex structure has been used as a constraint argument in other studies such as Sherer (1994), Rosenthal (1994).

⁷⁴This constraint addresses the same issues as Prince's (1990: 361) concept of **Harmony** within the foot, but differs in the more limited range of evaluation results offered here.

(5.12)

	<u>FtForm</u>
(s _m)	÷
(s _{mm})	÷
(s _m s _m)	÷
(s _{mm} s _m)	÷
(s _m s _{mm})	*
(s _{mm} s _{mm})	*

What this constraint penalizes is not trimoraic feet, as did the *s_{mmm} of Sherer (1994), because as will be seen below (§ 5.2), the foot (s_{mm} s_m), here equal in status to (s_m s_m), is acceptable and necessary to account for a number of attested forms. Adding this constraint to the hierarchy yields the correct result for words like *respóndent*:

(5.13)

/re-spond-ent/	Min-2	Non-Fin	Stem-Ft-R	FtForm	Lapse-s	Stem-Ft-L
☞ re-(spón)-(dent)					sss	
(ré-son)-(dent)				!*		*

The replacement of a monolithic **FtBin** by three constraints, each representing an aspect of foot structure conflated into **FtBin**, allows for the separate ranking of each component constraint and a more fine-tuned approach, retaining the insights behind **FtBin** while allowing for the attested data to be properly accounted for under OT.

5.1.2 Vowel lengthening as compensatory lengthening

At this point, **FtBin** has been adequately replaced by the three constraints **Min-2**, **Lapse-s** and **FtForm**. However, much more needs to be said about the nature of the lengthening process constrained above (§ 4.3.1) by *m, for if this new constraint hierarchy is applied as is to the case of *reprísal*, the following, incorrect, tableau results:

(5.14)

/re-pris-al/	Min-2	Non-Fin	StemFtR	FtForm	Lapse-s	StemFtL	*m
re-(prĩ)-sal					!sssss		mmmm
re-(pris-al)		!*	*		sss		mmm
☞ *(ré-pri)-sal					sss	s	mmm

The extra mora inserted into the stem syllable of *reprisal* results in two other unstressed, unfooted syllables, incurring many more **Lapse** violations than the competing forms. Yet promoting **Stem-Ft-L** in the hierarchy to eliminate *(ré-pri)-sal would then yield an incorrect optimal candidate for *résident*, i.e., *resídent. The difference between the two types still cannot be represented in the current constraint hierarchy; the principal contrast thus far between this hierarchy and the one presented in § 4.3.1 is that it yields the correct surface form for the /-ent/ type rather than the /-al/ type of suffixed word.

The two types of suffixation being discussed here differ structurally only in the weight of the suffix syllable. Words ending in /-ent/ have a heavy final syllable, which can be footed in itself, while those ending in /-al/ and its allied suffixes have an unfootable (by itself) final monomoraic syllable. The constraints **NonFin** and **Stem-Ft-R** both prevent the final light syllable from being footed with the main-stressed stem in the /-al/ cases, while the heavy /-ent/ can be safely footed without violating either. This difference provides the basis of the solution to the contrast in behavior of the two types. Part of this solution is based on the idea of compensatory lengthening as seen in Zec (1994), McCarthy & Prince (1990a, b), Hayes (1989). Here, it will be proposed that the lengthenings seen in words like *reprisal*, *tóne*, *sónant* can actually be separated into two different classes, one representing true mora addition into the candidate, while the other is an instance of compensatory lengthening.

Historically, it is clear that English words of the type *tóne* owe their long vowels to compensatory lengthening following the loss of a final surface [-ə], which has been proposed above to still be present underlyingly in the grammar. Rather than conceiving of this lengthening as resulting from a mora being inserted into the representation, it will instead be proposed here that the mora in question is “borrowed” from the suffix:

(5.15)

s	s
	\
m	m
	/

to n<ə>

That is, the number of moras in the optimal candidate /tón/ is identical to that seen in the faithful (but suboptimal) candidate */tónə/. No additional violation of *m is incurred by this structure; put another way, the number of moras in this representation is the number dictated by the **Moraic Prominence** constraint. The two competing candidates differ crucially in that the stem, limited to the morpheme /ton/ by suffixation constraints, is bimoraic only in /tón/:

(5.16)

/ton-ə/	Min-2	Non-Fin	StemFtR	FtForm	Lapse-s	StemFtL	*m
(t _ó m _n -ə _m)		!* ⁷⁵	s				mm
(t _ó mm)n-ə _m					sss		!mmm
☞ (t _ó mm)n-<ə>					sss		mm
(t _ó m)n-ə _m	!* ⁷⁵				sss		mm

The monomoraic stem of *(t_ó)nə is fatally light, while *(tónə) violates **NonFin** and **Stem-Ft-R**.⁷⁵ The suffix /-ə/ is eliminated from the morphological stem (and thus the foot, by **Stem-Ft-R**) by the high-ranking suffix subcategorization constraint Align(Suf_{al}, L; Stem, R), /-ə/ being classed as an al-type suffix; that is, /-ə/ must align to a stem and so cannot be part of that stem (§ 4.3.2). Thus, the stem, which will coincide with the word's single foot, can only consist of the morphological root {ton}. The onsetless nature of the suffix ensures that the stem syllable must be open, its final consonant acting as onset to the suffix syllable, so only a candidate with a long root syllable can yield the bimoraic foot demanded by **Min-2**. The structure with mora augmentation, *(t_ómm)n-ə_m, which would have been the optimal form

⁷⁵Note that a candidate *(t_óm_nm)ə would violate the highly ranked **Final Margins** constraint (4.17); this is one factor preventing the mora from linking to the root-final consonant.

under the constraint hierarchy suggested above in (4.33), is less optimal than the winning $(t\acute{o}_{mm})n-\partial$ due to its added mora, violating *m.

The optimal candidate $(t\acute{o}_{mm})n-\partial$ does violate a number of faithfulness constraints, which consequently must be ranked lower than those in the hierarchy shown above. However, it is relevant to discuss these constraints in the context of the grammar as a whole. They include both **Max-Segment** (or some similar faithfulness constraint ensuring that segments are parsed into prosodic structure) and what Zec (1994: 4) calls **Structural Coherence**, which requires moras to be filled locally. Since well-documented compensatory lengthening processes (see McCarthy & Prince 1986, 1988, 1990a,b) display shifts like those seen above in (5.15), such processes must be representable in an OT framework. Zec's (1994) **Moraic Prominence** constraint (§ 4.1.2) allows for moras to be projected by, yet not parsed to, their sponsoring segments. In terms of the formal expression of such constraints using the No-Intervening framework, this is analogous to the alignment of segment to mora, with violations:

(5.17) **m-Coherence:** NI-RL(Seg, m, Seg)

This effectively states that every segment not aligned with a mora will incur a violation. Taken in conjunction with **Moraic Prominence**, which enforces the number and configuration of moras and margins projected into the representation, and a necessary parallel constraint governing syllable margins rather than moras (**Margin Coherence**),⁷⁶ the effect of these constraints will be the familiar alternation of onsets and rimes seen in syllables, the common alternating pattern generally being the most optimal:

⁷⁶Due to the slightly different nature of the Margin, which can encompass more than one segment, the formulation of **Margin Coherence** would differ from **Moraic Coherence**. A plausible formulation would be NI-RL(Seg, M, m).

(5.18)

/CvCvC/	MorPrm	m-Coh	M-Coh
☞ MmMmM C vC v C		CCC	v v
MmmM M C vC vC		!CCCv	v v
MmmmM C vCvC	!C	CC	vCv
MmMMM C v C v C	!v		v

Moving a mora's alignment out of sequence with its projecting segment will incur violations of **m-Coherence**, as some segments will then intervene between the deprived segment and the nearest mora:

(5.19)

/ton-ə/	m-Coh
(t _o m _n -ə _m)	tn
(t _o mm)n-ə _m	tn
(t _o mm)n-ə	!tnə

However, since **m-Coherence** is low-ranked in English, this violation does not prevent the form with the shifted mora from surfacing as the optimal candidate when appropriate.

In the case of a suffix like /-ə/, the segment involved, the final schwa, never surfaces anyway, due to the general deletion of final /-ə/ (§ 4.14). However, for words suffixed in /-əl/, such as *reprisal*, this analysis requires that the suffix vowel /a/ likewise fails to align to a mora:

(5.20)

s s
| \ \
m m \
| / |
to n<a>l

Again, the stem (restricted morphologically to {ton}) becomes bimoraic by associating the mora correspondent to the suffix vowel into the stem. The number of total moras in the candidate remains minimal at two, avoiding further violations of *m. The suffix syllable, deprived of its mora in this representation, displays on the surface either an epenthetic [ə] or a syllabic sonorant, the result of /l/ remaining linked to the vestigial final syllable. All of the members of the /-al/ group display a reduced, unstressed suffix syllable. It is difficult to say whether this indicates an unparsed vowel segment, or a vowel segment parsed to a margin (or perhaps without direct association to a subsyllabic), because in English there is little difference between weak footed moras (e.g., (pál)tal) and unparsed moras (e.g., ma(líg)nant); both show reduced vowels. Since reduced vowels are defined as those which show no vowel quality, there is not much to suggest whether the surface [ə] one may posit in such instances is a reflex of the vowel (since its crucial identifying features are not parsed) or of a mora-less syllable nucleus.

For the contrast between forms like *reprisal* and *résident*, the understanding of the former as showing compensatory lengthening by incorporating the stranded final mora rightfully belonging to the suffix provides a solution to the problem. A constraint **NoOrphans**, which prohibits unfooted word-final moras, would favor a candidate incorporating the suffix mora into the stem foot, which requires two moras in any case:

(5.21) **NoOrphans:** NI-R(MWd, Ft, m)

This states that no moras intervene between the right word edge and the right foot edge. Placed ahead of **Lapse-s**, this is sufficient to optimize the correct candidate for *reprisal*:

(5.22)

/re-pris-al/	Min-2	NonFin	StemFtR	NoOrph	Lapse-s	StemFtL	*m	m-Coh
☞ re _m -(prí _{mm})-sal					ssssss		mmm	rprsal
re _m -(prí _{mm})-sa _m l				!m	ssssss		mmm	rpral
re _m (prí _{ms} -a _m l)		*!	s		sss		mmm	rprsl
(ré _m -prí _m)-sa _m l				!m	sss	s	mmm	rprsl

Borrowing the necessary mora from the suffix again would defoot it and incur more **NoOrphans** and **Lapse** violations. For cases such as this, the addition of an extra mora not required by **Moraic Prominence** to satisfy foot binarity in the stem foot results in the optimal candidate.

5.2 Mora augmentation and the stem

The constraint hierarchy just developed needs to be further extended to account for a number of other structural types found among the /-ent/ and /-al/ forms. For example, while forms with heavy prefixes of the type *(com)(prí)sal* are correctly predicted by the above constraint hierarchy, attested outputs like *cóncident* would incorrectly be rejected in favor of **(con)(fí)(dent)*, and the unconstrained ability to add moras to the representation (thus far constrained only by the low-ranking **m*) could yield many unattested successful forms. However, for cases like *sónant*, where the stem is monomoraic with no other means of expansion, the possibility of moraic augmentation in the optimal candidate is necessary. In fact, it is only in such cases, when the footable domain is a discrete monomoraic morpheme, that any unusual moraic adjustment (of either type, compensatory or augmented) occurs. To eliminate incorrectly predicted forms like **con(fíd)ent*, and to rule out mora addition in most cases, the constraint **m* (§ 4.1.3) will be promoted within the existing hierarchy:⁷⁷

(5.25)

/con-fid-ent/	NonFin	StemFtR	NoOrph	Lapse-s	*m	StemFtL	m-Coh
(com ⁿ m)-(fí ^{mm})-(de ^m n ^{mt})					!mmmm		cfdt
(com ⁿ m)-(fí ^{mm})-den ^{mt}			!m	sss	mmmm		cfdet
☞ (có ^m n ^m -fi ^m)-(de ^m n ^{mt})					mmmm	s	cfdt
(có ^m n ^m -fi ^m)-(de ^m n ^{mt})		!s		ss	mmmm	s	cfdt

The crucial constraint here is **m*, which eliminates the lengthened **cncident* due to its extra mora. The optimal form, *cóncident*, is an example of a word in which the foot type (s_{mm} s_m) is acceptable (§ 5.1.1), indeed preferable within the context of this approach to (s_{mm})s_m, which would both incur **Lapse-s**

⁷⁷Note that in this position, the **m* constraint will not be able to eliminate desirable augmented forms such as *sónant* (5.24).

violations and leave the stem syllable outside the main-stressed foot, violating **Stem-Ft-R**.⁷⁸ At this point, the constraint **Stem-Ft-L**, which was useful in the simplified constraint hierarchy offered in the preceding chapter, is no longer significant and will be dropped from the discussion of relevant constraints.⁷⁹

As was suggested above, the unconstrained addition of moras to feet, whether by augmentation or compensatory “borrowing”, would produce many incorrect forms. For example, bisyllabic monomorphemic forms such as *dígital* and *órbita* show no lengthenings, although forms like *dígital and *orbital would avoid a violation of **NoOrphans**, unlike the attested forms; a similar lengthening might be expected for the same reasons in heavy stems like that seen in *inférna*, i.e., *inférna. Furthermore, no lengthening is ever seen in any stems of two syllables or greater, even when they present unfooted syllables. Thus *ma(líg)nant* surfaces with an initial unfooted syllable rather than the less Lapsed *mālignant, and we see *co(lós)sal* (with geminate penult) rather than *cōlóssal. Turning to other types, the same situation is seen in *calámit* and in unsuffixed words like *aráchnid*, *appéndix*. In fact, even when relatively high-ranking constraints such as **NonFin** are violated, bare stems still fail to show lengthening as an alternative way of satisfying those constraints, e.g., we see *(có)met*, *(brí)gand* instead of alternatives like *(cǒ)met or *(brī)gand. The only cases where lengthening does appear (apart from those discussed above, i.e., monomoraic stems like *sǒnant*) is in forms like *(rè)(bútt)al*, *(dè)(fénd)ant*, *(è)(vápo)rate*, where the lengthened item is a monomoraic affix, and the lengthening is optional.⁸⁰

This suggests a high-ranking constraint which maintains the moraic coherence of stems with two or more moras supplied by **Moraic Prominence**. Such a constraint could be formalized as follows:

(5.26) **Stem-Coherence:** NI(m, Stem, m)

⁷⁸This is of course a theory-internal argument about abstract structure, as the surface forms for both types would be identical.

⁷⁹While this constraint remains in the hierarchy, it is understood as ranked so low as to have no significant effect on the choice of optimal candidate. Thus it need not be included in the subsequent discussion, as is the case with most constraints.

⁸⁰Similarly, we have words like *(vǎlor)(iz)(ǎt)ion*, where the lengthened item is a monomoraic suffix (in this case /-iz-/) and the lengthening is also optional. This will be discussed in § 6.2.

This states that every mora is aligned to either edge of the stem. For each intervening mora, a violation will be incurred. Under this definition (which allows for alignment to either edge, rather than both), monomoraic and bimoraic stems will both satisfy this constraint (since there are two edges), while any further moras will incur violations:

(5.27)	<u>Stem</u>	<u>Stem-Coh</u>
	/m/	÷
	/mm/	÷
	/mmm/	*m
	/mmmm/	*mm
	(etc.)	

Ranking this constraint below **Moraic Prominence** will force each stem to contain only the moras projected by **Moraic Prominence**. Any further moras, such as those added in violation of *m, or “borrowed” in violation of m-Coh, will fatally violate **Stem-Coh**. The only case where this will not be true is where the stem is monomoraic, as a bimoraic augmented stem will not violate **Stem-Coh**, and will incidentally satisfy **Min-2**, yielding the effects seen in the English data. It is assumed hereon that the constraint **Stem-Coh** is an undominated (except by **Moraic Prominence**) member of the English constraint hierarchy, and candidates which violate it, such as *inférnal, *cómet, *orbítal, will not be presented in the following tableaux, as no English words of this type, exhibiting vowel alternation that would suggest an underlying short vowel, ever surface. Multimoraic monomorphemic stems showing long vowels (e.g., *páint*, *dánger*, *líbel*) never exhibit vowel alternation, but rather always show the lexically long stem vowel in all forms.

If surface long vowels were all regarded as underlyingly long, we’d expect to find long vowels anywhere in the word, and see alternations in various positions. Instead, what the data shows is long vowels predominantly appearing in monosyllabic morphemes of the shape /CVC/, and being especially prevalent at the right edge of the word, preceding affixes. Vowel alternation involving long vowels also occurs in stems that appear in these contexts. This, combined with the rarity of long vowels in multimoraic roots

and the similar distribution of long vowels at the right edge of finally-stressed unsuffixed verbs and adjectives, suggests that the majority of these long vowels are not present in the lexical entry, but are rather the result of the morphologically conditioned lengthening process discussed above, enforced by the position of the root within the stem constituent, and the presence of the suffix /-ə/, which appears on the right edge of many stems of this type.

5.2.1 Constraint domains

To derive the correct results from the **Stem-Coh** constraint proposed above, it needs to be slightly reformulated, and a new concept and notation introduced for the Alignment/No-Intervening constraint family, that of the **constraint domain**:

(5.28) **Stem-Coh:** NI(Stem: m, Stem, m)

This states that for every mora **in the domain of a stem**, no mora intervenes between that mora and the stem. This limitation is important, because otherwise moras outside the stem (such as suffixal moras) would incorrectly incur violations of this constraint. With the domain now specified in the constraint, such moras will not be evaluated with regard to this constraint.

Constraint domains offer a further refinement of the notation for expressing formal constraints, but adding them to the notation does not add further complexity to OT, because domains were in fact implicit (but ambiguously so, as was the case with alignment violation) in the theory already. For example, when an alignment constraint was presented, e.g., Align(Ft, R; PrWd, R) or Align(s, R; Ft, R), it was only evaluated in terms of the feet or syllables in a particular word. However, in an utterance, there could be other syllables or feet, for example in the following word, or in following clitics, inflectional endings, or “level II” suffixes, which could incur unwanted violations of such constraints.⁸¹ By limiting the OT tableau to the

⁸¹The failure of “level II” suffixes to foot, proposed in § 4.2.1, can be explained by limiting **Lapse-s** to the prosodic word. Thus, syllables outside the prosodic word do not incur violations of this constraint and do not need to be footed. Requiring feet to align with the right edge of the prosodic word, for example with some kind of **Edgemost** constraint, would actually prevent these syllables from being footed, as footing them would incur violations.

word, the domain for these constraints was being implicitly limited to the Morphological or Prosodic Word.⁸²

Likewise, when a constraint is relevant for cross-word effects, such as the cross-word rhythmic effects seen in English (e.g., the well-known case of rhythmic stress adjustment, for example *thirteen mén*), the domain of the constraint would be a supra-word phrasal constituent. It should be clear that it is not appropriate for constraints to apply outside their domains in some cases. Expressing the domain explicitly in the formalization specifies precisely how the constraint will operate. Certain correspondence constraints (McCarthy & Prince 1995) also implicitly refer to domains, as the correspondence governed by such constraints is evaluated only over a certain constituent type (e.g., the reduplicant), and not others (e.g., the base or stem). For the previous constraints offered herein, the constraint domain happens to be the next higher morphological category, the Morphological Word, rather than the Morphological Stem as in this case. Promoting **Stem-Coh** into the hierarchy of significant constraints as a undominated (except by **Moraic Prominence**) constraint restricts the possibilities for intrusive lengthening in the stem to footable monomoraic morphemes.⁸³

There is one further restriction on lengthening in words, which is a stronger restriction than on the lengthening within stems. In English, suffixes are never augmented outside of stem constituents, even when they are monomoraic. For example, a candidate *(in)(fér)(nāl) would be more optimal than the correct (in)(fér)nal on the basis of the current constraint hierarchy (since **Lapse-s** is then not violated at all), but such forms never surface. Suffix morphemes show lengthening rarely, and only when they become “stem-ized” due to subcategorization constraints, e.g., ({vic}){(tǒri)}ous (lengthened by CiV lengthening, § 5.2.2), where the suffix becomes part of stem constituent as a result of the addition of /-ous/, which requires a stem on the immediate left.⁸⁴ Thus, it is necessary to propose a faithfulness

⁸²It should be clear that members of any constituent hierarchy can act as constraint domains; that is, both morphological and prosodic constituents can be constraint domains.

⁸³Note however that neither **Stem-Coh** nor **m-Coh** will prevent true compensatory lengthening *within* the stem resulting from either segment deletion or shift (see § 5.2.2).

⁸⁴Another example might be *adjectíval*, with /-iv/ forming a foot of its own due to subcategorization constraints imposed by /-al/. This form, along with the similar *sùbstantíval*, are however problematic due to the irregular stress seen in their unsuffixed counterparts (see § 6.6).

constraint **SuffixMoras**, which prevents material to the right of the stem (i.e., suffixes) from acquiring any more moras than those projected by **Moraic Prominence**:

(5.29) **SuffixMoras:** NI-R(PrWd: PrWd, Stem, m)

This states that no moras intervene between the right edge of each prosodic word and the right edge of some stem. By ranking this constraint below **Moraic Prominence**, the number of moras appearing to the right of the rightmost stem constituent will be restricted to the number projected by **Moraic Prominence**. This constraint will also encourage all “borrowable” moras to shift into the stem (reducing the number to the right of the stem), hence this constraint must also be ranked below **Stem-Coh**, or moras will shift into the stem even when the stem is already bimoraic:

(5.30)

/re-pris-al/	MorPrm	StemCoh	Suf-m	NonFin	StemFtR	NoOrph	Lapse-s	*m
☞ re _m -(prí _{mm})-sal							ssssss	mmm
re _m (prí _{mm})-(sa _{mm} l)			!m					mmmm
re _m -(prí _{mm})-sa _m l						!m	ssssss	mmmm
(ré _m prí _m)-(sa _{mm} l)			!m					mmmm
re _m (prí _m s-a _m l)				*!	s		sss	mmm
(ré _m -prí _m)-sa _m l						!m	sss	mmm

In the rest of the work, candidates which violate **SuffixMoras** will ordinarily not be presented in the tableaux.

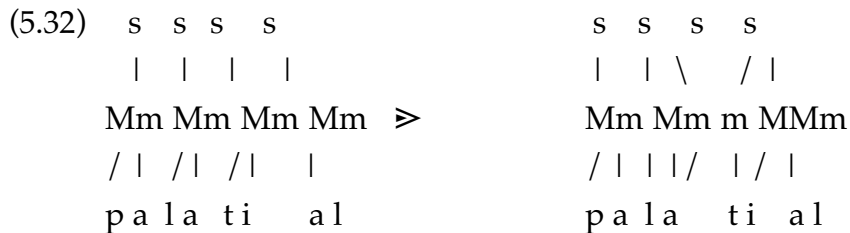
5.2.2 Compensatory lengthening within the stem

One well-known lengthening process in English is so-called CiV lengthening (Chomsky & Halle 1968: 47). In such words, a light open syllable is lengthened when followed by a sequence of /i/ and a non-high vowel.

(5.31) canáđian managérial advisórial colónial
 státion palátial arábia comédian

The vowel /i/ itself in such a position, however, is not lengthened, e.g., *official*, *militia*. Furthermore, if the syllable in question is closed, lengthening does not occur, e.g., *provincial*, *adverbial*. The appearance of a long vowel under the stated conditions, however, is nearly universal throughout the data.

These facts suggest an adjustment inside the stem, that is, a mora internal to the stem is shifted to the lengthened syllable. The most likely source of the shifted mora is the vowel /i/ itself, which appears before a vowel of higher sonority and so can be conceived of as linking to the otherwise empty onset position of the following syllable. This is clear enough when the consonant preceding the /i/ is palatalized, as in *palātial*. The palatal place features of the /i/ can be understood as linking to the onset, the result surfacing as [š]. The syllable belonging to the /i/ has no surface instantiation; its mora has shifted to the preceding syllable. In derivational terms, this could be understood as a process:



The “underlying” third syllable loses its mora to the second, and its onset constituent to the last syllable. Even for words in which palatalization does not occur, and the /i/ appears to retain its syllabic character (e.g., *memōrial*), there is no reason why the structural analysis on the right of figure (5.32) should not be valid, i.e., the surface syllable structure of *memōrial* would be s(me_m)-s(mo_{mm})-s(ria_{ml}), /ri/ forming the onset of the third surface syllable. Under such an explanation, the number of moras within the stem does not change (continuing to satisfy **Stem-Coh**), only the arrangement of moras within the stem is different.

Of course, in an OT account, a serial explanation is not possible. Instead, the same insight can be captured by proposing a contrast between the way moras are projected by **Moraic Prominence**, and the way they are associated to segments by other constraints. This contrast is implicit in Zec’s (1994) approach to compensatory lengthening in various languages. Here,

although the /i/ projects a mora, it fails to link to that mora.⁸⁵ The constraint governing this process is likely to be **Onset** (Smolensky & Prince 1993), which demands that all syllables have an onset. It is the demand for an onset segment which takes the /i/ away from its syllable and mora:

(5.33)

/palati-al/	MorPrm	StemCoh	Onset	NonFin	StemFtR	NoOrph	Lapse-s	*m
☞ pa _m (lá _{mm})-tia _m l						*	ssssss	mmmmn
pa _m (lá _{mm})(ti _m -a _m l)		!* *	*		*		SSS	mmmmnm
pa _m (lá _{mm} ti _m)-a _m l		!* *	*			*	ssssss	mmmmnm
pa _m (lá _m ti _m)-a _m l			!* *			*	ssssss	mmmmn
(pá _m la _m)(ti _m -a _m l)			!* *	*	*			mmmmn
(pà _m la _m)(tí _m -a _m l)			!* *	*	*			mmmmn

When /i/ joins the onset of the final syllable, the **Onset** constraint is satisfied. Additionally, the fact that the stem segments /t/ and /i/ are now outside the head foot is not relevant, as constraints like **Stem-Ft-R** refer to whether a syllable intervenes between the stem edge and the foot edge. So long as the misaligned constituent is only an onset, no violations of such constraints will occur. For candidates where /i/ remains in the stem foot, **Onset** will always be violated.⁸⁶

For words like *advérbial*, we might expect a similar structure, namely *ad(vé_mr_{mm})bial*, where the extra mora links to the consonant /r/ and thus no detectable surface effect occurs. However, it appears that something besides the need for an onset is interacting with the constraints given thus far; otherwise, words like *milítia* cannot be explained. The vowel /i/ itself is never lengthened in the CiV environment, although the **Onset** constraint in the hierarchy given above would demand lengthening. Thus, the **Onset**

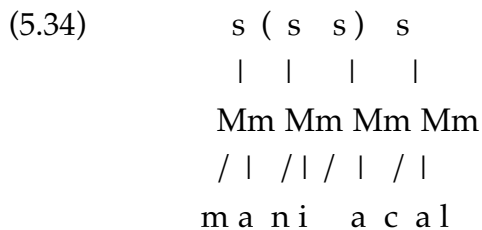
⁸⁵It will be proposed in § 7.5.2 that what are typically referred to as segments should actually be accounted for with two different constituents, morphological segments (or phonemes), which are part of the morphological hierarchy, and prosodic segments, which are part of the prosodic hierarchy. These roughly correspond to the items comprising “input” and “output” strings in Correspondence Theory (McCarthy & Prince 1995). Following this, the behavior of /i/ here could be explained via the projection of a mora by the morphological segment /i/, but its failure to associate to the corresponding prosodic segment /i/, which instead links to the syllable onset.

⁸⁶Note that this account also assumes that multiply linking /i/ to both the onset and mora constituents is dispreferred; see (5.35) below.

constraint is not sufficient to trigger the mora shift, and it appears that there is a further requirement that the shifted mora move to a more sonorous segment than /i/ itself, that is, only to the vowels /a,e,o/. Also, the mora will always shift to the left, never to the right. These further conditions suggest that more constraints are relevant to the behavior of moras in this context.

The most obvious reason for the mora to apparently shift to the left is the need for the foot to remain bimoraic, enforced by the high-ranking **Min-2**. A form *(pàla)(tǵǵǵ) would not only violate **Non-Finality**, but would place the head foot outside the stem. In most words that display CiV lengthening, the /i/ in question is on the right edge of the stem, often as part of an /-ity/ class suffix like /-ion/ or the /-i-/ extension extending many stems further suffixed by /-al/ suffixes (i.e., forms in /-ial/, /-ious/, /-ian/, /-ior/, etc.). Thus, the mora projected by the /i/ optimally aligns within the stem, and within the foot (which will align to the stem by **Stem-Ft-R**), even in candidates whose /i/ segment itself leaves the foot to form an onset. Likewise, in CiV lengthening cases when the /i/ is not stem-final, it is in foot-final position in what is often the only plausible foot in the word, e.g., (mǎni)ac, which can be understood as {(mǎ)nǵac}.

However, when /i/ occurs in a CiV position, but is not foot-final, a very different effect is seen, similar to that seen when other vowels become adjacent, e.g., manǵacal. The structure of such forms can be understood as follows:



The stressed foot (nǵa) is understood as containing only two moras; the apparent lengthening of the stressed syllable reflects the linking of the /i/ vowel to both the mora of the head syllable and the onset position of the succeeding weak syllable. Whatever causes diphthongization in long vowels, where a vowel segment linked to two subsyllabic constituents (in the case of long vowels, moras) surfaces as a diphthong, the same effect can be seen when such segments are likewise doubly linked to a mora and following

onset, at least when the vowel is stressed. A similar structure was suggested previously (§ 4.1.5) to explain final vowel lengthening. The segmental constraints behind the diphthongization will not be explored in this work, but will be left to future research.

Forms like *maníacal* offer one example where it is more optimal to multiply link /i/ to two subsyllabics than to delink its mora. Multiply linking the segment still satisfies the high-ranking **Onset**, and indeed all words where vowels are contiguous in this way show this apparent lengthening which indicates onset-filling, e.g., *théater*, *Bóas*, *créate*, *páean*. Thus, the constraint against multiple linking of a segment to two subsyllabic constituents must be fairly low-ranked. Such a constraint might be formulated as follows:

(5.35) **NoMultiLink** NI-R(Seg, m, M)

This states that the right edge of a segment should align to the right edge of a mora, with no margin constituent intervening. A segment multiply linked to a mora and a margin constituent (but not one linked to two mora constituents, as in a long vowel) would incur a violation.

Another instance where this constraint appears to be violated is in words like *milítia*, where the structure is best understood as *mi(líti)ḡa*, the /i/ being linked to both the final mora in the foot and the onset of the last syllable. A likely explanation for the failure of the preceding /i/ to lengthen is that the mora, to escape the segment that projected it, requires any segment it associates to to have higher sonority than the projecting segment. This might be understood as a constraint aligning moras to the feature [+low] when possible:⁸⁷

(5.36) m->[+low] NI(Foot: m, [+low], Seg)

Preventing a mass migration of moras to segments like /a/ at the expense of all other moraic segments would be faithfulness constraints ensuring that each segment has a subsyllabic constituent to correspond to, for example the

⁸⁷The interaction of such a constraint and parallel ones taking other features (such as [+high], [+son] and [+obs]) as arguments might form the formal basis of the sonority-based system of mora association.

constraint **Moraic Coherence** (5.17). The vowel /i/, however, cannot draw off the mora, since it is of the same sonority as the foot-final /i/ offering the additional mora. The same would be true of any moraic consonant found in a closed syllable preceding the /i/. Thus, only vowels containing the feature [+low], in open syllables within the same foot can undergo this type of compensatory lengthening:⁸⁸

(5.37a)

/palati-al/	Onset	NonFin	StemFtR	NoOrph	Lapse-s	m/low	m-Coh	*Multi-Link
☞ pa _m (lá _{mm})-tí _a m _l				*	ssssss		pltil	
pa _m (lá _m ti _m)-í _a m _l				*	ssssss	!m	pltl	*
(pá _m la _m)(tí _m -í _a m _l)			!*			m	pltl	*
(pà _m la _m)(tí _m -í _a m _l)		!*	*			m	pltl	*

(5.37b)

/militi-a/	Onset	NonFin	StemFtR	NoOrph	Lapse-s	m/low	m-Coh	*Multi-Link
☞ mi _m (lí _m ti _m)-í _a m				*	ssssss	mmm	mlt	*
mi _m (lí _m ti _m)-a _m	!*			*	ssssss	mmm	mlt	
mi _m (lí _{mm})-tí _a m				*	ssssss	mmm	!mlti	
(mí _m li _m)(tí _m -í _a m)			!*			mmm	mlt	*
(mì _m li _m)(tí _m -í _a m)		!*	*			mmm	mlt	*

For cases where the /i/ is not foot-final, the superordinate constraints governing foot placement override these constraints:

(5.38)

/maniac-al/	Onset	NonFin	StemFtR	NoOrph	Lapse-s	m/low	m-Coh	*Multi-Link
☞ ma _m (ní _m í _a m)c-a _m _l				*	ssssss	m	mncl	*
(má _{mm})(ní _a m _c -a _m _l)			!*				mncl	
ma _m (ní _m a _m)c-a _m _l	!*			*	ssssss	m	mncl	
(má _m ni _m)(í _a m _c -a _m _l)			!*			m	mncl	*
(mà _m ni _m)(í _a m _c -a _m _l)		!*	*			m	mncl	*

⁸⁸For English, this means /a, e, o/, e.g., *paláti-al*, *amnēsia*, *emōtion*. Words with orthographic /u/ in this position always surface with the triphthong [ɪuɪ] (e.g., *confúsi-on*), which shows the same quality regardless of the following environment (e.g., *confúse*, *refúsal*).

Thus the only instance in which CiV lengthening can take place is when a stressed open syllable with a low or mid vowel is followed by the weak member of the foot, a syllable with the nucleus /i/. If the head syllable of the foot is closed, no candidate expanded with the /i/'s mora will succeed. The phenomenon of CiV lengthening also gives further evidence for the gemination proposed in § 3.3.2, as words with such geminates, e.g., *remisión*, *potássium*, fail to show CiV lengthening, which is best explained by regarding the head syllable as closed and already bimoraic due to the geminate, i.e., *(mís),(tás)*.

It might be asked why /u/ does not cause a similar lengthening, for example in words like *grádual*, *cásual*, *contínual*. However, the segment /u/ in such cases is not simply the vocalic correspondent of the glide /ɥ/, but rather shows a complex features set which includes palatal and velar elements. The palatal element often surfaces in the onset of /u/'s syllable, as in the words above. Both the palatal and velar features are evidently not able to align to a given subsyllabic at the same time, hence this /u/ cannot simply join to an onset and give up its mora in the way the simpler /i/ can. For such cases, the round and velar features of /u/, on the right edge of the segment, can act as the onset of the following syllable, while the palatal features on the left edge can act as the onset of the /u/ syllable itself, the mora still remaining within the foot. In this sense, /u/ is always multiply linked to mora and margin constituents.

5.3 Extrametricality and word-final syllable effects

It is relevant at this stage to return to the unaffixed words and to reexamine them under the expanded constraint hierarchy arrived at above. While extrametricality effects had previously been handled by **NonFin** and **Edgemost**, it will be necessary to further explain why certain unusual foot shapes can occur at the right edge of the word. For example, the simplified tableau offered above in (4.4) suggested that words of the shape /LH/ should appear with final stress, and indeed a small majority of all bisyllables (53% as in table 3.20) do. However, most of these words are verbs and adjectives

which are to be understood as suffixed in /-ə/,⁸⁹ and the percentage of nouns showing this pattern is smaller (42%). Of these, again a majority are deverbals of the type *revérse*, *accláim*, and so are actually suffixed deverbal forms of the shape /L(H)-ə/. The prevalent outcome for unaffixed words of this kind, once the schwa-affixed forms are removed from consideration, is actually /(LH)/, as in *légend*, *brígrand* (which would need to be accounted for in any case). Such a violation of **NonFin** is expected in bisyllables with light initials, such as (*cómet*), because it is not possible to build a stressable foot from such stems without including the final. However, the candidates /(LH)/ and /L(H)/ both equally violate **NonFin**, which states that the head foot in the word does not appear finally in the word. Yet, the form /(LH)/, with initial stress, is clearly preferred to /L(H)/, although it violates **FtForm** (a candidate /(L)H/ would be precluded by **Min-2**).

The solution to this problem lies in refining the constraint hierarchy with regard to word-final effects. The evidence suggests that there is actually a strong restriction on main stress occurring on the word-final syllable, and that a word-final weak (unstressed) syllable (in some cases dominating a deleted schwa) must always appear, except of course in monosyllables. This is not the same as restricting the final foot from coinciding with the word edge, the current definition of **NonFin**, since feet can be mono- or bisyllabic. This distinction can be addressed by proposing a constraint **NonFin(ś)**, which may be formalized as follows:

(5.39) **NonFin(ś):** Align(MWd: ś, R; s, L)

This states that every stressed syllable in the Morphological Word must have another syllable following on its right, and precludes the final syllable from ever being main-stressed (although it might be footed, e.g., the suffix /-ent/, which is always footed but stressless). Building on this, it is also possible to formalize the old **NonFin** as **NonFin(Ft')**:

(5.40) **NonFin(Ft')**: Align(PrWd: Ft', R; s, L)

⁸⁹For the words in question with final long syllables, e.g., *detáin*, *repéal*, *profáne*, *sublíme*, the final heavy syllable has the structure -VVC, suggesting a lengthened vowel, i.e., *de(táin)* from /de-tan-ə/.

This states that the head foot should misalign with the Prosodic Word by at least one syllable on its right edge. If no syllable intervenes between the head foot and the Prosodic Word, a violation of this constraint is incurred.

The preceding constraint, however, would be satisfied by any number of syllables which intervene between the head foot and the right edge of the word, so long as it is at least one. However, it is clear from the data that the head foot in English tends to be as close to the right side of the word as allowed for by the **NonFin** constraints. Above, the number of syllables which could intervene between the head foot and the right edge of the word was normally limited to one by **Edgemost**, which was presented above as an Alignment constraint (§ 4.1.1). This can also be re-formalized as a No-Intervening constraint, which penalizes syllables that intervene between the head foot and the edge of the word:

(5.41) **Edgemost:** NI-R(MWd: Ft', MWd, s)

Note that this new formulation of the **Edgemost** constraint refers to the Morphological Word rather than the Prosodic Word (see § 5.2.1). Applying this expanded constraint hierarchy to /brigand/ proceeds as follows:

(5.42)

/brigand/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	FtForm	Lapse-s
☞(brímgamnd)		*			*	
bri _m (gámnd)		*	!*			SSS
(brím)(gamnd) !*				S		
(brím)gamnd !*				S		SSS

This revised constraint hierarchy does not affect previous conclusions about other types, but does account for trochaic feet of the sub-optimal shape (LH) occurring at the right edge of the word.

This non-canonical foot can also occur elsewhere in the word. Again, the source of this situation appears to be the presence of a heavy syllable in an unstressable position. For example, when such a syllable appears before the main-stressed syllable, and a preceding light syllable is available, an (LH) foot

appears in the optimal candidate, e.g., *àlexándrine*, *vòluntéer*, *ànalgésia*, *sàturnália*, *tàrantélla*. This can be enforced by modifying the **NonFin(ś)** constraint presented above in (5.39) to specify that the syllable following the stressed syllable should be stressless:

(5.43) **NonFin(ś):** Align(MWd: ś, R; s[∞], L)

where the notation s[∞] indicates a weak syllable.⁹⁰ This constraint will have the following effect on words long enough to show adjacent potential stresses:

(5.44)

/aleksandrine/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	FtForm	Lapse-s
☞ (àm lémk _m)(sám _n n _m)-(drēm _m n)				S	*	
a _m (lémk _m sám _n n _m)-(drēm _m n)				S	*	!SSS
a _m (lèm _k k _m)(sám _n n _m)-(drēm _m n)			!*	S		SSS
a _m (lémk _m)(sám _n n _m)-(drēm _m n)				!SS		SSS

The adjacency of the two stressable syllables (e.g., *alèxándrine) results in one giving up its potential stress. Since main stress cannot move off the penult without violating **Edgemost**, it is the antepenult that is restricted from becoming the head of a foot. The same effect can be seen for prefixes, e.g., (rècon)(cíl)e. For trisyllables in this situation (e.g., (at)(lán)ta), the antepenult remains footed, as there are no other syllables to join this foot. Due to its proximity to the head foot, such a foot can appear unstressed, showing an optionally reduced vowel (see § 5.2), but it is nevertheless a foot. Regarding such a monosyllabic foot as stressless prevents the optimal candidate (at)(lán)ta, with no secondary stress on the initial, from fatally violating the **NonFin(ś)** constraint. Such stressless feet are further discussed in § 5.4.1.

⁹⁰Since constituent heads (i.e., strong members) can be referred to in constraints (e.g., McCarthy & Prince 1993a: 14), there is no cogent reason not to also allow reference to weak members. The binary relationship strong/weak can be considered parallel to right/left, both being oppositional features of prosodic constituents. The constraint given here is akin to Hung's (1994) **Rhythm** constraint, which implicitly refers to weak members through a grid notation.

Another word-edge phenomenon governed by the extrametricality constraints is seen in the treatment of some four-syllable stems, such as those seen in *américa*, *parámetro*. The most optimal candidate with regards to the **Lapse-s** constraint would be the unattested **(áme)(rica)*, which foots all syllables yet does not treat the word-final foot as the primary stressed foot, which would violate the principle of non-finality. However, the gap between the head foot and the word edge is two syllables, incurring two violations of **Edgemost**. The attested type *a(méri)ca* thus succeeds despite its **Lapse-s** violations:

(5.45)

/america/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s
 a _m (mé _m rí _m)ca _m				S	S	*	ssssss
(á _m me _m)(rí _m ca _m)				!ss			
(à _m me _m)(rí _m ca _m)		!*					

5.3.1 Monomoraic words revisited

At this point the revised hierarchy may be applied to the bisyllabic and trisyllabic unaffixed words discussed in chapters three and four. The expected stress patterns based upon underlying syllable weight will be laid out, and the minority patterns resulting from gemination and /-ə/ suffixation will also be presented. Note that the issue of prefixation will be deferred until the next section. The set of bisyllables which are expected to surface in true monomoraic stems is as follows:

(5.46)

/ss/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ (LL)		*						
(L)L	!*			S	S			SSS
☞ (H)L				S	S		S	SSS
(HL)		!*						
L(H)		*	!*					SSS
(L)H	!*			S	S		**	SSS
☞ (LH)		*				*		
☞ (H)(H)				S				
(HH)		!*				*		
(H)(H)		!*	*					

Note that gemination in true bisyllables will not be detectable, as the penult will be stressed regardless. Cases in which apparent final stress appears will always be examples of suffixation in /-ə/; where a stressed apparently light final is involved, both suffixation and gemination are to be understood, e.g., *baguette*. Such words are suffixed and so possess internal morphology; in the table below, the symbol 'H' is used to cover both ordinary heavy syllables and proposed geminates:

(5.47)

{ss}-ə	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ (LL)-ə							*	SSS
L(L-ə)		!*		S	S			SSS
(H)(L-ə)				!s	S			
☞ (HL)-ə							*	SSS
☞ L(H)-ə							*	SSSSSS
L(H-ə)		!*		S	S			SSS
(LH)-ə						!*	*	SSS
(H)(H)-ə				!s			*	SSS
(HH)-ə						!*	*	SSS
☞ (H)(H)-ə							*	SSS

These account for all finally stressed bisyllables, which must show a heavy penult, geminate or not. Due to the fact that the suffix in question is a single light syllable, the patterns which appear in schwa-suffixed bisyllables are quite similar to those seen in trisyllables. The optimal trisyllables are similar to those yielded by the earlier, primitive hierarchy explored in (4.3):

(5.48a)

{sss}	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ (LL)L				s	s		*	SSS
L(LL)		!*						SSS
L(L)L	!*							
(LLL)		!*						SS
☞ (LL)(H)				s				
(LL)(H)		!*						
L(LH)		!*				*		SSS
(H)(LL)				!ss				
(H)(LL)		!*						
☞ (HL)L				s	s		*	SSS
(H)(L)L	!*			s	s		*	SSS
(H)(LH)				!ss		*		
(H)(LH)		!*				*		
(H)L(H)				!ss				SS
☞ (HL)(H)				s				

Trisyllables with intrinsically light penults will always take initial stress, regardless of the weight of the initial or final syllables. Likewise, those with heavy penults will always take penultimate stress:

(5.48b)

{sss}	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ L(<u>H</u>)L				s	s		*	sssss
(<u>L</u> H)L				s	s	!*	*	SSS
L(<u>H</u> L)		!*						SSS
☞ L(<u>H</u>)(H)				s				SSS
L(H)(<u>H</u>)		!*	s					SSS
L(<u>HH</u>)		!*						SSS
(<u>L</u> H)(H)				s		!*		
☞ (H)(<u>H</u>)L				s	s		*	SSS
(<u>H</u>)(HL)				!ss				
(H)(<u>H</u> L)		!*						
☞ (H)(<u>H</u>)(H)				s				
(H)(H)(<u>H</u>)		!*	*					
(<u>H</u>)(H)(H)				!ss				
(<u>HH</u>)(H)				s		!*		

Trisyllables with any other pattern (for example, with final stress, or a stressed light penult) are understood as being suffixed in /-ə/. The optimal forms for these suffixed cases accord with the minority patterns seen for surface trisyllables. Again, the symbol 'H' refers to both typical heavy syllables and the proposed geminates:

(5.49a)

{sss}-ə	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
(<u>LL</u>)(H)-ə				!s			*	SSS
☞ (LL)(<u>H</u>)-ə							*	SSS
L(<u>L</u> H)-ə						!*	*	ssssss
(H)(<u>L</u> H)-ə				!ss		*	*	SSS
(H)(L <u>H</u>)-ə						!*	*	SSS
☞ (H <u>L</u>)(H)-ə							*	SSS
(<u>H</u> L)(H)-ə				!s			*	SSS
L(<u>H</u>)(H)-ə				!s			*	ssssss
L(H)(<u>H</u>)-ə			!*				*	ssssss
☞ (LH)(<u>H</u>)-ə						*	*	SSS
(<u>L</u> H)(H)-ə				!s		*	*	SSS
(H)(<u>H</u>)(H)-ə				!s			*	SSS
☞ (H)(H)(<u>H</u>)-ə							*	SSS
(<u>H</u>)(H)(H)-ə				!ss			*	SSS
(<u>H</u> H)(H)-ə				!s		*	*	SSS

Suffixed forms with heavy stem-final syllables will display final stress on the surface; underlyingly, these heavy stressed syllables are penults, as in the unsuffixed forms above in (5.48b). Suffixed forms with light stem-final syllables will show surface penultimate stress:

(5.49b)

{sss}-ə	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
(<u>LL</u>)(L-ə)				!s	s			
☞ L(<u>LL</u>)-ə							*	SSS
(LL)(<u>L</u> -ə)		!*		s	s			
(<u>LLL</u>)-ə							*	!sssss
(<u>H</u>)(LL)-ə				!ss			*	SSS
☞ (H)(<u>LL</u>)-ə							*	SSS
(<u>HL</u>)(L-ə)				!s	s			
(<u>H</u>)L(L-ə)				!s	s			SS

When the penultimate syllable in the stem is light, as in (5.49b) this results in a form which shows a stressed light surface penult, in contrast to the corresponding unstressed forms. When the penult is heavy, the surface stress pattern is identical to that seen in unsuffixed forms with the same stem type:

(5.49c)

{sss}-ə	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
L(<u>H</u>)(L-ə)				!s	s			SSS
(<u>LH</u>)(L-ə)				!s	s	*		
☞ L(<u>HL</u>)-ə							*	ssssss
(H)(<u>H</u>)(L-ə)				!s	s			
(<u>H</u>)(HL)-ə				!ss			*	SSS
☞ (H)(<u>HL</u>)-ə							*	SSS

As the crucial constraints for determining prosodic structure involve interactions between the final three syllables, monomoraic words of greater length should be expected to show the same head feet, with secondary stresses feet iteratively projecting from the left side of the head foot. Heavy syllables adjacent to stressed syllables should surface either as weak members of feet (LH), or as unstressed monosyllabic feet (H), as was outlined above in § 5.3.

The investigation into the relative frequencies of these stress patterns, presented in § 3.2, was relevant for determining both the proposed constraint

hierarchy and the lexically determined morphological structures which are provided to it as “input”. Presumably, the acquisition of the grammar works in the same way. OT constraint hierarchies usually yield up a single optimal candidate for a given structural type. Thus, if a language presents to the learner what appear to be different realizations for an apparently unitary type, the grammar must choose one as the unmarked case, the case in which the general constraint hierarchy applies and the “input” structure is transparent. It is sensible for the grammar to choose the most frequent pattern as its basic one, and this is what has been done here as well. Kager’s first pattern (that showing final syllable extrametricality) has been chosen as the basis for the development of the constraint hierarchy.

Using that constraint hierarchy, which addresses the majority of forms, as the strict foundation for the grammar, the remaining forms are able to be understood as containing further structures (gemimates, morphological structure) which allow them to also be accounted for using the same constraint hierarchy. The assumption behind the trends seen in § 3.2 is that the high percentages are indications of 100% mixed with structurally marked cases, while the low percentages are indications of 0% likewise mixed. The degree of mixture depends greatly on the unusualness of the structural marking, which ranged from low frequency structures, such as gemination plus suffixation of /-ə/ (Kager’s “idiosyncratic” third type), to relatively high frequency things such as the suffix /-ə/ itself (present in Kager’s second type, which he ascribed to verbs and adjectives generally and characterized by loss of final syllable extrametricality).

However, if some other pattern had been most frequent, or if such a pattern becomes the most frequent stress pattern of English, it would be expected that a reanalysis would take place of both the constraint hierarchy and the lexical forms which are fed into it. It is likely that this has happened numerous times during the development of English, and as can be seen from various alternant forms (§ 6.3, § 6.4, § 7.4.1), regularization and adjustment of lexical forms according to the current hierarchy are still taking place in the present-day dialects of English.

The preceding tableaux are applicable to stems consisting of a single morpheme. Some bisyllabic and trisyllabic words are however prefixed, and this internal morphological structure can have an effect on how words surface, especially with regard to mora augmentation, which is prohibited in

all the above forms by **Stem-Coh**, since they are stems with two moras or greater. Some of the effects which can be seen in prefixed words are discussed in the next sections.

5.4 The treatment of prefixes

From the preceding sections it should be clear that prefixes such as /re-/, /de-/, /pro-/ can surface in three different ways. For example, the suffix /re-/ can appear as [ré-], [ré-] or [rə-]. This is interpreted here (following the arguments presented above) as due to the underlyingly monomoraic structure of the prefix. The lexical form of such prefixes is of the type /re-/, with a short vowel. When such a prefix is part of a foot, in the head position, as in (*ré*)dent, the short vowel is stressed and surfaces with its full quality. When such prefixes are alone in the foot, they are lengthened as is permitted by **Stem-Coh**, and surface as long vowels, e.g., the noun (*rē*)ject. Finally, as was noted above in § 5.2, when such prefixes appear alone in a footable non-main-stressed position, they may optionally appear with long or reduced vowels, but never with short full vowel quality, as in (*rè*)(*prīs*)al ~ (*rə*)(*prīs*)al. The behavior of such prefixes depends entirely upon how it is footed and the status of its foot.

5.4.1 Prefixes in suffixed words

The treatment of prefixes in words suffixed in /-ent/ has been detailed above (§ 5.1.2), but the treatment of words in /-al/ needs to be slightly revised. Under the current constraint hierarchy (shown with relevant constraints only), the optimal candidate for the structure /re-pris-al/ is actually (*rē*)(*prīs*)al, a candidate that was not offered in previous tableaux (cf. 5.22):

(5.50)

/re-pris-al/	Min-2	Non-Fin(F')	StemFtR	NoOrph	Lapse-s	*m	m-Coh
☞ (<i>rē</i> _{mm})-(prī _{mm})-sal					sss	mmmm	rprsal
re _m -(prī _{mm})-sal					!sssss	mmm	rprsal
re _m -(prī _{mm})-sa _m l				!m	sssss	mmmm	
re _m -(prī _m s-a _m l)		*!	s		sss	mmm	rprsl
(<i>rē</i> _m -pri _m)-sa _m l				!m	sss	mmm	rprsl

Adding an intrusive mora to the prefix in the optimal candidate does not violate **Stem-Coh**, and produces the best candidate in terms of **Lapse-s** violations. However, there is an alternation between the prefix surfacing with a full long vowel, and with an unstressed schwa, e.g., *(rə)-(pr̥is)al*, which appears to be optional, perhaps related to rate of speech. Such optional alternations in vowel quality are seen in other unstressed feet, like the /-ent/ suffix (e.g., *(pr̥ési)(d̥ənt) ~ (pr̥ési)(d̥ent)*) and in forms like *(cóm)(pən)(sā)te ~ (cóm)(pen)(sā)te*. Heavy monosyllabic prefixes in these positions can also exhibit an alternation between full short quality and reduced quality, e.g., *(con)(gr̥és)sional ~ (cən)(gr̥és)sional*. Long vowels in such positions appear either in full or reduced forms, but never with full short vowel quality, e.g., *(còn)(nōt)(tǎ)tion ~ (còn)(nə)(tǎ)tion*. The behavior of monosyllabic prefixes in this manner is consistent across a variety of word types, the underlined vowels all appearing optionally long or reduced:

- (5.51) (pr̥o)(cés)sion (d̥e)(libe)(rǎ)tion (d̥e)(t̥éri)o(rǎ)te
 (pr̥e)(dómi)(nant) (pr̥o)(híbit)ive

In positions where the prefix is not adjacent to the stressed stem, it will always show full stress, e.g., *(pr̥è)(con)(c̥éi)ve*, where the vowel in the prefix /con-/ is reduced but the vowel in the initial syllable (*pr̥è-*) is footed and stressed, as it is not adjacent to a stressed syllable.

This again suggests the “stressless” foot, seen above (§ 5.3) in *(at)(lán)ta*, which optionally fails to display vowel quality in its single syllable. These feet would be predicted to appear at the word-edge, or between two other foot heads, as a result of **NonFin(ś)**. That is, the reduction in quality in a syllable is related to the adjacency to other stresses. For example, while the suffixes in *président*, *résident* may optionally appear with the full vowel [ɛ], those in *propónent*, *respóndent*, where the suffix follows the main stressed syllable, are always reduced. The prefixes which reduce always precede stressed syllables, and the reduced syllables in the other cases noted above are also adjacent to stressed syllables. These reduced feet should not be regarded as defooted but only as unstressed, and the optional reduction is in line with

that of other unstressed syllables.⁹¹ A further constraint is required to restrict the proliferation of these unstressed feet; otherwise, forms such as *a(lek)(sán)(drine) would surface as optimal candidates:

(5.52) **Lapse- s_{∞}** : NI(MWd: s_{∞} , $\acute{s}$, s_{∞})

This states that all unstressed syllables are adjacent to stressed syllables, and will be violated by *a(lek)(sán)(drine) but not (at)(lán)ta or the other examples given above. Stressless feet should only appear when enforced by the various **Non-Finality** constraints.

(5.53)

/aleksandrine/	Non-Fin(F')	Non-Fin($\acute{s}$)	Edgemost	StemFtR	Lapse- s_{∞}	FtForm	Lapse-s
$\text{a}_m(\text{l}\acute{\text{e}}_m\text{k}_m)(\text{s}\acute{\text{a}}_m\text{n}_m)-(\text{dr}\bar{\text{e}}_{mm}\text{n})$			S			*	
$\text{a}_m(\text{l}\acute{\text{e}}_m\text{k}_m)(\text{s}\acute{\text{a}}_m\text{n}_m)-(\text{dr}\bar{\text{e}}_{mm}\text{n})$			S		!*		SSS
$\text{a}_m(\text{l}\acute{\text{e}}_m\text{k}_m\text{s}\text{a}_m\text{n}_m)-(\text{dr}\bar{\text{e}}_{mm}\text{n})$			S			*	!SSS
$\text{a}_m(\text{l}\bar{\text{e}}_m\text{k}_m)(\text{s}\acute{\text{a}}_m\text{n}_m)-(\text{dr}\bar{\text{e}}_{mm}\text{n})$		!*	S				SSS
$\text{a}_m(\text{l}\acute{\text{e}}_m\text{k}_m)(\text{s}\text{a}_m\text{n}_m)-(\text{dr}\bar{\text{e}}_{mm}\text{n})$			!SS				SSS

In words where multiple unstressed syllables do appear adjacently, e.g., *óorigin*, higher ranking constraints such as **Non-Fin(F')** enforce these optimal forms despite the violation of the lower ranked **Lapse- s_{∞}** .

5.4.2 Multisyllabic prefixes and constituency

There are a number of multisyllabic prefixes, and although they are not as common as the monosyllabic ones mentioned above, many behave in a similar way. For example, prefixes like /omni-/ , /ambi-/ , appear to subcategorize for a stem on their right; they align their right edges to the left of the stem, and can become part of the stressed foot under the appropriate conditions:

⁹¹This can be accounted for by an optional constraint which enforces faithfulness for vowel quality features in unstressed syllables. A stronger constraint requiring quality features in stressed syllables is mandatory at all times.

(5.54) (àmbi)({dék}tr)ous (àm)(bí{va}(l)ence) (òm)(ní{po}(t)ent)

When the root is able to form a foot on its own, as in *àmbidéxtrous*, the prefix stands in its own secondary foot. But when the root is light, it requires the light final of the prefix to comprise the necessary bimoraic foot, as in *ambívalence* and *omnípotent*. Due to the presence of further syllabic material supplied by the prefix, the lengthening seen in the shorter form *pótent* is unnecessary in *omnípotent*.

(5.55)

/omni-pot-ent/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	Lapse-s	*m
☞ (ommm)(nímpom)- (temnmt)			S				mmmmmm
(òmmmnim)(põmm)- (temnmt)			S				!mmmmmm m
(ómmnim)(põmm)- (temnmt)			!ss				mmmmmm

More common among multisyllabic prefixes is the behavior seen by /inter-/, /super-/ and /hyper-/. These prefixes appear to behave as if they were morphological words in themselves, and indeed *súper* and *hýper* can appear as independent words. For example, in nouns like *ínterview*, *íntercom*, *súperman*, *súperstar* the main stress falls on the antepenult, rather than the penult, which should be heavy, since it is a closed syllable. By regarding the prefix as a morphological word, this behavior can be explained using the current constraint hierarchy:

(5.56)

/inter+view/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ (imnm)(temrm)] (viewmm)			S				
(imnm)(térm)] (viewmm)		!* *					
(imnm)(temrm)] (vIEWmm)	!* *		S				

The constraint **Non-Fin(ś)** prevents the main stress from falling on the final syllable in the given morphological word constituent; another syllable in the same morphological word must follow. Since [(in)(ter)] forms a morphological word constituent, its main stress cannot fall on its second syllable, which is morphologically word-final, despite its weight in that position. The final syllable of the prosodic word, which itself forms a second morphological word constituent, fails to take main stress because of the stronger restriction imposed by **NonFin(Ft')**, whose domain is the prosodic word. The final syllable in the prosodic word may not take main stress. Of the possible candidates, *interview* causes the least violations of the various non-finality constraints.

In most words, the second syllable of /inter-/ fails to be stressed, even when followed by a light, unfooted syllable, e.g., *interdépendent*, *intercollégiate*. The restriction on stress in that position remains, even when the stress in question is not the main stress in the word; it will always be the main stress within the morphological word constituent /inter-/, and that is the domain addressed by the constraint **Non-Fin(ś)**. There are a minority of words, however, in which prefixes of this kind are in fact not behaving like morphological words, e.g., *intécalary*, *intépolate*, *intépret*, *supérfluous*, *supérlicative*. For words in /inter-/, it is easy to parse an alternate structure prefixed in /in-/, i.e., /in-técal-ary/, /in-tépol-ate/, /in-tépret/. In other cases, if the sequences in question were interpreted as prefixes, the stems would be defective, e.g., */super-l-ative/. In other cases, the prefix is clearly behaving as a stem, and is further suffixed, e.g., *supérieur*, *intérieur*, *intérial*, *supérial*. Thus, the sequences in question do not necessarily subcategorize as morphological words, although in most cases they do; the process is lexical. Similarly, in a few cases prefixes like /ambi-/ appear to subcategorize as morphological words, e.g., *àmbisyllabic*. The question of how this subcategorization should be handled in an OT grammar will be addressed in § 7.1.1.

Words with multiple morphological word constituents provide further information for fleshing out the constraint hierarchy. In previous examples, words contained only one morphological word constituent, and so there was no issue regarding which morphological word would contain the main stress. In forms like *interview*, the rightmost morphological word, although it contains the “root” rather than the prefix, is restricted from taking

main stress by **NonFin(Ft')**, which applied over the domain of the prosodic word. But in other words where the rightmost morphological word consists of more syllables, there is no mechanism by which this choice can be made:

(5.57)

/inter-act-ive/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s
☞ (i _m ⁿ m)(tē _m r _m)] (á _m c _m)-ti _m v			SS			*	SSS
(i _m ⁿ m)(tē _m r _m)] (a _m c _m)-ti _m v		!* !	S			*	SSS
☞ (i _m ⁿ m)(tē _m r _m)] (à _m c _m)-ti _m v			SS			*	SSS

Clearly, in such cases the rightmost morphological word accepts the main stress. This implies a second **Edgemost** constraint, this one applicable to the domain of the prosodic word, rather than the morphological word:⁹²

(5.58) **P-Edgemost:** NI(PrWd: Ft', PrWd, s)

This constraint is ranked below the other relevant constraints; in previous cases it would have had no effect, as the **Edgemost** constraint, effective in the domain of the morphological word, would have always superseded it in words with a single morphological word constituent.

(5.59)

/inter-act-ive/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	P-Edge
☞ (i _m ⁿ m)(tē _m r _m)] (á _m c _m)-ti _m v			SS		*	SSS	S
(i _m ⁿ m)(tē _m r _m)] (a _m c _m)-ti _m v		!* !	S		*	SSS	SS
(i _m ⁿ m)(tē _m r _m)] (à _m c _m)-ti _m v			SS		*	SSS	!SSS

⁹²Many adjectives of this type show alternative forms which take stress on the initial in phrases, e.g., *interàctive télévision*. This is likely due to a constraint governing the adjacency of Ft' constituents in the domain of the prosodic phrase (e.g., Align(PrPhrase: Ft', R; Ft[∞], L)), which is ranked higher than the **P-Edgemost** constraint. In isolation such a constraint would be inapplicable, as only one Ft' constituent would be present in the phrase. Constraints governing supra-word constituents will not be discussed in this work apart from for reasons of clarification.

There is one further effect seen due to the presence of two morphological word constituents in such words. Bisyllabic stems, when surfacing alone in the prosodic word, are forced to eliminate their final syllable from the main foot whenever possible, in order to avoid violating **NonFin(Ft')**. However, when such stems make up the second morphological word component in a word, it is possible for the first morphological word, the prefix, to take the main stress, leaving the second stem free to be footed entirely. As a result, such words are stressed on the prefix; example include *súpermàrket*, *hýpermàrket*, *súpersòníc*.

(5.60)

/super-market/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	Lapse-s	P-Edge
☞ (sùmpemr _m)l (màmr _m)kemt					*		SSS
(sùmpemr _m)l (mámrm)kemt			!s	S	*	SSS	S
(sùmpemr _m)l (mámrm)kemt	!* !				*		S
(summ)(pémrm)l (mámrm)kemt		!* !	S			SSS	SS
(sùmpemr _m)l (mámrm)kemt			!s		*	SSS	SSS

While *(már)ket* is the most optimal simplex form, with a prefix that forms a morphological word it can appear as a secondarily stressed *-(márket)*. When stems are suffixed, as in *interpérsónal*, this effect, which is due to **NonFin(Ft')**, is removed since the stem is no longer word-final, and such suffixed forms show main stress on the final, root morphological word constituent, enforced by **P-Edgemost**. This series of constraints will also affect stress in compound words, and compounds where the final member forms a stem that would violate **NonFin(Ft')** are stressed on the first member, e.g., *vídeotape*, *políceman*, *élderbèrry*. Suffixed compounds take stress on the second member, e.g., *nymphomániac*.

Monosyllabic prefixes can also form morphological words, although again subcategorization for this appears to be lexical. Examples of this include *dē-contáminate* and *rē-creáte* (cf. *récreàte*; this is a minimal pair, subcategorization of the prefix being the only difference). Such suffixes tend to

have a specific meaning, rather than being virtually meaningless (i.e., /re-/ ‘repeat’, /de-/ ‘undo’, /pro-/ ‘in favor’). This can be understood as a lexical subcategorization, and these meaningful prefixes might even be viewed as separate lexical entries, subcategorizing for the morphological word, simply homophonous with their meaningless counterparts.

5.4.3 Prefixes in unsuffixed words

Prefixes are also involved in effects involving the right edge of the word, motivated by **NonFinality** constraints. As was discussed in § 3.3.4, there are some classes of nouns and adjectives which correspond to prefixed verbal stems of the type *con-dúct*. Some show the same stress pattern as the verb, e.g., *resúlt*, *efféct*, while others show stress on the prefix, e.g., *íncrease*, *rébound*, *récess*, *défect*, *préfix*. The former group clearly shares the same structure as the corresponding verb forms, the suffix /-ə/ being retained and accounting for final stress. The latter group differs in that the suffix /-ə/ is apparently not included in the nominal form, and such words show initial stress on the prefix due to **NonFin(Ft’)**. However, unlike the case in bisyllables formed from unitary stems, mora augmentation is possible because the prefix is not part of the morphological stem. Thus, lengthened forms are expected in both monomoraic prefixes and stems in such cases, without incurring violations of **Stem-Coh**:

(5.61)

<i>increase</i> /in-kres/	Min-2	Non-Fin(F')	Non-Fin(š)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ (í ⁿ m)-(krē ^m s)				s				mmmm
(í ⁿ m)-kre ^m s				s	!s	*	sss	mmm
(í ⁿ m-kre ^m s)		!*						mmm
(i ⁿ m)-(krē ^m s)		!*	*					mmmm
i ⁿ m-(krē ^m s)		!*	*				sss	mmmm

(5.62)

<i>decrease</i> /de-kres/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ (děmm)-(krēmm)s				s				mmmm
(děmm)-krems				s	!s	*	sss	mmm
(dēm-krems)		!*						mm
(dēmm)-(krēmm)s		!*	*					mmmm
(dēm-krems)		!*						mmm
dem-(krēmm)s		!*	*				sss	mmm

(5.63)

<i>defect</i> /de-fekt/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ (děmm)-(fēmkm)t				s				mmmm
(děmm)-fēmkm)t				s	!s	**	sss	mmmm
(dēm-fēmkm)t		!*						mmm
(dēmm)-(fēmkm)t		!*	*					mmmm
dem-(fēmkm)t		!*	*				sss	mmm

The optimal candidates in these cases would have been eliminated by **Stem-Coh** if they had formed in themselves complete stems, e.g., *brigand*. For longer stems, e.g., *dé-ficit*, *ré-primand*, and for multiply prefixed forms such as *dé-re-lict*, the larger foot domains preclude lengthening:

(5.64)

<i>derelict</i> /de-re-likht/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ (dēm-rem)-(limkm)t				s				mmmm
(děmm)-rem-(limkm)t				!ss			ss	mmmm
(děmm-rem)-(limkm)t				s				!mmmm
dem-(rēm-limkm)t		!*					sss	mmmm
(dēmm)-(rēmm)-(limkm)t				s				!mmmmmm
dem-(rēmm)-(limkm)t				s			!sss	mmmm

(5.65)

<i>deficit</i> /de-fisit/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ (dé _m -fi _m)si _m t				s	s	*	sss	mmm
(dē _{mm} -fi _m)si _m t				s	s	*	sss	!mmm
de _m -(fi _m si _m t)		!*					sss	mmm
(dē _{mm})-(fi _m si _m t)				!ss				mmm

In both cases, the adoption of the prefix into the head foot, combining with the initial light syllable of the stem, is the most optimal choice available.

There is another type of prefixed word which does not show these effects, but rather shows a short vowel in the prefix, e.g., *récord*, *próduct*, *próject*. It will be proposed that this type, which is much more restricted in distribution than the type of word described above, is treated as if both prefix and root formed a single stem, i.e., such forms are treated like *brígang*. How this different treatment of subcategorization and constituency in these forms can be lexically handled using OT will be discussed in § 7.1.1. Finally, there are word-initial sequences which are apparently monomoraic prefixes, but which never show lengthened or stressed alternants, e.g., *appráisal*, *avówal*, *aséssment*. These stand in contrast to forms like *apróve* ~ *àpprobátion*, *attést* ~ *àttestátion*, where the prefix may appear with vowel quality when stressed and can be given an underlying form /ad-/, assimilation taking place in the adjacent obstruents. While forms like *appráisal* could be given the same interpretation, it is simpler for the grammar to rather regard such forms as prefixed in /ə-/. The segment /ə/ is never lengthened regardless of moraicism, and also cannot be stressed.⁹³ Since such syllables never appear with any vowel quality, it is difficult to say how the grammar would be able to assign any other segment to their nuclei. This is further evidence for an underlying segment /ə/ within the grammar, beyond its role in the verbal and adjectival suffix /-ə/.

5.5 Summary of lengthening effects

Most of the lengthening effects described in this chapter are the direct result of morphological constituency. Feet which are restricted, for a variety of

⁹³The best explanation for this would involve head syllables demanding an association to place features, which /ə/ lacks.

reasons, to align with open, light syllables, are forced to become bimoraic due to the **Min-2** constraint. There are a variety of strategies for achieving this bimoraicity, but all produce the similar effect of a lengthened vowel in an otherwise monomoraic morpheme. The overgeneration of forms with lengthened vowels is prevented by **Stem-Coherence** and **SuffixMoras**, which ensure that such lengthening is licit only within monomoraic stems and prefixes (§ 5.2). This results in the attested distribution of arguably lengthened vowels, which are never seen in multimoraic stems or in unstemmed suffixes.

Vowel lengthening as seen in forms affixed with the /-al/ group of suffixes (e.g., *tónal*, *reprísal*) is enforced by both **Stem-Ft-R**, which aligns the foot to the right edge of the stem, and **NoOrphans**, which favors the association of the mora projected by the suffix to the stem, where it can be footed. Among the /-ent/ group, which subcategorize similarly to the /-al/ forms but are intrinsically bimoraic, lengthening occurs only when insufficient moras are present in the stem, e.g., *sónant*. When more material, such as a prefix, is present in the word, it can be incorporated into the stem foot, e.g., *résident*. The contrast between the segmentally similar forms *reprísal* and *résident* is due to the effect of **NoOrphans**, which does not apply to the latter as its suffix is footable (§ 5.1.2).

Monomoraic prefixes such as /re-/ , /pro-/ can also be lengthened when they occur alone in a foot. In some cases, constraints enforcing extrametricality effects such as **NonFin(Ft')** prevent stems from appearing within the primary stressed foot, and the prefix alone appears as the main stressed foot, e.g., the noun *défect*. In other cases, the prefix stands in a stem and morphological word constituent of its own due to its subcategorization, e.g., *rè-crēâte* 'create again', which can be contrasted with *récrēâte* 'refresh', where the prefix is part of a single morphological word constituent and fails to undergo lengthening since it forms a bimoraic foot with the stem syllable /cre/ (§ 5.4.2). The prefix can also lengthen where it forms a secondarily stressed foot, appearing to the left of the main stress, e.g., *rēvólt*, *dēbāte*, *prōvīde*. However, such feet are understood as stressless, and may appear optionally with reduced vowels, e.g., *rəvólt*, *dəbāte*, *prəvīde* (§ 5.4). This is due to the proximity of the main stress and the restriction of adjacent stresses enforced by **NonFin(ś)** (§ 5.3).

These lengthenings can be contrasted with another, well-known type of lengthening, so-called “CiV lengthening”. This takes place within the stem, and is the result of a mis-correspondence between moras projected by segments and moras which associate to segments. An unassociated mora is projected (for an /i/) by **Moraic Prominence**, and associates to the vowel of the preceding syllable in the optimal candidates, because the /i/ links to the onset of the following syllable due to the **Onset** constraint. This shift does not violate **Stem-Coherence**, as the mora count within the stem remains constant. This lengthening is different from the lengthenings described above as it is the result of interactions between prosodic, rather than morphological, constituents. Another purely prosodic lengthening effect is seen when two vowels appear in hiatus, and is also enforced by the **Onset** constraint (§ 5.2.2).

The morphologically conditioned lengthenings identified herein went previously undetected due to theoretical assumptions about the underlying lexical forms of these words, which were thought to contain long vowels. This assumption has led to the various proposals of shortening discussed in chapter 2, and the subsequent difficulties this approach posed for accounts of English stress. The distribution of long and short vowels in the English data, as presented in chapter 3, demands a new understanding of the underlying forms seen in “level I” words, and the OT account presented in chapters 4 and 5 provide an analysis which successfully treats both stress and vowel alternation. The next chapter applies the OT hierarchy arrived at here, as well as the conclusions drawn about underlying forms, to further cases of “level I” affixation and “stress retraction” in English.

6. Stress retraction and constituency

6.0 Retraction and OT

In the preceding chapter, most of the forms discussed took their main stress on one of the final three syllables, in accordance with Hayes' principle of extrametricality, now captured in an OT context by using various constraints. However, there are various suffixed types which can show their main stress further back in the word, or further back than expected purely on segmental grounds. These are the forms which have been described as showing "stress retraction" by Kager (1989) and others, as described in chapter three. As with previous cases of suffixation, these will be accounted for using subcategorization constraints and the constraint hierarchy arrived at in chapter five. In this chapter, a number of the retraction effects noted by Myers (1987), Kager (1989) and others are discussed and accounted for structurally, most prominently the retraction seen in the /-ate/ and /-ize/ series of suffixes (§ 6.1). The constraint hierarchy under discussion will be augmented to account for new phenomena, and further insight will be given into the role of morphological subcategorization in the prosodification of English words, as a number of affix types are treated in detail (§§ 6.2-6.6). Some contrasts between British and American pronunciations are accounted for by proposing slightly different constraint hierarchies for these two major dialects. Following the treatment of various affixes, some of the notable retraction effects discussed in the literature are addressed (§ 6.7), and finally the role of the non-Latinate vocabulary will be touched upon (§ 6.8).

6.1 Words in /-ate/ and /-ize/

Two sets of very common verbal affixes which show a consistently "retracted" stress pattern are /-ate/ and /-ize/. About 95% of such forms trisyllabic and longer show antepenultimate stress, regardless of the weight of the penult:

- | | | | |
|-------|-------------|-----------------|---------------|
| (6.1) | (hési)tate | (cándi)date | (hýphe)nate |
| | (désig)nate | (éx)(pur)gate | (pǒ)(ten)tate |
| | (déle)gate | (cón)(cen)trate | (démon)strate |

This effect is in direct contrast to the treatment of both unsuffixed words and words in the /-ent/, /-al/ or /-ic/ suffix groups. This type of footing is instead identical to that seen at word edges in unsuffixed words, and it will be proposed here that these suffixes in fact subcategorize for morphological words rather than morphological stems:

- (6.2) **al-Suffixation:** Align(Suf_{al}, L; Stem, R)
 ate-Suffixation: Align(Suf_{ate}, L; MWord, R)

The constraint **NonFin(ś)**, proposed above (5.39) to account for stress in words like *brigand*, will prevent the morphological word-final syllable (in this case the penult) from taking the main stress in the prosodic word. Thus, for cases like *désignate*, the following tableau will result:⁹⁴

(6.3)

/de-sign+at-ə/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm
 (dém-simgm)] +(nā _{mm} t)ə						*
(dē _{mm})-(simgm)] +(nā _{mm} t)ə				!s		
dem-(sí _{mm} g)] +(nā _{mm} t)ə			!*			
(dè _{mm} -simgm)] +(nā _{mm} t)ə				!s		*

Since the **Edgemost** constraint as formulated above in (5.41) aligns the head foot to the right edge of the morphological word (rather than the prosodic word), placing the head away from its edge (which in this case corresponds to the stem edge, the right edge of /-sign-/) will incur violations; however, the more highly ranked constraint **NonFin(ś)** prevents the penult (which is

⁹⁴As in preceding chapters, certain conventions will be used to mark morphological and prosodic structure in candidates. Feet will be surrounded by (), stems by { }, morphological words by [], prosodic words by (). Furthermore, ‘-’ may be used to divide stem syllables from affixes, while ‘+’ indicates a division between morphological words and affixes. The latter conventions are meant to further clarify notationally complex representations, as in (6.3).

final in the morphological word) from ever taking main stress.⁹⁵ Thus, the unusual presence of (LH) stems within the word, and the general antepenultimate stress seen in these forms, is explicable via suffix subcategorization, the morphological word boundary appearing before, rather than after, the level I suffixes in question.

For monosyllabic stems suffixed in /-ate/, some British pronunciations fail to show stress on the final stem syllable, e.g., *rotáte*, *castráte*, *vacáte*, *migráte*, as predicted by the current constraint hierarchy:

(6.4)

/migr+at-ə/	Min-2	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s	*m
(mī _{mm})]+(grā _{mm} t)e				s				SSS	mmmm
(mī _{mm})]+(grā _{mm} t)e			!*					SSS	mmmm
(mī _m)]+(grā _{mm} t)e !*			*					SSS	mmm
mī _m)]+(grā _{mm} t)e				s	!s			SSSSS	mmm
(mī _m)]+gram)tēn			!*	s	s		*	SSS	mmm

Setting the main stress on the suffix violates only the **Edgemost** constraint, while stressing the stem would violate the more highly ranked **NonFin(ś)**. Some dialects, however, including most American dialects, do not permit the main stress to fall outside the Morphological Word, and the stress is retained on the stem (i.e., *mīgrāte*). This suggests a constraint restricting the main stress to fall within the morphological word, **Ft' to MWd**, which outranks **NonFin(ś)** in those dialects. This constraint can be formally defined as follows:

(6.5) **Ft' to MWd:** NI(PrWd: PrWd, MWd, Ft')

This states that for each prosodic word, no head foot intervenes between the prosodic word and the morphological word. This will be satisfied only if the head foot is inside of some morphological word. In a position outside the

⁹⁵In the handful of words which show penultimate stress, e.g., *impréginate*, the morphological structure of such words is evidently different, and is parallel to that of the /-al/ words, i.e., the suffix subcategorized for the stem rather than the word. See § 7.1.1 for how such cases would be handled lexically within OT.

morphological word, it will intervene between a prosodic word edge and a morphological word edge, incurring a violation. Adding this constraint to the hierarchy does not affect previous results, as all optimal candidates among word types dealt with above would satisfy it. Monomoraic stems of the type seen in *migráte* would be typically lengthened as in the previous cases discussed, explaining the dearth of light stems of this kind (e.g., *nárrate*, a geminate stem). Note that as with the *sónant* type, the lengthening is no longer seen when the stem is extended by a prefix, e.g., *immigráte*.

The subcategorization of /-ate/ and /-ize/ for the morphological word rather than for the stem, as with the /-al/ and /-ent/ suffixes, suggests that the stress pattern seen in words with these suffixes should resemble that seen in their unsuffixed forms, and that suffixed words that are further extended with /-ate/ or /-ize/ would retain their stress patterns. For example, we find pairs such as *órient* ~ *órientate*, *nátional* ~ *nátionalize*, *èditórial* ~ *èditórialize*, *indivídual* ~ *indivídualize*. This is expected, because the role of the stem remains the same within both forms, the main-stressed foot unable to encompass the suffix /-al/, which must follow the stem due to subcategorization. However, to yield the correct forms, it must be recognized that the definition given to **Edgemost** above (5.41), which was based on previous derivational conceptions of extrametricality, is in fact insufficient to account for this (although this has no effect on the validity of previous tableaux). Previously, **Edgemost** stated that the head foot of a given morphological word preferred to align to the right edge of that word, and was only thwarted in classic extrametricality cases like (*óri*)*gin*, *re(vér)**sal*, by the higher-ranking constraint **NonFin(Ft')**. However, when the morphological word is not congruent with the prosodic word, as in the /-ate/ and /-ize/ forms, **NonFin(Ft')** will not apply to morphological word-final syllables which are not word-final in the prosodic word. Thus, nothing would prevent the incorrect form *[vi(sú-al)]ize. Instead, a new definition for the **Edgemost** constraint should be introduced, formulated as follows:⁹⁶

(6.6) **Edgemost:** NI-R(MWd: Ft', Stem, s)

⁹⁶The old formulation of the **Edgemost** constraint, which may be renamed **WdEdgemost**, is of course retained in the hierarchy, but must be ranked below the new **Edgemost**. **WdEdgemost** is relevant where more than one stem is present in a given morphological word; in such cases, the rightmost stem will be expected to hold the main stressed foot (see § 6.5).

This replaces the morphological word with the morphological stem as the second argument of the constraint. Thus, the head foot attempts to align to the right edge of the stem (rather than the morphological word). This correctly penalizes the misaligned stem in the morphological word constituent *[vi(sú-al)], and enforces the correct [(vísu)-al]. It is the consistent form of the stem throughout such complex words in English that produces, for most stems, the apparent uniformity of stress under affixation which inspired derivational theory (§ 1.4). Only when the stem itself is altered, for example by the addition of an /-ity/ suffix, which subcategorizes within the stem, is the apparently “basic” stress pattern of a given morpheme altered.

Changing the formulation of **Edgemost**, which had referred to the morphological word as both the domain and the second argument of the constraint, does not affect the results of the preceding tableaux. In most cases, the words being treated were unsuffixed and the right edges of the morphological word and stem were identical. For the suffixed cases discussed previously, e.g., *résident*, *reprísal*, the suffixes were both monosyllabic and standing at the right edge of the word; thus, any candidate *re(prísal) which satisfied the old **Edgemost** constraint by aligning to the right edge of the morphological word would have been eliminated by the higher-ranking **NonFin(Ft')**. Even in the /-ize/ and /-ate/ cases discussed above, e.g., *désignate*, the right edge of the stem was congruent to that of the morphological word. It is only in cases where more than one syllable intervenes between the stem and the prosodic word edge that the new formulation becomes necessary. Applying this new definition of **Edgemost**, the complex multiply suffixed forms in /-ate/ and /-ize/ can be accounted for:

(6.7)

/ visu-al+iz-ə/	Min- 2	Non- Fin(F')	Non- Fin(ś)	Edgemost	StemFtR	Lapse-s	*m
☞ (v _i m _{su} m)-a _m] +(l _i m _m z)-<ə>						SSSSS	mmmmmm
v _i m(sú _m -a _m] +(l _i m _m z)-<ə>				!s	s	SSSSS s	mmmmmm
(v _i m _{su} m)-(ǎ _m m)] +(l _i m _m z)-<ə>			!*	s		SSS	mmmmmm
v _i m(sù _m -a _m] +(l _i mmz)-<ə>				!s	s	SSSSS s	mmmmmm

Both **Edgemost** and **Stem-Ft-Rt**, which restrict the head foot to the right stem edge, enforce its position within the morphological word. Thus, words like *visualize* will have morphological words stressed identically to those seen in unaffixed forms such as *visual*. The fact that most of the given constraints, such as **NoOrph**, govern the morphological word means that the constituent relationships seen within morphological words will remain constant despite suffixation, e.g., *revítalize*, *mótorize*, as long as that suffixation subcategorizes outside the stem.

6.2 Extended forms of -ate

As was mentioned above in § 3.3.4, the suffix /-ate/ can be understood as consisting of an affix /-at/ extended by /-ə/, and other extended forms of this suffix are also attested, for example /-at-ion/, /-at-ive/, /-at-or/, /-at-ory/. Unextended /-at/ is seen in a small group of words such as *áternate*, *désignate*, *íntricate*, *immédiate*, all ending in [-ət] rather than [-eɪt]. These words are all nouns or adjectives, and many, but not all, have corresponding verbs in [-eɪt]. Most words of this type show the same subcategorization as verbal forms in /-at-ə/,⁹⁷ but the subcategorization seen in the other extended forms is less obviously straightforward.

The subcategorization of nominal abstracts in /-ation/ is identical to that seen in the verbs in /-ate/. As with other cases of /-ion/ suffixation, the stress in these forms is universally on the penult, e.g., *cèlebrátion*,

⁹⁷A small minority, e.g., *apóstate*, *insénsate*, and some variant forms such as *intrícate* show a different subcategorization, similar to that for /-al/ suffixes. See § 7.1.1 for treatment of this kind of problem.

rècommendation, désignation. This stress results from the stem-forming behavior of the suffix /-ion/, which like other suffixes of the /-ity/ group aligns its right edge, rather than its left edge, to the right edge of the stem (4.25). Thus, the /-ation/ suffix itself is the rightmost footed stem syllable in the word:

(6.8) (cèle)(bráti)on (rèco)(men)(dǎti)on (désig)(náti)on

On the basis of these forms, it is not immediately clear whether the suffix /-at/ subcategorizes for the morphological word (i.e., [{celebr}][{ation}], with two stems and two morphological words in a single prosodic word) or whether the subcategorization for /-ion/ overrides that, yielding either one morphological word and two stems (i.e., [{celebr}{ation}]) or one morphological word with one long stem (i.e., [{celebration}]). The primary indicator of such constituency, stress retraction, is not solely attributable, as in the verbal forms, to the antepenult's final position in a morphological word constituent. Since the penult syllable (*ǎt-*) takes main stress as the head of the rightmost stem foot, the antepenult can never be stressed, due to **NonFin(ś)**, regardless of morphological constituency. However, other constituency-influenced features apparent in certain words indicate that /-ation/ is indeed subcategorizing for a separate morphological word:

(6.9)

/de-sign+at-ion/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	FtForm	NoOrph	Lapse-s	*m
☞ (dè _m -si _m g _m)] +(ná _{mm} ti)o _m n					*	*	SSS	mmmmmm
(dè _{mm})-(si _m g _m)] +(ná _{mm} ti)o _m n			!s			*	SSS	mmmmmm
de _m -(si _m g _m)] +(ná _{mm} ti)o _m n		!*				*	SSSS s	mmmmmm

The sub-optimal candidate *(dè)(sig)(náti)on would surface if not for an **Edgemost** violation, incurred by the stressed prefixal foot (*dè*) standing two syllables from the edge of its stem; this foot is the head foot of its morphological word. Since the domain of the constrain **Edgemost** is the morphological word, the fact that the suffix forms the head foot to the entire

prosodic word is irrelevant to the local **Edgemost** violation in the leftmost morphological word /de-sign/.

More evidence for the presence of multiple morphological words in these forms is found in multiply suffixed forms. Words like *visualizátion*, *còdificátion*, *òrganizátion* suggest the following morphological structures:

(6.10) [{visu}al] iz [{ati}on] [{codi}fic][{ati}on] [{organ}] iz [{ati}on]

The initial secondary stress seen in each of these words is expected only if constraints are acting locally in the domain of an immediate morphological word, rather than if the morphological word was parallel to the entire prosodic word (otherwise *visuàlizátion, *codìficátion, *orgànizátion would be expected). The behavior of /-iz/ in such forms is also notable; it perhaps forms a third morphological word constituent (demanded by /-at/) and is at least a stem in itself. The two optional behaviors of the suffix, as a long vowel /-īz-/ or a reduced /-əz-/, result from its positioning directly before the main stress of the word, making it an unstressed foot. As a monomoraic morpheme, /-iz/ would need to be lengthened to form a foot in itself, but as an unstressed foot, it may optionally surface with a reduced vowel (§ 5.3). Subcategorization constraints prevent it from becoming part of another stem or word constituent.

Forms in /-ator/ also show subcategorization for the morphological word, and furthermore illustrate that the suffix /-at/ becomes part of a morphological stem that however stands outside the morphological word in these forms, i.e., /[{moder}][{at}or/. This is the identical structure that would be expected for verbs like *móderate*, i.e., /[{moder}][{at}ə/. However, in the tableau given previously for such forms (6.3), **Edgemost** was used to eliminate candidates with stress on the suffix. Since **Edgemost** as reformulated applies only within the morphological word, elimination of such candidates via **Edgemost** is not possible; instead, the constraint **Ft' to MWd**, introduced above in (6.5), plays a crucial role:

(6.11)

/moder+at-or/	Non-Fin(F')	Non-Fin(ś)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
☞ (mó _m dem) +(ră _{mm})t-o							SSS
mo _m (dé _m)+ra _m) t-o _m r		*!	*	S	S	*	SSSS s
(mó _m dem) +(rà _m t-o _m r					!s		
(mò _m dem) +(ră _{mm})t-o			!*				SSS

The form *mòderător, with stress outside the morphological word constituent, violates this constraint; otherwise, both this form and the attested *móderător* would be equally optimal. For completeness, it is appropriate for the tableau for *désignate* be reformulated:

(6.12)

/de-sign+at-ə/	Non-Fin(F')	Non-Fin(ś)	Ft to MWd	Edgemost	StemFtR	FtForm	Lapse-s
☞ (dé _m -síngm)] +(nă _{mm} t)ə						*	SSS
(dě _{mm})-(síngm)] +(nă _{mm} t)ə				!s			SSS
de _m -(síngm)] +(nă _{mm} t)ə		!*					SSSS s
(dè _m -síngm)] +(nă _{mm} t)ə			!*			*	SSS

As was the case with the /-at-ə/ forms, monosyllabic stems in /-at-or/ pattern differently in American and British English. This can be accounted for by proposing a different ranking of this **Ft to MWd** constraint in each of these major dialects. In British, the constraint is ranked higher than **NonFin(ś)**, thus we see forms like *dictător*, similar to *dictăte*:

(6.13)

UK: /dict+at-or/	Non-Fin(F')	Non-Fin(ś)	F'to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
(dí _m c _m) +(tằ _m m)t-o		!*					SSS
(dí _m c _m]+ta _m) t-o _m r		!*	*	S	S	*	SSS
(dí _m c _m) +(tằ _m to _m r		!*			S		
☞ (d _i m _c m) +(tằ _m m)t-o			*				SSS

In American English, the ranking is reversed, and we find *díctāte*, *díctātor*:⁹⁸

(6.14)

US: /dict+at-or/	Non-Fin(F')	F'to MWd	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s
☞ (dí _m c _m) +(tằ _m m)t-o			*				SSS
(dí _m c _m]+ta _m) t-o _m r		!*	*	S	S	*	SSS
(dí _m c _m) +(tằ _m to _m r			*		!s		
(d _i m _c m) +(tằ _m m)t-o		!*					SSS

Nearly all words in /-ator/ of more than three syllables show main stress on the preantepenult,⁹⁹ e.g., *móderātor*, *ínnovātor*, *cómmentātor*, *extérminātor*, as expected for a suffix that subcategorizes for a morphological word constituent.

6.3 The suffixes -ary, -ory and -atory

The extended suffix /-atory/ presents a complex picture, primarily due to the nature of the suffix /-ory/, which extends /-at/ in these cases. In British English, words with stems two syllables and longer present four

⁹⁸The apparent exceptions *crēāte*, *crēātor* behave as if analyzed as coming from a stem /creāt/, rather than being a suffixed defective stem /*cre/.

⁹⁹Exceptions which show different stress, e.g., *conspírador*, *prédador*, each of which have no corresponding verbal form in /-ate/, apparently show a different lexical subcategorization, perhaps that of the variant of /-at/ proposed in § 6.3.

different stress patterns; in American English, this is reduced to two, but there is on the surface little apparent reason for the distinction:

(6.15) <u>British</u>	<u>American</u>
[1,0,2,0,0]: ádulàtəry	
[2,0,1,0,0]: òbligátəry	
[1,0,0,0,0]: círculətəry	[1,0,0,2,0]: círculətòry
[0,1,0,0,0]: defámətəry	[0,1,0,2,0]: defámətòry

American forms fail to display a long suffix vowel /-āt/, while British forms never stress the suffix /-ory/. Words showing the first two patterns, which are simply inversions of each other, are stressed like *círculatòry* in American (e.g., *ádulatòry*). Most British words which stress the /-āt/ suffix may display both of these patterns, suggesting that these are actually a single pattern and, since these are adjectives, main stress can swap to the secondarily stressed foot under certain phrasal conditions, as in other kinds of adjectives (§ 5.4.2, fn. 24).

In American English, there is a correspondence between morphological structure and the stress type; most words showing initial stress are similar to *defámətòry*, consisting of prefix plus monosyllabic stem, while most words showing initial stress consist of single bisyllabic stems, as in *láboratòry*. This generalization can be extended to longer words, e.g., *hallúcinatòry*, *interrógatòry*. However, there are exceptions, such as *réspiratòry*, and some words may show both types of stress, e.g., *obligatory*. In British, there is less of a pattern, and some words may show as many as three types, e.g., *cónfiscàtory*, *cònfiscátory*, *confíscatory*. If /-ory/ was similar in behavior to one of the /-al/ suffixes, such as /-or/, we would expect the optimal candidates in all cases to show forms similar to [(cón)(fis)](càt)əry in all cases regardless of the length of the stem. We would also expect monosyllabic stems to never be stressed if prefixed (cf. *réspirātor*, *éstimāte*), and to be heavy if stressed (cf. *gýrāte*, *dǒnāte*, *díctāte*). However, many words like *exclámatory*, *decláratory* contradict this expectation, while only a minority (e.g., *mígratòry*, *consólatòry*) do not. Looking on to the suffix /-ory/ itself, while on the surface it resembles an /-al/, neither the data nor the current constraint hierarchy supports this supposition. To determine what the morphological structure of /-atory/

words might be, it is first necessary to investigate the /-ory/ words, as well as the similar forms in /-ary/.

The first difficulty in treating the words in /-ary/ and /-ory/ is that identifying them is not completely straightforward. It is not clear whether words like *victory*, for example, are /-ory/ words, or /-or/ words extended by the suffix /-y/. Another problem is the surface realization of the suffixes. In British English these always surface as [-əri], and in American this is also often the surface reflex. This makes these suffixes homophonous with the level two suffix /-ery/, which is usually not stressed (e.g., *bríbery*, *distíllery*); when it is, it is identical (in American) to the stressed output of /-ary/, e.g., *mónastèry*. However, despite the suffixes' stresslessness in British, it is still possible to see that underlyingly the British grammar still recognizes their different vowel qualities, in extended forms of certain words, e.g., *victórious*, *àntiquárian*, *àuditórium*. When looking at words suffixed with /-ary/ and /-ory/, a majority do look as though they follow the typical /-al/ suffix pattern, wherein stems are parsed into Hayesian feet, excluding the prefix:

(6.16)	(rǒ)tary	ad(vǐ)sory	(áudi)tòry
	(sén)sory	in(fír)mary	(líte)ràry
	(búr)glary	di(réc)tory	(cáte)gòry

However, there are a large number of words which fail to exhibit this pattern, failing to stress heavy stem-final syllables, even when such syllables are the only ones in the stem:

(6.17)	désultòry	prómontòry	répertòry	óffertòry
	ádversàry	mómentàry	cómmèntàry	
	vóluntàry	légendàry	sécondàry	
	prébendàry	frágmentàry	sédentàry	

While forms like *advísory*, *infírmory* have been offered as evidence for the cycle by Kiparsky (1982a, b), other equally relevant forms like *ádversàry* and *sálivàry* provide contradictory evidence.¹⁰⁰ The surface forms of words in /-ary/ and /-ory/ are not completely dependent either on their

¹⁰⁰There are also UK variants *salívary* and *advérsary*, but that is beside the point.

morphological structure, their syllable weight, or their relationship with any unsuffixed “base” form.

It is not possible, however, for /-ary/ or /-ory/ to subcategorize for morphological stems, as do the /-al/ and /-ent/ suffixes. If they did, we would expect prefixation to occur as with the /-ent/ suffixes, since /-ary/ and /-ory/ are also bimoraic. But apart from the single initially stressed form *éxcretòry*, prefixed forms like *advísory* rather resemble /-al/ suffixes in retaining stress on the lengthened monosyllabic stem. Furthermore, heavy stems that should retain stress, such as those of *ádversàry*, *prómontòry*, fail to do so. The behavior of such forms becomes comprehensible, however, if /-ary/ and /-ory/ are understood as subcategorizing for the morphological word, just like /-ate/ and /-ize/:

(6.18)

/ad-vers+ary/	Non-Fin(F')	Non-Fin(ś)	Ft to MWd	Edgemost	StemFtR	Lapse-s	*m
☞ (ám _d m)(ve _m r _m) ₊ (sà _m ry _m)				S			mmmmnm
(a _m d _m)(vé _m r _m) ₊ (sa _m ry _m)		!*					mmmmnm
(àm _d m)(ve _m r _m) ₊ (sǎ _m m)ry			!*	S		SSS	mmmmnm
(àm _d m)(ve _m r _m) ₊ (sá _m ry _m)	!*		*	S			mmmmnm

The fact that the stem coincides with the morphological word means that the stem-final syllable cannot be stressed, even if it is heavy, as was the case with the /-ate/ forms. This accounts for the words in (6.17); however, there are also forms with antepenultimate stress to be accounted for, e.g., *infírm*ary. The solution to this problem is in fact similar to Kiparsky’s, although for different reasons. Since /-ary/ and /-ory/ align themselves to morphological words, the morphological word constituents in question may be identical to forms which occur as independent words, i.e., [in{firm}ə], the morphological word constituent of *infírm*, with final suffixal schwa. If the definition of the constraint **Final Margins** (4.17) is altered to take the morphological, rather than the prosodic word as its initial argument, then schwa can be understood as unparsed in morphological word-final position, and forms like *advísory* may be accounted for:

(6.19)

/ad-vis-ə+ory/	Non-Fin(F')	Non-Fin(ś)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s	*m
 (a _m d _m)(v _i m _m)sə]							SSS	mmmmmm
(o _m r _y m				!s	s			mmmmmm
(a _m d _m)(v _i m _s ə _m)]								
(o _m r _y m						!*	SSS	mmmmmm
(á _m d _m v _i m)sə _m]								
(o _m r _y m								

The mora belonging to the unparsed schwa joins the stem foot; failure to do so would incur a violation of **NoOrphans**, whose domain is also the morphological word. Under this analysis, the primary difference between words like *infirmary* and *adversary* is the presence of a morphological word-final schwa in the former, which prevents the root syllable from standing in the unstressable final position. While it is straightforward to say that words like *legend* and *moment* aren't extended with schwa, while words like *advise* and *direct* are, no generalizations of this kind about paradigms hold completely; *adverse* may also appear with (or without, i.e., *adverse*) a schwa. The morphological structure of complex words such as these /-ory/ and /-ary/ cases is ultimately lexically determined, and is not strictly dependent upon the presence of similar morphological structures appearing elsewhere in a "paradigm" (see § 7.4.1).

One final thing should be said about the difference between the British and American realizations of /-ary/ and /-ory/. The American forms of the suffix are stressed when following an unstressed syllable,¹⁰¹ which indicates that they are part of the prosodic word. However, these suffixes are never stressed in British, and this suggests that in British they stand outside the prosodic word, i.e., that they subcategorize for the prosodic word on the left of the suffix. This also explains why stress remains on the stem even when it is monosyllabic, e.g., *rotary*, and never moves to the suffix as in *rotáte*.

Having understood /-ory/ as subcategorizing for the morphological or prosodic word, it becomes easier to interpret the /-atory/ forms. British-only forms with lengthened suffix vowel, such as *migratory*, have the structure

¹⁰¹This is likely to be enforced by a **Rhythm** constraint of some kind, although not **NonFin(ś)**, which only applies within the morphological word.

[(mī)](grā)tə)ory, the suffix subcategorizing for the prosodic word *migrātə*. Alternate forms like *migratory* fail to show the morphological schwa suffix; having a structure [(mī)]grat)ory, the suffix vowel /a/ can no longer be main-stressed, as it is final in the prosodic word and main stress would violate **NonFin(Ft')**. Due to its position, it is not even footed, and thus surfaces as one in a succession of unstressed syllables. However, since none are part of the morphological word, the constraint **Lapse-s[∞]** (5.52) would not be violated:

(6.20a)

UK: /migr+at-ə+ory/	Non- Fin(F')	Non- Fin(ś)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
☞ [(mī _{mm})](grā _{mm})t<ə> _m -o _m ry _m			*				SSS
[(mī _{mm})](grā _{mm})t<ə> _m -o _m ry _m	!*						SSS
[(mī _m grā _m)tə _m] -o _m ry _m	!*			S	S	*	SSS
[(mī _{mm})](grā _m t<ə> _m) -o _m ry _m	!*				S		

(6.20b)

UK: /migr+at+ory/	Non- Fin(F')	Non- Fin(ś)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
[(mī _{mm})](grā _{mm})t-o _m ry _m !*			*				
[(mī _{mm})]grā _m t-o _m ry _m	*					!*	SSS
☞ [(mī _{mm})]grat-o _m ry _m	*						SSS
[(mī _m grā _m)t-o _m ry _m !*		*		S			

The optimal candidate in the second case lengthens its stem by borrowing the orphaned mora from the suffix /-at/. Note that an otherwise optimal candidate *[(mī_{mm})](grā_{mm}t)-o_mry_m would be eliminated by the high-ranking constraint **SuffixMoras**, which prevents syllables outside the stem from undergoing mora augmentation.

British forms with longer stems would be expected to show initial stress as well as a long /-āt/ when suffixed with schwa (e.g., *círculātory*), as in the unextended forms (e.g., *círculāte*), which share the same prosodic word structure, and indeed such forms do surface:

(6.21a)

UK: /circul+at- ə+ory/	Non- Fin(F')	Non- Fin(š)	Ft' to MWd	Edgemost	StemFtR	NoOrph	Lapse-s	*m
[(ci _m r _m cu _m)] (lā _{mm})t<ə> o _m r _y r			!* S				SSS	mmmmun
☞ [(ci _m r _m cu _m)] (lā _{mm})t<ə> o _m r _y m							SSS	mmmmun
[(ci _m r _m)(cú _m)] la _m)t<ə>m o _m r _y m		!* S	* S	S	S	*	SSS	mmmmun

(6.21b)

UK: /circul+at+ory/	Non- Fin(F')	Non- Fin(š)	Ft' to MWd	Edgemost	StemFtR	NoOrph	Lapse-s	*m
[(ci _m r _m cu _m)] (lā _{mm})t o _m r _y r			* S					mmmmun
☞ [(ci _m r _m cu _m)] la _m t) o _m r _y r						*	SSS	mmmm
[(ci _m r _m)(cú _m)] la _m)t o _m r _y m		* S	* S	S	S			mmmm

The variant form of such words lacking the schwa shows the completely stressless suffix complex, i.e., *circulatory*. Words consisting of prefixed stems, e.g., *obligatory*, would also be expected to show this form, since the position of the stem as morphologically word-final prevents it from taking the main stress.

As was indicated above, the American forms in /-atory/ show only two stress patterns, neither of which allow for a long vowel in the /-āt/ suffix. The best explanation for this is that the schwa-extended /-atory/ forms are not present in American; otherwise we would expect forms like *mígratory.¹⁰² In American, all /-atory/ forms are of the type /+at+ory/, as in /migr+at+ory/:

¹⁰²A form *mígratory would be prohibited by the different ranking in American English of **Ft'** to **MWd**, see § 6.1.

(6.22)

US: /migr+at+ory/	Suf-m	Ft to MWd	Non- Fin(ś)	Edgemost	StemFtR	Lapse-s	*m
[(mímm)](grāmm) !* (tòmrym		*					mmmmmm
[(mímm)](grāmm) !* (tòmrym			*				mmmmmm
[(mímm)] (grāmto)mrym			*			!sss	mmmmmm
[(mím] gram) (tòmrym		!*	*	s	s		mmmm
☞ [(mímm)]gra (tòmrym			*			ss	mmmm

Since /-ory/ does not subcategorize for the stem constituent, as did /-ə/, the syllable /-at/ is not a stem and cannot be lengthened (§ 5.2.1). Monosyllabic stems will borrow the mora belonging to the suffix to achieve bimoraicism in the stem. Longer stems will produce results similar to the British forms:

(6.23)

US: /circul+at+ory/	Suf-m	Ft to MWd	Non- Fin(ś)	Edgemost	StemFtR	Lapse-s	*m
[(címrmcúm)] !* (lām)(tòmrym		*					mmmmmm
☞ [(címrmcúm)] lam (tòmrym						ss	mmmmmm
[(címrmcúm)] (lām to)mrym						!sss	mmmmmm
[(címrm)(cúm] lam) (tòmrym			!*	s	s		mmmmmm
[(címrm)(cúm] lam) (tòmrym				!s	s		mmmmmm

The same should be true of all forms with multisyllabic morphological words, and indeed there are such initially stressed forms based on prefixed monosyllabic stems, e.g., *obligatòry*, *respiratòry*, *deprecatòry*.

However, the most frequent stress pattern for prefixed monosyllabic stem cases, in both American and British English, is actually that seen in *obligatòry*, *explánatòry*, *declaratòry*. The presence of a short, stressed root vowel in these cases led Myers (1987) to propose that /-atory/ was a

“shortening” suffix, despite some exceptions seen in monomoraic unprefixated stems, such as *migratòry* (§ 2.2.1). Under the analysis proposed here, it is clear that the monomoraic stems of these words are being extended to include the affix /-at/, i.e., that the morphological structure of these words is e.g., [ex{plan-at}]ory, [de{clar-at}]ory, the stem containing two morphemes. The suffix /-at/, which in previous examples subcategorized for the morphological word, is in these cases subcategorizing within the stem, like an /-ity/ class suffix. This second type of subcategorization will be referred to as /-at₂/:

(6.24) **ity-Suffixation (-at₂):** Align(Suf_{ity}, R; Stem, R)

The fact that words like *obligatory* can appear with both stress patterns indiscriminately suggests that this is a case of two possible subcategorizations for the “same” affix. There is no structural reason why one variant or the other appears in a given word. It might be proposed that there is a second homophonous affix /-at₂/ which subcategorizes differently from the /-at/ seen in the /-ate/ verbs. Some conjecture about the distribution of this /-at₂/ can also be made; it appears primarily in extended forms of stems which do not have a corresponding /-ate/ verb, but rather make verbs or adjectives in /-ə/:¹⁰³

(6.25)	excíte	excítatory ¹⁰⁴
	consérve	conservatòry
	obsérve	observatòry
	comménd	commédatòry
	refórm	refórmatory
	excláim	exclámatòry
	expláin	explánatory
	infláme	inflámmatory
	decláre	decláratòry

¹⁰³There are of course exceptions, e.g., *confiscatòry*, *compensatòry*, *derogatòry*, on the one hand, indicating that the distribution is lexical.

¹⁰⁴This word is an example of the behavior of an underlyingly long monosyllabic stem, i.e., a stem with a lexically long vowel (/cít/) rather than a lengthened one. The long vowel always surfaces, e.g., *excitátion* and the /-atory/ form given here.

defáme	defámatòry
prepáre	prepáratòry
oblíge	oblígatòry

This suggests that for stems with corresponding verbs in /-ate/, the same subcategorization constraint is used to construct the extended form in /-atory/, while for other stems a different constraint, referring to /-at₂/, is used. That stems with both types of verbs (e.g., *oblíge*, *óbligatè*, *respíre*, *réspiratè*) can show both stress patterns is perhaps not accidental. Most longer stems do not show /-at₂/, but these also regularly show verbs in /-ate/. One interesting exception is the British pair *lábor* ~ *labóratory*, where the addition of /-at₂/ has made a stem of the /-or/ suffix. In terms of further extensions to /-at₂/, nominal abstracts in /-ion/ formed from such stems will be indistinguishable from those in /-ation/, since the /-ion/ suffix will construct the same type of stem in the optimal candidate:

(6.26)

/ex-plan-at ₂ -ion/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s	*m
 (è _m x _m plā _m) +(nǎ _{mm} ti) _{om} n					*	SSS	mmmmmm
(è _m x _m)(plā _{mm}) +(nǎ _{mm} ti) _{om} n					*	SSS	!mmmmmm
(e _m x _m)(plā _{mm}) +(nǎ _{mm} ti) _{om} n		!*			*	SSS	mmmmmm

Thus, the variant /-atory/ forms can be accounted for through alternate suffixation and morphological structure. A similar solution will be offered below for forms in /-ative/. The key factor in the optional stress contrast seen among these forms is selectional rather than morphological or prosodic; one set of stems combines with an affix which, although homophonous with /-at/, subcategorizes differently. The question of how such affixes are lexically selected will be explored in § 7.3.1; since the relationships outlined here are not strictly predictable and some words show multiple possible pronunciations, it is clear that this is a lexical process, and the optimal candidate for a given word will depend upon which “input” morphological form is used.

One final pattern among British /-atory/ forms still needs to be accounted for. These are alternate forms with longer stems nevertheless showing main stress on the suffix /-āt/, e.g., *circulātory*. The clearest explanation of these forms involves the subcategorization of /-ory/. This suffix demands the morphological word, as was seen in words like *prómontòry*. However, the suffix /-at/ also demands the morphological word, and in the American examples given above, only the subcategorization for /-at/ was satisfied, e.g., [[circul]]atory. Proposing two morphological word constituents for such forms, i.e., [[circul]][at]ory, is problematic; such a morphological word constituent [at] would have no stem, and it is unclear whether it should be required to have one. Another approach would be to limit the number of morphological word components allowed in a given prosodic word, and rank the subcategorization constraint for /-at/ above that for /-ory/ to yield the morphological structures given above.¹⁰⁵ If the tables were turned, and /-ory/ successfully subcategorized for the morphological word, the following tableau would result:

(6.27)

UK: /circul+at- ə+ory/	Non- Fin(F')	Non- Fin(ś)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s	*m
☞ [(címrmcúm) (lāmm)t<ə>] ómryr							SSS	mmmmun
[(címrmcúm) (lāmm)t<ə>] ómrym				!s			SSS	mmmmun
[(címrm)(cúm la_m)t<ə>m] ómrym					!s	*	SSS	mmmmun

This yields the alternate form *circulātory*. For other types, such as *migrātory* or the schwa-less versions of /-atory/ (which includes all American cases of

¹⁰⁵For words that seem to require two morphological words, like compounds or /-ation/ forms (§ 6.2), the ranking of subcategorization for morphological words by roots or affixes like /-ion/ would simply be higher than the constraint which restricts the number of MWords in the PrWord. The subcategorization for /-ory/ would be ranked below the same constraint.

this kind), the different subcategorization would make no difference in the optimal surface form.

(6.28)

UK: /circul+at+ory/	Non- Fin(F')	Non- Fin(š)	Ft to MWd	Edgemost	StemFtR	NoOrph	Lapse-s	*m
$[(\acute{c}i_m r_m c u_m)(l \acute{a}_{mm})t]) !^*$ o _m r _y n								mmmmn
$\Rightarrow [(\acute{c}i_m r_m c u_m)l a_m t])$ o _m r _y n						*	SSS	mmmm
$[(\acute{c}i_m r_m)(\acute{c}u_m l a_m)t]) !^*$ o _m r _y n				S	S			mmmm

Since most words with schwa-augmented forms of /-atory/ in British English can show either pronunciation, this suggests that the two subcategorization constraints for /-ory/ and /-at/ are not crucially ranked with respect to each other, and that one or the other may equally be enforced. Another possibility is that the subcategorization of /-ory/ for the morphological word is optional in British (the British /-ory/ also subcategorizes for the prosodic word), but when this constraint does appear, it takes precedence over the /-at/ subcategorization.¹⁰⁶ The presence of this constraint would also account for another optional type seen among the cases with stress on a long stem syllable, of which *explóratory*,¹⁰⁷ *respíratory* may be examples. Morphological structures like [re{spir}at]ory would yield this result, i.e., [re({spí_{mm})r}at]ory with mora borrowed from the /-at/ suffix. Since the stem is no longer final in the morphological word, it can be stressed; the resulting word structure is effectively that of the /-al/ forms.

6.4 Extended forms in /-at-iv/

The forms in /-ative/ present a complex situation similar to that seen with /-atory/. While words with monosyllabic stems all show initial stress, longer stems display three types of stress pattern, seen in both American and British English. Again, many words display more than one type, and the same

¹⁰⁶This depends largely on whether the alternate forms are dialectally different, or equally optional for a given speech community. The Celex database does not contain this information.

¹⁰⁷Unless this stem is underlyingly long in some individual's grammars, and short in others, as in the variants *cýclic* ~ *cýclic* seen between some speakers (§ 4.3.3).

words can show differing but overlapping stress patterns between the two dialects. The first pattern illustrated below shows a long /-āt/ syllable, while in the other two types that syllable is unstressed:

- (6.29) [1,0,2,0]: législàtive
 [1,0,0,0]: nóminative
 [0,1,0,0]: provócative

The second pattern presents three consecutive unstressed syllables, while the third puts stress on the stem syllable. In the case of monomoraic roots, this allows the underlying short vowel quality to surface. Myers (1987) also cited /-ative/ as a “shortening” suffix for this reason.

As was the case with /-atory/, the various forms in /-ative/ will be accounted for via alternate morphological structures. Many words show variant pronunciations, e.g., *cómmutàtive* ~ *commútative*, *législàtive* ~ *législative*, *quántitâtive* ~ *quántitative*, *cóntemplâtive* ~ *contémplicative*, *cónnotâtive* ~ *connótative*, and the distinction between contrasting morphological structures in such words must be lexical. Most words with the first pattern, for example, show variants with the second. The first pattern is what would be expected if /-at-ive/ subcategorized like its individual parts, i.e., [{quantit}]{at}iv:

(6.30)

/quantit+at-iv/	Non-Fin(F')	Non-Fin(ś)	F'to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
 (quám ⁿ mtim)] (tām ^m)-tiv							SSS
(quám ⁿ mtim)] (tām ^m -tim ^v					!s		
(quàm ⁿ mtim)] (tǎm ^m)-tiv			!*				SSS
(quám ⁿ m) (tim] ta ^m)-tim				!s	s		SSS

However, the majority of forms do not exhibit this structure, and of those that do, most also present alternate forms with the second stress pattern given above in (6.29), i.e., *quántitative*, with stress three syllables from the right edge. Such words have been cited as evidence of ternary feet in English

(Selkirk 1982b). They require an alternate structure, which is slightly different than that given above, and is not at all transparent with regard to the internal elements of the suffix, i.e., [{quantit}{at}]iv:

(6.31)

/quantit-at+iv/	Non-Fin(F')	Non-Fin(ś)	F'to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
(quám ⁿ m ^m tím ^v - tām] tím ^v					!s	*	SSSS
(quám ⁿ m ^m tím ^v - (tām] tím ^v	!*	*			s		
☞ (quám ⁿ m ^m tím ^v - (tām ^m)] tiv							SSS
(quám ⁿ m ^m tím ^v - (tām ^m)] tiv	!*						SSS
(quam ⁿ m ^m) (tím ^v -tām)] tím ^v					!s		SSS

The optimal candidate $[(quánti)(tā)]tīve$ is expected to show two possible surface forms. Since the second foot ($tā$) is in an unstressable position (due to **NonFin(ś)**), it must be an unstressed foot and can surface either with or without vowel quality, as in other unstressed feet seen in prefixes and suffixes (§ 5.4). In other words, forms showing the first two alternating types of stress can both be derived from such a structure; both surface forms are equally optimal. It is harder to explain the structure itself; nothing in the subcategorization of the elements /at/ and /iv/ seem to require a morphological structure $[_{at}]iv$, yet this structure yields the correct results under the current constraint hierarchy. Another subcategorization constraint specifically referring to /-ative/ would be necessary to impose this structure on the candidates.

The final stress pattern, that seen in *provócatīve*, is restricted almost entirely to stems made from prefixed monosyllabic roots. These form a stem with the /-at/ element of the suffix, suggesting the /-at₂/ proposed above for similar stems in /-atory/. The same distributional patterns apply to these roots. Such a subcategorization means that the root syllables will never lengthen, and thus /-ative/ gave the appearance of being a “shortening” suffix (§ 2.2.1). The structure of such forms would then be like $[com\{par-at_2\}iv]$ (or $[com\{par-at_2\}]iv$):

(6.32)

/com-par-at ₂ -iv/	Non-Fin(F')	F'to MWd	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s
 (com ^m m)-(pá ^m -ra ^m)t-i ^m v						*	SSS
(có ^m m-m-pa ^m -râ ^m t-iv)]				!s	s		
(cò ^m m-m-pa ^m -rá ^m t-iv)]	!*			s	s		

No forms with lengthened stem vowels could surface, due to the high-ranking constraint **Stem-Coh** (§ 5.2). Of course, underlyingly long vowels remain long here as elsewhere, e.g., *connōtative*, and underlyingly long roots can show both patterns (e.g., *cónnōtâtive*, *cóntemplâtive* ~ *contéplative*) while true monomoraic roots cannot (e.g., *compárativ* ~ **cómparâtiv* ~ **cómparative*). All monosyllabic roots show stress on the root syllable, e.g., *lúcrative*, *fórmative*, *vócative*.

All this suggests that the distribution of /-at₂/ might be not be purely lexical, as suggested above, but rather enforced by constraints interacting with the root type. It could be possible to propose that a number of ranked subcategorization constraints govern the suffix /-at/:

(6.33) Word subcategorization: Align(at, L; MWd, R) = [_]at

Stem subcategorization: Align(at, R; Stem, R) = { _ at }

These constraints conflict; as long as /-at/ remains outside the morphological word, it is not likely to form part of any successful stem constituent, as feet which cross the morphological word boundary will be dispreferred by a variety of the constraints in the current hierarchy, and perhaps others. But if /-at/ is forced into the morphological word by other constraints, it can then display its stem subcategorization. The interaction between root length and stem-formation would need to be explained: only monosyllabic roots join in this way with /-at/, while longer roots appear to remain in their own stems. Relatively low-ranking subcategorization constraints which interact with some prosodic constraints might be able to account for this; in any case, the relationships involved are complex and this is left to future investigation. A

more conservative account of these facts would rely on an allophonic /-at₂/, which has a clear if lexically enforced distribution among monosyllabic roots.

6.5 Other suffixes with word subcategorization

A number of other English suffixes also show the subcategorization for the morphological word seen in the cases reviewed above. This section will round out the discussion of such suffixes, highlighting the differences between them and true “level two” suffixes, which subcategorize for the prosodic word. One affix showing subcategorization for the morphological word is the nominal suffix /-ment/. Stems taking /-ment/ surface identically in words unextended by the suffix. The initial /m/ of the suffix fails to close preceding syllables, as it would if the suffix subcategorized for the stem:

- (6.34) báníshment méasurement mánagement
 góvernment wónderment séttlement

In other words, the presence of successive consonants which cannot form an onset (e.g., /rm/, /ʃm/, /lm/), due to the initial /m/ of the suffix, would be expected to lead to the projection of a mora (by **Moraic Prominence**) and a heavy penult. However, the penult in these forms does not receive the primary stress.

Furthermore, monomoraic stems which show compensatory lengthening due to /-ə/ suffixation in their verbal forms continue to show this lengthening despite that fact that the /m/ in /-ment/ should close the stem syllable (as above), making these syllables bimoraic, without any need for further lengthening:

- (6.35) concéalmment refínement abāsement

Note that under the derivational approach proposed by Myers (1987) (see § 2.2.3), which assumed underlying long vowels for such forms, these long vowels should have been shortened by “closed syllable shortening”, which was the central rule of the shortening theory. Here, subcategorization for the morphological word constituent will instead account for these surface forms:

- (6.36) [[baniš]]ment [[govern]]ment [[setel]ment]

[con{cel}ə]ment

[re{fin}ə]ment

[ə{bas}ə]ment

Under the current constraint hierarchy, in such cases the penult can never be stressed, due to **NonFin(ξ)**. Thus **gouvérnement* will never be optimal. For words in /-ə/, the root syllable is not in this constituent-final position, because the /ə/ syllable is. Schwa fails to be parsed into prosodic structure in morphological word-final position, as was laid out in § 6.3. Stems which never appear unsuffixed nevertheless adhere to the principles of word-formation enforced by the constraint hierarchy, e.g., *dócu*ment, showing the structure [[docu]ment].

Some interesting effects occur when /-ment/ interacts with other suffixes. A few words in /-ize/, such as *chástisement*, form nouns in /-ment/. The suffix /-ize/ also subcategorizes for the morphological word, and there appear to be alternate structures which can both appear optimally, [{chast}iz]ment and [{chast}{iz}ə]ment, resulting in the alternants *chástisement* and *chastísisement*. The lexically optional presence of /ə/ in the structure appears to account for the difference. In a few other words, an element /a/ appears on the right of the stem, producing an unusual stress pattern, e.g., *témperament*, *médicament*, with structures like [{temper}a]ment, which would produce the attested forms. When extended with further suffixes, such as /-al/, /-ment/ can form a stem in itself, e.g., [(ex)(pèri)(mént)al]. The suffix /-al/, in addition to requiring a stem, also must be inside a morphological word constituent, and the constraint requiring this evidently outranks the subcategorization of /-ment/ for the morphological word. A similar effect is seen verbs extended in /-ary/, which also compete with /-ment/ for the morphological word, e.g., *còmplimentary*, with the structure [com{pli}{ment}ə]ary. Alternate forms like *cómplimentàry* suggest that /-ment/ can also win this competition in some grammars. Verbs in /-ə/ can be formed from /-ment/ words, although any constraint which might subcategorize /-ə/ for the morphological word is evidently not ranked so high as that for /-al/, since the morphological word boundary apparently remains to the left of /-ment/:¹⁰⁸

¹⁰⁸Note that this implies that /-ə/ also fails to parse at the prosodic word boundary, as was originally proposed in § 4.1.5.

(6.37)

/com-pli+ment-ə/	Non-Fin(F')	Ft' to MWd	Non-Fin(ś)	Edgemost	StemFtR	NoOrph	Lapse-s
 (cómmmpli _m)] (me _m n _m)t-ə _r							SSS
(co _m mmp _m li _m)] (mé _m n _m)t-ə _r		!*					SSS
(co _m m _m)(plí _m m)] (me _m n _m)t-ə _r			!*				SSS

However, the presence of /ə/ requires that /-ment/ be stemmed, and the resultant vowel that surfaces is usually full (*cómpliment*, verb) rather than reduced (*cómplimənt*, noun). In the data in general vowels in stems appear less likely to reduce than those in affixes, outside the stem.¹⁰⁹

For noun/verb pairs like *ségment* ~ *segrént*, *frágment* ~ *fragmént*, it is not clear whether these words are being interpreted as forms in /-ment/, or as stems {fragment}, {segment}. In the latter case, /-ə/ suffixation in the verbal cases would yield the attested finally stressed verbal forms. In British dialects, the ranking of **Ft' to MWd** below **NonFin(ś)** would correctly yield *[(seg)](mént)ə* beside *[(cómpli)](ment)ə*, suggesting that the monosyllabic stem forms can be interpreted as suffixed in /-ment/.

(6.38)

/seg+ment-ə/	Non-Fin(F')	Non-Fin(ś)	Ft' to MWd	Edgemost	StemFtR	NoOrph	Lapse-s
(sémg _m)] (me _m n _m)t-ə _r		!*					SSS
 (semg _m)] (mé _m n _m)t-ə _r			*				SSS

The corresponding American forms, which should show initial stress (§ 6.1), usually do, and forms with final stress, such as *augmént*, *tormént*, can be understood as not suffixed in /-ment/, but as unitary stems (i.e., /augment/).

Another suffix which subcategorizes for the morphological word is /-able/. This suffix is similar in form and meaning to the /-ity/ class suffix

¹⁰⁹This kind of effect could of course be attributed to a constraint which enforces the surfacing of full vowels in strong syllables within the domain of the stem.

/-ible/, and the segmental structure of the latter element in these suffixes is apparently /bil/, as becomes clear in forms that are further suffixed:

(6.39) lègibíity àlterabíity vísibíity mèasurabíity

Since the initial syllable of the suffix is never stressed, it would be difficult to determine when a word is showing /-ible/ or /-able/ apart from the orthography. The best measure of this should be the subcategorization shown by the word in question. While /-ible/ forms part of the stem, as in all /-ity/ class suffixes, /-able/ seems to subcategorize for the morphological word:

(6.40) quótable cátable redéemable
 cháritable aménable inévitable

As with the other affixes of this kind noted in this chapter, /-able/ can combine with word constituents ending in /-ə/, e.g., [re{dem}ə]able, as well as unextended stems, e.g., [{čarit}]able. Forms which show apparent shortening appear to be regarded by the grammar as suffixed in the homophonous /-ibil/, e.g., flámmable, implá cable, formí dable (which can alternate with fórmidable), inapplí cable (also iná plicable). Confusion in the grammar is further exacerbated by the many words which could well be suffixed in either /-able/ or /-ible/, e.g., comméndable, deléctable. A further type occurs, as with the suffixes /-ary/, /-ory/, /-ment/ above, when the suffix /-ə/ is lexically not included in the suffixed morphological word constituent:

(6.41) [cóm{par}]able [irrè{voc}]able [irrè{fut}]able

Alternants such as *irrevó cable*, *irrefú table* show that the grammar may also construct “input” forms like [irrè{voc}ə]able. Reasons for this kind of alternation in the lexical forms will be further discussed in § 7.4.2. As with the other suffixes of this kind, the combination of two suffixes subcategorizing for the morphological word results in alternate surface forms, e.g., [ré{cogn}]{iz}əable ~ [rè{cogn}]{í z}əable. It might be asked why words in /-able/ do not surface with secondary stress on the suffix, i.e., *cháritàble, similar to *nécessàry*. The best explanation of this is that the suffix is segmentally /-əbil/, with initial schwa, and thus cannot be stressed.

Similar situations exist among the suffixes /-ist/, /-ism/, some cases of /-y/, and a number of other, less frequent suffixes. The distinction between suffixes subcategorizing for the morphological word, and those which subcategorize for the prosodic word (the level two suffixes), appears in stems which would interact with **NonFin(Ft')**, a high-ranking constraint which takes the prosodic word as its domain. An example is the alternation seen in *téléphone* ~ *téléphonist*. The form *téléphone*, consisting of two stems, the stemmed prefix {tele} and the stem {fon}, is unable to be stressed on the rightmost stem due to **NonFin(Ft')**:

(6.42)

/tele-fon/	Non-Fin(F')	Ft to MWd	Non-Fin(ś)	Edgemost	WdEdge most	StemFtR	NoOrph	Lapse-s
 (té _m le _m)-(fō _{mm})n				S	S			
(té _m le _m)-fo _m n]				S	S	!s	*	SSS
(tè _m le _m)-(fō _{mm})n	!*		*					
te _m (lé _m -fo _m)n]	!*					S		SSS

Any form stressed on the final two syllables will violate the non-finality constraints. However, the final stem shows a lengthened vowel due to the requirement for stems to be footed, and feet to be bimoraic. Adding a suffix to the right of the morphological word, but still within the prosodic word, shifts all syllables in the word one syllable further from the right edge of the prosodic word, the area affected by **NonFin(Ft')**. Thus, the prosodic structure given to the identical morphological word will be different:

(6.43)

/tele-fon-ist/	Non-Fin(F')	Ft to MWd	Non-Fin(ś)	Edgemost	WdEdge most	StemFtR	Lapse-s
(té _m le _m)-(fō _{mm})]-(ni _m s _m					!s		
(tè _m le _m)-(fō _{mm})]-(ni _m s _m			!*				
 te _m (lé _m -fo _{mm})](ni _m s _m t						S	SSS

The constraint **WdEdgemost**, which is the former **Edgemost** constraint from § 5.3, replaced above in § 6.1 in a refinement of the working constraint hierarchy, governs cases such as this, where two stems are present in the morphological word. Although a head foot on the right edge of both *{tele}* and *{fon}* would satisfy **Edgemost** (which only specifies that the head foot is aligned to the edge of some stem), **WdEdgemost** tries to align the head foot to the right edge of the morphological word. Although *{fon}* itself cannot take main stress due to **NonFin(ś)**, the foot (*léfon*) satisfies the various word-edge constraints even though it divides a stem. Apart from cases like this however, suffixes of this kind which subcategorize for the morphological word will appear to have little effect on the prosodic structure assigned to the morphological word constituents in question, maintaining the consistent “look” of suffixed stems in these cases.

6.6 Multiply suffixed words

This section looks briefly at how multiply suffixed words illustrate the interaction of the various subcategorization constraints presented above. Different suffixes present constraints which are differently ranked, with the result that the signature effect of various affixes fail to appear in some cases. A number of such cases were discussed above regarding the affixes subcategorizing for the morphological word, but similar effects are seen for affixes in the */-al/*, */-ent/* and */-ity/* classes.

The */-al/* class suffixes subcategorize for stems, which means that other suffixes can themselves be “stem-ized” when suffixed by this type.

(6.44)	încréméntal	coïncidéntal	moméntous
	tèrritórial	àctuérial	sensátional

Words like *moméntous* and *coïncidéntal* illustrate how */-ent/* can become stem-ized when followed by */-al/* class suffixes. Note that the */-ent/* suffix itself still requires a stem, so that the structures involved are [*{mom}{ent}ous*], [*coin{cid}{ent}al*]. The root stems behave as expected, augmenting when necessary to satisfy stem constraints (i.e., *mōméntous*), but remaining

Cases where /-al/ and /-ent/ suffixes are extended by /-ity/ class suffixes are more commonplace. In all cases, the subcategorization for the /-ity/ class suffixes to take a morphological word on their right and to form part of a stem with the morpheme on their left overrides all other subcategorization constraints, and the constraints governing these affixes must be the most highly ranked of the types discussed here:

(6.46) r`ecept`iv`ity              `abnorm`al`ity voc`al`ic

Again, the /-al/ class take stems on their left; the /-ity/ class suffixes force them to become part of stem that includes the suffix. The morphological word boundary, however, is at the right edge of the word, enforced by the subcategorization of the /-ity/ class. Thus the structures created are *[re{cept}{ivity}]*, *[[voc]{alic}ə]* rather than **[re{cept}{iv}ity]*, which would produce the incorrect result. The subcategorization of /-ity/ for a morphological word on the right outranks that of /-al/, and other constraints evidently enforce the restriction of a single morphological word constituent to the candidates. Only in certain situations, such as in compounds (§ 5.4.2), do constraints enforcing the presence of multiple word constituents allow such structures to surface.

Similar to the multiply suffixed words taking the /-al/ series are the stressed suffixes, such as /-ese/, /-ee/, /-eer/, as in *ch`in`e`s`e*, *c`ommand`e`er*, *d`ivorc`e`e*. Such suffixes will be understood as forming stems and lying within the morphological word. The simplest way to account for such structures is to regard these affixes as themselves suffixed in /-ə/. Thus, the structure of such words would be as follows:

(6.47) *[[chin]{es}ə]* *[com{mand}{er}ə]* *[di{vorc}{e}ə]*

The suffix /-ə/ subcategorizes for a stem constituent, as do the other /-al/ class suffixes. The stressed suffixes would also subcategorize for stem constituents, but would be suffixed themselves, producing a situation in which they were stressed. The stem syllables immediately preceding them would be unable to take stress by **NonFin(ś)**, and could appear either as unstressed feet or the unstressed members of bisyllabic feet, as in the examples above. Stems requiring feet would be augmented although not

stressed, as in *chīnĕse*. As with most unstressed feet that are nevertheless part of a stem, these stressless feet are not reduced, but display their long vowel. One interesting word in /-ese/ is *jápanese*, whose stress pattern conflicts with that of the simplex *japán*. The latter shows a stress pattern which would be ascribed to Kager's third, idiosyncratic group (§ 3.1.2). This suggests a morphological structure [*japann*ə], with both geminate and final suffixal schwa. Suffixing the stem {*japann*} with /-ese/ will produce the attested form, as the geminate syllable would fail to take stress due to **NonFin(ś)**, and the usually dispreferred foot type (*jápann*) would nevertheless be the most optimal. Another interesting form is *appointĕe*, which must have the prosodic structure [ə(poin)(tĕ)ə]. The prefix /ə-/, proposed above in § 5.4.3 to account for prefix syllables that are never stressed, fails to form a foot *(əpoint), even when this would be otherwise favored by the constraint hierarchy. Instead, the stem syllable {*point*} remains a monosyllabic foot, even though it is adjacent to the main stress.

6.7 Retraction

Kager (1989) and others have described a series of phenomena referred to as stress retraction, which was presented in § 3.1.4. Many words ascribed to this category are completely regular according to the general stress pattern as analyzed here, and were only described as retracted due to theoretical assumptions within the derivational framework. For example, words of the following kind, showing so-called Strong Retraction, were thought to warrant a final stress because of the weight or length of their final syllables. The main stress in such words, which actually appears on the antepenult, was thought to be retracted via a serial derivational process:

(6.48) ábsolùte	mánifèst	húrricàne
tácitùrn	páradìse	dífficùlt

Preventing the final syllable from receiving main stress, due to the extrametricality effect (enforced by **NonFin(Ft')**) yields the regular pattern [*(ś_{m(m)}ś_m)(ś_{mm})*] for most of these forms. The final feet in question, belonging to the stem constituent, may not take the main stress but are nevertheless footed, satisfying various constraints such as **Edgemost**, **Stem-Ft-R** and **Lapse-s**.

(6.49)

difficult /dificult/	Non-Fin(F')	Non-Fin(ś)	Edgemost	StemFtR	Lapse-s	FtForm	Lapse-s
☞ (dí _m fi _m)(cu _m l _m)t					S		
(dí _m fi _m)(cù _m l _m)t	!*						
(dí _m fi _m)cu _m l _m t		!s	S				SSS
di _m (fí _m cu _m l _m)t	!*					*	SSS
(dì _m fi _m)(cú _m l _m)t	!*	*					

The theories cited by Kager (1989) in his analysis did not have a mechanism by which a syllable could be footed and yet not automatically receive the promotion to head foot given by End Rule: Right (apart from retraction, which was applied afterwards to undo the effects of the End Rule). While the (potentially) full vowels of the final syllables in these forms prompted earlier theorists to posit secondary stresses, here these syllables are considered to be stressless feet (§ 5.3, § 5.4.1). The characteristic sign of stressless feet is the optional reduction of short vowels in closed stem syllables (e.g., *díffíc[ə]lt*); long vowels in the same position can also reduce in some dialects (e.g., *húrricane* [háríkən]).

So-called Weak Retraction similarly follows the usual group I stress pattern:

(6.50) e(léc)tròde sta(lág)mìte py(rámi)dòide

Again, the final heavy syllables seem to form a monosyllabic foot. Words belonging to this group show an set of infrequent suffixes such as /-oid, -ite, -ide/. A few word pairs, like *sphéroid* ~ *sphérical*, suggest that these words have a similar structure to those in /-ent/, enforcing the bimoraic stem in such cases via stem augmentation (i.e., $[[\{sfé\}(r)oid]]$). However, there are too few examples to determine the further behavior of these suffixes, and while /-oid/ is quite clearly a suffix, other endings like /-ite/ or /-ode/ which can be grouped into this category are morphologically less clear. Monomorphemic words ending in long vowels show similar stress patterns.

The words described as undergoing Sonorant Retraction are numerous, and this retraction mode appears to contradict the regular stress pattern, as an apparently heavy, closed syllable is not footed but is rather skipped over:

- (6.51) *hélminthòid* *cávalcàde* *árgentine* *hóttentòt*
 Háckensàck *pálimpsèst* *álgernòn*

These can be contrasted with words whose penults are closed by non-sonorants, as in (3.11). However, the fact that sonorants figure in the relationship recalls the solution to a different problem. There are a number of words like *álabàster*, *cátaclysm*, *régister* which appear to show unusual stress, disregarding a stressable penult and rather displaying main stress on the antepenult. All of these forms end in surface syllabic sonorants, and the best explanation of this type involves regarding this final “syllable” as a final margin constituent, structurally similar to an onset cluster (§ 4.1.3):

- (6.52) [(ála)(bastr)] [(káta)(klizm)] [(régistr)]

A similar explanation has been used to account for the apparent monosyllabicity (in the context of derivational theory, cf. § 6.3) of the suffixes /-ary, -ory/ by Chomsky & Halle (1968), Liberman & Prince (1977), Kager (1989), among others, who suggest an underlying structure like /-orɪ/, /-arɪ/ for these affixes. In phonological terms, syllabic resonants of the kind proposed in (6.52) are more plausible than such word-final syllabic glides. And if resonants in final position can be regarded as syllabic, there is no reason not to regard the resonants seen word internally in the same way:

- (6.53) [(hélmn)(Joid)] [(kávɫ)(kād)] [(árɯn)(tɪn)] [(hótn)tot]
 [(hákn)sak] [(páɫmp)(sest)] [(álɯr)(non)]

Almost all of these words have no internal morphological structure, and the internal “vowel” proposed for the penult in the traditional analysis of these forms has no distinctive vowel quality, but closely resembles the syllabic resonant in most cases. The reasons for proposing a syllable in this position in previous analyses were evidently due to orthography and the assumption that clusters of the form -CRC- could not appear word internally, although

-CR# could be proposed word-finally. The postulation of such lexical forms would help explain the contrast seen between words like *hóttentot* or *cávalcade* and *recóndite* or *appéndix*, which show the expected stress patterns for words with closed penultimate syllables. The latter two would thus have full vowels in their penults (e.g., /rekondīt/, /apendiks/), while the former are really bisyllabics (e.g., /kavlkād/, /hotntot/) with internal -CRC- clusters which surface as word-internal syllabic nasals.¹¹¹ There is no *a priori* reason to restrict lexical entries containing such internal consonant sequences, and this proposal provides an explanation of how such lexical entries could surface while at the same time accounting for the phenomenon of “sonorant retraction”. The idea that lexical entries must consist of nicely alternating and easily syllabifiable CV sequences, which mirror their surface forms, is just another assumption of derivational theory.¹¹²

Kager’s examples of so-called “long retraction” included words of the following type, cited above in (3.9):

(6.54)	péregri ⁿ àte	cátamar ⁿ	manípulatòry
	amélior ⁿ àte	húllaballòo	hallúcinatòry
	detérior ⁿ àte	rígamaròle	artículatòry

The types in /-ate/ and /-atory/ have been discussed above in (6.7) and (6.21). The /-ate/ forms are multiply suffixed, and the stem-subcategorizing suffixes (here /-or/ and /-ine/¹¹³) are unable to take the main stress due to their

¹¹¹There are a number of alternative ways to understand these forms. They could be seen as containing an underlying stem-internal /ə/, i.e., /kavəlkād/, /hotəntot/. This vowel is unable to take a position as the stressed syllable in the foot (§ 5.4.3), and so will always be the weak member of a foot when footed. Another possible interpretation would involve projecting a mora from the syllabic sonorants (by **Moraic Prominence**, as they appear between two less sonorant segments), which acts the nucleus of a weak foot syllable, i.e., (cám^vl_m)(cam^mde).

¹¹²On the other hand, any proposed lexical form which is not the best antecedent of the optimal surface candidate, but was rather expanded by default strategies like epenthesis (e.g., if /ktmrn/ was a plausible lexical entry for *cátamaran*), would be replaced in the lexicon by a more optimal underlying form which yields the same surface form (e.g., /katəməran/). This process is known as **Lexicon Optimization** (§ 7.1). The only reason unusual forms like /kavlkād/ would remain in the lexicon is that such a form best yields the surface form *cávalcade*, given the English constraint hierarchy.

¹¹³This is not a common suffix, but a suffixal identification is useful to explain similar words such as *díscipline* ~ *disciplinⁿary*.

position at the end of the morphological word. In such forms, the stress remains on the stem foot, which is thus one syllable further to the left than had there been no internal suffixation. The morphological word boundary is inside the prosodic word, and so the sequence of stressless syllables does not incur any fatal violations. The situation is similar in the /-atory/ words.

The remaining few words, which are monomoraic, are best explained through the presence of underlying /ə/ within the stem, i.e., /katəməran/, /huləbəlu/, /rigəmərōl/ (§ 5.4.3). These vowels surface with no quality and are plausibly not otherwise interpretable to learners. The unstressability of /ə/ will force the attested foot structures. The limited distribution of schwas in lexical entries means that there are not so many of these marginal forms. However, in OT all forms, including these, need to be accounted for, and the proposal of /ə/ by the grammar is the most effective way of dealing with these forms, as there is then no need to rework the entire complex grammar of foot-formation on the basis of a few infrequent and lexically unanalyzable forms.

6.8 The non-Latinate grammar

The primary focus of this work concerns the Latinate vocabulary of English, a set of words borrowed into English via French and Latin, and possessing many of the morphological and grammatical features of Romance languages. In many ways, the Latinate grammar can be regarded as a foreign sub-grammar grafted onto English. In this section, the “native” English grammar will be briefly discussed, as its words also need to be accounted for via the same constraint hierarchy. However, this is not the main focus of this study, and a detailed investigation into the native lexicon will be left for future investigation.

“Native” English words are quite different from Latinate ones; there is far less morphology, and most words are only one or two syllables long. There are certainly some vowel alternations to be seen, for example *wíld* ~ *wílderness*, or *kéep* ~ *képt*, *wíde* ~ *wíðth*. However, most such words or affixes show morphological idiosyncrasies, and do not display regular patterns. For example, the suffix /-th/ is also seen with the irregular alternation *lóng* ~ *léngth*, with stem allomorphy, while *éight* ~ *éighth* fails to present the shortening seen in *fíve* ~ *fíth*. The extension to *wíld* seen in *wílderness* is likewise unique to that word. On the other hand, the

alternation seen in *kēep* ~ *képt* appears regularly in a limited set of verbs which form their past tense in /-t/. Following the principles outlined above, most of these cases can be explained in the same way as the “shortening” cases from the Latinate vocabulary (§ 4.3.1). If the suffix /-t/ is understood as subcategorizing within the stem, then morphologically conditioned lengthening will not take place, since the addition of the suffix into the stem makes it bimoraic, rendering lengthening unnecessary, i.e., [{kep}ə] ~ [{kept}ə], [{wid}ə] ~ [{wid-J}ə].

However, this does not explain the mass of monosyllabic words of the form *hít*, which show what in the Latinate grammar would be considered a single light syllable. According to the current constraint hierarchy, we would expect such a syllable to be lengthened, but in reality about 28% of monosyllabic words show this structure. One solution to this problem would be to label all these stems as geminate-final (that is, they have lexical moras linked to the stem-final consonant). This might appear to be problematic, since geminates appear only in about 13% of the Latinate words, not 28%. Such a solution also appears to introduce complexity into the lexical entries involved. However, there are a number of arguments in favor of the geminate analysis. One is that many of these forms are historically geminate, such as *hít* (Old English *hittan*), *líck* (Old English *liccian*), and the underlying geminate structure (i.e., a lexical consonantal mora) can be understood as surviving into the synchronic grammar.

Another point is that arguments on the basis of lexical complexity fall down when regarding monosyllables with surface long vowels, such as *rēad*, *ēat*, *cōal*. These account for about 29% of monosyllabic roots, and if regarded as underlyingly long, would themselves need to be marked with a mora in the lexicon. Marking the geminates rather than the surface long vowel cases (which would then be regarded as /red/, /et/, /col/) with moras involves almost exactly the same number of stems, a bit fewer in fact. This latter interpretation also accounts for alternations like *kēep* ~ *képt*, *slēep* ~ *slépt*, without making forms like *ēighth* exceptions. Such words would be regarded rather as underlyingly long (i.e., {ēt}~{ēt-J} vs. {fiv}~{fiv-J}), as in the Latinate examples given above.

A second explanation, which seems less promising, would be to regard non-Latinate stems as subcategorizing directly for the prosodic word, avoiding the constraints which enforce final consonant extrametricality on the

morphological word. In such a situation, final consonants would be syllable-closing and moraic, hence words like *hít* would be bimoraic, and lengthening would only be seen in open syllables, e.g., *bé*, *gǒ*.¹¹⁴ It is likely that this second explanation poses a number of further difficulties, which will not be further explored here. The issue of how the non-Latinate vocabulary of English further behaves will have to be left to future research.

6.9 Conclusions about English stress patterns

In chapters 3-6, structures have been proposed to account for the various stress patterns seen in both complex and simplex words in English. The various alternations seen between related words, and the differing stress patterns seen across the vocabulary, have largely been accounted for by proposing various morphological structures for different word types, enforcing this primarily through affix subcategorization. Taking the “Latin” stress pattern showing final syllable extrametricality (Kager’s group I) as the basic stress pattern (§ 3.3), the general verbal pattern (group II) can be accounted for by the presence of a suffix /-ə/ (§ 3.3.4), while idiosyncratic examples of apparently light monosyllabic feet (group III) can be understood through the proposal of lexical moras linked to consonants, i.e., geminates (§ 3.3.2). Differing patterns seen among the various affixed forms have been connected to the different morphological subcategorizations demanded by these affixes (§ 4.2), and the resultant edge effects seen in these constituents (as seen in chapters 4-6).

Vowel alternation has been accounted for by reassessing the status of long vowels in morphologically predictable positions (§ 4.3). By analyzing the relative frequencies of syllable types throughout the lexicon, rather than concentrating upon a few unusual forms, the unexpectedly infrequent distribution of light syllables in monosyllabic roots as compared to multisyllabic roots could be identified and accounted for through the proposal of morphologically conditioned lengthening (§ 4.3.1). Vowels which failed to undergo “shortening” can now be understood as underlyingly long, while short vowel quantity can be seen as the unmarked case for most vowels in the lexicon (§ 4.3.3). This reassessment of the underlying forms for Latinate roots

¹¹⁴It is true that almost no words of the form **bé* appear in English, but this would also be the case under the explanation offered previously.

led to further clarifications, such as the otherwise inexplicable “retraction” and “shortening” seen in forms like *résident* (§ 5.1).

The preceding account of the general stress patterns of English has the advantage of being expressible via a single constraint hierarchy, which is necessary for a constrained Optimality Theory account. No previous treatment of English stress has been able to account for the variety of stress patterns in a uniform manner, or for contrasts in the treatment of segmentally identical forms without resorting to unstructured exception marking and extra-theoretical provisos, excluded from this approach. Here, idiosyncratic patterns have been accounted for by linking prosodic and morphological constituents, necessary for the overall analysis, to lexical items themselves. These tactics require no further complexity than traditional lexical entries and affixation rules, but allow for a coherent OT account of English stress and vowel alternation.

The above account relies on the lexicon as the storehouse of idiosyncratic information, both segmental and constituent-based. In other words, it is “underlying” morphological and prosodic structures which are responsible for the varying surface realizations afforded to segmentally similar candidates. These different arrangements of lexically specified constituents are necessary for the grammar to account for the many different stress patterns that the language learner is confronted with when acquiring the grammar of English. Each and every word must be an acceptable output, or put another way, a lexical form which can fit into the existing context of the grammar must be found for every word.

For most words, the correspondences between the structures required by the grammar and the logical structure of the word are consistent, and it is this regularity which allows for the correct grammar to be generated as it stands. However, not all correspondences between the prosodic structure and logical morphological structure of words will be regular. In some cases, an unusual word demands a structure which puts it at odds with other parts of the grammar, perhaps with related forms in the “paradigm”, or with general assumptions about word structure. Such words are commonly regarded as “exceptions”, and must be accounted for as well in the grammar. The next chapter treats the concept of exceptionality, and how it may be dealt with in Optimality Theory. In doing so, it also reassesses the role of the lexicon in

assigning morphological structures, and the relation of the lexicon to the rest of the grammar.

7. The Architecture of the Lexicon

7.0 Treating lexical exceptions

In the previous chapters, constraint hierarchies were proposed which accounted for the most general patterns seen in the relationship between word structure and word stress. The most common type, corresponding to Kager's (1989) first stress pattern, was accounted for by the application of the constraint hierarchy to simple strings of "input" morphemes. Other less common types, such as Kager's second and third groups, were accounted for by the introduction of further structure into the underlying representation, for example prosodic material (moras which accounted for syllable weight) or morphological material (the suffix /-ə/). For complex affixed words, subcategorizations were offered to enforce domains within which the "regular" patterns could then be seen. These proposals were able to account for a large majority of the attested forms, covering a variety of patterns. However, there remains a residue of attested forms which, without further explanation, would not appear to represent optimal candidates. These would be described in a traditional derivational system as "lexical exceptions", words whose unusual structure can only be accounted for special lexical entries which exempt them from otherwise general processes.

In this chapter, ways of accounting for such forms within the context of Optimality Theory will be explored, and conclusions will be drawn about the role of the lexicon in OT. Following a proposal by Russell (1995), it will be argued that the most theoretically sound way to approach the lexicon is to incorporate it into the constraint hierarchy itself, resulting in a unified system of representation for the entire grammar. Morphological selection will be re-examined in the context of Optimality Theory, whose general principles will be used to explicitly formalize the way in which "input" morphological structures are selected into the evaluation process, bringing this into line with the more familiar selection of prosodic constituents. The two major theoretical perspectives on morphological selection, the modern version of the generative model typified by Pinker (1991), and the network model of Bybee (1985, 1995) will be discussed and the assumptions behind traditional underlying forms clarified in terms of Optimality Theory.

The view to be developed herein follows a proposal of Russell (1995) that morphological selection can be represented using constraints within the hierarchy, just as McCarthy & Prince (1993a, b) have represented other traditionally lexical properties, such as affix subcategorization, with constraints (as was done in the preceding chapters). Proposing that not just the assignment of morphological structure, but that morphological selection itself is the result of high-ranking constraints, leaves open the possibility that in some languages these constraints may be low-ranked and may thus interact with prosodic constraints, as has been proposed by Russell (1995) to explain certain underspecification effects in Hua. Carrying through this proposal unifies the grammatical system, dispenses with an autonomous lexicon (but not its benefits) and represents the entire grammar through the common formalism of the constraint hierarchy.

In the following sections, this view of the lexicon is developed. The status of lexical exceptions is first examined (§ 7.1), and an OT solution accounting for structurally exceptional forms is proposed (§ 7.1.1), which leads to questions about the status of the “input” string within OT (§ 7.1.2). Issues of complexity and economy in representation, and a discussion of the major current perspectives on morphology and the lexicon follow (§ 7.2), as well as a re-evaluation of the assumptions underlying them (§ 7.3). A third, semantic constituent hierarchy is proposed, for use with lexical selection constraints within OT (§ 7.3.1), an idea which is further developed using Russell’s concept of constraints as the lexicon (§ 7.3.2, § 7.3.3). This proposal is shown to retain the power and expressiveness of previous approaches to morphology and the lexicon, while avoiding their pitfalls (§ 7.4). It further allows for the formal determination of complexity and economy in the grammar (§ 7.4.2), and an evaluation metric (§ 7.4.3). After the OT principle of the universality of constraints is reassessed in the context of this approach (§ 7.4.4), some proposals for future research are put forth, regarding the formalization of constraints and the treatment of segmental and post-lexical effects.

7.1 Exceptional forms

There are various reasons why forms may be regarded as exceptional, but they all revolve around assumptions about the word’s lexical entry. For example, the form *vanílla* is exceptional only in the sense that an assumption exists regarding the relationship of its surface segmental form to its

underlying form. Since on the surface, with regard to segmental quality and duration, *vanílla* consists of three light syllables, a correspondingly light underlying lexical entry /vanila/ is proposed. This is a reasonable assumption, because it holds for many other words, such as *pámela*, *óorigin*. Being such a word, it should also show the stress pattern of *pámela*, yet it does not. However, it is the surface stress of *vanílla* itself that is indicative of the intrinsic weight of the penult, even if that weight is not also expressed by segmental quality or duration (as it is in the “long” vowels). Because the correspondence between the extra mora and its surface realization is not transparent, the surface form *vanílla* can be regarded as “exceptional”. Nevertheless, proposing a lexical form /vanil_{ma}/, using structures (in this case, lexical moras) necessary for other words (like those with lexically long vowels), allows for the proper evaluation of candidates against the constraint hierarchy, rather than simply marking the word as exceptional within the grammar of English. The lexical form is **prespecified** with prosodic structure which allows the correct optimal form to be arrived at by the grammar (Inkelas, Orgun & Zoll 1994).

The proposal (by the grammar) of a structure like /vanil_{ma}/ to account for the attested spoken form *vanílla* would be an example of **Lexicon Optimization**, the calculation of the simplest plausible lexicon from the language as it is learned, a process which each speaker must undergo. Lexicon Optimization, introduced by Prince & Smolensky (1993: 192) demonstrates that the relationship between surface and lexical forms as defined by Optimality Theory is bi-directional, i.e., OT principles can be used to determine the lexical entries from the surface forms as well as vice versa. Such a mechanism is necessary to explain language acquisition. Inkelas (1994: 6) explains Lexicon Optimization as follows:

“Of all possible underlying representations that could generate the attested phonetic form of a given morpheme, that particular underlying representation is chosen whose mapping to phonetic form incurs the fewest violations of highly ranked grammatical constraints.”

In other words, the lexical entry which incurs the least violations in producing the known surface form, given a known constraint hierarchy, will be stored in the lexicon. Lexicon Optimization can be conceived of as a specific instance of **grammar optimization**, the process by which a particular

grammar is acquired and fine-tuned to best account for all utterances in a particular language.

Given a system showing a variety of patterns, the best way to prosodically account for the stress of *vanilla* is to propose an underlying geminate, a strategy which also accounts for a sizable class of such words. Other alternatives, such as changing the general constraints governing stress, would have a greater impact on the lexicon, demanding unusual lexical forms for a larger number of words, for example, words of the type *óigin*. More radical changes, such as weakening OT to allow multiple constraint hierarchies in a single grammar (so-called “cophonologies”; see § 1.4) is an even more problematic move, because it damages one of the central insights of OT, that a single ranked constraint hierarchy can be used to model the entire grammar of a language (rather than just selected tractable data, as in Lexical Phonology). The approach taken here is that it is theoretically preferable, when dealing with exceptional forms, to propose lexical structures containing elements which are already part of the grammar (such as prosodic or morphological constituents) than to weaken a central tenet of the theory itself. Under the concept of Lexicon Optimization, the grammar will construct the most plausible lexical “input” forms for unusual words, utilizing the structures available to the grammar. Such lexical items will, when evaluated by the constraint hierarchy, yield the attested form, just as is the case with all “regular” words.

Besides proposing lexical prosodic structure for exceptions, it has also been shown how specifying morphological structure in an input form can be used to yield the correct candidate. A sizable set of words showing apparently final main stress can be accounted for by proposing a suffix, /-ə/, which shares many characteristics of suffixes in the /-al/ group. By proposing this structure, the grammar allows for a large class of words to be handled by a single constraint hierarchy, where otherwise some kind of “cophonology” divided vaguely according to syntactic category would have had to be proposed (and this correspondence between stress and category was not very strong, as was shown in § 3.3). Words described here as possessing this suffix are “exceptional” in the sense that, again, the suffix does not manifest itself segmentally, but only through its effect on stress, and in some cases, vowel length.

Other stress patterns have also been explained by morphological subcategorization in the input form. The behavior of words in the /-ic/, /-al/ /-ate/, and “Level II” suffix groups differs due to the different subcategorization frames of those suffixes. Since the behavior of such suffixed words is quite consistent across each category, these have not been generally regarded by theorists as “exceptions” but rather as different word types. In fact, one of the foundations of Lexical Phonology is the division of the grammar into “strata” or “levels” based primarily upon generalizing over a series of affixal subcategorization types. For such suffixed words, it is generally acknowledged that their “input” candidates do not simply consist of segments, but that their morphological structure is also relevant, since their behavior is predictable according to their suffixation rather than their segmental structures alone. Yet, when such morphologically complex words do not behave as expected (as is the case for a small minority), they also need to be reasonably accounted for in the grammar.

Above in § 5.4.3, it was noted that although most nouns formed from a prefixed monosyllabic stem show surface forms like *défect*, *préfix*, *increase*, there is a residue of forms similar to *récord*, *próduct*, *réfuse*. Structurally, they are entirely predictable with regard to their segmental form, and show the normal stress pattern seen in stems like *brígang*. What is exceptional about them is their interpretation as unitary stems, when they are clearly prefixed and prefixed stems usually show a different behavior. Another example of this kind of exception is *imprégnate*. Its penultimate stress is perfectly regular for a monomorphemic stem or /-al/ suffixed word. It is only the fact that words in /-ate/ generally show antepenultimate stress, due to the subcategorization of /-ate/ for Morphological Words rather than stems, which makes *imprégnate* an exception.

7.1.1 Lexical exceptions and the constraint hierarchy

Here, it will be proposed that words like *récord* and *imprégnate* have lexical inputs like $[{\text{re-cord}}]_{\text{Stem}}\text{MWd}$ and $[{\text{im-}}\{\text{pregn}\}_{\text{Stem-ate}}\text{MWd}]$, despite the usual behaviors of these morphemes (which lead us to expect $[{\text{re-}}\{\text{cord}\}_{\text{Stem}}\text{MWd}]$ and $[{\text{im-}}\{\text{pregn}\}_{\text{Stem}}\text{MWd-ate}]$). How this will be achieved under OT will follow directly from the expression of subcategorization in general within OT. Examples of subcategorization constraints were seen in § 4.2.2:

- (7.1) al-Suffixation: Align(Suf_{al}, L; Stem, R)
 ic-Suffixation: Align(Suf_{ic}, R; Stem, R)

Here, the notation 'Suf_{al}' generalizes the first argument of the constraint over the /-al/ set of suffixes, but in a strict OT constraint hierarchy, each suffix in the group would require its own individual constraint (although not necessarily crucially ranked in relation to each other), stating its relationship to the category Stem:

- (7.2) Align([al], L; Stem, R)
 Align([or], L; Stem, R)
 Align([ous], L; Stem, R) (etc.)

That is, part of the “input” form evaluated by the constraint hierarchy is prespecified by these subcategorization constraints, of which at least one is present for each affix.

In the accounts offered in chapters 4-6, it was never made explicit how lexical items were mapped to the Stem constituent; it was clear from the usual assumptions of phonology that forms like /ton/, /limit/, /malign/, /sign/ were stems, just as it was clear that /pre-/, /de-/, /re-/ were prefixes and /-ent/, /-ize/, /-ic/ were suffixes. Following the logic behind subcategorization constraints such as (7.1) above, it is not unreasonable to propose that each root morpheme also requires a constraint mapping it to a superordinate Stem constituent:

- (7.3) Align(/ton/, R; Stem, R)
 Align(/limit/, R; Stem, R)
 Align(/malign/, R; Stem, R) (etc.)

If this is accepted, it becomes clear that the entire inventory of morphemes in the language is present in the constraint hierarchy itself. Any other way of regarding this leads to a situation in which affixes are listed explicitly in the constraint hierarchy, while root morphemes are rather automatically mapped from the lexicon into Stems. Although maintaining a distinction between affixes and roots in this way may seem in some ways to be appealing, in logical terms a “magic pipe” between the lexicon and the Stem category

amounts to the same thing as a set of constraints like those in (7.3), and in fact such a set is a better explanation of the situation in languages where all roots do *not* simply map to a single category.¹¹⁵ Having individual constraints allows for the accounting of unusual behavior in a way consistent with the principles of OT.

Taking this further, it can be seen that the entire “input” structure can have its morphological structure specified in the constraint hierarchy. Thus, given an “input” structure /de-fect/, the subcategorization for /fect/ will align it to a morphological stem, while the prefix /de-/ will subcategorize for such a stem on its right. The constraints enforcing this higher morphological structure, ranked higher than the constraints which add prosodic structure to the representation, will ensure that only candidates with the correct morphological structure [de-{fect}_{Stem}]_{MWd} will survive:

(7.4)

/de-fect/	A([fect], R; Stem, R)	A([de], R; Stem, L)	Non-Fin(F)	Non-Fin(ś)	Edgemost
☞ [dĕ{-fect}]					
[[{dĕ-fect}]]		!*			
[[{dĕ}fect]	!*				

The violation of the prefix subcategorization constraint by *[[{dĕ-fect}]] removes the possibility of a structure resembling the *brígand* type of monomorphemic stem. However, in OT all constraints, even subcategorization constraints, are violable. For the few “exceptional” words of the type *récord*, there must be a superordinate constraint of the following kind dominating the general subcategorization constraint for /re-/:

(7.5) Align([re-cord], L/R; Stem, L/R)

Applying only to the morphemes underlying the noun *récord*, this constraint overrides any subordinate constraints which might favor a different structure (e.g., *[re{cord}]). There is nothing exceptional about the structure of words like *récord*, apart from their interpretation as simplex rather than prefixed

¹¹⁵For example, there are languages with several declensions or conjugations for each syntactic type, each showing particular morphology, stem extensions or vocalisms.

stems. By introducing a constraint like (7.5) into the grammar, a lexical idiosyncrasy can be captured using the same methods adopted for general subcategorization.

However, there is an unanswered question left in this solution, which is more clearly seen in another type of exception found in the lexicon. These involve words which, although clearly related, cannot be accounted for by a single unitary morphemic “root”, and so cannot be as easily explained away. For example, the pair *zéal* ~ *zéalous* shows a vowel alternation which is not regular for words in /-ous/; there should not be any “shortening” effect seen with this suffix. Since the great majority of words in /-ous/ do not show such an alternation (e.g., *fámous*, *griévous*, *vénous*), it is difficult to link the stem of *zéalous* directly to the /zel/ or /zēl/ underlying *zéal*, without unusually including the suffix in the stem. An alternative explanation is to posit a prosodically distinct stem /zell/, with a geminate final consonant, which combines with the suffix in this case. Further cases for allomorphy are seen in pairs like *retáin* ~ *reténtive*, where the alternating vowels /ā~e/ are not simply prosodic variants, or *destróy* ~ *destrúction*, where the allomorphs have considerably different segmentism. Finally, there are pairs like *knífe* ~ *kníves*, *wífe* ~ *wíves* where the corresponding segments differ only by a distinctive feature, a variation apparently conditioned by the morphological context. Notably, other sets of words exist where this alternation is not seen in the same context, e.g., *óafs*, *chíefs*, *bríefs*.

For such forms, it is necessary not only to propose an unusual morphological structure for the “input” form, but to propose different root morphemes for the various connected forms. This is not representable in the same way as the constraint offered above for *récord*, because it is the very selection of the relevant morphemes which is involved. While there is again nothing prosodically unusual about structures like /de-struk-tion/ or /wīv-z/, it is the imperfect relationship between these forms and related ones like /de-stroy/ and /wif/ which complicate the grammar and make these words “exceptional”. In the absence of a prosodic or morphological constraint explanation, only idiosyncratic lexical selection can account for these words, whose underlying semantic roots must then be understood as connected to multiple parallel morphemes. Any action of constraints governing the morphological or prosodic structure of these forms is pre-empted by the selection of an idiosyncratic “input” form. While the constraints that have

been offered up to now in this analysis of English allowed for a variety of relationships between the “input” string and various morphological and prosodic constituents, there has been no mechanism offered which governs or modifies the choice of the “input” string itself, which has been taken as a given, supplied in an unspecified way by an autonomous lexicon.

7.1.2 Underlying forms and “input” forms

Such cases, wherein semantically and segmentally related words require multiple underlying forms, raises questions about just what the “input” string in the OT paradigm is meant to represent and how it should be understood, and calls for a brief discussion of the concept of lexical forms within derivational phonology and morphology. Inherent to all the theoretical approaches to phonology and morphology discussed above, including Optimality Theory as described by Smolensky & Prince (1993), is the concept of **generation**. In traditional generative terms, underlying forms consisting of segmental strings undergo rules that change them into phonetic output. This theoretical perspective is continued in Optimality Theory by the characterization of the morphological constituents as “input”, and the representation of those morphological constituents harmonically aligned to prosodic constituents as “output”. The formalization of the relationship between input and output forms is the focus of the grammar, whether it be rule- or constraint-based. Such a grammar thus depends fundamentally on the input forms, which can be determined only in the context of the interdependent relationships between phonology, morphology and the lexicon described in § 1.2. This dependence on the input form has been codified even more strongly in more recent developments in OT such as Correspondence Theory (McCarthy & Prince 1995).

These input forms, once arrived at on the basis of the data, are understood in all these theories as residing in a lexicon. This lexicon, usually conceptualized as a list, effectively links structural morphemes with semantic lexemes. Within an OT framework, this presents a heterogeneous view of the grammar, where the lexicon supplies the desired “input string” to *Gen* on the basis of unspecified semantic processes, *Gen* expands this representation by superimposing upon it various combinations of prosodic constituents, and the constraint hierarchy evaluates these candidates, yielding up the correct optimal surface form. Thus, while the evaluation process in Optimality

Theory is novel, the process of lexical selection is regarded as similar to that seen in derivational theories. This allows for the apparent plausibility of serial derivation in OT, which was argued against in § 1.5.

The presence in the data of exceptions to otherwise phonologically regular correspondences between stems, such as the *récord* or *retáin* case noted above in § 7.1.1, suggests that certain lexemes correspond idiosyncratically to a series of morphemic forms, against the “regular” pattern wherein each individual lexemic item is associated with a single morpheme in a predictable combinatory manner. Such exceptional cases illustrate how the choice of “input” form can impact on the evaluation process and thus on the role of the constraint hierarchy itself, which is fed by the result of the lexical selection process. In the strong form of OT advocated here, all surface forms must be produced using the constraint hierarchy, putting a great deal of the explanatory power for exceptional forms into what is under current OT (as well as in derivational theories) an autonomous, relatively unstructured lexicon.

Thus, it is relevant to further investigate the process of morphological selection, i.e., the selection of the “input” form from the lexicon and its “delivery” to *Gen*. In most OT analyses, *Gen* has been used to introduce various combinations of prosodic constituents, members of the prosodic hierarchy, onto the “input” morphological structure, and has had little role in choosing different morphemic parses of the input forms, although the morphological hierarchy seems to otherwise parallel the prosodic hierarchy in structure. Subcategorization constraints have been used to account for the behavior of certain affixes, as shown above (§ 4.2.2, § 6.1), and the proposal for subcategorizing all morphemes above in § 7.1.1 formalizes every morphological subcategorization in terms of the constraint hierarchy, and allows it to govern the choice of the “input” morphemic structures of candidates as well as their prosodic structures. In other words, *Gen* can produce candidates which vary in morphemic structures as well as prosodic structure, and it is high-ranking subcategorization constraints in the hierarchy which eliminate the “incorrect” structures, whether they be descriptively irregular (e.g., *[{défer}], *[{inject}]), or, as in the case of exceptions, the expected regular forms (e.g., *[{récord}], *[{in{fám}ous}], *[{ím{pregn}}ate]).

Going one step further, such constraints could be used to choose not only the morphological subcategorization of the morphemes (as in *récord*, *imprégnate* or *infamous*), but also the selection of the “input” morphemes themselves, on the basis of the desired semantic constituents or **lexemes**, accounting for the variant root morphemes seen in words like *zéalous*, *retention*, *destruction*, or the alternant lexical structures which can appear for certain /-atory/ words (§ 6.3). That is, the “input” should be understood as purely semantic, and constraints linking semantic lexemes to morphological constituents are used to select the morphemes composing candidates as well as to morphologically subcategorize and prosodify them. It is such constraints whose high ranking would enforce the elimination of otherwise regular candidates such as **zéalous*, **retantive*, **destróytion*.

7.2 Complexity and economy in generative theory

Accepting such a proposal requires getting past the traditional model of the autonomous lexicon, seen as the repository for all idiosyncratic material in the language, and the unitary morpheme, naturally linked to its semantic content or lexeme. To do so, it is necessary to further examine the theoretical motivation for the notion of the “input” lexical form, modified by grammatical processes to yield varying surface forms across a range of words. The traditional motivation for proposing generative phonological effects derives from the presence of surface contrasts in such “related” words. The role of the generative process in all generative theories, including Optimality Theory, is to change a unitary, “basic” morpheme into all of its surface manifestations, on the basis of regular and predictable environmental conditioning, whether these be expressed as rules or constraints.

This need for generating multiple instantiations from a single prototype is motivated by a principle of **economy**, or the striving for **least complexity**. One major problem with this approach, however, is that no principled way to measure economy and complexity within generative theory has ever been generally accepted. The principle of economy in early generative work was focused upon generalizing the rules as far as possible, with the result that the complexity found in real linguistic data was removed from the grammar proper and encapsulated in the lexical entry. By adding complexity to the lexical entries, such as additional non-surfacing phonemes

or diacritic markings, the rules themselves could be made more general, elegant and economical.

However, this economy in the formulation of rules is **not** identical to a general principle of economy, which might be called the economy of expression. In other words, the economy of rules is not equivalent to the economy of the grammar as a whole, which would minimize complexity throughout the system, rather than in just one of its subcomponents. The lack of a formalization for the lexicon, which tended to simply consist of a list of all “unpredictable” elements, allowed for its concomitant increased complexity to pass unregarded. The lack of an evaluation metric, and the association of the principle of economy only with streamlined rules (without regard to the resulting over-complex lexical entries) has been a consistently problematic feature of traditional generative analyses. One example of this is the case of *níghtengale* and *míghtily* noted above (§ 1.4.1). Chomsky & Halle (1968) preferred to postulate underlying forms with a velar fricative /x/, rather than marking them as exceptional to one of their shortening rules. Lexical Phonology, on the other hand, preferred to emphasize what was regarded as a fundamental distinction between affixed (“derived”) and non-affixed (“basic”) words, again to avoid similar exception marking. That there is no formal way to judge the costs and benefits of either interpretation highlights the fact that there is no accepted evaluation metric in the field, only impressions and intuitions.

Lexical Phonology attempted to limit rule-based economy by introducing the counter-assumption that one should avoid “complicating” the lexicon. Instead, the rules were complicated by being made to apply less generally (§ 1.4.2). Indeed, there are no concrete methods in derivational theories for measuring the complexity of either rules or the lexical representations. For example, one measure of lexical complexity might involve adding up the number of morphemes in the lexicon. Natural languages, however, present very large lexicons containing many different lexical items, many of which share the same meaning. In terms of its social function the existence of large lexicons appears to be encouraged rather than restricted. New words are constantly being coined, some from entirely new morphemes, and many for concepts which already have names. The lexicon does not seem to have an upper size limit, nor does possessing a large mental lexicon appear to degrade the performance of one’s language ability.

Chomsky & Halle (1968) suggest in their use of the principle of economy that the crucial element is not the number of morphemes, but the number of morphemes needed to characterize “related” words. Single morphemes are used to generate as many related forms as possible, regardless of the complexity of the rules or lexical items required. In a language ideally conforming to this pure generative model, all word-structure would be predictable, each word ultimately linked to a single identifiable morpheme. Yet all natural languages show a robust subset of “irregular” forms which specifically do not conform to this ideal, the mastery of which is essential for acceptance into the native speaking community. Sometimes, regular words can be adopted into these irregular categories, for example the dialectal verb paradigm *drag* ~ *drug*. It has also been shown, for example in the case of Maori, that grammars do not always take the most abstract, generative route available when presented with related, alternating words (Hale 1973, Kiparsky 1982d: 62-3). While the regularization of irregular forms shows that this kind of economy plays a role in natural language, it appears to be only one of many competing influences on the grammar.

Bybee (1995) criticizes the generative model, which she refers to as the “dual-processing model”, for deriving regular forms by rule from base morphemes while irregular forms are simply listed in the lexicon, suggesting “two highly distinct types of processing”, which she claims is inconsistent with the psychological evidence. In the generative model, this concept of a “base” form arose from the intuitively appealing idea that longer words are composed of shorter units, which are enlarged via affixation. For cases like *instrument* ~ *instrumental* ~ *instrumentality* or *walk* ~ *walked* ~ *walking* the concatenative derivational explanation happens to appear to be straightforward. However, this approach breaks down for sets like *strategy* ~ *strategic* ~ *strategian* ~ *stratagem*, *imperi-ous* ~ *imperi-al*, *experi-ence* ~ *experi-ment*, which present no clear unaffixed “base” form.¹¹⁶

The source of this apparent problem is the conflation of the “base” morpheme with the unaffixed prosodic word, which implies that the prosodic word is likewise a category of basic semantic meaning and morphological constituency. Delinking the structural, segmental morpheme

¹¹⁶For example, the compilers of the Celex wordlist (Baayen, Piepenbrock & van Rijn 1993), always chose one such suffixed form as “basic”, listing it, suffix included, as a unitary base morpheme.

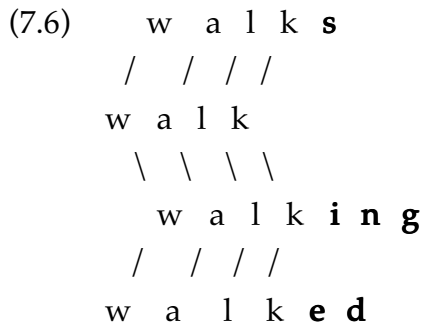
from its prosodic manifestations essentially removes this problem. In other words, both the problem of overgeneralizing rules at the expense of lexical entries for the sake of “economy”, and the problems of quantifying lexical complexity in order to alleviate it, are epiphenomena of the rule-based generative conception of phonology and morphology. The common element shared between the words above is a morpheme or string of segments (e.g., /strateg/), which should be regarded as a root-level constituent regardless of the fact that the morpheme itself never appears in an unsuffixed prosodic word. The subcategorizations for this morpheme in each prosodic word are, as in all words, idiosyncratic and recorded in the lexicon, i.e., in the “lexical part” of the constraint hierarchy.

7.2.1 Bybee’s network model of morphology

In the OT analysis of English stress and vowel alternation presented previously, the prosodic differences seen between “base” and “derived” words have been linked to a contrast in morphological structure, governed by the subcategorization behavior of affixes (§ 4.2.2, § 6.1, § 7.1.1). This provides a conception of the lexical or “input” item as arranged into morphemic units which may be combined with each other and map to morphological and prosodic constituents. Yet the very concept of discrete, recombinable morphemes has by no means been accepted by all theoretical frameworks. Bybee (1995: 1) points out that linguists have argued every conceivable point of view between the spectrum of Chomsky & Halle’s (1968) completely concatenative rule-based system, wherein all words are reduced to sets of unitary morphemes, to Vennemann’s (1974) completely lexical system, in which all words, regular or not, are recorded “whole” in the lexicon. Bybee (1995) argues in favor of her own network-style system (Bybee 1985) in contrast to the “dual-processing” generative model of Pinker (1991). It will be argued here that while these approaches emphasize supposedly competing viewpoints, each represents a facet of a complex, interconnected process that necessarily contains elements of both perspectives.

Bybee claims that the lexicon does not contain so-called “base” morphemes strung together by rules, but rather has access to all surface forms, whose internal interrelations are available to the entire system. Words are thus listed as whole units, with connections characterized by **lexical strength** linking the phonemes common to different forms of the “same”

word. This concept of lexical strength is tied to notions of frequency and the observations that “regular” patterns are common throughout the lexicon, while irregular words tend to be idiosyncratic in form but show high individual word frequencies. High frequency phoneme strings, such as affixes, will have high lexical strengths, and can thus extend their links to further words in cases of innovation or regularization. Particular paradigms, e.g., that of the verb *walk*, are indicated in Bybee’s network representations by links between the phonemes seen in the stem *walk* in all its various forms. The links between the stem phonemes will also highlight the contrast between themselves and the links between affixal phonemes in forms like *walks*, *walked*, *walking*, *walker*.



This method of linking also allows Bybee to model non-concatenative relations, such as that seen in *sing* ~ *sang* ~ *sung* ~ *song*. The phonemes /s,n,g/ can be linked together in the paradigm, signifying the stem, while the vowels will connect to similar alternating vowels in *ring* ~ *rang* ~ *rung* and other sets. Thus, minority patterns will also be represented by lexical links, although these links will not be as strong as those connecting majority patterns. Bybee represents the strength of lexical links via the thickness of the lines used in diagrams like (7.6).

The dialectal tendency for structurally similar words to “regularize” to one of these minority patterns, such as in *bring* ~ *brang* ~ *brung* or *drag* ~ *drug*, can also be accounted for in this model. These patterns of regularity across the lexicon are referred to as **type frequency**. In many languages, there are competing groups of sub-regular patterns such as these, which sometimes increase in number and in any case remain robust. Bybee claims that the generative model is not well-equipped to account for these kinds of relationships.

There is also another sort of frequency, known as **token frequency**, which refers to the frequency of individual words in a language. Bybee notes that “irregular” forms tend to have a high token frequency, making up for the small number of lexical links between such forms with the strong lexical presence of the forms that do occur. High token frequency can maintain very irregular morphological relationships, such as the suppletion seen in *go* ~ *went*. Bybee suggests that the high token frequency of the past tense form *went* (as opposed to the present form *wend*) allowed it to break the weak links of its old paradigm and instead act as part of the paradigm of *go*, with which it has no phonemic correspondences. Both type and token frequency contribute to the lexical strengths of the connections between the forms in the lexicon and influence the construction of new entries.

7.2.2 Form over content

There is some striking evidence to illustrate that the generative principle of economy of stems is, as Bybee maintains, simply one competing tendency in the lexicon. There is a set of high-frequency English words which are evidently identified and incorporated into the grammar on the basis of their surface melodic (phonemic) form, rather than according to obvious semantic and morphological links assumed to form the basis of paradigmatic organization in both generative theory and Bybee’s system. For these words, the shape of their surface realizations requires that their corresponding morphological constituents are subcategorizing differently from the morphemes found in other words of their semantically related paradigms, although it would be more “economical” in generative terms to produce all the words in question from the same morpheme.

For example, most trisyllabic prefixed verbs whose lexical stems happen to end in phonetic [-eɪt] or [-aɪz] take on the stress pattern of proper verbs in /-ate/ and /-ize/ (e.g., *cónfiscàte*, *mésmerìze*), regardless of the fact that many of these words are neither morphologically nor historically members of those categories:

(7.7)	súpervise	sùpervision (not *sùpervisación, etc.)
	círcumcise	cìrcumcisión
	récognise	rècognition
	télevise	tèlevision
	ímprovise	(impròvisación)

The shape of the semantically connected nouns to these verbs indicate that these words are based on morphologically (and historically) prefixed monosyllabic roots which simply happen to have the morphological structure /Ciz/ and /Cat/, i.e., *súpervise* is morphologically /super-vise/, *círcumcise* is /circum-cise/, *ímprovise*¹¹⁷ is /im-pro-vise/. Most of these forms show a morphological paradigm parallel to prefixed stems like *explóde* ~ *explósion*. However, for the verbal forms, the presence of a rime /-iz/ at the end of the word has been enough to suggest to the grammar that these forms are suffixed in /-ize/, despite the fact that this interpretation complicates these paradigms (in generative terms) and introduces stem allomorphy. For example, a stem /circumc/ as abstracted from *círcumcise* forms no other words, and if regular would have an abstract noun *cìrcumcización. Historically a prefixed form of the root /cis/, also seen in *excíse*, *círcumcise* would be stressed like *cìrcumvént* if it were not incorrectly interpreted as suffixed in /-ize/.

Further data suggests that surface similarity is not the only criterion for such an interpretation. Simply the presence of a segment sequence /-at-/ at the end of an analyzable morpheme is apparently enough to warrant the classification of these words as suffixed in /-ate/:

(7.8)	supérative
	rérelative
	tránslator, tránslate

These words are, historically, prefixed forms of a Latinate stem /lat/ (compare *emóitive*, *divísive*, *indecísive*, *sùpernórma*). In these cases, the grammar has,

¹¹⁷ The derived form *impròvisación* indicates the regular nominal structure expected in true /-ize/ words, e.g., *stàndardización*, *gèneralización*.

at some point, added morphological complexity to the lexicon by introducing a novel stem which is irreconcilable with the other members of its paradigm (we expect *relâte* ~ **relâtive* like *emôte* ~ *emôtive*).¹¹⁸

That this interpretation is ahistorical suggests that a morphologically regular paradigm has in each case been changed (in generative terms) to a morphologically irregular one simply because the words in question were melodically interpretable as being part of a productive pattern of suffixation. This type of reinterpretation is not at all predicted by the generative system of constructing the lexicon according to the economy of stems principle, and although Bybee's system can better represent the links between the phonetic realizations, it is awkward to denote the paradigmatic links in these cases, because each sequence /iz/ in forms like *sûpervise* would then be multiply linked (to both stems and suffixes) in the network. Only a logical separation between structural morphemes and underlying semantic constituents, and the means of gracefully accounting for such complexity through superordinate selection and subcategorization constraints, can model these interesting forms cleanly and consistently.

7.3 Morphology and the lexicon

Although in Bybee's system there are no morphemes *per se*, "even though words entered in the lexicon are not broken up into their constituent morphemes, their morphological structure emerges from the connections they make with other words in the lexicon," (Bybee 1995: 5-6) and Bybee speaks of "parallel sets of phonological and semantic connections." Bybee's model is not incompatible with that offered thus far in the OT analysis presented here, but differs in that it seems to require an overt listing of each (prosodically defined) word in the lexicon. This is the result of a theoretical conflation similar to that seen in generative theory, where the prosodic and morphological words were conflated. Bybee is instead conflating the semantic denotation of a given word with its morphological correspondent, through the medium of the prosodic word as the minimal lexical unit.

¹¹⁸On the other hand, forms like *órátor* may be analyzed as based upon a single stem {orat}, despite the forms *orâte*, *orâtion*. Alternatively, this can be another case of the stem-subcategorizing form of /-at/ (§ 6.3). Either explanation demonstrates the potential idiosyncrasy of lexical selection and subcategorization.

It is possible to propose, within the context of OT, a third, semantic hierarchy, which maps to the prosodic and morphological constituent hierarchies, and which would be subject to constraints governing the lexical selection of morphemes, the correspondence between morphemes and semantic structures.¹¹⁹ Bybee's approach lists each (prosodically defined) word as a single entry with both morphological and prosodic components, linked to other such words according to prosodic, morphophonemic and semantic criteria. While this system is capable of modeling many of the morphological patterns and regularities seen in natural language, it fails to capture all the relationships expressible via configuring three separate hierarchies, prosodic, morphological and semantic, using Optimality Theory. By formally delinking the independent concepts contained in these hierarchies from representations that typically conflate them, a more comprehensive, expressive system can be arrived at.

A similar conflation of structures is found at the basis of the criticisms of the traditional lexicon offered by Golston (1995, 1996). Golston claims that OT and its generative ancestor theories make a fundamental claim about prosodification, referred to as the "Theorem of Impossibility" (Golston 1995: 1):

(7.9) Every underlying form is an impossible surface form and vice versa.

What Golston means is that since "underlying" or "input" forms are prosodified during evaluation (or derivation), they must not be prosodified in the lexicon and are thus in themselves impossible surface forms. Golston (1996: 1) then takes this a step further to claim that this means "speakers cannot store things that they can say and vice versa." Golston's assumption here expresses a misconception about underlying representations arising from the "input/output" model inherited from generative theory. When regarding the underlying form as the closest possible copy of the surface form (following the usual conceptions of Lexical Phonology and reflected in OT by the concept of Faithfulness), it is simple to regard the "stored" underlying

¹¹⁹A rough outline of how this might work is presented in § 7.3.1; a detailed examination of such a semantic hierarchy is however beyond the scope of this work.

representation as simply a (pronounceable) variant of the surface word. The problem of conceptualizing an underlying form lacking prosody lies behind theories, such as those of Vennemann (1974) or Bybee (1985, 1995), which demand that the lexicon store fully prosodified words.

Rather than being viewed as the stored form of a word, the traditional “input” form is understood here as a set of morphemic constituents lexically selected by their corresponding semantic constituents, which comprise the “true” input form. Prosodification of these morphemes involves the selection of various prosodic constituents from a set of prosodic structures, which interact with the morphological constituents and allow for the expression of the features in the morphemic constituents into acoustic output, or speech. As spoken natural language is the apparently unitary product of this triple linking, it is difficult to conceptualize or represent the separate elements of this process outside the realm of abstract theorizing. When Golston claims that the existence of a completely independent, unprosodified morphological constituent requires that “speakers cannot store things they say”, he is crucially assuming that this “storage” is equivalent in full to the morphological component, and that the form the stored item should take must be “pronounceable”. Under the approach set out here, every word is understood as having associated with it a set of prosodic, morphological and semantic constituents, representing a decomposition of the associated surface form into three types of structures. None of the individual constituents in the three hierarchies are independently utterable; requiring them to be so is similar to asking someone to utter a mora, a foot or a stem. It is only in the context of all three hierarchies that an utterance is possible.¹²⁰ The constraint hierarchy, and thus the grammar, is the formal description of the interaction of these three hierarchies, yielding the attested language (and thus its surface forms).

The approach to morphological selection taken here continues this line of reasoning: the lexicon, which may be conceptualized (following Bybee) as a network relating semantic, morphological and prosodic constituents, is argued to contain all information necessary to describe the correspondences between sound and meaning in the language. However, while Bybee’s

¹²⁰The necessity for semantic, morphological and prosodic constituents is of course applicable only to verbal communication. In a signed language, a gestural hierarchy might be understood to replace the prosodic hierarchy.

approach attempted the same thing by overtly listing every word in the language as an independent quasi-prosodic/semantic/morphological unit, this is not necessary here, although the same desired effect, the recording of all relevant relationships in the lexicon, results. Instead, every component of the language, i.e., the different constituent members of the three hierarchies, appear as arguments to constraints in the hierarchy. Correspondences between constituents play a similar role as links in Bybee's networks, but the ranking and violability of constraints allows for constituents to be understood as discrete while surfacing in a variety of ways (e.g., coalescence, templatic morphemes, alternations). Instead of simply representing links between segmentally similar prosodified surface words, the constraints in question represent not only the connections between semantic constituents and their selected morphemes, but also the subcategorization relationships seen between morphemes, and the relationship between morphemes and prosodic constituents.

In Bybee's system, the strength of the links between words corresponded to issues of frequency. Here, frequency also plays a similar role, but this is represented in terms of constraint ranking rather than simply indicating which relationships are more frequent. Very common prosodic patterns, for example, will be governed by a series of prosodic constraints. The crucial ranking of such constraints above others, which would yield a different result under another ranking, is determined by the patterns most frequently encountered in the language. The advantage of the OT approach proposed here is that such a constraint hierarchy covers the entire language, and affects every form, while Bybee's links are purely local and describe only the properties of some group of words, apparently unrelated to others. The constraint hierarchy accounts for the entire range of constraints, from those which affect all words to those which affect only a few, by ranking and violability. Conversely, words with idiosyncratic behaviors may require (for example) selectional constraints that are more highly ranked than the general constraint which would normally disallow the form. The high ranking of such a constraint is again linked to frequency, and it is precisely such idiosyncratic forms which are expected to have high token frequencies, represented here by a high-ranking selectional constraint.

Both Bybee's network approach and the OT solution offered here are able to represent the fact that within a language, while there are large sets of

data which conform to broad, regular patterns, there are also smaller sets that conform to minor patterns, and other sets or items that directly contravene the general trend. In a suitably explanatory linguistic theory, the grammar (and its associated lexicon) must be able to robustly account for all these types. Bybee's system arbitrarily uses the (prosodically defined) linguistic word as the primitive category and only allows indirect access to morphemes; semantic content is linked to segmental identity and it is difficult to see how meaningful prosodic constituents would be represented, divorced from their segmental contexts. The OT solution allows for the theoretical separation of the linguistic word into three constituent hierarchies, whose members' relationships define the entire grammar, and which expresses, through the constraint hierarchy, simultaneous relationships between every constituent in the language rather than just local relationships between small groups of forms. Bybee's networks are a useful way of visualizing some of the relationships between words which must be captured by an explanatory grammatical theory, but the OT constraint hierarchy can capture both the various morphological relationships modeled in Bybee's networks and the complete range of inter-constituent relationships, from the very general (e.g., stress and syllabification) to the very particular (e.g., allomorphy, idiosyncratic subcategorization), which cannot be modeled simply by connections between structurally "related" words in a network.

7.3.1 Morphological selection using the constraint hierarchy

While a full account of the semantic hierarchy is beyond the scope of this work, the section presents a brief outline of how the semantic hierarchy might interact with the morphological hierarchy to model lexical selection. This semantic hierarchy can be imagined as roughly parallel in form to the morphological system; using this, it is possible to illustrate how the theoretical confusion regarding "base" and "derived" words, so central to the disagreement between generative theory and the network model, could arise. The contrast between "base" and "derived" words, as well as the difference between regular and irregular morphology, are shown to simply be the result of differing constituent structures. Morphological selection as an interaction between semantic and morphological constituent hierarchies will be illustrated in the following rough sketch of English past tense alternation,

exemplified here by the regular forms *walk* ~ *walked* and the irregular suppletive paradigm *go* ~ *went*.

Involved in the simplified discussion of these relations are four lexemes, which will be represented in curly braces: {go}, {walk}, {Present} and {Past}, and four morphemes, /go/, /went/, /walk/ and /-ed/. These lexemes are not identical with the English words used here to represent them, but rather represent the semantic content underlying the corresponding labels.¹²¹ The four morphemes are purely structural entities made of segments, and as can be seen, are not isometric with the four lexemes. A simplified model of the lexicon would contain the following links (represented by a colon) between the lexemic and morphemic constituents:¹²²

- (7.10) {walk} : /walk/
 {Past} : /-ed/
 {{go} {Present}} : /go/
 {{go} {Past}} : /went/

The lexeme for the concept “walk” happens to be isomorphic (in that it is unitary) with the corresponding morpheme /walk/, as is the case with the regular past tense marker /-ed/ and the semantic concept of past tense. The {Present} lexeme, however, is not singly linked to any morpheme. Neither is the lexeme for “go”, as the morpheme /go/ is linked only to a complex locus referring to two lexemes: both the concept {go} and the present tense lexeme in combination. A similar situation exists for the past tense morpheme /went/. When a speaker wishes to express the complex concept “go + Past”, a compositional correspondence of individual lexeme to individual morpheme is not available, because there is no morpheme linked to {go} on its own. This could be expressed as the violation of a constraint which might be formulated in the following manner:

- (7.11) Link(Lexeme, Morpheme)

¹²¹No further decomposition of such lexemes, while plausible, will be entertained in this work.

¹²²In the diagrams found in this section, the braces ({}) represent semantic constituents, rather than stem constituents as in previous chapters.

The functor **Link** simply indicates some type of correspondence between the two constituent types used as arguments in the constraint. Such a constraint is enforced by lexical selection; when the lexeme in question completely corresponds to a morpheme selected as part of a candidate utterance, the constraint is satisfied for that candidate. The principle, however, is the same as that seen for the prosodic and morphological constraints discussed previously (§ 7.1.1): the structure of material in one hierarchy enforces the selection of material in another.

For a regular form, such as *walked*, the optimal morphemic candidate /walk-ed/ satisfies the constraint, each lexeme showing a one-to-one correspondence with a morpheme. For the semantic input {{go} {Past}}, however, the optimal morphemic candidate /went/ will incur a violation of this constraint, because this one-to-one correspondence fails to hold. By contrast, a candidate /go-ed/ could satisfy this constraint. That /went/ is the correct form is seen in the lexicon:

(7.12) {{go} {Past}} : /went/
 {{go} {Present}} : /go/

According to this lexicon, the match of /go-ed/ to {{go} {Past}} should not be possible; while the lexeme {Past} could successfully correspond to the morpheme /-ed/, the failure of {go} to correspond to any single morpheme should ensure a violation. The optimal candidate /went/, however, would be selected by the network. Just like a bound morpheme (e.g., [ceive]) in the framework of Inkelas (1989), the lexeme {go} appears to be subcategorized for combination with a tense lexeme, and can never correspond by itself to a single morpheme.

Rather than evaluating these constraints via comparison to an autonomous lexicon, as was just done, it is instead possible to use mechanisms already present in Optimality Theory, parallel to the treatment of morphological subcategorization (§ 1.3.5, § 4.2.2, § 6.1), to integrate the lexical network presented above into the constraint hierarchy itself. This has the advantage of unifying all grammatical relationships under a single framework, rather than handling completely parallel processes in entirely different ways (with little logical motivation for such a split). Just as specific morphemes could be given as arguments to constraints (McCarthy & Prince

1993a), the lexical network can be represented as a series of constraints taking lexemes as their universal arguments:

(7.13) Link({{go} {Past}}, /went/)

Ranked above the more general constraint Link(Lexeme, Morpheme), this constraint ensures that /went/ surfaces, in spite of its violation of the latter constraint. The candidate /go-ed/, while satisfying Link(Lexeme, Morpheme), violates the higher-ranked constraint (7.13) and thus is less optimal than /went/. For regular forms, no high-ranking constraint prevents the compositionality of, e.g., {walk} and {Past}. Following the more general version of the lexemic correspondence constraint (7.11), the lexemes seen in {{walk}{Past}} can correspond to the morphemes /walk-ed/. The morphological ordering and subsequent prosodic expression of the combined form result from the morphological subcategorization and other constraints contained in the remainder of the constraint hierarchy. This constraint-based view of the lexical selection of irregular forms via high-ranking selection constraints is akin to the Elsewhere condition of Kiparsky (1982a, b), which was presented in the context of a derivational, rule-based approach.

7.3.2 Specific and general constraints

The type of constraint seen in (7.13) will be referred to herein as a **specific constraint**, and takes as its argument a specific complex constituent rather than a general category of constituents. The usual type of constraint (e.g., 7.11) which takes a non-specified constituent as an argument will be referred to as a **general constraint**. Specific constraints capture what would traditionally be termed lexical information. The suffixal subcategorization constraints given previously (e.g., 6.2) are also examples of specific constraints. This concept of using both specific and general constraints in the hierarchy has the added advantage of capturing Bybee's notion of token frequency, and expressing it not as a separate notion of lexical strength, represented as a thick line or boldface type, but by the relative ranking of specific constraints in the constraint hierarchy. Conversely, type frequency would be represented by the relative ranking of general constraints. In this way, all these relationships can be represented via the same mechanism, the constraint hierarchy.

Frequent irregular forms would require high-ranking specific constraints which maintain their irregularity in spite of (lower-ranked) general constraints which the successful candidate violates. Once an irregular form becomes infrequent enough that the status of its corresponding constraint is not recognized during language acquisition, and it falls below the general constraint that would otherwise govern it, the form would become “regularized”. For example, the mechanism for forming a regularized compositional form **goed* exists in the grammar, as it parallels *walked* in every way but one. Only the relatively high ranking of the specific constraint $\text{Link}(\{\{\text{go}\}\{\text{Past}\}\}, / \text{went} /)$ prevents the candidate **goed* from surfacing, and it is the wrong relative ranking of this very specific constraint, or its absence, that allows children at some point in their development to produce such forms. The various arguments (developmental, psycholinguistic) in favor of the network model proposed by Bybee (1995) are equally applicable to the OT system proposed here, because in fact the network model of the lexicon described above, with its nodes and links, is logically equivalent to the constraint hierarchy, which can equally represent all the relations present in such a network.

Thus the case of “irregular” patterns can be accounted for, as Bybee suggests, by a unified rather than a “dual” process. Constraints linking particular lexemic structures to specific sets of morphemic constituents must necessarily be more highly ranked than the corresponding general constraints in order to yield the “irregular” forms. For regular cases, no such high-ranking constraints are present in the grammar, and the compositional linking of the two constituents succeeds. The relative rankings of these constraints directly reflects Bybee’s notion of lexical strength, but in a formal, unified way. The general constraints capture the most general regularities, those exemplified by high type frequency in Bybee’s terms, while the more specific constraints’ high ranking corresponds to and derives from their high token frequency.¹²³ Across the range of regular to irregular, there are many sub-regularities which will be governed by degrees of less general constraints,

¹²³In the case of irregular sets of words, such as the strong verbs in English, coequal constraints governing such a set will be reinforced by the frequency of some members of the set (e.g., *drive* ~ *drove*), rather than by all members, as some may have low frequencies but maintain themselves on the strength of the group (e.g., *strive* ~ *strove*).

which will prevent some or all of the smooth alignments found in the general cases from surfacing.

7.3.3 Constraints as the lexicon

Using the complex lexeme {{go} {Past}} as an argument to a constraint specifying its morphemic analog is wholly parallel to constraints used in OT to account for subcategorization effects (e.g., McCarthy & Prince 1993a). This type of constraint may at first appear to be very powerful; most of the constraints usually seen in the OT literature are of the type referred to here as “general” constraints. However, such specific constraints are formally and functionally identical to more general constraints, and they consist, like the others, solely of relationships between linguistic constituents, members of the three hierarchies. Furthermore, specific constraints are no more powerful than traditional lexical entries, and they are more constrained than lexical entries because they must conform to the general principles for OT constraints.

Indeed, the type of specific, complex information captured by specific constraints is usually confined to the lexicon, whether it be represented as a list or a networked structure. Following suggestions of Russell (1995) and Hammond (1995),¹²⁴ however, it is possible to understand these specific constraints as equivalent to the lexicon, the network of relationships between constituents seen in approaches like that of Bybee (1995) being precisely representable as a hierarchical set of constraints. This allows the lexicon to be represented with the same formalism as the constraints themselves. Since the lexemes, their corresponding morphemes, and their subcategorization frames are all represented in constraints present in the hierarchy, there is no need for an autonomous lexicon. Furthermore, as part of the constraint hierarchy, the lexical information is subject to ranking, while the information in the traditional lexicon of derivational theory is has generally not been constrained by any strong theoretical principles.

Conceptually, this idea is akin to Kiparsky’s (1982a) proposal that lexical entries are specific identity rules. In other words, each distinct semantic entry is governed by at least one constraint (and in some cases, more than one)

¹²⁴Golston (1995) has also proposed representing lexical entries as specialized bundles of constraints, although in a slightly different context than that presented here.

associating it with morphological and prosodic constituents such as segments, moras, and morphological categories. Besides removing an independent, unstructured lexicon and its separate process of morphological selection from the theory, Russell's approach allows a more flexible treatment of issues such as underspecification. Rather than proposing lexical stems with defective phonemes when underspecification or allomorphy appears to occur, these phenomena can be accounted for by the ranking of certain constraints. Russell (1995) applies this approach to an account of grammatically conditioned ablaut in the Papuan language Hua.

In Hua, complex ablaut patterns appear to require different lexical forms for a series of person markers in the verbal system. However, Russell resolves this problem by representing the lexical entries of these person markers as constraints that rank low enough to participate in feature-changing phonology. As a result, these forms can be accounted for with unitary morphemes whose canonical segmental forms are sometimes overridden by superordinate constraints demanding the appearance of certain features in certain contexts. In such cases, a competing candidate which violates the lexical constraint and thus shows a somewhat different segmentism than the canonical morpheme can nevertheless surface as the most optimal candidate, because the candidate possessing the canonical morpheme has violated a higher-ranking prosodic constraint. This view of the lexicon allows for underspecification while still respecting the status of both the segment and the morpheme. As Inkelas (1994: 5) has argued, "Underlying representation *has* to matter, underspecification *is* necessary to capture three-way contrasts, and grammar-blind principles of underspecification cannot be right."

For English, unlike Hua, the very specific "lexical" constraints governing morphological selection for specific lexemic constituents will generally be found near the top of the constraint hierarchies, and will provide the morphemes usually presented as the "input" to the function *Gen* in the Optimality Theory framework. For a language without much underspecification, these may continue to be informally conceived of as part of a "given" lexicon. However, in a complete linguistic theory, only the semantic intentions of the speaker can be the true input to *Gen*. As long as typical OT constraint tableaux are understood as omitting the top levels which select the morphemes used as input to the lower section of the hierarchy, they remain

accurate (only) as descriptions of the relationships between the morphological and prosodic hierarchies. However, it must be kept in mind that, as Russell (1995) has shown, it is possible for “lexical” constraints selecting morphemes to be relevant at a low enough level in the constraint hierarchy that they must compete with the more usual prosodic and morphological constraints.

In English, there are a number of cases where this low ranking of selection constraints can be used to account for otherwise irregular correspondences. For example, the contrast between plurals like *wíves*, *líves*, *léaves* and *óafs*, *bríefs*, *chíefs* was noted above in § 7.1.1. The voicing of the stem-final spirants might be the result of a constraint requiring voicing in this position, which we can call **VoicedSpir**.¹²⁵ Opposing that constraint would be the specification for the segment (or phoneme) /f/, which would involve a constraint specifying that /f/ is voiceless:

(7.14) **Voiceless-f:** NI(/f/, [-voice], [voice])

This states that for every /f/, there are no other [voice] features intervening between [-voice] and the boundaries of the segment, i.e., the segment is filled with the feature [-voice]. This would be just one of the constraints defining the segment or phoneme /f/, and /f/ as a morphological constituent is defined by its relationship to such features. Ranking this specification for /f/ above **VoicedSpir** will prevent the /f/ in a selected input string from ever voicing in this position. Such is the case for the regular *óafs*, which is selected, as usual, by a lexical selection constraint located relatively high in the constraint hierarchy.

(7.15a)

oafs	Link({oaf}, /ōf/)	Voiceless-f	VoicedSpir	Link({wife}, /wif/)
☞ /ōf/-s			*	
/ōf[+voice]/-z		!*		
/ōv/-z	![-voice]			

¹²⁵The formal details of this and other constraints at the level of segmental phonology will not be delved into here, but will be left for the sake of brevity to further research.

The tableau presented in (7.15a) is different from others presented previously in this work, because it shows candidates which do not conform segmentally to the “underlying form”, which is selected by the lexical selection constraint **Link**({oaf}, /ōf/), which connects a morpheme /ōf/ to the concept ‘oaf’. The candidate */ōv/-z represents a substitution of the segment /v/ for the segment /f/; such candidates will be eliminated by the appropriate specification (with /f/) found in the **Link** constraint.¹²⁶

The candidate /ōf[+voice]/-z, however, represents something slightly different. This candidate contains the segment /f/, so it is morphologically sound and does not violate the **Link** constraint. However, the /voice/ feature of this /f/ is in this candidate /+voice/, rather than the expected /-voice/. Such an aberration should be regarded as parallel to other permissible mis-alignments such as the unstressed heavy syllable seen in *brígang*; there are constraints which try to restrict such structures, but they are violable and can be overridden by higher-ranking constraints. In this case, the constraint which eliminates such faulty /f/ segments is **Voiceless-f**, which is part of the definition of the segment itself. Since this constraint outranks **VoicedSpir**, the segment /f/ as enforced by superordinate selection constraints will always occur with the expected features, even in voicing environments. However, one can propose that at an earlier stage of English, **VoicedSpir** outranked **Voiceless-f**. In such a situation, /f/ would always become voiced in the environments defined by **VoicedSpir** (as was once the case), and would only surface with the correct voicelessness as defined by **Voiceless-f** in positions where **VoicedSpir** did not apply.

Since in Modern English **VoicedSpir** is ranked below **Voiceless-f**, and *oafs* selects an /f/ in its stem, the candidate /ōf/-s succeeds. While this candidate violates **VoicedSpir**, the competing candidates are eliminated by more highly ranked constraints. However, for words like *wíves*, the situation is different. Here, proposing that these words have selection constraints which are ranked below **VoicedSpir** allows for the voicing alternation seen in *wífe* ~ *wíves*. In such a case, the candidate /wíf/-s, with an input string containing /f/, would be rejected in favor due to a violation of **VoicedSpir**.

¹²⁶It is possible to propose a whole series of candidates with incorrect segmentism (perhaps with every possible combination of segments), and this is one way to conceptualize how the segments found in candidates are produced by *Gen*. All of these candidates, apart from the ones with the “correct” segmentism, will be eliminated by high-ranking lexical selection constraints.

For the semantic unit {wife}, the **Link** constraint which associates it with an /f/ is ranked lower than **VoicedSpir**, and so fails to maintain the /f/ in the optimal candidate. While the segment definition constraint **Voiceless-f** eliminates candidates which contain irregular /f/ segments, the segmentally different /wīv/-z succeeds, as the /v/ segment satisfies **VoicedSpir** (and has nothing to do with **Voiceless-f**).

(7.15b)

wives	Link({oaf}, /ōf/)	Voiceless-f	VoicedSpir	Link({wife}, /wīf/)
/wīf/-s			!*	
/wīf[+voice]/-z		!*		
☞ /wīv/-z				[-voice]

The presence of the /v/ segment does violate the lexical selection constraint, but does so minimally (the difference between the candidate and the canonical morpheme is a single feature, [voice]). Other segmentally different potential candidates (e.g., */wīg/-z, */wīb/-z, */wīp/-s, etc.) would all incur more (or more highly ranked) violations.¹²⁷ **VoicedSpir** is presumably not applicable to the environments found in the singular /wīf/.

To summarize, the low ranking of the selection constraint for the root /wīf/ allows for the normal selection of the phoneme /f/ in this root to be overridden, and a variant root morpheme /wīv/ to be selected as the “input string” for the stem of the plural wīves. A similar approach might be used to account for past-tense vocalism in the strong verbs,¹²⁸ and other cases in which underspecification provides an effective explanation.

In such cases, the frequency of the irregular forms (and thus the “lexical strength” of the irregular class) is reflected in the high ranking of the constraint governing the defining feature of the irregular class (in the example above, **VoicedSpir**) relative to the ranking of the lexical selection constraints for the words involved. Here, the lower ranking of the

¹²⁷The details of how this would work are left to further research.

¹²⁸For example, the alternations seen in *run* ~ *ran*, *shine* ~ *shone*, could be accounted for via a subcategorization constraint aligning such verbal roots to a preterite morpheme consisting of a vowel subcategorized to appear within the stem. Being ranked higher selectionally than the stem vowel, it replaces it in the input candidate. Regular preterites in /-ed/ would not subcategorize for this morpheme.

Further lexical alignment constraints define the reduplicant's prosodic configuration (in this case prefixal) and ensure that it properly corresponds with morphological material found in the stem:

$$(7.18) \quad \begin{array}{l} \text{Align}(S[\text{Past Tense}], L; \text{PrWd}, L) \\ \text{Align}(S[\text{Past Tense}], L; \text{Stem}, L) \end{array}$$

This set of constraints, in combination with others expected in such a grammar, will yield a word with a reduplicant prefix syllable.

7.4 Economy and morphological selection

The two supposedly competing morphological models discussed in Bybee (1995), the “dual-process” generative model and Bybee’s network model, simply emphasize different aspects of the morphological system. Generative theory focuses upon the abstraction of regular and predictable relationships which help to organize the huge amount of data inherent in natural language, localizing much of that explanatory power in the individual morphemic units and the rules that would allow them to regularly concatenate. However, its explanatory power is not suited to account for another aspect of language, which not only tolerates but seems to value what would typically be termed “complexity”. In every grammar, there will be a number of words for which more than one morphological stem needs to be proposed, in direct contravention to the generative principle of economy, and forms like these require a principled representation in any theory, rather than being simply labeled as “exceptions” and “irregularities”.

Bybee’s network approach better accounts for the many sub-regularities and irregularities seen in natural language, but her representations, in which morphemes appear only in the context of networked units composed of mixed prosodic/semantic/morphological elements (“words”), fail to express the complete range of relationships available to the grammar as described above and exploited using the mechanisms of Optimality Theory. Bybee’s theory models the links between morphological and semantic constituents postulated here, but in conflating morphemes, prosodic words and lexemes, restricts the possibilities for the possible range of interhierarchical interactions (for example at the phrasal level) and fails to account for the potential

misalignment of constituents which produce so many of the prosodic effects explicable via Optimality Theory.

Once the notion that words or morphemes must correspond to “basic” meaningful units can be dispensed with, the generative and network paradigms can be seen as each better representing one of two competing but coexisting tendencies in natural language. Both systems attempt to describe a series of relationships between surface form and meaning, while organizing those sets of relationships in quite formally different ways. Using the sets of hierarchies and the Optimality Theoretic approach adopted here, all logical distinctions between generative theory and Bybee’s network theory fall away. Once the lexicon and the rule-based system of traditional generative derivational theory are fully replaced by Optimality Theory, all surface forms can be represented in an identical manner: through the constraint hierarchy. Representing the lexical network by a set of ordered constraints has the added advantage of retaining the explanatory power of the prosodic and morphological hierarchies, absent from Bybee’s system, and providing for the unification of a series of previously disconnected notions, like lexical strength and lexical connections, in a single schema, the constraint hierarchy. Furthermore, the system of violable constraints allows for the explanation of effects which only surface in certain situations, that is, the minority patterns. These are enforced by what neither previous theory could account for, the lower-ranking constraints, which only apply in cases where more than one candidate survives to the point where that constraint can have an effect.

Returning to the exceptional types discussed in § 7.1, how they can be accounted for in the constraint hierarchy should now be clearer. General patterns of high frequency (Bybee’s “type frequency”) result from the application of general constraints upon “input” morphemes of a “general” form, i.e., those chosen by general selection constraints. Exceptional types such as *vanilla*, with additional lexical prosodic structure in the “input” form, also conform to the general prosodic constraints but require an additional (or expanded) high-ranking specific selection constraint enforcing the association of a “lexical” mora to particular morphemes. Words with irregular morphological subcategorization in the “input” form, such as *record*, also satisfy general prosodic constraints, but violate the general morphological constraints governing the behavior of prefix morphemes. Such cases also require a high-ranking specific constraint, this time a subcategorization

constraint referring to both the prefix and root, to enforce the attested surface form. In other words, forms like *récord* require a specific constraint selecting a stem {record} to link with the nominal form, which must outrank the usual selection of the morphemes /re{ }/ and {cord} for this lexeme.¹²⁹

Because morphemes are simply subcategorized, ordered groups of segments, there is no intrinsic difference between prefixes and roots under this approach. It is the association of semantic content to morphemes, and the subcategorizations given to them, which determine the grammatical contrast between affixes and roots in a given language. In some contexts stem or affix morphemes might not be associated with any discrete semantic denotation, e.g., the morphemes seen in *receíve*, a word whose meaning would be enforced by the lexical selection constraint **Link**({receive}, /re-ceive/) rather than through an analysis of its constituent parts (as could *post-war*, where meanings are plausibly associated with the individual morphemes). The association of /re-/ (which can in some cases be associated with a meaning) and /ceive/ (which is by itself meaningless) obtains the meaning ‘receive’ solely due to the preceding **Link** constraint. In principle all morphemes are meaningless in themselves, and only the contrast between morphemes which appear in a single lexical selection constraint (and thus appear to maintain a consistent denotation in all cases, such as /post-/ or /war/) and those which can appear in many such selection constraints (e.g., /-ceive/, /re-/, /ob-/, etc.) creates the impression of meaningful versus meaningless morphemes.

Since morphemes are introduced into the representation by selection constraints, the same model can be extended to the question of allomorphy. The default case for related words showing a consistently identifiable root is to link that word’s lexeme to a single root morpheme. However, for semantically related words whose root morphemes cannot be structurally identical under any possible ordering of the prosodic and morphological constraints in the hierarchy (e.g., *destróy* ~ *destrucción*), the attested forms can be arrived at by proposing distinct specific selection and subcategorization constraints for each contrasting lexemic instance, similar to the way synonyms must be regarded. Thus, there would be two lexical selection

¹²⁹Whether this specific constraint overrides the usual subcategorization for /re-/ or simply presents a different root /record/ (and whether these alternatives actually differ) is best left for future research.

constraints for the lexeme {destroy}, one appearing in conjunction with a verbalizing lexeme, the other with a nominalizing lexeme. Likewise, multiple alternate realizations for the same word, as was seen in /-atory/ words which presented two possible “input” structures (§ 6.3), imply two selection constraints referring to the same lexemes, which are not crucially ranked. This would yield two equally optimal outcomes for the words in question.

For all these cases of specific lexical constraints, it is the relative frequencies of such words (Bybee’s “token frequency”) which allow these additional, non-general constraints to be maintained by the language learner in their personal constraint hierarchies. When these constraints drop down below the corresponding general constraints in the hierarchy and lose their effect, historical regularization occurs. Such cases can be seen across the lexical types, for example, *cómbative* (loss of extra consonantal mora enforced by specific selection constraint), *cýclíc* [siklik] (loss of vocalic mora), *próduce* (loss of exceptional subcategorization needed for variant *próduce*), *mélted* (loss of exceptional selection and subcategorization constraints which yielded *mólten* as the past participle of *mélt*), and *élongate*, which can also be pronounced *elóngate* with stem-level subcategorization like [*im{prégn}ate*]. A number of other /-ate/ forms once regularly surfaced showing such a subcategorization (e.g., *demónstrate*), but the number has dwindled in the past century as the specific constraints enforcing this are evidently dropping out of speakers’ constraint hierarchies, and the usual subcategorization of /-at/ for the morphological word is enforced.

Strong variant minority patterns, such as those seen in the strong verbs or in plural like *wíves*, also show strong token frequencies for members of their types. When words showing such patterns are considered frozen, the variant “input strings” could be considered to also be listed for each member of individual paradigms by high-ranking selection constraints. However, if the various forms can still be modeled by the interaction of active constraints, this can be achieved by positing prosodic constraints which outrank the (low-ranking) lexical selectional constraints, effectively governing the selection of the input string (or some of its constituent parts) by the prosodic constraints, rather than the usual, opposite directionality which selects prosodic constituents on the basis of the morphemic “input”.

Above, as well as in § 1.5, it was suggested that “input” strings should be considered simply as subcategorized morphemes, members of the morphological hierarchy, consisting of segments.¹³⁰ Above in this chapter, it was proposed that all of these morphemes are linked arbitrarily (i.e., lexically in the traditional sense) to semantic lexemes by specific selection constraints. If morphemes are defined only as subcategorized segment sequences, then morphemes of similar shape are comparable from a purely structural point of view. Thus, “input/output correspondence”, a central concept of Correspondence Theory, can be understood as correspondence between members of the prosodic and morphological hierarchies, while “output/output correspondence” involves correspondence among members of the prosodic hierarchy, and “input/input correspondence” refers to correspondence between members of the morphological hierarchy. The introduction of a third, semantic hierarchy, which is the true “input” hierarchy, reduces the “basic” status of the morphological constituents as found in Correspondence Theory, but allows for influences in both directions and subjects morphological constituents to the same limitations which prosodic constituents suffer in OT. This balances the theory, rather than emphasizing the role of the “input” string and maintaining the lexicon as a separate, vague, yet instrumental component of the grammar.

7.4.1 Identifying English morphemes

The preceding account of English grammar in Optimality terms depends heavily upon the more limited definition of morphemes as purely structural entities, lexically selected and subcategorized by selectional constraints. The identification of such morphemes by the language learner constructing a constraint hierarchy would then be straightforward. When morphemic correspondence (the identification of similar or identical segmental sequences between words) coincides with a semantic or

¹³⁰One way to conceptualize this is to imagine a series of Alignment constraints, aligning a segment to the edge of a lexeme. Ranking these constraints would produce an ordered series of segments (e.g., /cat/ is formed by Align({cat}, R;/t/, R), Align({cat}, R;/a/, R), Align({cat}, R;/k/, R), ranked in that order). A general constraint mapping lexemes to morphological root constituents would then provide the reference point between the edges of abstract lexemic categories and concrete morphological and prosodic constituents. The traditional lexicon is in effect the same thing; this simply defines such a list of ordered and classified segments in terms of OT mechanisms.

grammatical correspondence, as in traditional paradigms, the combined entity, corresponding to the traditional meaningful morpheme, may be further compared to its prosodic realization.

Nevertheless, two structurally similar forms which are not semantically or grammatically related can still be expected to show similar relationships in regard to their prosodic behavior, insofar as that prosodic behavior is not conditioned by other, semantic or grammatical, factors. Thus, when discussing morphemes in structural terms, we only need to regard their phonological instantiations. For words with virtually meaningless prefixes, like /ab-/, or prefixes that only carry identifiable meaningful content in some cases, like /re-/ or /de-/, each instance of the prefix can be regarded either as “the same” prefix morphologically, or as a series of structurally identical prefixes with different semantic denotations. The results, in terms of their prosodic behavior, will be identical, so long as all the instantiations of the affix in question subcategorize to form the same type of morphological constituent.

Similarly, less than meaningful roots like /-ceive/, appearing across words like *receive*, *conceive*, *deceive*, can be understood to either be unitary morphemes or sets of structurally identical morphemes. Since these identical morphemes undergo the same unusual stem change in the nominal abstract (i.e., *recéption*, *decéption*, *concéption*), it is quite possible (and simpler) for the system to regard the link as belonging to the structural morpheme rather than to the set of links holding the meanings and structures together. The separation of structure and meaning in this model allows for all possible patterns of combinations to be considered by the system. Thus, an investigation of the phonology of English requires searching for examples of two kinds of correspondences, purely structural ones relating the morphological forms to their prosodic structure, and paradigmatic ones, which help to suggest what single morphemes with multiple links (i.e., those found in “related” words) should look like, if they are intended to yield the attested forms through purely general constraints. In a theory where morphemes are purely structural, every case of a discrete¹³¹ structure /re/ can

¹³¹Here “discrete” does not necessarily mean concatenative, but rather an instance in which a morphological constituent /re/ presented to the constraint hierarchy could account for the prosodified surface form, regardless of whether a correspondent phonetic sequence [re] is identifiable in that surface form.

be associated with a “prefix” morpheme /re/, and additional constraints will only be necessary for the different subcategorizations (e.g., *récord* vs. *réject*) that /re/ can undergo. The subcategorization of highest type frequency will be modeled in the general constraint governing this prefix, while any other cases will need specific subcategorization constraints to be represented. As was seen in § 7.2.2, general subcategorization constraints for affixes can be so strong that during Lexicon Optimization they override normal selectional constraints and demand new, ahistorical, unparadigmatic lexical forms.

That morphemes can be regarded in purely structural terms, divorced from their meanings, should be apparent from the attested set of English words. In most complex Latinate words, the combination of morphemes is not semantically or grammatically predictable, and there is a need for overt subcategorization to account for the construction of such words. Affixes like /-al/, /-ous/, /-ary/, /-ive/ all form adjectives, for example, but the particular meanings which result, although relatable across words sharing a particular structural stem, can be idiosyncratic. The most consistent grammatical or semantic relationship between forms, the closest thing to a “paradigm” among the Latinate words, is that of verbs in /-at-e/ with nominal abstracts in /-at-ion/ and adjectival abstracts in /-at-ive/. For most Latinate words, the choice of affix needed to form a particular semantic modification to the root is completely arbitrary, the result of Latin (or Indo-European) word-construction processes no longer semantically transparent to the grammar.

The same arbitrariness sometimes extends to the choice of multiple stems. For example, the verb *compóse* is semantically relatable to the nouns *compónent* and *còmposítion*, which have related but different and structurally unpredictable meanings. Both forms also show irregular allomorphy, the former in the root, that latter in a stem extension preceding the suffix /-ion/. We similarly find the verb *propóse*, with related nouns *propónent*, *pròposítion* and *propósal*. The difference in shades of meaning seen in such cases is clearly not a function of the suffix, but is completely lexical, that is, linked to particular lexemes. The presence of the root /pos/ supplies little content to the meanings of the words, and the suffixes, although carrying some semantic content, cannot yield a consistent semantic denotation that could provide a compositionally predictable meaning. The common meaning seen in each of the two sets of words is apparently

determined by the prefixes, although these have no meaning in themselves and only present the meanings in question in combination with the meaningless root. All these cases would be conceived of herein as governed by superordinate lexemic constraints, which link lexemes to combinations or morphemes, and illustrate the clear need for separation of morphemic structure from meaning.

7.4.2 Economy in the constraint hierarchy

It has been assumed here that representing the greatest number of forms with a single general selection or subcategorization constraint, rather than with individual but virtually identical constraints, is more economical. In this sense, selection constraints choosing a unitary root to underlie multiple words are more general than selection constraints introducing idiosyncratic roots accounting for only one member of a paradigm. Of course, English contains a number of paradigms whose member forms fail to accommodate a single underlying morpheme. That some stem allomorphy should be found is not surprising in a language like English, which has borrowed not only words but entire morphological systems from a number of other languages. For example, some forms in /-ive/ and /-ion/ show irregular stem allomorphy that cannot be accounted for in prosodic terms. An example such as the pair *jóin* ~ *júnction*, which could be regarded as semantically and grammatically parallel to *inscríbe* ~ *inscríption*, cannot be precisely related in the same manner, the latter pair being distinguished only prosodically, by a moraic contrast. There is a principled historical reason for this: *jóin* is a borrowing from Old French, *joindre*, while *júnction* is purely Latin, from *junctiō*, *junctionis*. Both forms originate in Latin *jungere*, 'join', which is why both their phonological forms and meanings are similar. But *jóin* could not be consistently "generated" by any "base" form /jung-/ that one might posit for *júnction*, and it is likewise counterproductive to posit a new vowel which produces the alternation [oi] ~ [ʌ] to account for this word, for example, although this has been proposed (by Halle 1977). Other irregular cases, such as *retáin* ~ *reténtive*, *decéive* ~ *decéptive* have a similar history, while in *propél* ~ *propúlsive*, the culprit is a synchronically irregular example of Latin morphology (infinitive *pellere*, participle *pulsus*) which has been taken over in its entirety into English.

As proposed here, the appropriate way of treating these irregular forms would be through specific selection constraints introducing these idiosyncratic morphemes as “input” forms. Attempts to account for exceptional words purely through prosodic constraints would result in some words (e.g., *vanílla*) requiring prosodic constraints incompatible with more sizable sets of words. Such constraints would be impossible to propose without suggesting multiple constraint hierarchies or “cophonologies”, a tactic which would greatly weaken OT (although this has been proposed by McCarthy & Prince 1993a, Itô, Mester & Padgett 1994). The modification of lexical entries with further structure in this way may be understood as prespecification (Inkelas, Orgun & Zoll 1994). However, this term implies that other entries are not pre-specified; this is not the case, since the selection of all morphemes is prespecified in the lexical part of the grammar, and the subcategorization for “default” morphemes will be prespecified by general constraints associating a given morpheme’s segments to morphological and prosodic constituents. It is only those minority forms which fail to conform to the majority generalization that require specific constraints enforcing prespecification of prosodic and morphological structure, idiosyncratically associated to a particular lexeme.

Under Lexicon Optimization (§ 7.1), the proposed lexical entry which incurs the least violations in producing the surface form, given a known constraint hierarchy, will be stored in the lexicon. While the usual interpretation of Lexicon Optimization assumes that the morpheme in question will have a single abstract underlying form that will yield all attested surface phonetic forms (under the usual generative assumption), the representation of lexical entries as selection constraints rather than as items in a list allows for a change in the definition of Lexicon Optimization that will also admit cases of multiple stems and allomorphy. Instead of choosing a particular underlying representation, Lexicon Optimization is defined by the insertion of specific (lexical) constraints into the constraint hierarchy which account for the surface forms while incurring the fewest violations over the particular evaluation, and all evaluations in the language. Lexicon Optimization will only propose specific constraints when there is no more efficient way of accounting for a given word under the general constraint hierarchy.

At this point it is relevant to ask how this efficiency can be measured, and what the cost to the grammar would be of introducing, for example, a new vowel /i/ which surfaced as [ɪ] and alternated prosodically with [oi]? And could one then posit alternate forms of [ɛ] which alternated with /ē/ rather than /ī/? Adding additional structures must add additional complexity in the form of additional constraints. On the other hand, each irregular word adds another constraint to the hierarchy as well. To assess what mechanisms the grammar will choose to bring least complication to the constraint hierarchy, it is necessary to have an evaluation metric with which to judge complexity and economy.

7.4.3 Evaluation metrics and OT

If the lexical listing approach of Vennemann (1974) were the only theory of morphology available, the description of the grammar of English would look very much like the descriptive classification of English words seen in the Appendix. However, as in Bybee's (1985, 1995) method of listing networks of forms, while some generalizations can be captured in this way, others are missed. Approaches like Bybee's or Vennemann's focus too strongly on the semantic/morphological side of the system, while the generative approach overly focuses upon the morphological/prosodic side. In the version of OT presented here, the phenomena of "irregular" allomorphy and suppletion were accounted for by specific "lexical" constraints, constraints which are in any case necessary both for linking lexemic constituents to the morphemes which project their featural content into the prosody (Russell 1995), and for specifying the subcategorization of affixes and stems (Inkelas 1989, McCarthy & Prince 1993a).

The traditional concept of "regular" patterns would be here applied to cases in which general constraints outrank any corresponding specific, lexical constraints; such specific constraints would then have no effect, and would not need to be specifically recorded in the hierarchy. Looked at from the perspective of language acquisition (as "Lexicon Optimization"), the general constraints are defined by the most frequent plausible constrained relationships seen in the language. Under this approach, "regular" morphology involves those surface words whose prosodically optimal candidates do not violate general selection constraints, while irregular

morphology denotes the set of optimal candidates whose success was ensured only by superordinate specific selection constraints.

To explain an alternation of limited distribution by positing an additional general constraint, not necessary to account for most forms, will require evidence of a pattern in the data, rather than simple stem allomorphy. Since there does not seem to be, contrary to the usual generative assumption, a strong restriction against multiple stems, proposing a general constraint to account for only a few forms requires both positive evidence of a pattern and the lack of numerous counterexamples. The rigorous theoretical framework defined by the OT constraint hierarchy means that new general constraints inserted into the hierarchy by the grammar (or equally, by the theorist trying to formalize the grammar) may very well have a greater effect than simply accounting for the desired idiosyncratic forms. Because general OT constraints apply to every utterance in the language, the promotion of additional general constraints for the sake of handling a few unusual forms might inadvertently spoil previously valid evaluations for hundreds more, eliminating as sub-optimal the attested surface forms. Unlike rules that can be added to the grammar, marked to apply only to certain words, general constraints cannot be added just to fill in some gaps. Each applies over the entire set of all candidates, and its effects must be taken into account.

Utilizing the notion of general and specific constraints, it is possible to define an evaluation metric, and to explain why optimization of the grammar will produce general and specific constraints based on type frequency patterns. Specialized constraints applying only to a small, arbitrarily defined set of words will have to be expressed as a series of specific constraints, while true general constraints governing constituents will appear only once in the hierarchy (per constituent relationship). Representing an alternation seen across many words by specific constraints would be quite costly with regard to the number of individual constraints, but generalizing produces a single constraint which accounts for the same data. However, if an alternation cannot be properly represented by the introduction of a new general constraint into the hierarchy without incorrectly changing the evaluation of many other words, then it is far less costly, again in terms of the overall number of constraints, to provide specific constraints for the minority.

Such an evaluation metric formalizes the treatment of generalizations seen across languages and expressed in diachronic processes such as regularization. An explanation suggested above for regularization is the loss of a specific superordinate selection constraint from the hierarchy, the particular constraint being reranked below its corresponding general version and losing all independent force. This effective loss of a specific constraint from the hierarchy simplifies the grammar in a quantifiable way, by one constraint. Beyond the question of generating all the correct forms, a proposed grammar formulated entirely as a constraint hierarchy can thus be evaluated for economy according to how many relevant constraints (constraints whose violations affect the selection of any optimal candidates) are necessary. The degree of irregularity is directly represented in such a grammar by the presence of superordinate specific constraints, and the means for its regularization can be seen as the descendance (and thus effective elimination) of those specific constraints when they fall below their corresponding general constraints in ranking.

However, it is not being proposed here that the goal of the grammar is to reduce the absolute number of constraints in a language; this would lead to the same paradox as the traditional generative model, since we would expect wholesale regularization on that basis. Rather, the grammar selects the minimal constraint hierarchy *which will account for the attested forms of the language*. The goal of the language learner is to join the communicative community, not simplify its way of communicating. Specific constraints will be enforced by the lexical strength of the words which require them; only when the words in question are no longer part of the target set of attested words (due to changes in word usage and other social factors) will the constraints required to maintain them be dropped, simplifying the grammar. In the same way, for sociolinguistic reasons, new specific constraints can enter the grammar with foreign or innovated words, further complicating the grammar. From competing models of the target language, the simplest is striven towards which will accurately model the language. All such competing models must include ways of representing lexically strong irregular forms, and grammars which fail to capture such forms would be rejected due to coverage failure.

7.4.4 The universality of constraints

The issue of representing traditional lexical properties using specific constraints brings up the issue of the universality of constraints, one of the principles of Optimality Theory offered by McCarthy & Prince (1994: 3). They propose that all constraints are found in every language, and it is simply the rankings of constraints which differentiate languages. This is likely to be true in regard to the general constraints. However, very specific constraints, such as **Free-V** (Smolensky & Prince 1993: 101) or subcategorization constraints referring to particular morphemes, like **Align-um** (McCarthy & Prince 1993a: 22-4) need not be understood as appearing in every language, and proposing that they do weakens the universal claim itself, as it suggests that every possible morpheme in every possible language is overtly present in the constraint hierarchies of all languages.

Since every specific constraint is an instantiation of the corresponding general constraint, the existence of specific constraints does not contravene the universality of constraints. Specific constraints ranked below their corresponding general constraint would never have any effect and thus can be regarded as being absent from the particular language. Only those specific constraints which are more highly ranked than their general counterparts will impact upon the grammar, and will be overtly recorded in the constraint hierarchy of a given language. The constituent arguments for specific constraints are found elsewhere in the grammar and these constraints are simply the encoding of complex, lexical information required anyway by all linguistic theories, put into a form that allows them to be ranked within the constraint hierarchy and evaluated uniformly in the same manner as the general constraints. The grammar can be understood as consisting of a “vocabulary” of all the constituents found in the three hierarchies, prosodic, morphological and semantic. Relations between all members of these hierarchies can be represented in constraints, covering varying degrees of complexity, but only those relevant to producing output need to be overtly expressed in the constraint hierarchy for a given language. Of course, all constituents used in a given language will be represented in the constraint hierarchy in at least some general constraints.

The identification of specific and general constraints put forward here can be related to a proposal by Kiparsky (1994) that “each constraint (PARSE,

FILL, SPREAD, etc.) has at least two versions: one holding generally over structure of a particular type (segments, vowels, place feature, etc.) and one (or more) holding specifically of the marked structure of that type” (Inkelas 1994: 10). This effectively states that every constraint type can appear in general or specific form. Kiparsky claims further that no constraint can refer to unmarked feature values, a claim which has ramifications for his theory of markedness. Here, no such generalizations are required. OT allows for any structure which is part of the three linguistic hierarchies to appear as an argument to constraints; however, constraints which would restrict what linguists interpret to be the most “unmarked” structures are likely to be those ranked lowest in the constraint hierarchy, thus the apparent universal absence of such constraints (if this is truly the case).¹³²

This interpretation of the universality of constraints means that the universal content of any grammar will consist of the general constraint types available. Following the evaluation metric proposed above, the complexity of a given grammar can then be measured in terms of how many specific constraints are overtly listed in the constraint hierarchy. The simplest grammar for a given language will be the one which uses the least number of specific constraints to produce the correct output; the tendency to reduce the number of specific constraints will always be in conflict with the need to produce output that meets the standards of the linguistic community, and these specific constraints will be reinforced by the token frequency of the words they support. While the task of deciding precisely how to count the number of distinct specific constraints will be left to future research, it is likely that identical constraints covering a set of words, e.g., the subcategorization constraints for the /-al/ prefixes, can be grouped (as they would be in the hierarchy, in a single column, since they are not crucially ranked) as one constraint. To do otherwise would run counter to the insight that the generality of application for constraints should be maximized, which accounts for the choice of general constraints in Lexicon Optimization.

¹³²While OT allows for any configuration of constituents in its constraints, this does not predict that all configurations will be common or that they will occur at random. There are certainly tendencies for certain configurations of constraints to be very common cross-linguistically. However, OT does allow a mechanism by which very “marked” structures which nevertheless surface can be accounted for.

7.5 Some proposals for future research

It is not possible to explore here all the ramifications of the modified OT offered in this study. For example, discussion of segmental phonology has been largely avoided, not because it cannot be handled within this framework, but because it is outside the scope of this work and would demand an equivalent amount of text to do it justice. One goal of the preceding chapters was to illustrate that many long-standing assumptions about the English lexicon were incorrect, and that through a thorough comparative analysis of the data over a number of parameters, a simpler paradigm for English lexical forms could be arrived at. These forms could then be used as “input” for an Optimality analysis which would yield the correct surface forms. Another goal was to show that the OT framework could be expanded in order to robustly handle lexical exceptions through the constraint hierarchy, rather than reverting to concepts such as serial derivation, cyclic effects, cophonologies, exception marking or other devices of derivational theory, to yield a streamlined, logically coherent model of the grammar of natural languages. This final section offers some further proposals for refining OT, without carrying them out to their logical conclusions. That will be left to future work.

7.5.1 The formulation of constraints

The concept of the universality of constraints offered above provides the framework for a very constrained linguistic theory. However, the question of whether an OT system is constrained and explicit then lies in the choice of constraints. Most constraints appearing in current OT studies, like the rules, conditions and parameters which preceded them in derivational theories, are descriptive stipulations rather than formal expressions. They refer to hierarchical constituents in a number of ways, have numerous manners of incurring violations, and in some cases possess inordinate power, yet all are represented similarly, through a “nickname” and a column in a constraint hierarchy. One extremely powerful type of constraint is the negative formulation, usually symbolized by ‘*’. This deceptively simple representation could be used to restrict any constituent or combination of constituents, potentially producing linguistically impossible structures.

In this work, an attempt has been made to introduce formalized constraints where possible, using more explicit versions of McCarthy &

Prince's (1993a) Generalized Alignment, and the No-Intervening formalism of Ellison (1995) and Zoll (1996). Constraints such as **Lapse-s** and ***m**, although given short nicknames for the sake of brevity, were presented formally as well, and their violation conditions were made explicit. These formalizations refer to constituent edges, and present a template within which any constituent category could be inserted into the domain, universal or existential argument positions. It will be briefly argued here that one goal of OT should be to limit all constraints to those which may be formalized using these and similar templates, and that unstructured constraints which do not have clear violation conditions, which may result in impossible forms, or which are allotted inordinate powers, should be removed from the theory.

It is a corollary of the above argument that all relevant constraints then can be represented using formal descriptions. Alignment constraints have been used to represent iterative footing (with the arguments Foot and PrWd), extrametricality (PrWd and Syllable), cyclic effects (Foot and Morpheme categories), morphological subcategorization (Stem and Affix) and syllabification (Foot and Syllable) in studies such as McCarthy & Prince (1993ab), Cohn & McCarthy (1994). Yet, it is not possible to represent some relevant constraints with the Alignment formalism. One example of a such a necessary constraint is **FtBin**, or **Min-2** under the interpretation of Green & Kenstowicz (1995). As a stipulative, constraint, it is easy to describe what **FtBin** does, but it cannot be formalized using Generalized Alignment as presented in McCarthy & Prince (1993a), because the alignment of foot to syllable or mora using edges can always be satisfied with a single syllable or mora rather than the required two (i.e., $\text{Align}(\text{Ft}, \text{L}, \text{s}, \text{L})$ & $\text{Align}(\text{Ft}, \text{R}, \text{s}, \text{R})$ or vice versa). The definition of **FtBin** seems to require counting, a theoretical mechanism which is otherwise avoided in OT (cf. McCarthy & Prince 1993a: 8).

However, **Min-2** can be represented by using the concept of constraint domains, introduced in § 5.2.1. By limiting a No-Intervening constraint to apply across the foot alone, the situation behind **Min-2** can be not only captured, but it becomes clear why two is a significant number in this context, but there is no language which requires (for example) three moras per foot:

(7.19) **Min-2:** NI(Foot: m, m, Ft)

This states that in across a foot, no foot edge intervenes between the edge of a mora and another mora. Note that the lack of a specified edge indicates that this constraint can be satisfied by success on either edge.¹³³ This constraint requires every mora to be adjacent to one (but not necessarily more than one) mora to be satisfied, thus enforcing foot-binarity. The reason why two is the “magic” number is due to the two-dimensional nature of prosodic constituents - they only have two relevant edges. Changing the constraint by using either edge (or both edges) as arguments would have the effect of producing what Hayes (1995) calls the “unbounded foot”:

(7.20) Unbounded Foot: NI-L(Foot: m, m, Ft)
 NI-R(Foot: m, m, Ft)
 NI-LR(Foot: m, m, Ft)

All these constraints state that every mora in the foot must be adjacent to another mora within the foot on one or both edges. To minimize violations of this kind of constraint, all moras in the word will be incorporated into the foot; only the edgemost moras in the word will incur violations. (Note that the same relations could be expressed using the syllable instead of the mora, yielding bisyllabic or unbounded feet.) Formally expressing the **Min-2** constraint in this manner not only accounts for the binarity effect in an explicit way, but also accounts for the limited range of foot types (binary or unbounded) seen across natural languages. It is to be hoped that formally defining all constraints in this way will both lead to further insights of this type and eliminate analyses which weaken the theoretical robustness and credibility of OT.

7.5.2 Feature-changing and post-lexical effects

One final issue to be addressed here involves the interface between phonology and phonetics, as understood in an OT framework. The surface output used to crucially assess the identity of optimal candidates during Lexicon Optimization is the phonetic form taken by structures expressed

¹³³Alternatively, the constraint can be understood as having an edge argument (L OR R), rather than the (L AND R) seen in constraints like **Lapse-s**.

through the prosodic hierarchy. All the constituent boundaries and subconstituent members are, following McCarthy & Prince's (1995) concept of Correspondence, determined by the segmental tier, which acts as the grounding level of phonological structure. However, if the segments (or **phonemes**) which make up morphemes are regarded as morphological units, rather than at the bottom end of the prosodic hierarchy, the relation of the terminal boundaries of the prosodic and morphological hierarchies must be clarified. It will be proposed here that morphological phonemes are not identical to the constituents on the segmental tier, with which they are often conflated in the usual representations of the prosodic hierarchy. "Output" segments are to be understood as the prosodic correspondents of "input" phonemes (a morphological category), and as in usual interpretations of OT these correspondences need not be perfect.

If the prosodic hierarchy is defined as governing all speech sound output and providing, via the various levels of constituents, all of its characteristics, then the true terminal elements of the prosodic hierarchy must be phonetic features.¹³⁴ Features also terminate the morphological hierarchy, traditionally being the subconstituents of phonemes. However, it would again be a mistake to regard these features as identical in both hierarchies, and two types of feature, prosodic (phonetic) and morphological (phonological) should be understood. While the correspondences between prosodic and morphological features will be clear in many cases, morphological features will always be more abstract than their phonetic realizations, which can vary widely. Allowing again for violable faithfulness relationships between correspondents in the two hierarchies, so-called "post-lexical" effects should be accountable using a single constraint hierarchy, with no need for any "post-lexical level". Such effects would involve constraints which take prosodic constituents (such as feet, syllables, moras, margins, segments or features), and not morphological phonemes or features, as their arguments. Thus their effects will depend only upon the position of (for example) segments relative to other prosodic structures (e.g., syllables, feet), and will not take the morphological affiliations of the segment into account.

¹³⁴These can be understood as abstract elements for notational purposes, but they would directly correspond to measurable acoustic phenomena.

The account of English phonology and morphology presented above in this study illustrated how the various “level-ordering” effects suggested by derivational theories could be accounted for by subcategorization constraints and the correspondence and alignment of morphological categories with prosodic ones. The familiar distinction between levels I and II rested on differing subcategorizations between the affix and the Prosodic Word. But post-lexical effects in derivational theory are different, in that they are defined as applying to all segments, regardless of constituency. This phenomenon does not require any kind of ordering in OT, as a post-lexical level would imply, but involves recognizing what constituents are being referred to as the “output” string. The output is usually represented in terms of surface segments, but these tend to be idealized phonological segments, rather than real phonetic realizations.

While such output segments are typically treated as unitary items which can be counted and used to delineate constituent boundaries, the reality is less clear-cut, as can be seen from the word *compatible*:

- (7.21) Surface: k^həmp^hæɾəbəl
 Phonemes: kom-pat-ibil

The correspondence between the segments and surface sounds listed is certainly one-to-one (satisfying alignment constraints), and one could state which phonemes would belong to a prefix /kom-/ or a suffix /-ibil/, but the surface phones themselves do not map exactly to their underlying idealized phonemic forms. The feature [aspiration] is part of the morphological (or phonemic) structure of voiceless stops in English, but the /t/ in *compátible* has lost this feature, as well as much of its voicelessness and stop qualities, being reduced to the flap [ɾ]. The vowels /o/ and /i/ are quite distinct phonemically, but here both surface as [ə], a neutral central vowel. The issue of featural faithfulness arises in such contexts, and becomes more acute with a word like *edition*. The related form *edit* suggests this verb ends in a /t/, but neither that nor the first segment of the suffix /-ion/ is in evidence:

- (7.22) Surface: ədɪʃən
 Segments: edit-ion

Instead, a segment /š/ not appearing in the “input” has replaced both /t/ and /i/, creating a contrast between the number of segmental units in the “input” and “output” forms. These kinds of discrepancies have usually been regarded as resulting from phonological and post-lexical sound changes. While they can be accounted for by the selected parsing of certain segmental features due to various OT constraints, the issue of how this might affect the delineation of super-segmental constituents remains.

Reinterpreting the OT “output” string as an ordered set of prosodic segments containing phonetic features allows for a more flexible understanding of surface sound patterns. The prosodic segments may correspond to the morphological phonemes, but can dominate prosodic feature matrices which may not possess correspondents for all morphological features. In this way, the output form of prosodic segments may differ dramatically in terms of which phonetic features are present as correspondents to the morphological features, and what those correspondents are, but it may still be completely faithful on the segmental level. Conversely, segments may be unfaithful in correspondence, but all of their features might be faithful.

A diagram of the type of hierarchical structure which can account for both “feature-changing” and “postlexical” phonological effects proposed here could appear as follows, the example below being the noun *fly*:

(7.23)	Ft	Prosodic Foot
	s	Prosodic Syllable
	\	
	Ons m m	Prosodic Nodes
	/	
	X X X X	Prosodic Segments
	[][][][]	Prosodic Features
	{ } { } { }	Morphological Features
	f l i	Morphological Phonemes
	/ /	Morphological Root
	{ }	Morphological Stem

The role of the prosodic features can be exemplified by the case of vowel quality in the various English dialects. Surface vowels in all dialects are always different from the underlying forms proposed, e.g., morphological /e~ē/ surfaces in American English as [e~ij]. In other dialects, such as British, Australian or South African English, the realization of vowel quality (and thus the precise phonetic correspondents) for these phonemes differs widely, but all dialects show faithful correspondence between segments and some set of features. It is simply the identity of the prosodic features which varies. The uniformity of morphological features (in most cases) across these dialects can be proposed as the reason why speakers of these different dialects can for the most part understand each other, once the corresponding prosodic and morphological features and segments are identified.

The two-dimensional, unitary nature of segment-level constituents allow single phoneme or segment edges to be defined, while the multiple, unordered nature of features allows for the incomplete parsing of some or all features into the prosodic hierarchy. Thus, postlexical effects like flapping, vowel reduction and the like can be represented on the prosodic hierarchy in the same parallel prosodic representation without affecting constraint arguments referring to the unitary segments or phonemes themselves. For example, the prosodic segment flap [D] can correspond to a morphological phoneme /t/. For constraints with morphological arguments this segment will act like a /t/, while for those governing prosodic arguments, it would be a flap. Flap will appear when some constraint associates the prosodic segment flap with morphological /t/, while the aspirated stop segment [t^h] and the glottal stop segment [ʔ] will be associated with /t/ by different constraints which reference different constituent environments. Under this interpretation neither flap, glottal stop nor aspirates are phonemes of English, present in lexical selection constraints, but they *are* segments, associations of prosodic features, which may correspond to the phoneme /t/.¹³⁵

In this model, the only difference between lexical and post-lexical sound changes would be that the former will be governed by constraints

¹³⁵Similar to morphemes, phonemes would be defined by a series of constraints which place the relevant morphological features within phonemes. One such constraint has been illustrated above, as **Voiceless-f** (7.14). Phonemes themselves are just labeled morphological constituents; it is the feature content of the phonemes which influences their segmental and prosodic behavior.

using higher morphological categories than the phoneme as arguments. The entire relationship between the “underlying” morphological structure and the phonetic surface form can thus be represented in a single representation arrived at via a single parallel process, the optimal evaluation of the constraint hierarchy.

7.5.3 Conclusion

The representation of the lexicon via the constraint hierarchy allows for a completely unified system based on a single mechanism, constraint evaluation. This mechanism provides an evaluation metric for proposed grammars and a uniform framework for representing all kinds of linguistic relationships. It can model diachronic relationships of regularization (through constraint shift up and down the ranked hierarchy), irregularity, regularity, type and token frequency, majority and minority patterning, and language acquisition. The proposed limitation of constraints to formally defined expressions further limits the mechanisms available to the grammar to constituent members of the three constraint hierarchies. The model of language offered here is maximally constrained in that only prosodic, morphological and semantic constituents, formally expressed in the constraint hierarchy, may participate in the formal description of the grammar. Using such a constrained system, the strengths and weaknesses of a grammar proposed to account for a particular language may be accurately measured according to its success in producing all the attested output forms using the least number of relevant specific constraints. It is hoped that these proposals will be of interest to those doing further work in this field.

Appendix - Stress patterns in suffixed words

A.1 The computational study and data

This appendix presents some of the results of the computational study which underlies this research, focusing on so-called level one suffixation in English. The goal of the study was to correlate the relationship between stress and syllable weight across various morphologically-defined word types. The undertaking of an extensive new study of the corpus of English was a conscious decision not to use the data sets which have appeared in works on English phonology since Chomsky & Halle (1968), but to investigate anew the status of those words, and look at the general patterns seen across the entire lexicon. The data used in the study derives primarily from a specific machine-readable corpus, the Celex Lexical Database of English developed at the Max Planck Institute in Nijmegen (Baayen, Piepenbrock & van Rijn 1993). This list contains 52,446 lemmas (words regarded as basic forms, subsuming inflectional and other variants) derived from 160,594 word-forms drawn from the COBUILD corpus of the University of Birmingham. This database should be considered large enough to serve as a reasonable model of the mental database of the native speaker, and includes all words that occur with any frequency in Modern English. This database contains pronunciation, syntactic and morphological and word frequency information. The pronunciations are British; most of these correspond phonologically to American pronunciations; when they clearly do not, as in the case of words in /-atory/ and a few other sets, distributions for American English pronunciations have been calculated separately, based on the Pronlex dictionary distributed by the Linguistic Data Consortium (Comlex 1995).

To process these databases, I designed and programmed a number of tools written in Prolog and implemented on a Macintosh PowerBook. The first tool was designed to convert wordlists of any shape, in any language, into a Prolog database consisting of structured objects. This flexible design was especially important as various corpora were used at different stages of the research, until the Celex corpus was settled upon. The second tool allowed for complex searches of such converted databases according to morphological and phonological criteria including stress and syllable weight patterns, and the generation of contrastive tables organized around the various patterns that were encountered. Thus, using the front-end

application, I could collect the set of all words suffixed in e.g., /-al/ or /-ent/, the set of all words with three syllables, or words with certain orthographic sequences. For such subsets, I could then create tables outlining the various stress or weight patterns presented, or the morphological structures, complete with information about the percentages encountered and the frequencies of the words. The program also allowed me to view the words in the chosen subsets, create new subsets based on that information, and combine or and contrast various subsets. Information drawn from those tables will be presented below.

In terms of numerical details, homonymous words with different syntactic categories are counted as different words; variant pronunciations of a word have been counted individually only when the contrast is relevant to the feature being compared. Thus the variant pronunciations *combátive* and *cómbative* will only count as two entries when stress pattern is being compared; when syllable weight patterns are being compared both will be regarded as part of the same entry. Most of the percentages cited below are calculated on the basis of part to the whole, and indicate what percentage of a certain subset of the lexicon contains a certain feature, regardless of the frequencies of individual lexemes. The pronunciation data from the corpus was normalized, since the Celex database presented a mixture of phonetic (post-lexical) and phonological detail. The phonetic detail was abstracted to arrive at a purely phonological interpretation for the entries. Morphological information was also normalized, as the morphological theory evidently used by the Celex compilers reflected only one of the many possible approaches to English morphology. The data presented in these tables is meant to be theory-neutral as far as possible, and is based solely on the segmental and stress information from the corpus. None of the theoretical conclusions reached in the main body of this study have been impressed upon the data here; this data rather informed those conclusions.

A.2 Presentation of the data

The transcriptional conventions used here are designed to simplify the presentation of the data. Data appearing in brackets represents the phonetic representation found in the Celex corpus. This representation, similar to traditional dictionary pronunciations, will be replaced in the tables by a quasi-phonemic representation for a number of reasons. One is the contrast between what are certainly corresponding phonemes (e.g., /əɹ/: [ø:]~[ɹ]) found in British and American pronunciations. Another is the presentational simplification of

presenting vowels and diphthongs that participate in the vowel quantity alternations that this investigation focuses upon with the same symbols, distinguished solely by length markings. The system used is as follows :

(A.1)

/a/	=	Br. [æ, a], Am. [æ]	/ā/	=	[eɪ]
/e/	=	[ɛ]	/ē/	=	[iɪ]
/i/	=	[ɪ]	/ī/	=	[aɪ]
/o/	=	[ɑ]	/ō/	=	Br. [əʊ], Am. [oʊ]
/u/	=	[ʌ]	/ū/	=	[juu]
/ʊ/	=	[ʊ]	/uʊ/	=	[uʊ]
/ə/	=	[ə]	/aʊ/	=	[aʊ]
/or/	=	Br. [ɔ:], Am. [ɔr]	/oʊ/	=	Br. [ɔ:], Am. [ɔʊ]
/ər/	=	Br. [ə:], Am. [r]	/oɪ/	=	[ɔɪ]
/ar/	=	Br. [ɑ:], Am. [ar]			

The vowels seen in the left column will be generally described as “short” while those on the right will be described as “long”. The variation between British [æ] and [ɑ] was considered to be for the most part conditioned by the following consonant, and these are merged in the notation as /a/, which behaves phonologically like a short vowel in all cases despite the phonetic length of the British back variant, e.g. *pást* as [pá:st]. The convention /ū/ for the triphthong [juu] has been chosen for its stressed instantiation despite the fact that this phoneme does not always appear as (or behave phonologically as) a long vowel, and when reduced surfaces as [jə]. The unstressed form of this triphthong will be written as /u/. In general, /ə/ shows up in the quasi-phonemic notation to indicate a reduced vowel. Syllabic sonorants are also indicated with a leading /ə/, e.g., /əl/, /ər/, /ən/, /əm/. Primary and secondary stresses are marked with the acute and breve as usual.

Morpheme boundaries will be noted inside the quasi-phonemic notation with hyphens, e.g., *dènotátion* /dè-nōt-áč-ən/. Stress patterns will be described using a bracketed list, e.g., [2,0,1-0]. Only primary, secondary and zero stress will be indicated in this descriptive formalism, which in this case indicates a word with penultimate stress and initial secondary stress. The hyphen will be used to indicate the suffix boundary, as many patterns within stems remain consistent, regardless of the weight of the suffix. In most tables, example forms are given simply with

orthography, long vowels being marked with the macron, and stresses marked with the breve and acute, as in the main chapters. The same conventions outlined above apply to the marking of long vowels in orthography. The remainder of the appendix is a presentation of the distribution of stress patterns in the major suffix classes amongst the Latinate or “level one” suffixes. The classifications are based on a combination of stress placement and morphological criteria such as the behavior of prefixes and roots in combination with the suffixes in question. Information on the distribution of stress patterns in unsuffixed words has been presented in chapter three of the main text.

A.3 Suffixes like /-al/

The suffixes grouped here include the monosyllabic suffixes /-al/, /-or/, /-ive/, /-ous/, /-us/, /-um/, /-ar/, and /-an/. All are vowel-initial, and the suffix vowel is always reduced. The underlying vocalism of some of these suffixes may be inferred from the following forms which show further suffixation:

(A.2)	tonálicity	/tōn-ál-iti/	formálicity	/form-ál-iti/
	tutórial	/tūt-ōr-i-əl/	èditorial	/èdit-ōr-i-əl/
	lùminócity	/lūmin-ós-iti/	pompócity	/pomp-ós-iti/
	àjectíval	/ājikt-ív-əl/	sùstantíval	/sùbstənt-ív-əl/ ¹³⁶
	pùritánical	/pūrit-án-ik-əl/		

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0] and are about evenly divided between heavy (CVC-) and long-vowel (CVV-) roots. A minority shows a light stem (CV-):¹³⁷

(A.3)	CVV-	43%	tónal	/tōn-əl/	lócal	/lōc-əl/
			rázor	/rāz-ər/	mótor	/mōt-ər/
			fámous	/fām-əs/	héinous	/hān-əs/
			plósive	/plōs-iv/	mótive	/mōt-iv/
			lócus	/lōk-əs/	mínus	/mín-əs/

¹³⁶But note that these examples for /-ive/ show an irregular stress pattern and are of uncertain structure.

¹³⁷Hereon, the initial **C** of syllable structures like **CVCC-** are understood to stand for any syllable onset. The final cluster of the root, which forms the onset of the suffix syllable, is likewise not included in the representation of the root syllable.

CVC-	44%	sérum	/sé-rəm/	dátum	/dát-əm/
		róman	/róm-ən/	págan	/pág-ən/
		fórmal	/fórm-əl/	fráctal	/frákt-əl/
		véndor	/vénd-ər/	cáptor	/kápt-ər/
		nérvous	/nérv-ər/	víscous	/vísk-əs/
		áctive	/ákt-iv/	cáptive	/kápt-iv/
		díscus	/dísk-əs/	cámpus	/kámp-əs/
		díctum	/díkt-əm/	quántum	/kwónt-əm/
CV-	13%	gérman	/jérm-ən/	úrban	/órb-ən/
		mámmal	/mám-əl/	érror	/ér-ər/
		zéalous	/zél-əs/	férrous	/fér-əs/
		pássive	/pás-iv/	mássive	/más-iv/
		phállus	/fál-əs/	státus (alt.)	/stát-əs/
		véllum	/vél-əm/	álum	/ál-əm/

About 13% of the forms show a short light syllable in the stressed foot, contrary to expectations about foot structure. A number of these forms have long variants, e.g., *státus* /stát-əs/ ~ /stăt-əs/, *lócus* /lók-əs/ ~ /lók-əs/, *dátum* /dát-əm/ ~ /dăt-əm/.

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

(A.4) [1,0-0] 45%

[0,1-0] 55%

[1,0-0]

CVV-CV-	14%	séasonal	/séʒən-əl/	fúneral	/fúnər-əl/
		áuditor	/óʊdɪt-ər/	cóunselor	/káunsəl-ər/
		dángerous	/dǎŋjər-əs/	lúminous	/lúmin-əs/
		sécretive	/sékrət-iv/	nútritive	/nútrit-iv/
		úterus	/útər-əs/	túmulus	/túmɪl-əs/
		stádium	/stádi-əm/	ópium	/ópi-əm/
		púritan	/púrit-ən/	lútheran	/lújər-ən/
CVC-CV-	27%	órbital	/órbɪt-əl/	cárdinal	/kárdɪn-əl/
		cónqueror	/kónkər-ər/	émperor	/émpər-ər/
		cáncerous	/kánsər-əs/	déxterous	/dékstər-əs/
		púrposive	/pérpəs-iv/	sénsitive	/sénsɪt-iv/

CV-CV-	59%	cálculus	/kálkjəl-əs/	íncubus	/ínkjəb-əs/
		máximum	/máksim-əm/	péndulum	/péndjəl-əm/
		méxican	/méksik-ən/	pártisan	/pártiz-ən/
		pívotal	/pívət-əl/	nóminal	/nómin-əl/
		éditor	/édit-ər/	mónitor	/mónit-ər/
		vénomous	/vénəm-əs/	óminous	/ómin-əs/
		pósitive	/pózit-iv/	ádditive	/ádit-iv/
		stímulus	/stímjəl-əs/	tétanus	/tétən-əs/
		plátinum	/plátin-əm/	bárium	/mónit-ər/
		áfrikan	/áfrik-ən/	véteran	/vétər-ən/

There are also a few words (less than 1%) with unexpectedly heavy but unstressed penults:

(A.5)	CVC-CVC-	súbstantive	/súbstənt-iv/
	CV-CVC-	góvernør	/gúvərn-ər/
		ádjektive	/ájəkt-iv/
		cávernous	/kávərn-əs/

Of the second, slightly more numerous pattern with penultimate stress, a large majority of these forms show heavy penultimates. These stems are either bisyllabic morphemes with final heavy syllables, or prefixed monosyllables:

(A.6)

[0,1-0]

s-CVV-	41%	cathédral	propósal	equátor	advísor
		explósize	desírous	cohésive	prognáthous
		papýrus	detrítus	factótum	angólan
s-CVC-	53%	triúmphal	revérsal	precúrsor	detéctor
		moméntous		horréndous	
		vindíctive	defénsive	olýmpus	prospéctus
		ugándan	subúrban	moméntum	addéndum
s-CV-	16%	colóssal	rebúttal	proféssor	abéttor
		opréssive	sucéssive	collóssus	narcíssus
		flagéllum	errátum	tibétan	moróccan

The general stress distribution noted above is paralleled by the longer forms. Again, words with light penults show the final stress pattern [1,0-0] while words with heavy penults show [0,1-0]:

(A.7) Trisyllabic stem forms:

[0,1,0-0] 52%

s-CVV-CV-	levíathan		
s-CVC-CV-	impépersonal	prepósterous	impérvious
s-CV-CV-	oríiginal	medícinal	ambássador
	compétitor	carnívorous	constítutive
	compétitive	eméritus	esóphagus
	alúminum	currículum	américan

[2,0,1,0] 37%

è-s-CVV-	ànecdótal	ìntertríbal	rèappráisal
	lòcomórive	ìndecísive	ìgnorámus
	àrborétum	pròtozóan	
è-s-CVC-	bènefàctor (alt.)	ìntercéptor	àmbidéxtrous
	sèmicónscious	ìnexpénsive	èucalýptus
	ìnterrégnum	rèferéndum	mànufàcture
è-s-CV-	rètrogressive	strèptocóccus	

[1,0,2,0] 11%

è-s-CVV-	súpervìsor	àrchetýpal
è-s-CVC-	chíropràctor	bènefàctor
è-s-CV-	prédecèssor	

[1,0,0,0] 0 %

è-s-CV-	spíritual
---------	-----------

Four syllables:

[2,0,1,0,0] 91%

è-s-CVV-CV-	sùbcutáneous		
è-s-CVC-CV-	ìnterpépersonal	phòtosénsitive	
è-s-CV-CV-	àrithmétiqueal	àboríiginal	ìnsectívorous
	sùpercílious	ìnterlócutor	prìmogéniture
	tìnnabulum	hìppopótamus	còsmopólitán

[0,2,0,1,0] 5%

s-è-CVV-	insècticídal	tyrànnosáurus	elìzabéthan
----------	--------------	---------------	-------------

s-è-CV-		desiderátum		
[2,0,0,1,0]	4%	tèperaméntal	ràdioáctíve	sèmicondúctor
		pèritonéum		
[1,0,0,2,0]	0%	sémicondùctor (alt.)		
<u>Five syllables:</u>	[2,0,0,1,0,0]	ìnfinitésimal	héterogéneous	
		àntedilúvian	vèterinàrian	
	[0,2,0,1,0,0]	ecclèsiástical		
	[2,0,2,0,1,0]	ìntercòntinéntal		

As the number of syllables in the word increases, instances of penultimate stress drop dramatically. Most examples are actually cases of multiple suffixation such as /-ent-al/ or /-ment-al/, such as *environméntal*, *intercòntinéntal*. Secondary stresses usually appear on alternating syllables to the left of the main stress. Words which fail to conform to this generalization are infrequent and tend to be morphologically complex.

A.4 Suffixes like /-ent/

The members of this group, which includes /-ent/, /-ant/, /-ence/, /-ance/, /-ency/ and /-ancy/, show a stress pattern that is identical with that of the preceding group. That is, words with heavy penults receive penultimate stress, while words with light penults show antepenultimate stress. The underlying segmental forms of the suffixes, which tend to surface as /-ənt/ or /-əns(i)/, are revealed by further suffixation, e.g., *còntinéntal*, *prèsidéntial*. The principal difference between the two sets, which suggests a separate grouping, is morphological. Whereas in the /-al/ grouping monosyllabic long vowel stems retain their form when prefixed, e.g., *revíve* ~ *revíval*, many stems in the /-ent/ group, with apparently similar structure, fail to show the stressed long vowel seen in the unsuffixed form, and instead display a bisyllabic foot with short vowels, e.g., *resíde* ~ *résident*. This alternation is not seen with heavy stems, e.g., *respónd* ~ *respóndent*.

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0] and are divided between heavy and long-vowel forms, as well as a sizable minority of light, short stem forms:

(A.8)	CVV-	35%	cláímant	mígrant	stúdent	pótent
			guídance	crédence	régency	vácancy
	CVC-	46%	sérvant	sólvent	sentence	prégnancy
	CV-	18%	érrant	pléasant	présent	pénance
			éssence	clémency	cúrrency	

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

[1,0-0] 41%

[0,1-0] 58%

Of the first set, both prefixed and unprefixed stems are seen, in contrast to the pattern seen in the /-al/ group. Most stem forms show a final light syllable:

(A.9)

[1,0-0]

CVV-CV-	7%	véhemence	cóuntenance	précedence (Br.)
CVC-CV-	33%	áspirant	ábstinent	sómnolence
		órdinance	méndicant	túrbulent
		cómpetency	córpulency	túrbulency
CV-CV-	59%	ínnocent	óccupant	hésitance
		résident	ássonance	réticence
		hésitancy	vírulancy	résidency

Of the second pattern with penultimate stress, the vast majority of these forms show heavy penults. These stems are either prefixed heavy monosyllables or, much less commonly, bisyllabic stems with heavy second syllables:

(A.10)

[0,1-0]

s-CVV-	29%	obéissance	compláinant	impédecence
		compónt	clairvóyance	applíance
		malféasance	excítant	accóuntancy
		infréquency	cohérecy	
s-CVC-	50%	triúmphant	deféndant	abúndance
		transcéndent	convérgence	astríngency
		contíngency	insólvcy	ascéndancy

		malígnancy		
s-CV-	21 %	admíttance	abhórrent	fluoréscent
		propéllant	recurrence	inclémency
		exígency	transpárecy (alt.)	discrépency

There are a number of forms here, such as *compláinant*, which are obviously prefixed long-vowel monosyllables and thus violate the generalization applied above to the /-ent/ group. Many of these show unusual allomorphy in their paradigms, for example:

(A.11) compôse~compônent survéy~survéillance obéy~obéisant

A few words allow for both realizations, e.g., *súbsident*, *subsídent*, *áspirant*, *aspírant*. Many others are words with open syllables in the penult, such as *convéyance*, *applíance*. Of the forms that show the unexpectedly stressed light penult, most prefixed cases correlate with verb forms that show this light stressed stem, e.g., *recúr*, *recúrrent*, *admít*, *admíttance*, *propél*, *propéllant*.

In the longer words, forms with light penults usually show the stem stress pattern [1,0-] while words with heavy penults show [0,1-]. Multiply prefixed stems show the same tendency to reduce long-vowel root monosyllables; many [0,1,0-0] forms are prefixed bisyllabic stems:

(A.12)

Trisyllabic forms:

[0,1,0,0]	52%	protúbérance	commúnicant	coíncidence
		recógnizance	incómpetent	ambívalent (alt.)
		bellígerent	omnípotent	signíficance
		bellígerency	omnípotency	irrélévancy
		incómpetancy	resíliency	
[2,0,1,0]	48%	àntecédence	ìncohérence	dìsinféctant
		còrrespóndent	èquidístant	jùrisprúdençe
		òmniprésent	ìncandéscence	àmbiváìlent (alt.)
		ìntermíttent	còrrespóndency	ìneffíciency

Four syllables:

[2,0,1,0,0]	44%	dìsinhéritance	ìnsigníficance
[2,0,0,1,0]	56%	sùperabúndant	sùperinténdent

The high percentage of forms with the pattern [2,0,0,1,0] is due to the presence of the prefix /super-/, which takes stress on its initial syllable regardless of the stress seen in the stem.

A.5 Suffixes like /-ic/

This group is of central interest to a study about vowel length alternation in English, because the suffixes to be listed here are intimately involved in two of the principal length-based alternations: vowel shortening (Myers 1987) and CiV lengthening (Chomsky & Halle 1968). The suffixes grouped here include the monosyllabic suffixes /-ic/, /-ion/ and /-ule/, and the bisyllabic suffixes /-ity/, /-ible/, /-ify/. To these may be added the i-extended suffixes, /-i-al/, /-i-or/, /-i-ous/, /-i-us/, /-i-um/, /-i-ar/, /-i-an/, /-i-ent/, /-i-ence/, /-i-ant/, /-i-ance/, and the combination /-ul-ar/. All are vowel-initial. Words in this group almost all take their main stress on syllable preceding the suffix. In most cases this places stress on the antepenult, except where the suffix is monosyllabic and the stress then falls on the penult. The initial suffix syllable is always light, making the surface stress pattern of these words similar to that of the above groups.

A.5.1 Shortening suffixes

The “shortening” suffixes placed in this group are /-ic/, /-ity/, /-id/, /-ish/, /-ible/, /-ify/, /-ule/, and its extension /-ul-ar/. All begin with high vowels. Words belonging to this set overwhelmingly show stress on the syllable preceding the suffix. Among bisyllabic stems, a very small amount, around 1% of the words involved, show antepenultimate stress; most of these are /-ic/ forms :

(A.13)

cátholic	árabic	chóleric	bíshopric	rhétoric
lúnatic	héretic	ársenic	pólitic	túrmeric
éligible	négligible	intélligible	córrigible	impóverish
mólecule	réticule	mínuscule	véstibule	rídicule
aríthmetic (noun)		ántarctic (Br. alt.)		Hómeric (Br.)

Most of these stems tend to show regular stress patterns when they are further suffixed, e.g., *polítical*, *herétical*, *cholérical*, *rhétorical*, *àrithmétique*, *moléculaire*, *ridículous*. None of the longer stems show this exceptional pattern.

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0(0)] and are divided between heavy, long-vowel and short-vowel forms:

(A.14)

CVV-	19%	scénic	seísmic	phóbic	próbité (alt.)
		déity	crúidity	féasible	áudible
		rísible	crúcible	únify	crúcify
		nótify	códify (alt.)	glórfify	túbular
CVC-	33%	céntric	tóxic	étnic	dígnity
		dénsity	láxity	sénsible	fléxible
		tángible	fórtify	mágnify	fálsify
		fúrnish	vánquish	ámpule	cápsule
		vírgule	pústule	ínsular	círcular
CV-	48%	cónic	tónic	mímic	quálicity
		vánity	sánity	crédible	vísible
		édible	módify	vílify	týpify
		clárfify	chérish	vánish	glóbule
		nódule	gránule	pópular	régular
		tábular	títular		

Here, while many words with long vowel stems in their “bare” forms do show a short vowel, e.g., *cóne~cónic*, *tóne~tónic*, a sizable minority of about 20% retain their long vowel, e.g., *báse* /bás/ ~ *básic* /bás-ik/, *scéne* /sén/ ~ *scénic* /sén-ik/, although some show regularized alternants like /sén-ik/, *cýclic* /síkl-ik/ ~ /síkl-ik/.

Words showing two syllables preceding the suffix overwhelmingly show the identical pre-suffixal stress pattern, apart from the few exceptions noted above.

(A.15)

[0,1-0(0)]

s-CVV-	8%	aphásic	phonémic	acóustic	stratéic
		gratúity	obésity	credúlicity	aménity (alt.)
		impúity	deféasible	dedúcible	redúcible
s-CVC-	38%	romántic	iámbic	semántic	ecléctic
		indémnity	divérsity	etéernity	modérnity
		rotúndity	profúity	divérsify	ídentify
		exémplify	eléctrfify	indémnify	contéptible
		convértible	convíncible	submérsible	accómplish
		extínguish	avúncular		

s-CV-	54%	demónic	robótic	errátic	duplicité
		necéssity	longévity	sonórité	docilité
		polárité	fatálité	pompósite	demýstify
		beátify	humídify	persónify	transmógrify
		solídify	syllábify	admíssible	compátible
		illégible	divísible	astónish	demólish
		binócular	spectácular	vernácular	

Many of the forms with long penults show hiatus, e.g., *prosáic*, *trocháic*, but a few are notable because their unsuffixed stems did not show a long vowel, for example, *stratéigic* /strətég-ik/ ~ *strátegy* /strátəji/, *systémic* /sistém-ik/ ~ *sýstem* /sístəm/.

For the trisyllabic stem forms, the distribution is even more skewed towards light syllable stems:

(A.16)

[2,0,1-0(0)]

è-s-CVV-	9%	ànalgésic	pàraplégic	òrthopédic	pròmiscúity
		òpportúnity	sùperflúity	àmbigúity	spòntanéity
		ìnsecúrité	còntinúity	ìndeféasible	ìnexháustible
è-s-CVC-	18%	hìerárchic	dialéctic	hýpodérmic	ùniversité
		tàcitúrnity	ùnifórmity		cònfraternité
		nònconfórmity	ìnestrúctible		còmprehénsible
è-s-CV-	73%	àutomátic	pàtriótic	èconómic	èquanímité
		sòlidárité	èlastícité	vòlatílité	sèrendípité
		ìnfidélité	ànonýmity	ìndivísible	cònsanguínité
		dìsestáblish	ìnexpresible	ìncompátible	pèrpendícular

Words of more syllables all show penultimate stress, but vary in the placement of the secondary stress. Many are multiply suffixed:

(A.17) Four syllables:

[2,0,0,1,0(0)]	37%	àgoraphóbic	pèripatétic	pòlytheístic
		chàracterístic	hòmodenéity	sìmultanéity
		dìscontinúity	vàriabílité	mànageabílité
		sèparabílité	nàvigabílité	pràcticabílité
		ìnstrumentálité	ùniversálité	spìrituálité

	còntradistínguish		
[0,2,0,1,0(0)] 63%	enthùsiástic	kalèidoscópic	hermàphrodític
	advèsabíltiy	excítabíltiy	accèptabíltiy
	predìctabíltiy	orìgináltiy	convèntionáltiy
	famìliáritiy	particuláritiy	convèrtibíltiy
	inflèxibíltiy	perfèttabíltiy	incòntrovértible
	incòmprehénsible		

For multiply suffixed forms, the difference in stress appears to often be morphologically influenced, with stress falling on the initial syllable of multisyllabic prefixes, e.g., /pèri-pàtét-ik/, /pòli-Jē-íst-ik/ vs. /en-Jùsi-ást-ik/, /dē-tèrmin-íst-ik/, /pàtèrn-əl-íst-ik/, although this does not always hold, e.g., *monópolistic* /monò-pol-íst-ik/.

The few words with five syllables preceding the suffix are multiply suffixed forms in /-ity/:

(A.18)

[2,0,0,0,1,0,0]:

/-ive/ + /-ity/: ràdioactívtiy

[0,2,0,0,1,0,0]:

/-able/ + /-ity/: inèducabíltiy incòmparabíltiy invìolabíltiy
insèparabíltiy

/-ible/ + /-ity/: incòrrigibíltiy inèligibíltiy

[2,0,2,0,1,0,0]:

/-able/ + /-ity/: ìnterchàngèabíltiy ìnapplicabíltiy

/-al/ + /-ity/: hòmosèxuáltiy cònfidèntiáltiy

/-ible/ + /-ity/: còmprehènsibíltiy ìndivìsibíltiy ìrrespònsibíltiy

There are a few longer words, such as *incònsequentiáltiy* and *hèterosèxuáltiy*.

A.5.2 /-ion/ and the CiV lengthening suffixes

Words of this group also take stress on the syllable preceding the suffix, and are further characterized by hiatus within the suffixal material. The frequent suffix /-ion/ shares this categorization with a set of suffixes typified by a glide /-i-/ extended by /-al/-type suffixes. Consequently, the environment for Chomsky & Halle's (1968) rule of CiV lengthening is created, and most low and mid vowels in open syllables do in fact appear long in this position. The suffix /-ion/ rarely

appears unextended, (e.g., *région*, *medállion*), and the vast majority of these forms are nominal abstracts in /-t-ion/ or /-s-ion/, formed from the set of Latinate verbs which made their adjectives in /-ive/, which was often similarly extended to /-t-ive/ or /-s-ive/. Only a few verbs, like *opíne* and *rebél*, form their verbal abstract by directly suffixing /-ion/ to form *opínion* and *rebéllion*. Forms in /-at-ion/ are treated separately (§ A.8.1).

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0] and are divided between heavy, long-vowel and short-vowel forms: (A.19)

CVV-	22%	région	nátion	mótion	úunion	áuuction
CVC-	52%	scánsion	súction	áction	sánction	
CV-	26%	pínion	stállion	pássion	rátion (alt.)	

All words showing two syllables preceding the suffix show the identical penultimate stress pattern. Among the bisyllabics all three syllable types are found in the stressed penult. Most stem forms are prefixed monosyllables:

(A.20)

s-CVV-	36%	cohésion	emótion	complétion		
		inflátion	perhéllion	eróssion		
s-CVC-	48%	projéction	diménsion	compúnction		
		concéption	recúrision			
s-CV-	28%	batállion	vóllition	audítion	contrítion	opínion
		compánion	frúition	regréssion	rebéllion	

For trisyllabic stems, all forms show penultimate stress with the pattern [2,0,1,0]:

(A.21)

š-s-CVV-	48%	dìsillússion	dìstribússion	rèssolússion
š-s-CVC-	22%	màledíction	gènufléction	cìrcumscripción
		rètropséction	sàtissáfaction	rèprodúction
š-s-CV-	29%	àbolítion	àdmonítion	dècompréssion
		rètroggréssion	cìrcumcísion	

Words of more syllables all show penultimate stress. They vary in the placement of the secondary stress:

(A.22)

Four syllables:

[2,0,0,1,0]	cìrcumlocútion	ìntercommúñion	rèdistribútion
	còntradistínction	ànimadvérñion	ìnterposítion
[0,2,0,1,0]	incòmprehénsion	elèctrocútion	derèquisítion
	irrèsolútion		

Most of these forms can be characterized as prefixed forms, and the difference in stress appears to often be morphologically influenced, multisyllabic prefixes taking stress on their initial syllables. Some words show alternants with both kinds of stress, e.g., *dissatisfáction*, *dissàtisfáction*. The words in /-i-/ plus /-al/-type suffix show parallel stress patterns.

A.6 The bisyllabic suffixes /-ary/, /-ory/

In American English these suffixes surface as [-èri], [-òri] when stress does not fall on the immediately preceding syllable, and as [-əri] when stress occurs immediately before the suffix. In British English both suffixes appear as [-əri], which is homophonous with the stress-neutral suffix /-ery/. When suffixed, however, the quality of the underlying vowel is expressed:

(A.23)

àuditórium	/ð̣dit-óri-əm/	victórious	/vikt-óri-əs/
àntiquárian	/àntiḳ-ári-ən/	àdversárial	/àdvərs-ári-əl/

When stress patterns are cited, the American pattern with the secondary stress noted will be used. The corresponding British pattern is identical, but lacks the secondary stress on the suffix.

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0,0] and are, like the monosyllabic-suffix forms above, primarily divided between heavy and long-vowel forms:

(A.24)

CVV-	20%	príory	rótary	óvary	
CVC-	35%	réctory	sénsory	quándary	búrglary
CV-	45%	mémory	mámmary	glóssary	gránary

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

[1,0,2,0] 65%

[0,1,0,0] 35%

Of the first set, the stems are mostly unprefixed and most show a final light syllable:

(A.25)

[1,0,2,0]

CVV-CV-	15%	áuditòry	díetàry		
CVC-CV-	26%	tránsitòry	ántiquàry	árbitràry	éxcretòry
CV-CV-	48%	státutòry	cátègòry	plánetàry	litéràry

However, there are a number of cases in which the antepenult is heavy but unstressed:

(A.26)

désultòry	prómontòry	répertòry	óffertòry
ádversàry	mómentàry	cómmèntàry	
vóluntàry	légendàry	sécondàry	
prébendàry	frágmentàry	sédentàry	

These make up 11% of the initially stressed stems.

Of the second pattern with antepenultimate stress, a majority of these forms show heavy antepenults. These stems are either bisyllabic morphemes with final heavy syllables, or prefixed monosyllables:

(A.27)

[0,1,0,0]

s-CVV-	37%	provísory	delúsory	centénary (Br.)	salívary (Br.)
s-CVC-	47%	olfáctory	diréctory	infírmory	exémplary
		frágmentary (Br. alt.)			
s-CV-	15%	accéssory	promíssory (Br.)	centénary (Am.)	
		coróllary (Br.)			

The stress distribution noted above is paralleled by the longer forms. Again, words with light penults usually show the final stress pattern [1,0,2,0] while words with heavy penults show [0,1,0,0]:

(A.28)

Trisyllabic forms:

[0,1,0,2,0]	48%	prohíbitòry	contríbutòry	heréditàry
		contémporàry		
[2,0,1,0,0]	48%	còntradíctory	sùpervísory	sàtisfáctory
		ànnivérsary	rùdiméntary	
[1,0,0,2,0]	4%	dísciplinàry		

Four syllables:

[2,0,1,0,2,0]	cìrcumlócutòry	ìnterplánetàry	dòmícíliàry
---------------	----------------	----------------	-------------

A.7 The suffixes /-ate/ and /-ize/

These vowel-initial suffixes are by far the most common verbal suffixes in English, and they are the only long-vowel suffixes with any significant distribution in the lexicon. Words in this group that have more than two syllables in the stem tend to keep the main stress two syllables from the suffix. This means that the penult will rarely be stressed regardless of its weight. There are a number of differences between British and American pronunciations of these words, and in some cases separate tables will be given for each of these major dialects.

A.7.1 /-ate/

This may be actually seen as two suffixes, one forming verbs and always showing the long vowel form /-ā́t/, the other forming nouns and adjectives and usually showing an unstressed /-ə́t/, although some nouns and adjectives with /-ā́t/ as a variant or as the only form do appear. The verbal suffix has a far wider distribution and will be treated first.

Most words with this suffix avoid penultimate stress, as was noted above. One distinction between British and American usage is the treatment of words with monosyllabic stems. The Americans do stress the penultimate in this case,¹³⁸ e.g., *nárrate*, *rótate*, while most British verb forms instead stress the final suffix syllable. A few British words show both types, e.g., *mandate*, *migrate*; and some

¹³⁸Forms like *equáte*, *reláte*, *diláte*, *defláte*, are of course not verbs in /-ate/ but prefixed stems /e-quá́t/, /re-lá́t/, /di-lá́t/, /de-flá́t/, and they maintain final stress in American. The word *creáte*, with its unusual stem, has evidently been interpreted in American as a monomoraic stem /krēá́t/.

verbs, like *fíltrate*, *láctate*, *hýdrate*, never show final stress. Many of these have homonymous nouns or adjectives:¹³⁹

(A.29) Forms showing only [1,0] in British : 26%

CVV-	58%	hýdrāte	lígāte	próbbāte	áerāte
CVC-	42%	láctate	pálpate	fíltrate	

Forms showing only [0,1] in British: 55%

CVV-	67%	pūpāte	mūtāte	nōtāte	
		ōrāte	placāte	rotāte	
		lōcāte	gŷrāte	donāte	
CVC-	13%	fixāte	dictāte		
CV-	20%	crēmāte	narrāte	negāte	ōrāte (alt.)

Forms showing both [1,0] and [0,1] in British: 19%

CVV-	20%	mīgrāte	/mígr-āt/ ~ /migr-át/
CVC-	80%	castrāte	/kástr-āt/ ~ /kastr-át/
		frustrāte	/frústr-āt/ ~ /frustr-át/
		mandāte	/mánd-āt/ ~ /mand-át/
		pulsāte	/púls-āt/ ~ /puls-át/

In American, the situation is vastly simplified by the overwhelming presence of initial stress:

[1,0]: 93%

[0,1]: 7%

Of the verbs listed above, only *créate* and *negáte* have final stress in American. As was mentioned previously, the unusual stem of *créate* allows for a monomoraic interpretation, leaving *negáte* as the only case of /-ate/ with this stress. There is also a regular variant *négate* in American.

The distribution of regular forms is as follows:

¹³⁹The Celex database does not specify whether these variants are dialectal, or if they are alternate pronunciations in the same dialect.

(A.30)

CVV-	56%	mútāte	nótāte	gýrāte	órāte
CVC-	36%	cástrāte	frústrāte	díctāte	
CV-	8%	nárrāte	plácāte (alt.)		

Words showing two syllables preceding the suffix tend to show an antepenultimate stress pattern in both American and British.

[0,1,0] 5%

[1,0,0] 95%

Of the regularly stressed forms, many types of syllable structure combinations may be seen:

(A.31)

[1,0-0]

CVV-CVC-	2%	défalcāte	bífurcāte		
CVC-CVC-	4%	cómpensāte	cóncentrāte	ínculpāte	órchestrāte
CV-CVC-	4%	énervāte	prómulgāte	démonstrāte	désignāte
CVV-CV-	12%	hýphenāte	fúmīgāte	náuseāte	ísolāte
		áutomāte	óvulāte (alt.)	détonāte (Br. alt.)	
CVC-CV-	29%	cálculāte	íntegrāte	ádvocāte	éxcavāte
		émbrocāte	púnctuāte		
CV-CV-	48%	póllenāte	válidāte	súblimāte	sálivāte
		rénovāte	cómmutāte	ínnovāte	

A small minority with heavy penults shows penultimate stress:

(A.32)

[0,1-0]

imprégnāte	infíltrāte	defálcāte
erúctāte	incúlcāte	demónstrāte (alt.)
incárnāte	ínculpāte	remónstrāte

All of these words have regular alternants, although some, such as *impregnate*, are not current in American usage, while American shows *elóngate*, regular in British.

Most trisyllabic stems also show penultimate stress, although there a handful of words with other patterns:

(A.33) Trisyllabic forms:

[2,0,1,0]	1%	rèincárnāte (alt.)	òriéntāte (alt.)			
[1,0,0,0]	2%	òrientāte	étiolāte	álienāte	óxygenāte	péregrināte
[0,1,0,0]	97%					

The few forms showing penultimate stress all have alternants. Three of the forms with initial stress have /i/ plus hiatus in the following unstressed syllables. The distribution of the regular forms is as follows:

(A.34)

s-CVC-CVC-	1%	reíncarnāte (alt.)		
s-CVV-V-	7%	expátriāte	abbréviāte	anníhilāte
		hallúcināte	recúperāte	
s-CVC-CV-	24%	extérmināte	incárcerāte	annúnciāte
		emáncipāte	intércalāte	intéropolāte
		habítuāte		
s-CV-CV-	67%	cōágulāte	eváporāte	humíliāte
		rēiterāte	assáassināte	recíprocāte
		extrápolāte	intérrogāte	

Words show penultimate stress regardless of the position of prefixes, e.g., *intéropolate*. Almost no words with heavy penults are evident. Most stressed long vowels inside roots are instances of /ū/, words with hiatus, or CiV lengthening. Open monosyllabic prefixes freely alternate between long and reduced forms.

(A.35) Four syllables:

[0,1,0,0,0]	18%	améliorāte	disórientāte	rēórientāte
		detértiorāte		
[2,0,1,0,0]	82%			
š-s-CVC-VC-	6%	òvercòmpensāte		
š-s-CVV-V-	24%	ìntermédiāte	dìssassóciāte	mìsapprópriāte
š-s-CVC-V-	29%	còntraíndicāte	dìfferénciāte	còsubstántiāte
š-s-CV-V-	41%	dècontámināte	cìrcumnávīgāte	rèhabílitāte
		rècapítulāte		

Here, all the main-stressed long vowels in roots are in positions of CiV lengthening. Stressed open prefixes tend to show long vowels.

A.7.2 The nominal/adjectival suffix /-ate/

This suffix surfaces as both /-āt/ and /-ət/. The forms showing a full long vowel in the suffix will be treated first. Of the few monosyllabic roots, only a British variant *cognāte* shows a difference in pronunciation between American and British, unlike the case with the verbs.

(A.36)

[0,1]: 14%

CVV-	33%	irāte		
CVC-	66%	ornāte (alt.)	cognāte (Br.alt.)	

[1,0]: 86%

CVV-	29%	lūnāte	hýdrāte	nítrāte	próbāte
CVC-	47%	órnāte	díctāte	fíltrāte	cógnāte
		mágnāte			
CV-	24%	pínnāte	légāte	quádrāte	

A number of the initially stressed forms correspond to the initially stressed verbs of British. A majority of these forms have alternants with the reduced-vowel suffix.

Most bisyllabic stems show antepenultimate stress:

[0,1,0] 17%

[1,0,0] 83%

The few forms with penultimate stress show in most cases a heavy penult.

(A.37)

s-CVV-	16%	inchóāte			
s-CVC-	67%	apóstāte	insénsāte	incárnāte	intéstāte
s-CV-	16%	apéllāte			

All of these show variants with the unstressed suffix, as do those with antepenultimate stress:

(A.38)

CVV-CVC-	3%	pótentate		
CVC-CVC-	3%	cóncentrate		
CV-CVC-	6%	mágrate	désignate	
CVV-CV-	12%	déviante	cýclamate	néonate

CVC-CV-	34%	cónjugate cándidate	óbdurate ínchoate	ádvocate
CV-CV-	42%	ággregate réprobate	ácetate	délegate

The effects seen here are similar to those seen in the verbs.

Most trisyllabic stems also show antepenultimate stress, although there a handful of words with other patterns:

(A.39) Trisyllabic forms:

[2,0,1,0]	6%	rèincárnāte		
[1,0,0,0]	6%	pátriarchāte		
[0,1,0,0]	88%			
s-CVV-CV-	7%	invíolāte	expátriāte	
s-CVC-CV-	43%	permánganāte bicárbonāte	invértebrāte	procónsulāte
s-CV-V-	50%	pontíficāte precípitāte	agglómerāte retículāte	barbítuāte

Words showing the reduced vowel suffix /-ət/ include most of the previous words and quite a few others. All monosyllabic stem forms show initial stress:

(A.40)

CVV-	35%	nítrate	prímate	clímate	príivate
CVC-	37%	phósphate mándate	súlfate mágnate	fíltrate	
CV-	28%	prélate	pálate	légate	quádrate

Bisyllabic stems are divided between penultimate and antepenultimate stress:

[0,1,0] 13%

[1,0,0] 87%

Many forms with penultimate stress show alternate forms with antepenultimate stress. Many are pronounced this way only in British.

(A.41)

s-CVV-	18%	inchóate	emírate (alt.)		
s-CVC-	55%	apóstate	insénsate	incárnate	intéstare

s-CV- 27% apéllate consúmmate (Br.) intrícate (Br. alt.)

All of these show variants with the long-vowel version of the suffix, as do those with antepenultimate stress:

(A.42)

CVV-CVC-	1%	pótentate			
CV-CVC-	5%	mágristrate	désignate		
CVV-CV-	12%	déviante	dúplicate	númerate	ópiate
CVC-CV-	33%	ádvocate	cándidate	dóctorate	
		éstime	ínchoate		
CV-CV-	58%	ággregate	délegate	prédicate	
		ánimate	gráduate	réprobate	

Most trisyllabic stems also show antepenultimate stress, although there are a handful of words with other patterns:

(A.43) Trisyllabic forms:

[2,0,1,0]	1%	rèincárnate			
[1,0,0,0]	2%	pátriarchate	lícentiate (alt.)		
[0,1,0,0]	97%				
s-CVV-CV-	14%	invíolate	expátriate	inébriate	appróprate
s-CVC-V-	28%	sophísticate	invértebrate	procónsulate	
		bicárbonate	penúltimate		
s-CV-V-	44%	pontíficate	conglómerate	barbítuante	
		precípitate	degénerate	artícuate	
		legítimate			

None of these forms have heavy penults. Many of them show nominal alternants with /-āt/, more commonly as British variants than American, but there is no apparent structural reason why some words have the long suffix variant and others do not (cf. *agglómerate*, *conglómerate*, the former of which has a long variant in British). Longer words, such as *bàccaláureate*, *àntepenúltimate*, *dispropórtionate*, and various /in-/ negations of words from the above set, all show antepenultimate stress. None has a heavy penult.

A.7.3 /-ize/

The verbal suffix /-ize/ also shows very few words with penultimate stress, most of which have regularly stressed variants. Words showing two syllables preceding the suffix tend to show an antepenultimate stress pattern in both American and British. Only a small minority, each showing a heavy penult, displays penultimate stress:

(A.44)

[0,1,0] 2%		affranchise	enfranchise	disfranchise
		amortize	aggrandize	
[1,0,0] 98%				
CVV-CVC-	1%	súpervise		
CVC-CVC-	4%	górmāndize	círcumcīze	éxorcīze
		éxercīze	ádvērtīze	bástardize
		módnērīze		
CV-CVC-	1%	sólemnīze	récognīze	
CVV-CV-	25%	móisturīze	plágiarīze	bówdlerīze
		ídolīze	hýbridīze	stábilīze
		mótorīze	násalīze	clímatīze
CVC-CV-	32%	cómpromīze	ímprovīze	órāgnīze
		púlverīze	dógmatīze	fértilīze
		hármonīze	súbsidīze	óxidīze
CV-CV-	36%	télevīze	cívilīze	cólonīze
		críticīze	fóssilīze	sátirīze
		týrranīze	ímmunīze	pástēurīze

One of the most interesting features of this set is the way in which words that are not historically or even synchronically (from a strictly morphological point of view) suffixed with /-ize/ follow the stress pattern associated with this group. As prefixed monosyllabic roots of the form /Cis/, we expect final stress ([2,0,1]) in the following words:

(A.45)

círcumcīze	cómpromīze	récognīze
télevīze	ímprovīze ¹⁴⁰	súpervīze

¹⁴⁰While its form suggests a prefixed root, this verb does show a regular nominal *improvisation*.

Note the nominal abstracts *circumcísion*, *télevision*, *sùpervísion*, *récognítion*, versus the regular type *immunizátion*.

A large number of trisyllabic stems show initial stress. Most of these are multiply suffixed, and retain the stress pattern of the suffixed form:

(A.46) Trisyllabic forms:

[2,0,1,0]	1%	pròpagándize	dìsenfránchise		
[1,0,0,0]	34%	vísualize	trívialize	cánnibalize	cápitalize
		chàracterize	cátégorieze	óxygenize	sécularize
[0,1,0,0]	65%				
s-CVV-CV-	25%	anaésthétize (Br.)	demóbilize	deódorize	
		idéálieze	compúterize	revítalíze	
		acclímatize			
s-CVC-CV-	33%	demáagnetize	románticize	extérnalize	
		colléctivize	evángelize		
s-CV-CV-	42%	ecónomize	anthólogize	apólogize	
		homógenize	capítalíze (Br. alt.)	legítimize	
		demócratize	políticize		

Multiply suffixed forms which do show antepenultimate stress are those that are prefixed, like *revítalíze*. For four syllables, the pre-antepenultimate stress actually outnumbers the penultimately stressed forms.

(A.47) Four syllables:

[0,1,0,0,0]	63%				
s-CVV-CV-	21%	extérieurize	matérialize	memórialize	
s-CVC-CV-	14%	concéptualize	indústrialize		
s-CV-CV-	65%	famíliarize	américanize	anáthematize	
		legítimatize	partícularize	políticalize	
		demíltarize	denátionalize		
[2,0,1,0,0]	36%				
s-CVV-CV-	40%	àttitúdinize	rèvolútionize		
s-CVC-CV-	60%	phòtosénsitíze	sèntiméntalíze	còmpartméntalíze	
[1,0,0,0,0]	0%	spíritualize			

For some speakers, words with hiatus (e.g., *famíliarize*) are pronounced with fewer syllables, i.e., [fəmiljəraɪz], effectively converting these to three-syllable stems with antepenultimate stress. Longer words all have pre-antepenultimate stress, e.g., *institúționalize*, *internátionalize*, *indivídualize*.

A.8 The extended forms of /-ate/

Stems combined with the productive verbal suffix /-ate/ go on to combine with further suffixes to generate verbal nouns in /-ion/, agentive nouns in /-or/ and adjectives in /-ive/ and /-ory/. These forms are interesting in that the suffix /-ate/ does not always receive stress or show its long vowel, and these variations are not phonologically predictable.

A.8.1 /-ation/

Like most forms of the /-ic/ group, words in the suffix /-ion/ strictly show their main stress on the syllable preceding the suffix, and forms in /-ation/ are no exception. All are given main stress on the verbal suffix, i.e., /-ǎʃən/, with the expected palatalization of the suffix-final /t/. The placement of the secondary stress is practically the only criterion for variation among the hundreds of forms in /-ation/. As with /-ate/, syllables adjacent to the stressed suffix are never stressed, regardless of their weight. Not all forms in /-ation/ have corresponding verbs in /-ate/; there are quite a few that add /-ation/ directly onto an unsuffixed verbal stem, as well as other stems with no related verbal form.

Monosyllabic root forms all show the pattern [0,1,0]:

(A.48)

CVV-	55%	mūtǎtion	nōtǎtion	gȳrǎtion	foundǎtion
		lōcǎtion	migrǎtion	striǎtion	ōvǎtion
CVC-	30%	castrǎtion	frustrǎtion	dictǎtion	filtrǎtion
		formǎtion	plantǎtion	damnǎtion	
CV-	15%	narrǎtion	cessǎtion	negǎtion	
		orǎtion (alt.)		cremǎtion (Br.)	

For words with two syllables preceding the suffix, all show a stress pattern [2,0,1,0], but many types of stem syllable structure combinations may be seen:

(A.49)

CVV-CVC-	3%	dèfalcǎtion	bìfurcǎtion	dèformǎtion
		èlongǎtion	fòmentǎtion	hìbernǎtion

		ùsurpáti <u>on</u>		
CVC-CVC-	8%	còmpe <u>ns</u> áti <u>on</u>	cò <u>nc</u> ent <u>ra</u> ti <u>on</u>	cò <u>nd</u> ensáti <u>on</u>
		ì <u>mp</u> ortáti <u>on</u>	ì <u>nc</u> arnáti <u>on</u>	trà <u>ns</u> portáti <u>on</u>
		pì <u>gm</u> entáti <u>on</u>	ò <u>rc</u> hestráti <u>on</u>	
CV-CVC-	5%	dè <u>m</u> onstráti <u>on</u>	à <u>da</u> ptáti <u>on</u>	à <u>nn</u> exáti <u>on</u>
		dè <u>si</u> gnáti <u>on</u>	ì <u>ll</u> ustráti <u>on</u>	là <u>m</u> entáti <u>on</u>
		lè <u>gi</u> sláti <u>on</u>	rè <u>fo</u> rmáti <u>on</u>	
CVV-CV-	9%	hý <u>ph</u> enáti <u>on</u>	fù <u>m</u> igáti <u>on</u>	dè <u>no</u> táti <u>on</u> (alt.)
		mò <u>t</u> iváti <u>on</u>	ì <u>so</u> láti <u>on</u>	dè <u>vi</u> áti <u>on</u>
		dè <u>h</u> ýdráti <u>on</u>	dè <u>no</u> táti <u>on</u>	è <u>v</u> ocáti <u>on</u>
		vì <u>o</u> láti <u>on</u>		
CVC-CV-	24%	ì <u>nt</u> egráti <u>on</u>	cò <u>nf</u> lagráti <u>on</u>	è <u>xc</u> aváti <u>on</u>
		è <u>x</u> planáti <u>on</u>	ì <u>ns</u> piráti <u>on</u>	cà <u>lc</u> ináti <u>on</u>
		còmputáti <u>on</u>	trà <u>ns</u> migráti <u>on</u>	
CV-CV-	50%	à <u>be</u> rráti <u>on</u>	rè <u>ve</u> láti <u>on</u>	dè <u>fa</u> máti <u>on</u> (alt.)
		sà <u>li</u> váti <u>on</u>	cèlebráti <u>on</u>	sà <u>lu</u> táti <u>on</u>
		à <u>vo</u> cáti <u>on</u>	à <u>pp</u> robáti <u>on</u>	dè <u>pu</u> táti <u>on</u>
		prò <u>vo</u> cáti <u>on</u>	rè <u>no</u> váti <u>on</u>	

With trisyllabic stems we find a contrast between two principal stress types, based upon the difference in secondary stress placement:

[2,0,0,1,0] 48%

[0,2,0,1,0] 52%

Of the first group, the forms fall into a few basic categories. One is those stems which showed unusual stress in their /-ate/ verbs:

(A.50)

rè <u>in</u> car <u>na</u> ti <u>on</u>	ò <u>ri</u> entáti <u>on</u>	è <u>ti</u> oláti <u>on</u>
à <u>li</u> enáti <u>on</u>	ò <u>xy</u> genáti <u>on</u>	pè <u>re</u> grináti <u>on</u>

To this may be added *mìs*cegenátion, *mà*nifestátion and the infrequent *tè*rgiversátion. Other words show multiple suffixation, the majority of which are in /-ize/:

(A.51)

/-ify/ + /-ation/:	cò <u>di</u> ficáti <u>on</u>	nò <u>ti</u> ficáti <u>on</u>
/-ment/ + /-ation/:	à <u>rg</u> umentáti <u>on</u>	dò <u>cu</u> mentáti <u>on</u>
/-ize/ + /-ation/:	pù <u>lv</u> erisáti <u>on</u>	ò <u>rg</u> anizáti <u>on</u>

Another set shows bisyllabic prefixation, composed either of two monosyllabic prefixes or a single bisyllabic prefix:

(A.52)

rèpresentation	rècommendation	dìsinclination
còeducation	ìnterrelation	mùltiplication

The distribution of the second set with the pattern [0,2,0,1,0] is as follows:

(A.53)

CV-CVC-	1%	admìnistración		
CVV-CV-	13%	expàtriación	abbrèviación	annihilación
		hallùcination	recùperación	
CVC-CV-	17%	extèrminación	incàrceración	annúnciación
		emàncipación	intèrcalación	intèrpolación
CV-CV-	69%	coágulation	eváporación	humíliación
		rēiteration	assássination	recíprocación
		extràpolación	intèrrogación	habituación

The distribution seen here is very similar to that of the corresponding verbs in /-ate/. Words with four syllable long stems also show a range of possibilities for secondary stress; most items in the first two groups are extended stems in /-ize/ and /-ify/ and their secondary stress is directly related to the stress seen in the verbal form.

(A.54)

[2,0,0,0,1,0]:	38%	cápitalización	cháacterización	hòspitalización
		fictionalización	ràtiocinación	dèclassificación (Br.)
[0,2,0,0,1,0]:	43%	acclimatización	collèctivización	demócratización
		demóralización	elèctrificación	īdèntificación
		syllàbificación		
		amèlioration	detèrioración	disòrientation
[2,0,2,0,1,0]:	18%			
š-s-CVC-CVC-	5%	òvercòmpensación		
š-s-CVV-CV-	10%	dissassòciación	mìsappropriación	
		èxcommùnicação		
š-s-CVC-CV-	30%	còntraindicación	mìspronùnciación	
		differèntiación	cònsustàntiación	

š-s-CV-CV-	55%	dècontàminátion	cìrcumnàvigátion
		rèhabilitátion	rècapìtulátion
		tìnnàbulátion	

The last group consists of stems without multiple suffixation. The few attested longer forms show a similar distribution:

(A.55)

[0,2,0,0,0,1,0]:	denàtionalizátion	partìcularizátion
	demìlitarizátion	
[2,0,2,0,0,1,0]:	òversìmplificátion	phòtosènsitizátion
[2,0,0,2,0,1,0]:	ìntercommùnìcátion	

A.8.2 /-ator/

The agentive suffix /-or/ combines with /-ate/ to form nouns that show a stress pattern similar to that of verbs in /-ate/. For monosyllabic roots, there is once again a difference between American and British pronunciation. Most British forms show penultimate stress, while most American ones show antepenultimate stress.

(A.56)

CVV-	43%	crēátor	vībrátor	cūrátor
CVC-	43%	dictátor	testátor	
CV-	14%	narrátor		

The only British form with initial stress is *órator* /ór-ət-ər/, which shows final stress in its unsuffixed form. In American, all of these forms show initial stress, as did the corresponding verbs. Of words with bisyllabic stems, only *conspíator* shows antepenultimate stress. This word notably has no corresponding verb in /-ate/. All the rest show a pattern [1,0,2,0], wherein the suffix shows secondary stress and a long vowel.

(A.57)

CVC-CVC-	1%	álnèrator		
CV-CVC-	9%	démonstrátor	prómulgátor	cómmèntátor
		íllustrátor	législátor	
CVV-CV-	12%	dúplicátor	rádiátor	áviátor
CVC-CV-	22%	cálcúlátor	árbitrátor	pérrètrátor
		éxcavátor	réspirátor	

CV-CV-	56%	décoràtor	générateur	élevàtor
		imitàtor	navìgàtor	gládiàtor
		cómmutàtor	ínnovàtor	éducàtor

The distribution is similar to that seen in the verbs in /-ate/. Trisyllabic stems show the pattern [0,1,0,2,0] except for the multiply suffixed word *tòtalizàtor*.

(A.58)

s-CVV-CV-	17%	expròpriàtor	accúmulàtor	negótiàtor
s-CVC-CV-	13%	extérminàtor	emàncipàtor	impérsonàtor
s-CV-CV-	70%	eváporàtor	accéléràtor	incíneràtor
		oríginàtor	refrìgeràtor	intérrogàtor

A.8.3 /-atory/

The adjectival and nominal suffix /-atory/ does not simply show the same stress pattern as the forms in /-ator/, but rather shows an unexpected variety of stress patterns. It is one of the combinations in which British and American stress differ noticeably. As with /-ory/, that part of the suffix only bears secondary stress under certain conditions in American; it is always unstressed in British.

For monosyllabic roots, there are two stress patterns.

[0,1,0,0]: 25%

[1,0,0,0]: 75%

The first type is seen only in British:

(A.59)

[0,1,0,0]

CVV-	66%	migràtory	plācàtory	rōtàtory	nūgàtory
CVC-	33%	mandàtory	gustàtory		

All these words may appear with variants showing the second pattern; all American words of this form follow the second pattern:

(A.60)

[1,0,0,0]

CVV-	36%	mígratòry	féudatòry	rótatòry	láudatòry
CVC-	19%	hórtatòry	mándatòry	púrgatòry	sígnatòry
CV-	45%	crématòry	óratòry	lávatóry	prédatóry

Bisyllabic stems show four stress patterns, only two of which appear in American:

[1,0,2,0,0]: 14%

[2,0,1,0,0]: 17%

[1,0,0,2,0]: 11%

[0,1,0,2,0]: 57%

The first two patterns are seen only in British; the second is the inversion of the first. Many words show more than one pattern.

(A.61)

[1,0,2,0,0]:

CVC-CVC-	17%	cónfiscàtory	
CVC-CV-	17%	réspiràtory	
CV-CV-	67%	ádulàtory	cómminàtory
		déprecàtory	óbligàtory

[2,0,1,0,0]:

CVC-CVC-	25%	còmpensátory	cònfiscátory	màsturbátory
CVC-CV-	25%	cìrculátory		
CV-CV-	49%	àdulátory	àpprobátory	

In American, these forms all belong to the other two patterns; in British many of these words appear in both groups.

(A.62)

[1,0,0,2,0]: 25% (Am.)

CVC-CVC-	6%	másturbatòry (Am.)	
CVC-CV-	11%	réspiratòry	
CV-CV-	83 %	ádulatòry (Am.)	cómminatòry
		déprecatòry	óbligatòry (Am. alt.)
		cìrculatòry	láboratòry (alt.)

[0,1,0,2,0]: 75% (Am.)

s-CVV-	11%	consólatòry (alt.)	explóratòry (alt.)	
		appróbatòry (Am.)	respíratòry (Br. alt.)	
		excítatòry		
s-CVC-	33%	conservatòry	confíscatòry	observatòry
		comméndatòry	refórmatoròry	
		compénsatòry (Am.)		
s-CV-	56%	exclámatòry	explánatòry	inflámmatòry
		decláratòry	defámatòry	obligatòry
		derógatòry	preparatòry	

labóratòry (Br. alt.)

A number of stems with long vowels in the unsuffixed form appear with a short vowel, e.g., *expláin* ~ *explánatòry*. Most of the forms showing the last pattern do not form verbs in /-ate/ but rather suffix /-atory/ to a bare verb stem. A few forms, like *explóratory*, have a long variant, and the British variant of *labóratory* is stressed on the antepenult as if this were a prefixed monosyllable rather than a bisyllabic stem. It may be useful to summarize the various British and American types:

(A.63)

British:

[1,0,0,0]	círculatory	/sérkjəl-ət-əri/
[2,0,1,0]	cìrculátory	/sèrkjəl-ăt-əri/
	cònfiscátory	/kòn-fisk-ăt-əri/
[1,0,2,0]	cónfiscàtory	/kón-fisk-ăt-əri/
	réspiràtory	/ré-spər-ăt-əri/
	óbligàtory	/ób-lig-ăt-əri/
[0,1,0,0]	respíratory	/rə-spír-ət-əri/
	oblígatory	/əb-líg-ət-əri/

American:

[1,0,0,2]	círculatòry	/sérkjəl-ət-òri/
	réspiratòry	/ré-spər-ət-òri/
	óbligatòry	/ób-lig-ət-òri/
[0,1,0,2]	confíscatòry	/kən-físk-ət-òri/
	oblígatòry	/əb-líg-ət-òri/

Trisyllabic stems also show a variety of stress types, which often show up in the same word. As before, the patterns showing stress on /-ăt-/ do not appear in American, and these words take on other patterns:

(A.64)

[0,1,0,2,0,0]:	elúcidàtory	antícipàtory	concíliàtory	congrátulàtory
[0,2,0,1,0,0]:	discrìminátory	hallùcinátory		retàliátory
[2,0,0,1,0,0]:	ànticipátory			
[0,1,0,0,2,0]:	hallúcinatòry (alt.)	congrátulatòry		discríminatòry
	recríminatòry	depréciatòry		concíliatòry
	retáliatòry	antíciatòry (Am.)		

[2,0,1,0,2,0]: ìnterrógatòry

The last pattern, which was the most common among trisyllabic stems, has only one representative here. A summary of the types would be as follows:

(A.65)

British:

[0,1,0,2,0,0]:	antícipàtory	/antísip-àt-əri/
	concíliàtory	/kən-síli-àt-əri/
[0,2,0,1,0,0]:	retàliátory	/rə-tàli-á-t-əri/
[2,0,0,1,0,0]:	ànticipátory	/ántisip-ət-əri/
[0,1,0,0,0,0]:	concíliatory	/kən-síli-ət-əri/
	retáliatory	/rə-táli-ət-əri/
[2,0,1,0,0,0]:	ìnterrógatory	/ìntər-róg-ət-əri/

American:

[0,1,0,0,2,0]:	antícipatòry	/antísip-ət-òri/
	concíliatòry	/kən-síli-ət-òri/
	retáliatòry	/rə-táli-ət-òri/
[2,0,1,0,0,0]:	ìnterrógatory	/ìntər-róg-ət-òri/

Also notable is the further suffixed form *còmbinatórial* ; other forms in /-atorial/ show patterns similar to the words in /-atory/.

A.8.4 /-ative/

Like /-atory/, the addition of /-ive/ to stems in /-ate/ produces an adjective or noun. Again, various types appear which often apply to multiple alternate forms of individual words, and disagree between American and British English. All monosyllabic roots show antepenultimate stress:

(A.66)

CVV-	19%	lúcrative	cáusative		
CVC-	16%	fíxative	fórmative	hórtative	
CV-	64%	nárrative	négative	vócative	frícative

Bisyllabic stems show three stress patterns, all of which appear in both American and British, but not always across the same words. Many words show multiple types in any case:

[1,0,2,0]: 18%

[1,0,0,0]: 15%

[0,1,0,0]: 67 %

(A.67)

[1,0,2,0]:

CVC-CVC-	2%	cóntemplàtive		
CV-CVC-	3%	íllustràtive	législàtive	
CVV-CV-	5%	cúmulàtive (Br.alt.)	rúminàtive (Br. alt.)	
CVC-CV-	9%	índicàtive (Br. alt.)	quántitativa	
CV-CV-	81%	cópulàtive (Br. alt.)	gèneràtive (Br. alt.)	
		mèditativa	ímitativa	pénetrativa
		cónnotativa	quàlitativa	cómmutativa

[1,0,0,0]:

CV-CVC-	8%	íllustrative	législative (Br. alt.)	
CVV-CV-	18%	cúmulative	rúminative	péjorative (Br.)
CVC-CV-	4%	quántitative (Br. alt.)		
CV-CV-	70%	cópulative	gènerative	figürative
		décorative	spéculative	nóminative

[0,1,0,0]:

s-CVV-	18%	inchóative	accúsative	commútative
		connóttative		
s-CVC-	35%	altérnative	consérvative	contéplative
		affírmative	infórmative	demónstrative
		prevéntative	supérative	
s-CV-	46%	provócative	compárativ	indícative
		evócative	impérative	

Here, many stems with a short vowel are seen that show a long vowel in their unsuffixed forms, although a few alternatively show long vowels as well. What generally distinguishes the last set from the others is that most (but not all) of these stems form suffixless prefixed verbs, while the first two sets are primarily made up of stems that tend to form verbs in /-ate/.

Trisyllabic stems show a similar distribution:

[0,1,0,2,0]: 38%

[0,1,0,0,0]: 43%

[2,0,1,0,0]: 18%

(A.68)

[0,1,0,2,0]:

s-CVV-CV-	25%	accúmulative (Br. alt.)	agglútinative (Br. alt.)
		appréciative (Br. alt.)	
s-CVC-CV-	8%	intéruptive (Br. alt.)	
s-CV-CV-	67%	alléruptive (Br. alt.)	authóritative
		significative (Br. alt.)	commémorative (Br. alt.)
		corróborative (Br. alt.)	degénérative (Br. alt.)
		commúnitive (Br. alt.)	

[0,1,0,0,0]:

s-CVV-CV-	22%	accúmulative	agglútinative	appréciative
		recupérative	remúnérative	vitúperative
s-CVC-CV-	15%	intéruptive	détérminative	subórdinative
s-CV-CV-	63%	alléruptive	commémorative	cōopérative
		significative	corróborative	

[2,0,1,0,0]:

è-s-CVC-	60%	àrguméntative	ùndemónstrative	rèprésentative
è-s-CV-	40%	interróguative		

In this case, American English appears to favor the variant without stress on the suffix, although there are some forms which retain variants or even prefer it. As was noted above, the morphological issue of whether a stem formed its verb in /-ate/ or not appears to be directly linked to whether the antepenult received stress, although this is not a hard and fast rule:

(A.69)

	<u>[1,0,2,0]</u>	<u>[1,0,0,0]</u>	<u>[0,1,0,0]</u>
générate	généruptive	généruptive	
législate	légisruptive	légisruptive	
compáre			compáruptive
consérve			consérruptive
índicate	índicative		índicative

A.9.1 /-ment/

This suffix is parallel in behavior to /-ate/ and /-ize/ in some ways. When verbal, it takes the main stress in bisyllabic words, but reverts to secondarily

stressed status when the word is trisyllabic or greater. In nominals, /-ment/ is only suffixed when the initial syllable of the word is light.

Words in /-ment/ may be divided into two groups, nouns (80%) and verbs (20%), of which the former greatly outnumbers the latter. The verbs show a final stress primarily absent from the nouns and may be treated first:

(A.70) Monosyllabic stems:

[0,1] : 84%

CVV-	11%	fōmént		
CVC-	55%	augmént	fermént	fragmént
		segmént	tormént	
CV-	33%	cemént	lamént	fomént

[1,0] : 16% ségment (alt.) cómment

Bisyllabic stems show the pattern [1,0,2], wherein the final syllable shows a full rather than a reduced vowel. There are a few words, like *cómplément*, *cómpliment* and *régiment* which show variants with the pattern [2,0,1]. All such verbs have light penults.

(A.71) Bisyllabics:

cómplimènt	cóplèment	dócumènt
ístrumènt	íplèment	órnamènt
régimènt	súplèment	

There is also the longer verb *expériment*.

Nouns with monosyllabic stems are primarily initially stressed:

[0,1] : 6%

[1,0] : 94%

The few nouns, like *cemént*, *lamént*, which show final stress are also verbs. Other nouns that have corresponding verbs are initially stressed, e.g., *áugment*, *férment*, *ségment*, *tórmement*. These all have heavy initial syllables, in contrast to the finally stressed nouns.

(A.72)

CVVCC-	8%	óintment	áilment	báilment
CVVC-	21%	básement	cásement	móvement
CVV-	19%	páyment	móment	ráiment

CVC-	44%	férment	figment	frágment
		tórment	pígment	gárment
		shípmént	ségment	júdgement
CV-	8%	clément	cómmént	

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

[1,0,0] : 33%

[0,1,0] : 67%

Structurally, almost none of the latter group shows a light penult, the sole exception being the form *inclément*, a prefixed form of the adjective *clément* seen above. However, there are a number of words with antepenultimate stress which do have heavy penults, such as *bánishment*, *chástizement*, *mánagement*, *báfflement*. In most cases, words with free stems appear to retain the same accentuation as their unsuffixed counterparts. A great many words with penultimate stress show prefixed monosyllabic stems.

(A.73)

[1,0,0] : 24%

CVC-CVV-	5%	fránchisement	chástisement (alt.)	
CV-CVC-	21%	mánagement	góvernment	bánishment
		séttlement	wónderment	
CVV-CV-	6%	múniment	nútriment	rúdiment
CVC-CV-	25%	árgument	ármament	éxcrement
		fírmament	íplement	ínstrument
CV-CV-	42%	dóccument	détriment	élement
		filament	mérriment	

[0,1,0] : 76%

CVVCC-	10%	annóuncement	annóintment	arráangement
CVVC-	37%	repláccement	concéalmént	confínement
		consígnment		
CVV-	10%	repáyment	emplóymént	endówmént
CVCC-	19%	adváncement	bombárdment	compórtment
		enchántment	enfórcement	
CVC-	19%	conférment	entrápment	fulfíllment
		asséssment		
CV-	5%	inclément		

Four syllable words show a similar pattern, but there are a few forms with unusual initial stress:

(A.74)

[0,1,0,0]:	61%			
[2,0,1,0]:	32%			
[1,0,0,0]:	2%	témperament	líneament	médicament
[1,0,2,0]:	5%	óverstàtement	únderstàtement	ádvèrtisèment
		úndergàrment	réconcìlement	

The last pattern is an inversion of the expected [2,0,1,0]. Again, words with antepenultimate stress tend to have light penults, end in liquids or be multiply suffixed.

(A.75)

s-CVV-CVC	17%	disàblement	enlíghtenment	entítlement
		acquírement	ennõblement	envíronment
s-CVC-CVC-	15%	disfránchisèment	enfránchisèment	advèrtisèment
		abándonment	encírclement	entánglement
		bewílderment		
s-CV-CVC-	31%	acknówledgement	embárrassment	embéllishment
		encóuragement	devélopment	embítterment
		imprísonment		
s-CVC-CV-	16%	disármament	reármament	présèntiment
s-CV-CV-	21%	embódiment	impédiment	arbítrament
		expériment	predícament	

There are apparently no words of the form s-CVV-CV- in this group. The word *impédiment* is notable in that the stem seen in *impéde* appears short. Of the words with penultimate stress, most are prefixed monosyllabic stems:

(A.76)

è-s-CVVCC-	10%	prèarràngement	rèarràngement	dìsappóintment
è-s-CVVC-	38%	prèordáinment	rèinstàtement	dìsengágement
		èntertáinment	rèalígnment	
è-s-CVV-	21%	òverpáyment	ùnemplóyment	dìsendówment
		dìsagrèement		
è-s-CVCC-	24%	rèimbúrsement	rèinfórcement	dìsconténtment

rèadjústment
 è-s-CVC 6% dìsintérment

Among longer words, penultimate stress is not found, although some words do show preantepenultimate stress again. Most of the words with penultimate stress actually show a heavy penult, and are in fact prefixed versions of stems seen earlier.

(A.77)

[0,1,0,0,0]: 25% accómpañiment impóverishment
 [2,0,1,0,0]: 75% hèredítament dìsembárrassment dìsentánglement
 dìsestáblishment dìsillúisionment ùndernóurishment

A.10 /-ist/ and /-ism/

As in the /-ment/ forms, for /-ist/ and /-ism/ words, stress appears to be related to the morphological type of the stem being suffixed. Words that are already suffixed tend to maintain that stress pattern, although there are exceptions.

Forms showing a monosyllabic stem are universally stressed with the pattern [1-0,(0)]; forms with light stems make up about a seventh of the group.

(A.78)

CVV-	48%	cúbist	cýclist	pápist	stýlist
		déism	núdism	rácism	áutist
CVC-	39%	ártist	báptist	léftist	línguist
		márxism	séxism		
CV-	13%	céllist	drúggist	fáscist	chémist
		tráppist	fáscism	sóphism	

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

[1,0,0] 88%

[0,1,0] 12%

Unlike the words in the /-al/ group above, many of these words showing antepenultimate stress nevertheless have heavy penults:

(A.79)

CVV-CVV-	2%	égōtist	áthēist	éuphūism	júdaism
CVC-CVV-	5%	jíngōist	pánthēist	híndūism	árchāism
CV-CVV-	5%	égōist	éssāyist	hérōism	

CVV-CVC-	6%	scíentist	sólipsism	sólipsist (alt.)	
CVC-CVC-	2%	ádventist	éxorcist	éxorcism	
CV-CVC-	3%	cóolumnist	sólipsist	ánarchist	módernism
CVV-CV-	25%	éucharist	bícyclist	cháuvínist	díarist
		hédonist	mótorist	sócialist	fátalism
		réalism	vócalism	níhilism	
CVC-CV-	19%	áctivist	álchemist	ársonist	pórttraitist
		fórmalism	sýmbolism	prágmatisim	
CV-CV-	34%	óculist	bígamist	bótanist	clássicist
		móralist	sátirist	strátegist	clássicism
		féminism	másochism	fétishism	cómmunist

Notable are the short penults seen in *sátirist*, *strátegist*, as compared to *sátíre* and *stratéigic*. A number of words have alternate forms with quantity fluctuations. Also notable is the shortening seen in *héroism*. Many of the words with heavy antepenults that are skipped have similarly stressed stems in isolation, e.g., *cóolumn*, *góvern*, *ánarchy*. There are also a number of stems that are already suffixed with /-al/ and /-ent/ type suffixes, e.g., *scíentist*, *vócalist*, *áctivist*.

Words with penultimate stress are in a minority. Almost all show heavy penults:

(A.80)

CVV-	53%	elítist	cartóonist	extrémist	escáapist
		machínist	deféatism	idéalism	surréalism
CVC-	33%	confórmist	resérvist		transvéstism
CV-	13%	libréttist	narcíssist (Br.)		narcíssism (Br.)

Very few forms show a stressed light antepenultimate syllable. Again, many stems are similar to their unsuffixed forms, e.g., *deféat*, *extréme*, *cartóon*, *confórm*.

Words with trisyllabic stems show a range of stress patterns:

Trisyllabic stems:	[0,1,0,0]	59%
	[2,0,1,0]	13%
	[1,0,2,0]:	7%
	[1,0,0,0]	20%

Words with antepenultimate stress have light penults:

(A.81)

s-CVV-CV-	17%	anaésthétist	podíatrist	salvátionist	beháaviorist
		illúSIONist	revívalist	beháaviorism	idéaLism
s-CVC-CV-	26%	projéctionist	numísmatist	románticist	
		misánthropist	patérnalist	obstrúctionism	
		somnámbulism	evángelism		
s-CV-CV-	57%	seísmólogist	ecónomist	phonólogist	
		expréssionist	ventríloquist	suprémacist	
		teléphonist	saxóphonist (alt.)	empíricism	
		fanáticism	antágonism	metábolism	

While most words appear to have the same accentuation as their unsuffixed forms, a few, like *teléphonist* and alternate British pronunciations of *capítalist* and *saxóphonist*, do not. Fewer words show penultimate stress, most of which have heavy penults:

(A.82)

è-s-CVV-	32%	pàrachútist (alt.)	òpportúnist	àbsolútist
		òpportúnism	pòlythéism	àbsentéeism
		mònothéism		
è-s-CVC-	50%	nònconfórmist	òbscurántist	ìrredéntist
		pròpagándist	òbscurántism	ìrredéntism
è-s-CV-	18%	viólínist	clàrinétist	

Words like *viólínist* follow the stress of the simplex *viólin*, despite the fact that the main stressed syllable is light. A few words show the inverse pattern, with initial stress; all have heavy penults:

(A.83)

é-s-CVV-	67%	párachùtist	mónothèist	mótorcÿclist
		mánicùrist	sáxophònist	ópportùnism (alt.)
		ábsolùtism	phárisàism	
é-s-CVC-	33%	táxidèrmist	métallùrgist	

Words with the pattern [1,0,0,0], with no secondary stress, primarily show light syllables in the penult.

(A.84)

ś-s-CVC-	13%	prótestantism			
ś-s-CV-	77%	náturalist	rátionalist	fúncionalist	cápitalist
		séparatist	fávoritism	álcoholism	púritanism
		rélativism			

Many of these are light bisyllabic stems extended in /-al/ suffixes, e.g., *cápitalism*, *rélativism*, but the suffix /-ate/ appears to show a similar treatment in *séparatist*, similarly the suffix /-ant/ in *prótestantism*. Similar patterns are seen for longer words:

(A.85) Four syllables:

[2,0,1,0,0]	89%	sèmifínalist	cònservácionista	trànscondéntalist
		ànthropólogist	sàdomásochism	àntisémítism
[0,1,0,0,0]	5%	sensácionista	colónialist	impérialism
		matérialism		
[2,0,0,1,0]	3%	càricatúrist (alt.)	violincéllist	ànthropomórphism
[1,0,0,2,0]	1%	cáricatùrist		
[1,0,0,0,0]	2%	spíritualist	spíritualism	

Most longer words are multiply suffixed and follow the stress patterns of the singly suffixed form.

(A.86) Five syllables:

[2,0,1,0,0,0]	àgricúlturalist	còngregácionista	còstitúcionista
	ìnternácionista	èvangélicism	
[2,0,0,1,0,0]	bàcteriólogist	mìcrobiólogist	phýsiothérapist
	pàleontólogist	àutoeróticism	
[0,2,0,1,0,0]	bactèriólogist (alt.)	collàborácionista	envìronmèntalism
	elèctromágnètism		

Six syllables: authòritàrianism totàlitàrianism humànitàrianism

Seven syllables: làtitùdinàrianism

A.11 /-able/

This suffix appears to be similar in realization to the /-ic/ type suffix /-ible/, but it maintains a different stress pattern which generally corresponds to words

with the orthographic representation ‘-able’. The suffix surfaces as /-əbəl/, but the underlying form /-abil/ is suggested by extended forms in /-ity/, e.g., *chàngeability* /čànĵ-əbíl-iti/. Forms showing a monosyllabic stem are universally stressed with the pattern [1,0,0] and shows all types of stems:

(A.87)

CVVC-	5%	chǎngeable	cóuntable		
CVV-	29%	breákeable	équable (alt.)	quótable	cápable
CVC-	31%	sólvable	thínkable	pálpable	
CV-	25%	lívable	flámmable	stóppable	próbable
		sýllable			

As can be seen, there is practically no distinction regarding the distribution of the different stem types, apart from those with extra-heavy stems, which are generally rare in English. Non-Latinate stems combine freely with /-able/, which is still derivationally productive.

Words showing two syllables preceding the suffix show two stress patterns with the following distribution:

[1,0-0,0] 31%

[0,1-0,0] 69%

Of the first set, most show a light antpenultimate syllable:

(A.88)

CVC-CVC-	2%	cóomfortable			
CV-CVC-	3%	démonstrable (Br.)		lámementable (Br.)	
CVV-CV-	10%	fávorable	nótticeable	réasonable	víolable
		súperable			
CVC-CV-	31%	ádmirable	áctionable	ánswerable	éxecrable
		márketable	sérviceable	fórmidable (alt.)	
		tránsferable (alt.)			
CV-CV-	53%	ápplicable	séparable	pálatable	váluable
		mánageable	créditable	énviable	cháritable
		préférable	végetable		

Quite often, words with final stress fail to maintain that stress in the suffixed form, e.g., *préférable*, *tránsferable*, *ádmirable*, the latter also showing no trace of the long vowel.

Of the second pattern with antepenultimate stress, a majority of these forms show heavy antepenultimates. Many of these stems are prefixed monosyllables, but a few are bisyllabic but stressed on a light antepenult:

(A.89)

s-CVVC-	4%	exch ^h angeable	surmóutable	accóutable
s-CVV-	40%	implá cable (Br.)	incá pable	amén able
		insá tiable	maintá inable	advís able
		unspé akable		
s-CVC-	38%	impré gnable	deléct able	accépt able
		transpó rtable	dispéns able	
s-CV-	18%	formí dable (alt.)	implá cable (alt.)	impró bable
		transfé rable (alt.)	hospí table	

Note the alternate pronunciation of *formí dable*, as well as *hospí table*, which are stressed on light syllables. Prefixed verbs in this set, unlike the previous, are seen to maintain their stress patterns.

The stress distribution noted above is paralleled by the longer forms. Again, words with light antepenults usually show the final stress pattern [2,0,1,0] while words with heavy penults show [0,1,0,0]. Words multiply suffixed with /-ify/, /-ize/ and similar long suffixes show the inverted pattern [1,0,2,0]:

(A.90) Trisyllabic forms:

[0,1,0-0,0]	52%			
s-CVV-CV	15%	indú bitable	decí pherable	
s-CVC-CV-	35%	invúl nerable	extí nguishable	uncón scionable
		incó mparable		
s-CV-CV-	50%	iné vitable	uné nvable	abó minable
		irré futable (alt.)		
		inví olable	consí derable	irré vocable (alt.)
[2,0,1-0,0]	34%			
è-s-CVVC-	8%	ìnterch ^h angeable	ùnaccó utable	
è-s-CVV-	62%	ìncónsól able	cèrtifí cable	dìsagrée able
è-s-CVC-	17%	ìnelúct able	ìncontést able	
è-s-CV-	12%	ìnapplicá ble	ìnhospí table	ùnforgétt able
[1,0,2-0,0]	14%			
é-s-CVV-	73%	nó tifiable		
é-s-CVC-	12%	quántifí cable	únbeliè vable (alt.)	

ś-s-CV-	15%	clássifiable	récognizable (alt.)	réconcilable
		mónosyllable		

While many monosyllabic stems retain their stress, others, as in alternate pronunciations of *irréfutable*, *irrévocable*, lose theirs.

(A.91) Four syllables:

[2,0,1,0-0,0]	indefatigable	indecipherable	inextinguishable
	uninhabitable	irrecoverable	
[0,1,0,2-0,0]	irreconcilable (alt.)		
[2,2,0,1-0,0]	irreconcilable (alt.)		
[2,0,0,1-0,0]	biodegradable		

Bibliography

- Anderson, S. (1985). *Phonology in the Twentieth Century*,
Chicago: U. of Chicago Press.
- Aronoff, M. (1976). *Word Formation in Generative Grammar*.
Cambridge, MA: MIT Press.
- Baayen, R. H., R. Piepenbrock & H. van Rijn (1993). *The CELEX Lexical Database (CD-ROM)*. Philadelphia, PA: Linguistic Data Consortium, University of Pennsylvania.
- Bauer, L. (1983). *English word-formation*.
Cambridge: Cambridge University Press.
- Beard, R. (1995). *Lexeme-Morpheme Base Morphology*.
Albany, NY: SUNY Press.
- Benua, L. (1995). Identity effects in morphological truncation.
in Beckman, Dickey & Urbanczyk, eds., *Papers in Optimality Theory*
(pp. 77-136). Amherst, MA: GLSA.
- Bird, S. (1990). *Constraint-based Phonology*.
Doctoral dissertation, University of Edinburgh.
- Bird, S. & T. M. Ellison (1992). One level phonology.
Research paper EUCCS/RP-51, University of Edinburgh.
- Booij, G. (1981). Rule ordering, rule application and the organization of
grammars. *Phonologica* 1980, 45-56.
- Booij, G. & J. Rubach (1984). Morphological and prosodic domains in Lexical
Phonology. *Phonology Yearbook* 1, 1-27.
- Borowsky, T. (1986). *Topics in the Lexical Phonology of English*,
Doctoral dissertation, University of Massachusetts, Amherst.
- Borowsky, T. (1989). Structure preservation and the syllable coda in English.
NLLT 7: 145-166.
- Brame, M. (1974). The cycle in phonology: Stress in Palestinian, Maltese &
Spanish. *LI* 5.1: 39-60.

- Browman, Catherine & Louis Goldstein (1988). Some notes on syllable structure in articulatory phonology. *Phonetica* 45: 140-155.
- Burzio, L. (1993). English stress, vowel length, and modularity. *Journal of Linguistics* 29: 359-418.
- Bybee, J. (1985). *Morphology*, Philadelphia: John Benjamins.
- Bybee, J. (1988). Morphology as lexical organization. in Hammond & Noonan, eds., *Theoretical Morphology*. New York: Academic Press.
- Bybee, J. (1995). Regular morphology and the lexicon. *Language and Cognitive Processes* 10.
- Chomsky, N. (1967). Some general properties of phonological rules. *Language* 43.1: 103-128.
- Chomsky, N. & M. Halle (1968). *The Sound Pattern of English*. New York: Harper & Row.
- Clements, G.N. (1985). The problem of transfer in nonlinear morphology. *Cornell Working Papers*, vol. 7.
- Clements, G.N. (1985). The geometry of phonological features. *Phonology Yearbook* 2: 225-252.
- Clements, G.N. (1990). Vowel height as a hierarchical feature. Handout, Cornell Linguistics Circle, Oct. 2.
- Clements, G.N. (1988). The role of the sonority cycle in core syllabification. in Kingston & Beckman, eds., *Papers in Laboratory Phonology 1*, Cambridge, MA: Cornell University Press.
- Clements, G.N. & S. R. Hertz (1996). An integrated model of phonetic representation in grammar. Ms., Cornell University and University of Paris III.
- Clements, G.N. & E. Hume (1995). The internal organization of speech sounds. in Goldsmith, ed., *The Handbook of Phonological Theory* (pp. 70-114). Cambridge, MA: Blackwell.
- Clements, G.N. & S.J. Keyser (1983). *CV Phonology: a Generative Theory of the Syllable*. Cambridge, MA: MIT Press.

- Cohn, A. (1989). Stress in Indonesian and bracketing paradoxes.
NLLT 7: 167-216.
- Cohn, A. & J. McCarthy (1994). Alignment and parallelism in Indonesian prosody. Ms., Cornell University, Ithaca, NY, and University of Massachusetts, Amherst.
- Cole, J. (1995). The cycle in phonology.
in Goldsmith, ed., *The Handbook of Phonological Theory* (pp. 70-114).
Cambridge, MA: Blackwell.
- Comlex (1995), *The COMLEX English pronouncing lexicon*.
Linguistic Data Consortium, University of Pennsylvania.
- Crowhurst, M. & M. Hewitt (1995). Directional footing, degeneracy, and alignment.
Rutgers Optimality Archive 65.
- Ellison, T. M. (1995). Phonological derivation in Optimality Theory.
Rutgers Optimality Archive 75.
- Freidin, R. (1978). Cyclicity and the theory of grammar.
LI 9.4: 519-549.
- Fudge, E. (1969). Syllables.
Journal of Linguistics 5: 253-286.
- Fudge, E. (1987). Branching structure within the syllable.
Journal of Linguistics 23: 359-377.
- Fudge, E. (1984). *English Word Stress*.
London: Allen & Unwin.
- Golston, C. (1995). Against syllabification.
Rutgers Optimality Archive 95.
- Golston, C. (1996), Direct OT: Representation as pure markedness.
Rutgers Optimality Archive 71.
- Green, T. & M. Kenstowicz (1995). The lapse constraint.
Rutgers Optimality Archive 101.
- Hale, K. (1973). Deep-surface canonical disparities in relation to analysis and change: an Australian example. In T. Sebeok (ed.), *Current Trends in Linguistics*, vol. 9: *Diachronic, Areal and Typological Linguistics* (pp. 401-458). The Hague: Mouton.

- Hale, M. & C. Reiss (1997). Grammar Optimization: The simultaneous acquisition of constraint ranking and a lexicon. Rutgers Optimality Archive 231.
- Halle, M. (1977). Tenseness, vowel shift and the phonology of back vowels in Modern English. *LI* 8: 611-625.
- Halle, M. & M. Kenstowicz (1991). The free element condition and cyclic versus noncyclic stress. *LI* 22.3: 457-501.
- Halle, M. & K. P. Mohanan (1985). Segmental phonology of Modern English, *LI* 16: 57-116.
- Halle, M. & J.-R. Vergnaud (1978). Metrical structures in phonology. Ms., MIT.
- Halle, M. & J.-R. Vergnaud (1987). *An Essay on Stress*. Cambridge, MA: MIT Press.
- Hammond, M. (1984). *Constraining metrical theory: A modular theory of rhythm and destressing*. Doctoral dissertation, UCLA.
- Hammond, M. (1995). There is no lexicon! Rutgers Optimality Archive 43.
- Harris, J. (1985). Spanish diphthongization and stress: a paradox resolved. *Phonology Yearbook* 2: 31-45.
- Hayes, B. (1981). *A Metrical Theory of Stress Rules*. Bloomington, IN: The Indiana University Linguistics Club.
- Hayes, B. (1982). Extrametricality & English stress. *LI* 13.2: 227-276.
- Hayes, B. (1985). Iambic and trochaic rhythm in stress rules. *Proceedings of the Berkeley Linguistic Society* 11.
- Hayes, B. (1986). Inalterability in CV phonology. *Language* 62.2: 321-352.
- Hayes, B. (1987). A revised parametric metrical theory", *NELS* 17: 274-289.
- Hayes, B. (1989). Compensatory lengthening in moraic phonology. *LI* 20. 2: 253-306.

- Hayes, B. (1995). *Metrical Stress Theory*.
Chicago: University of Chicago Press.
- Hung, H. (1994). *The Rhythmic and Prosodic Organization of Edge Constituents*.
Doctoral dissertation, Brandeis University.
- Inkelas, S. (1989). *Prosodic Constituency in the Lexicon*,
Doctoral dissertation, Stanford University.
- Inkelas, S. (1994). The consequences of optimization for underspecification.
Rutgers Optimality Archive 40.
- Inkelas, S., O. Orgun & C. Zoll (1994). Exceptions and static phonological
patterns: cophonologies vs. prespecification.
Rutgers Optimality Archive 124.
- Itô, J. (1989). A Prosodic Theory of Epenthesis.
NLLT 7: 217-259.
- Itô, J. & A. Mester (1994). Reflections on CodaCond and Alignment.
Phonology at Santa Cruz 3: 27-46.
- Itô, J., A. Mester & J. Padgett (1994). NC: Licensing and underspecification in
Optimality Theory. Rutgers Optimality Archive 38.
- Jensen, J. T. (1990). *Morphology*,
Philadelphia: John Benjamins.
- Kager, R. (1989). *A metrical theory of stress and destressing in English and Dutch*,
Dordrecht: Foris.
- Kahn, D. (1976). *Syllable-based generalizations in English Phonology*.
Doctoral dissertation, MIT. Published 1980, by Garland Publishers, N.Y.
- Kaisse, E. & P. Shaw (1985). On the theory of Lexical Phonology.
Phonology Yearbook 2: 1-30.
- Kean, M. (1974). The strict cycle in phonology.
LI 5.2: 179-203.
- Kiparsky, P. (1973). 'Elsewhere' in Phonology.
in Anderson & Kiparsky, eds., *A Festschrift for Morris Halle*,
New York: Holt Rinehart and Winston.
- Kiparsky, P. (1979). Metrical structure assignment is cyclic.
LI 10.3: 421-444.

- Kiparsky, P. (1981). Remarks on the metrical structure of the syllable.
in W. Dressler, ed., *Phonologica*, vol. 3. Innsbruck: Innsbrucker
Beiträge zur Sprachwissenschaft.
- Kiparsky, P. (1982a). Lexical Morphology and Phonology.
in I.-S. Yang, ed., *Linguistics in the Morning Calm*,
Seoul, Hanshin Publishing.
- Kiparsky, P. (1982b). From Cyclic Phonology to Lexical Phonology.
in van der Hulst & Smith, eds., *The Structure of Phonological
Representations.*, vol. 1 (pp. 131-175). Dordrecht: Foris.
- Kiparsky, P. (1982c), "Word-Formation and the Lexicon",
Mid-American Linguistics Conference, 3-29.
- Kiparsky, P. (1982d), *Explanation in Phonology*.
Dordrecht: Foris.
- Kiparsky, P. (1985). Some Consequences of Lexical Phonology.
Phonology Yearbook 2: 85-138.
- Kiparsky, P. (1994). Remarks on markedness. Trilateral Phonology Weekend
II. University of California, Santa Cruz.
- Lahiri A. & J. Koreman (1988). Syllable weight and quantity in Dutch.
Proceedings of the 7th WCCFL: 217-228.
- Lahiri, A. (1991). Abstractness in Lexical Representations.
Handout, University of Edinburgh talk.
- Leben, W. (1973). *Suprasegmental Phonology*.
Doctoral dissertation, MIT.
- Levin, J. (1985). *A Metrical Theory of Syllabicity*.
Doctoral dissertation, MIT.
- Liberman, M. & A. Prince (1977) On stress and linguistic rhythm".
LI 8.2: 249-336.
- Maddieson, I. (1985). Phonetic cues to syllabification.
in V. Fromkin, ed., *Linguistic Phonetics*. Orlando: Academic.
- Marchand, H. (1969), *English Word Formation*.
Munich: C.H. Beck'sche Verlagsbuchhandlung.

- Mascaró, J. (1976). *Catalan phonology and the phonological cycle*.
 Doctoral dissertation, MIT. Dist. by the Indiana U. Linguistics Club,
 Bloomington, Ind.
- Matthews, P.H. (1974). *Morphology*.
 Cambridge: Cambridge University Press.
- McCarthy, J. (1981). A prosodic theory of nonconcatenative morphology.
LI 12.3: 373-418.
- McCarthy, J. (1983). Prosodic organization in morphology.
UMOP 8: 61-75.
- McCarthy, J. & A. Prince (1986). Prosodic Morphology.
 Ms., University of Massachusetts, Amherst, and Brandeis University.
- McCarthy, J. & A. Prince (1988). Quantitative transfer in reduplicative and
 templatic morphology. In Linguistic Society of Korea, ed., *Linguistics
 in the Morning Calm*, vol. 2 (pp.3-35). Seoul: Hanshin Publishing.
- McCarthy, J. & A. Prince (1990a). Foot and word in prosodic morphology: the
 Arabic broken plural. *NLLT* 8: 209-283.
- McCarthy, J. & A. Prince (1990b). Prosodic Morphology and Templatic
 Morphology. In Eid & McCarthy, eds., *Perspectives on Arab Linguistics
 II*, Philadelphia: John Benjamins.
- McCarthy, J. & A. Prince (1992). Constraint interaction in Prosodic
 Morphology and Phonology. Handout, Cornell University talk.
- McCarthy, J. & A. Prince (1993a). Generalized Alignment. Rutgers Optimality
 Archive 7.
- McCarthy, J. & A. Prince (1993b). Prosodic Morphology I: Constraint
 interaction and satisfaction. Ms., University of Massachusetts,
 Amherst, and Rutgers University.
- McCarthy, J. & A. Prince (1994). The emergence of the unmarked: Optimality
 in Prosodic Morphology. Rutgers Optimality Archive 13.
- McCarthy, J. & A. Prince (1995). Faithfulness and reduplicative identity.
 in Beckman, Dickey & Urbanczyk, eds., *Papers in Optimality Theory*
 (pp. 249-384). Amherst, MA: GLSA.
- Mester, R. A. (1994). The quantitative trochee in Latin.
NLLT 12: 1-61.

- Milliken, S. (1988). *Protosyllables: A Theory of Underlying Syllable Structure in Nonlinear Phonology*. Doctoral dissertation, Cornell University.
- Mohanan, K.P. (1982). *Lexical Phonology*. Doctoral dissertation, MIT.
- Mohanan, K.P. (1985). Syllable structure and lexical strata in English. *Phonology Yearbook* 2: 139-155.
- Mössner, L. (1978). *Morphonologie*. Tübingen: Niemeyer.
- Myers, S. (1987). Vowel shortening in English. *NLLT* 5, 485-518.
- Myers, S. (1991). Persistent rules. *LI* 22.2: 315-344.
- Nanni, D. (1977). Stressing words in -ative. *LI* 8: 752-763.
- Nespor, M. & I. Vogel (1982). Prosodic domains of external sandhi rules. in van der Hulst & Smith, *The Structure of Phonological Representations*, vol. 1. Dordrecht: Foris.
- Nespor, M. & I. Vogel (1983). Prosodic structure above the word. in Ladd & Cutler, eds., *Prosody: Models and Measurement*. Berlin: Springer-Verlag.
- Nespor, M. & I. Vogel (1986). *Prosodic Phonology*. Dordrecht: Foris.
- Oehrle, T. (1972). The stress pattern of bisyllabic words in English. Paper presented at NELS 3, Amherst, MA.
- Padgett, J. (1995). Feature Classes. in Beckman, Dickey & Urbanczyk, eds., *Papers in Optimality Theory* (pp. 385-420). Amherst, MA: GLSA.
- Paradis, C. (1988). On constraints and repair strategies. *The Linguistic Review* 6: 71-97.
- Pater, J. (1994). Optionality and Optimality: Syllable weight effects in English secondary stress. Ms., McGill University.

- Pater, J. (1995). On the nonuniformity of weight-to-stress and stress preservation effects in English. Rutgers Optimality Archive 107.
- Pinker, S. (1991). Rules of language. *Science* 253: 530-535.
- Pinker, S. (1994). *The language instinct: the new science of language and mind*. London: Penguin.
- Pinker, S. & A. Prince (1991). Regular and irregular morphology and the psychological status of rules of grammar. *Proceedings of the 17th Annual Meeting of the Berkeley Linguistics Society* (pp. 230-251). Berkeley, CA: Berkeley Linguistics Society.
- Poldauf, I. (1984). *English Word Stress*. New York: Pergamon Press.
- Poser, W. (1982). Phonological representation and action-at-a-distance. in van der Hulst & Smith, *The Structure of Phonological Representations*, vol. 2. Dordrecht, Foris.
- Prince, A. (1983). Relating to the grid. *LI* 14: 19-100.
- Prince, A. (1990). Quantitative consequences of rhythmic organization. *Chicago Linguistic Society* 26.
- Prince, A. & P. Smolensky (1993). Optimality Theory: Constraint interaction in generative grammar. Ms., Rutgers University & University of Colorado, Boulder.
- Rialland, A. (1991). L'allongement compensatoire : nature et modèles. in Laks & Rialland, eds., *L'Architecture et la Géométrie des Représentations Phonologiques*. Paris: Editions du C.N.R.S.
- Rosenthal, S. (1994). *Vowel/Glide Alternation in a Theory of Constraint Interaction*. Doctoral dissertation, University of Massachusetts, Amherst.
- Rubach, J. (1984). Segmental rules of English and cyclic phonology. *Language* 60.1: 21-54.
- Russell, K. (1995). Morphemes and candidates in Optimality Theory. Rutgers Optimality Archive 44.

- Sainz, S. (1988). A noncyclic analysis of English word stress.
Working Papers of the Cornell Phonetics Laboratory 3: 1-82.
- Sainz, S. (1992). *A Noncyclic Approach to the Lexical Phonology of English*,
Doctoral dissertation, Cornell University.
- Scobbie, J. (1991). *Attribute Value Phonology*.
Doctoral dissertation, University of Edinburgh.
- Selkirk, E. (1980). The role of prosodic categories in English word stress.
LI 11.3: 563-605.
- Selkirk, E. (1982a). *The Syntax of Words*.
Cambridge, MA: MIT Press.
- Selkirk, E. (1982b). The syllable.
in van der Hulst & Smith (eds.), *The Structure of Phonological Representations*, vol.2. Dordrecht: Foris.
- Selkirk, E. (1984), *Phonology & Syntax*.
Cambridge, MA: MIT Press.
- Sherer, T. (1994). *Prosodic Phonotactics*.
Doctoral dissertation, University of Massachusetts, Amherst.
- Siegel, D. (1974). *Topics in English Morphology*.
Doctoral dissertation, MIT.
- Stampe, D. (1972). *How I Spent My Summer Vacation*.
Doctoral dissertation, University of Chicago.
- Stanley, R. (1967). Redundancy rules in phonology.
Language 43.2: 393-436.
- Steriade, D. (1987). Locality conditions and feature geometry.
NELS 17: 595-617.
- Vennemann, T. (1974). Words and syllables in natural generative phonology.
In *Natural Phonology Parasession*, pp. 346-374. Chicago: Chicago Linguistic Society.
- Waugh, L. (1992). Constraints on iconicity in the lexicon.
American Journal of Semiotics, 9.1: 7-48.

- Woods, A., P. Fletcher & A. Hughes (1986). *Statistics in language studies*.
Cambridge: Cambridge University Press.
- Yip, M. (1987). English vowel epenthesis.
NLLT 5: 463-484.
- Zec, D. (1988). *Sonority constraints on prosodic structure*.
Doctoral dissertation, Stanford University.
- Zec, D. (1990). Closed syllables in Lithuanian.
FLSM: 317-330.
- Zec, D. (1994). Coda constraints and conditions on syllable weight.
Ms., Cornell University.
- Zec, D. & S. Inkelas (1990). Prosodically Constrained Syntax.
in Zec & Inkelas, eds., *The Phonology-Syntax Connection*.
Chicago: University of Chicago Press.
- Zoll, C. (1994a). Subsegmental parsing: Floating features in Chaha and
Yawelmani. Rutgers Optimality Archive 29.
- Zoll, C. (1994b). Anchors away: a unified treatment of latent segments and
floating features. Ms., University of California, Berkeley.
- Zoll, C. (1995). Conflicting directionality. Ms., University of California,
Berkeley and University of Iowa.
- Zoll, C. (1996). *Parsing Below the Segment in a Constraint-Based Framework*.
Doctoral dissertation, University of California at Berkeley.